

Monolithic Refractories

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Monolithic Refractories

A Comprehensive Handbook

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*Answer Technology Inc.,
Illinois, USA*

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To Bakul, my beloved companion, counselor and advisor who devoted an enormous amount of her time to the book. Without her help, it would have been impossible to get the book to publication. I am ever indebted to her for her unselfish sacrifice.

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PREFACE

Traditionally, refractories are considered a unique industrial product used primarily in basic industries such as iron, steel, glass, cement and pottery. Being complex multi-component systems in most cases, they are difficult to predict than other materials using simple scientific methods and principles. It is quite an unknown material to the general public and also to other industries.

During the last two decades, a significant amount of attention has been paid to refractory ceramics as results could be predicted fairly accurately depending on the product and application. Although not much progress has been made in brick making, except in MgO–C bricks, significant progress has been made in monolithic refractories in materials, products and applications in the last two decades. The unique advantage of monolithic refractories is its continuous body. Without any joints, it provides better structural integrity and can be installed with relative ease requiring labors in less numbers and skill.

At present, there is a dearth of literature on monolithic refractories, except for a few company-sponsored books which are biased towards their own products. Thus, this book provides a basic understanding of the principles of monolithic refractory formulation, manufacture and application. The topics covered include the selection

of raw materials, product formulation for specific applications, and the principles behind the effective use of the products during application.

The book is arranged in chronological order, beginning with raw materials and proceeding to the relative order of use. The book is organized according to the methods of application of refractories, namely, castables, pumpables, plastics, ramming mixes, gunning mixes, mortars, coatings and dry vibratables. The chapter on castables deals with binders used in recent years for castable formulations which include calcium aluminate cements, hydrated alumina, and colloids, with their specific characteristics. The advantages and disadvantages of the different binders and their methods of application are also discussed in detail. The sections on ramming and gunning mixes deal with both conventional and the recently developed binder systems and their special behavior. Mortars, coatings and dry vibratables are dealt with using specific applications. Their advantages and the appropriate selection of the particular products are examined. Apart from the product area, the book also deals with the manufacture of monolithic mixes and pre-cast shapes, the evaluation and testing of refractory properties, system design, and application since the performance of monolithic refractories depends vastly on the proper installation and heating procedure before use. All these considerations are dealt with in this book to ensure that the refractory performs as desired.

Since refractories come into contact with different types of slag, a chapter is devoted to the discussion of slag reactions, especially those monolithic refractories in the blast furnace cast-house troughs, steel ladles and tundishes that come into contact with slags. The book was compiled using various sources, such as, books, periodicals, personnel communications and most recently, applied work in the field.

It is impossible to thank every individual who helped in the writing of this book. Nevertheless, some of them deserved special mention for their invaluable help rendered in its preparation. Dr.

Bakul Banerjee counseled me and provided proper direction, editorial suggestions, and assistance in producing the manuscript in its final format. I thank Mr. K. N. Singh of Bethlehem Steel Corporation, Dr. George MacZura and Mr. William Vance of Alcoa Industrial Chemicals, Mr. Leonard Krietz of Plibrico, Mr. Anthony Kojundic and Dr. Bjorn Myhre of Elkem Chemicals, and Dr. David Lankard of Lankard Laboratories for their technical suggestions. I also express my appreciation for the permission given by Mr. Akira Nishikawa of Plibrico, Japan Company Limited to reproduce a major portion of the topics on phosphate and silicate raw materials and refractory corrosion/erosion. My hearty thanks and appreciation go to Dr. Amit Dasgupta who thoroughly edited the manuscript within a short period of time. Lastly, I thank my daughters, Paroma and Reshoma who put up with me during the tenuous preparation of the manuscript.

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INTRODUCTION

Monolithic refractories differ from refractory bricks in that they are not shaped and fired before use, although the chemical and physical properties exhibit similar and sometimes better characteristics. The major advantage of monolithic refractories is that they hardly need any high-energy requirements for shaping and prefiring, thus saving on considerable amount of manpower and energy required. The specific advantages of monolithic refractories over firebricks are:

- Since no shaping or firing is involved, the manufacturing and delivery times are shorter, and less inventory is required by the user.
- Since no firing is involved, a significant amount of energy is saved and expensive pollution-control devices for furnace exhaust gases are not needed.
- The building of efficient furnaces often involve larger, complicated shapes to achieve the desired performance which needs costly skilled labor and takes longer time to complete. Monolithic refractories, on the other hand, involve simplified construction for complicated structures. They are easier to install and takes a shorter time to complete. Monolithic refractories can be applied using various methods depending on the type of application.

2 Monolithic Refractories

- Monolithic refractory linings are usually thinner than the corresponding brick structures. They are more durable since building with bricks involves the use of mortars to join the bricks together, thus creating weak points in the structure. Monolithic refractories can be built around supports and may not need to be self-supporting in applications such as thick-brick arch linings.
- Firebrick linings undergo larger thermal expansion compared to monolithic linings and thus can cause structural deformation or spalling. Monolithic linings are supported by fixed structures that operate as restraints against movement due to thermal changes. Thus, monolithic refractories provide longer durable linings.
- Monolithic refractories are available in different forms according to their method of application techniques and properties. Depending on the specific application, they can be formulated with different raw materials and grain sizes. Different methods have been developed for specific installation procedures so that the refractories can be gunned, poured, troweled, rammed, patched or injected.
- Monolithic refractories can be repaired locally without disturbing the whole structure, thus saving material, time and labor.
- Monolithic refractories are an ideal choice for installing gas-tight walls and, hence, are more suitable to build pressure vessels operating at high pressures and temperatures.
- Monolithic refractories are categorized according to their mode of physical condition and method of application.

The general categories of monolithic refractories are defined on the basis of their methods of use in a variety of applications:

Castable refractory. These refractories are supplied as dry mixed materials that need to be combined with water or other specified liquid and are mixed together at the application site. The material is applied by pouring where it finds its own level or by pouring

followed by vibration for making the material to flow and find its own level. Conventional castables are usually poured whereas low and ultralow cement castables are poured followed by vibration.

Pumpable refractory. This kind of refractory consists of a dry material that is mixed with water or other recommended liquid on site and is installed by a pump. The application is quicker since the mixed material is transported directly to the placement site by the pump instead of transferring through a hopper. Pumpable refractories are self-flowable and does not need to be vibrated for flow; and hence require less labor and time.

Injectable refractory. These are also a special kind of pumpable refractories but injected directly (commonly known as grouting) inside the furnace specifically for repair purposes, e.g., blast furnace walls in the bosh, lower and upper stacks.

Plastic refractory. As the name implies, these are made in a pliable form that can be shaped in any desired way. This form of refractory is ready to use and does not need to add any other ingredient on site. Depending on the use, these are either installed by hitting with a mallet or by pneumatic rammers to conform to the area and shape where they are installed. Depending on the nature of the bonding system, careful dry out methods need to be followed to avoid unwanted deformities by rapid heating. Plastic refractories may also be gunned but the kneaded material needs to be either shredded or supplied in a granular form for installation by special gunning equipment.

Ramming refractory. Such refractories are made in granular or pelletized form with all ingredients added prior to installation. They are installed by using ramming equipment, such as, shovel tamping or pneumatic ramming. The refractory material is dumped into the location of application in certain amounts so that the rammed thickness is not more than 100 to 125mm in layers until the space is filled up forming a solid body.

Gunning refractory. These refractories come in dry granular or pelletized forms to facilitate flow of the material through hoppers. They are installed by the use of a gunning equipment where the solid material is transported pneumatically through hoses and then the liquid, usually water, is added at the application nozzle. The success of the application depends on the skill of the installer since the water must be adjusted and added at the nozzle depending on the flow of the dry material.

Dry vibratable. These refractories are supplied as dry premixed materials and are installed dry by vibrating with a flat-plate vibrator. The compacted material hardens by heating at low temperatures near the heated surface whereas the inside still remains unbonded providing insulating effects to the refractory body.

Refractory mortars. Mortars can come either in dry form and are mixed with water on site to appropriate trowelling consistency, or can be provided premixed in appropriate consistency that are applied by trowelling. They are usually used to put bricks or other refractory shapes together.

Coatings. Refractory coatings usually come as suspended slurries that are applied as a thin layer by a spray gun and are heated to form a solid coating.

During the past two decades, the monolithic-refractory industry has seen a series of developments in material compositions with aggregate materials and bonding systems. Also new methods of applications helped them substitute brick applications in a variety of uses. The following developments stand out for having increased the market share of monolithic refractories in various applications:

- Although plastic refractories have been used for several decades, the plastics gunning opened up new application areas. In remote places often direct application of plastics is not

possible. Also, in bitterly cold areas sometimes conventional gunning is not feasible because the availability of regular flow of water is difficult. Gunning plastic refractory solved these problems providing good properties and convenience. It also provides quick installation inside furnaces and rotary kilns with increased ease and convenience.

- The development of low- and ultralow- cement castables (placed by vibration) opened up a variety of previously unused applications of monolithic refractories. Their properties are as good and, in most cases, exceed the properties of bricks saving in time, labor and money for the user. Low and ultralow cement castables have been applied in blast furnace cast-house troughs, slag and iron runners. They are applied in variety of furnaces, steel and transfer ladles, rotary kilns and tundishes. They are also used in various kinds of small and large precast shapes like tundish pocket-blocks, dams, weirs, baffles, impact pads, ladle-impact pads and well blocks, various kinds of burner blocks and other similar applications.
- The use of cement-free bonding systems by replacing calcium aluminate cement with sol-gel bonding constituted another step in the progress of monolithic refractories. The sol-gel bonding system produces a self-flow material, and provides significant advantages over low and ultralow cement castables with superior physical and chemical properties. Also, they can be installed by pumping which shortens the installation time. The compositions do not contain any chemically bonded water, and hence the drying and firing times are considerably decreased.
- Since the development of pumpable no-cement compositions, low and ultralow cement compositions have been modified by careful grain-size adjustment, so that they can also be installed by pumping. Such installation process by pumping eliminated some of the problems associated with low and ultra-low

cement compositions and thus increased their field of application.

- New type of low-cement castables are formed from the application of refractory materials such as alumina-magnesia spinel either as sintered or fused forms. These castables are completely free from SiO_2 and hence are used to make ultra-clean steel. Also, they are used in high temperature and high erosion areas because of their high strength at high temperatures.
- The latest development in this field is the application of low and ultralow cement and cement-free compositions by spray gunning or "shotcreting". This process combines gunning and low-cement castable refractories thus providing the convenience of gunning application approaching the properties of low-cement castable compositions. Although the current application of this technology is limited to some specific areas, it has strong potential with respect to various other uses in the near future.

CHAPTER I: RAW MATERIALS

Introduction

Monolithic refractories are mechanical mixtures of refractory materials that serve as aggregates, binders, mineralizers or other properties that are desired in the product. The combination of ingredients produces physical and chemical reactions, both when installed in-place of application and when used in making shapes as preformed bodies. The performance of each monolithic refractory depends on the raw materials that are used in the formulation of the monolith. As a result, the raw materials play a significant role in the finished product and clearly deserve special attention.

Alumina

The majority of monolithic refractories are dominated by high alumina or alumina-silicate materials. High-alumina refractory aggregates that are used in the formulation of monolithic refractories are composed of tabular alumina, white and brown fused alumina, calcined bauxite, calcined or sintered mullite and mullite-dominated alumina-silicates containing more than 50% alumina. High-purity alumina materials are also used as calcined or highly reactive forms in the matrix for bonding phases. The characteristics and appropriate use of these need special attention in the formulation of monolithic refractories.

Tabular alumina

Tabular alumina is made from aluminum hydroxide derived from bauxite by the Bayer process. The aluminum hydroxide from the Bayer process is pelletized into balls about 2.5 cm in diameter. After going through the drying process, these balls are fed into the top of a shaft kiln and are gradually heated as they go down the furnace. The aluminum hydroxide is heated and shrunk at high temperature (around 2035°C, just below the fusion point of alumina) with or without the addition of additives (i.e., MgO and B₂O₃). Hexagon- tablet-shaped alumina crystals are formed and which then grow to large sizes (40 to 200 μm). High-temperature sintering helps the large crystals to plug up the pores, and densifies close to the theoretical density.

The soda content is an important ingredient in the tabular alumina formation. The excellent thermal-volume stability and thermal shock characteristics of tabular alumina are attributed to the large crystals with closed spherical pores entrapped from recrystallization of the submicron alpha-alumina crystals during rapid sintering. Since tabular alumina is normally used in refractories for extremely high temperatures, the amount of soda present plays an important role in the manufacturing process.

Calcined and reactive alumina

Both calcined and reactive alumina play an important role in monolithic refractories. Several varieties of calcined alumina are available in a broad range of particle sizes. The selection of a particular kind of calcined alumina depends on the rheological effect of the desired form of the particles, particularly in castable refractories. The size of the refractory particles plays an important role in the flowability, setting and ultimate chemical reaction in the refractory body.

Calcined aluminas serve multiple purposes in different types of monolithic refractories. It reacts with the fine silica portion of the matrix because of its fine particle size. Together, they form mullitic

phases that offer both high refractoriness as well as stronger bonds. Depending on the fineness of the alumina and the silica particles (e.g., silica fume and super-ground alumina), the mullitic reaction can take place far below the theoretical temperature of mullite formation as indicated in the phase diagrams. The mullite formation can start as low as 1200°C as evidenced by x-ray diffraction.

During the formation of the mullite, a transient vitreous phase is formed which shows a slight shrinkage in the matrix. The matrix soon starts to expand with increasing temperature. The beginning of the vitreous phase may vary depending on the reactivity of both the alumina and the silica particles and, hence, the temperature may vary. It has been well demonstrated that precast shapes made from low cement castables either do not deform or deform much less under load when high fired prior to use to a temperature of 1370°C^{1,2} which proves that the deformation is either eliminated or significantly reduced when the vitreous state of reaction takes place by prior high firing.

Calcined alumina offers plasticity in plastics, better rheological properties in mortars and plays a significant role in the flowability either under vibration or by itself (self-flow). Calcined alumina in a variety of sizes plays a prominent role in the self-flow characteristics of castables. With particle size ranging from 0.1 to 20 µm, it plays an important role in filling the voids in castable structures, thus reducing the empty spaces remaining to be filled with water.

Calcined alumina is available in various sizes — some having broad distribution and some narrow. It is also available in regular grinds and in super-ground forms. Super-ground calcined alumina have quite small ultimate particle size. However, they also often agglomerate due to high surface energy and do not break down during operation in mixes. The result is larger particles than expected, sometimes causing unwarranted problems. Calcined alumina manufacturers have recently begun adding small quantities of

dispersing agents to the super-ground alumina to produce the desired particle sizes rather than agglomerates.

White fused alumina

White fused alumina is produced by melting calcined alumina at above 2040°C in an electric arc furnace. Although this is the pure form of fused alumina, the products available on the market are more porous and as such, the bulk density is lower than other alumina raw materials. It has high refractoriness, abrasion resistance as well as chemical inertness, but has not been found to be suitable or cost effective for most refractory applications. Table 1 shows the typical chemical analysis of a white fused alumina.

Table 1. Chemical analysis of white fused alumina.

Components	%
Chemical analysis,	
Al ₂ O ₃	99.64
SiO ₂	0.06
TiO ₂	0.00
Fe ₂ O ₃	0.03
CaO	0.03
MgO	0.07
Na ₂ O	0.15
K ₂ O	0.02
Bulk density, g/cc	3.58
Apparent porosity, %	7.8

Brown fused alumina

Brown fused alumina is produced by melting calcined bauxite in the presence of carbon to reduce the iron and silicon content. Brown fused alumina is available in a wide range of purity levels varying in silica, titania and iron contents. Table 2 gives the range of brown fused alumina available on the market.

Table 2. Range of chemical composition of brown fused alumina.

	I	II	III	IV
Al_2O_3	94.3	96.3	97.8	97.2
SiO_2	1.1	0.56	0.49	0.54
TiO_2	2.45	2.66	1.22	1.63
Fe_2O_3	1.5	0.1	0.23	0.29

Brown fused alumina may be used up to a temperature of 1650°C. Above that temperature, the grains begin to bloat possibly due to the presence of aluminum titanate. It is an extremely dense material with a bulk density of 3.94 to 3.97 g/cm³ and is practically nonporous. The porosity is due to the voids left while the melt cools down in the furnace. Brown fused alumina is most widely used for refractory applications in the blast-furnace cast house — in the trough, iron and slag runners and in all forms as castables, pumpables, ramming and gunning mixes. It has the required range of refractoriness, high abrasion resistance and good resistance to blast furnace slag.

Calcined bauxite

The use of calcined bauxite is perhaps most prevalent in refractory compositions since it provides a high percentage of alumina at a reasonable price. Bauxite ore is found as two different alumina hydrate minerals, — gibbsite, a trihydrate ($\text{Al}_2\text{O}_3 \cdot 3\text{H}_2\text{O}$), and diaspore/boehmite, both monohydrate ($\text{Al}_2\text{O}_3 \cdot \text{H}_2\text{O}$). Bauxite may contain different proportions of boehmite and gibbsite. The chemical compositions of different bauxites are shown in Table 3.

The bauxite used for producing calcined alumina or metallic aluminum is rich in iron and is never used for refractories in the Bayer process. Bauxite used for refractory applications is high in alumina content and low in impurities. Low-cement castables made with bauxite as the main ingredient often show shrinkage after firing to temperatures above 1500°C.

Table 3. Chemical composition of raw bauxite minerals.

Composition, %	Boehmite	Gibbsite	Diaspore
Al ₂ O ₃	58.1	60.1	72.3
SiO ₂	2.10	4.1	7.15
Fe ₂ O ₃	25.0	2.82	2.60
TiO ₂	2.67	2.65	3.22
CaO	0.10	0.10	0.20
MgO	0.10	0.20	0.20
L.O.I.	11.8	29.9	14.3

Although all three varieties of the hydrates form alumina by calcining, the resultant calcined alumina products differ in their characteristics because of the impurities associated with the minerals. All the hydrates convert into alpha alumina when heated to the temperature range 1000–1200°C.

There is a significant difference between the bauxites available from South America and China, — the two main suppliers in today's market. The South American bauxites (mainly gibbsite) are lower in titania and alkalis but are lower in density. The Chinese bauxites (mainly diaspore) are much higher in titania and somewhat higher in alkalis. Calcined South American bauxites vary in density from 3.15 to 3.28 g/cc whereas calcined Chinese bauxites may have densities as high as 3.50 g/cc.

Andalucite

Andalucite is a naturally occurring alumina-silicate mineral that is isomorphic with sillimanite and kyanite. Andalucite occurs in nature as large orthorhombic single crystals and are found in gray, greenish, reddish or bluish colors. The main source of andalucite has been South Africa but it is also mined in the former USSR, France and Spain. The highest grade of andalucite is found in the western region of the Transvaal province of South Africa. Along with sillimanite and kyanite, it is an important mullite former used as raw material in

refractories. It dissociates into mullite and silica at 1400°C and is associated with some expansion (about 6–7%). Andalusite has been a good source of alumina-silicate refractories for economic reasons and because of its high refractoriness particularly after conversion to mullite. Due to its extensive use in recent years, the availability is somewhat limited at present. One has to be careful in designing a monolithic refractory using andalusite as the main aggregate, since the expansion may not be acceptable particularly for high temperature applications.

Kyanite

Kyanite has the same chemical composition as andalusite and sillimanite but differs in crystal structure and physical properties. It is the heaviest among the isomorphous minerals and has a sp.gr. of 3.5–3.7. It is found in India, Africa and the Appalachian region in the United States. Kyanite from India, the best quality available in the world, is different in character from American kyanite, with a native occurrence as surface boulders. Virginian kyanite is associated with quartz, from which it must be separated after grinding down to a workable mesh. Virginian kyanite is not available in grain sizes larger than 35 mesh while Indian kyanite can be calcined in lump form which makes an excellent grog. Indian kyanite often shows higher than theoretical amounts of alumina due to its association with corundum in the ore. The use of kyanite is quite extensive in both shaped and monolithic refractories particularly due its high thermal expansion (about 22–23%) during conversion to mullite. The expansion compensates for the shrinkage occurring from clay minerals used in a lot of monoliths. It is mostly used in low alumina and alumina-silicate compositions in all types of monolithic compositions.

Sillimanite

Sillimanite, the third isomorphous mineral, has a similar chemical composition but ranks between andalusite and kyanite in density. It is

converted to mullite and silica at a much higher temperature than the other isomorphism (1535–1545°C) with an associated expansion of 12–13%. Its use in monolithic refractories is much more limited than either andalucite or kyanite. Chemical compositions of the three isomorphous minerals — andalucite, kyanite and sillimanite — are given in Table 4.

Table 4. Chemical compositions of andalucite, kyanite and sillimanite.

Composition, %	Andalucite	Kyanite	Sillimanite
Al ₂ O ₃	54.0 – 61.3	56.1 – 61.3	61.1 – 71.1
SiO ₂	37.4 – 43.8	37.4 – 41.0	22.5 – 31.5
Fe ₂ O ₃	0.48 – 1.70	0.52 – 0.60	0.50 – 1.62
TiO ₂	0.25 – 0.30	0.10 – 1.30	0.90 – 3.00
CaO	0.10 – 0.20	0.10 – 0.20	0.06 – 0.10
MgO	0.11 – 0.20	0.20 – 0.25	0.05 – 0.08
K ₂ O	0.10 – 0.20	0.02 – 0.08	0.05 – 0.10
Na ₂ O	0.08 – 0.10	0.02 – 0.05	0.06 – 0.08
L.O.I.	0.10 – 0.20	0.25 – 0.45	1.32 – 5.62

Clays

Kaolinitic clays

Kaolinitic clays include minerals like kaolinite, meta- and hydrated halloysite, nacrite and dicktite. These minerals have the same chemical formula $\text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2 \cdot 2\text{H}_2\text{O}$ except halloysites that have more water molecules. When heated, kaolin changes into meta kaolin and finally converts into mullite at around 1200°C. The dehydration process of clays are endothermic reactions. From differential thermal analysis, different clays can be identified from the position, size and shape of their endothermic and exothermic peaks. Chemical compositions of kaolinitic clays are given in Table 5.

Table 5. Chemical compositions of kaolinitic clays.

Composition, %	Kaolin	Halloycite
Al ₂ O ₃	35.3 – 38.8	36.2 – 37.8
SiO ₂	42.5 – 48.1	45.3 – 46.8
Fe ₂ O	0.32 – 1.10	0.15 – 1.15
TiO ₂	0.20 – 1.50	0.10 – 0.35
CaO	0.30 – 0.42	0.20 – 0.98
MgO	0.05 – 0.20	0.20 – 0.34
K ₂ O	0.15 – 0.45	0.20 – 1.10
Na ₂ O	0.10 – 0.70	0.02 – 0.08
L.O.I.	13.9 – 14.5	12.4 – 16.8

Bentonite

Montmorillonite is the principal constituent of bentonite. Like pyrophyllite, montmorillonite (Al₂O₃. 4SiO₂. 6H₂O) is a three-layer mineral in which part of the Al³⁺ ions are replaced by Mg²⁺ ions while Ca²⁺ and Na⁺ ions are absorbed between the layers. When dried, the water molecules between the layers are removed. When the mineral absorbs water again, water molecules get between the layers causing them to expand. Calcium bentonites absorb less water than sodium bentonites. Chemical compositions of two bentonites are given in Table 6.

Table 6. Chemical compositions of bentonitic clays.

Composition, %	Na-bentonite	Ca-bentonite
Al ₂ O ₃	15.4	15.0
SiO ₂	70.4	66.2
Fe ₂ O	2.00	2.30
TiO ₂	0.25	0.35
CaO	1.78	2.21
MgO	2.25	4.95
K ₂ O	0.21	0.3
Na ₂ O	1.89	0.45
L.O.I.	4.92	8.00

Bentonite (montmorillonite-containing) clays are the most commonly used among the raw clays. The use of specific clays depends on the specific composition of each monolithic refractory. For plastic compositions, clays containing higher amounts of montmorillonite (bentonite and bentolite) are used because of their plastic behavior whereas, for gunning and ramming mixes, producers now use clays containing lower amounts of montmorillonite. Raw clays play an important role in the formulation of plastics, ramming, gunning and mortars. These clays offer workability, adhesion characteristics and refractoriness at higher temperatures by converting to mullite. The compositions are sometimes adjusted so that the shrinkage associated with the calcining of the clays is compensated by kyanite or similar materials.

Calcined clays

A wide variety of clays are now used in the composition of monolithic refractories varying from high-alumina kaolinitic clay to lower alumina plastic (bentonite) clays. Kaolinitic clays in the calcined form provide the major source for almost all categories of lower-alumina monolithic refractories. The kaolinitic clays are sometimes mixed with low-grade bauxitic clays and then calcined into high-grade dense grains that are widely used. These grains are dense and possess low porosity and high refractoriness. The alumina content of these grains varies in the range of 47–70% with porosities in the range of 3–6%, thus providing a wide range of compositional varieties. The chemical composition and physical characteristics of these high temperature calcined clays are given in Tables 7 and 8.

Mullite

Mullite ($3\text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2$) has a specific gravity of 3.15. It softens at 1650°C and melts with dissociation into corundum and silica at 1810°C . Although a very stable compound, mullite is rarely found in

nature except on the isle of Mull on the west coast of Scotland (from where it derives its name).

Table 7. Chemical composition of high temperature calcined clays.*

Type	I	II	III
Chemical Analysis, %			
Al ₂ O ₃	46.8	58.6	68.8
SiO ₂	50.1	37.8	26.8
Fe ₂ O ₃	1.85	1.13	1.22
TiO ₂	0.95	2.21	2.82
CaO	0.04	0.06	0.06
K ₂ O	0.09	0.07	0.08
Na ₂ O	0.09	0.04	0.05
P ₂ O ₅	0.09	0.10	0.11

Table 8. Physical characteristics of high temperature calcined clays.*

Bulk Density, g/cc	2.62	2.80	2.88
Porosity, %	3.6	3.2	3.5
1600°C Reheat change, % Volume	3.1	0	0
Mineralogy:			
% Mullite	65	77	87
% Glass	20	23	13
% Cristobalite	15	trace	-
Max. recommended temperature for use	1600°C	1650°C	1700°C

(*Courtesy of C. E. Minerals, Andersonville, GA).

Synthetic mullite is produced by sintering kyanite, bauxitic kaolin and bauxite (in the correct Al₂O₃ to SiO₂ ratio) and a mixture of kaolin and alumina in rotary kilns. Kyanite, andalucite and sillimanite convert to mullite upon heating in varying degrees of volume change. Most sintered mullites are not pure mullite, but a mixture of mullite, glass and cristobalite, the amount of each depending on the chemical and

mineralogical states of the starting materials. Pure synthetic sintered and fused mullite is produced in electric arc furnaces by fusing the right proportions of pure Al_2O_3 and SiO_2 . The extent of fusion determines the state of the sintered and fused mullite.

Mullite is a unique refractory material with excellent thermal shock resistance, hot-load bearing characteristics, and high refractoriness. Mullite is formed in all alumina-silica refractory materials heated to sufficiently high temperatures. The amount of mullite formed depends on the chemical analysis and the heat treatment received.

Relatively pure bauxite with 70% Al_2O_3 content converts to mullite at a temperature of around 1650°C whereas 60% Al_2O_3 containing bauxitic kaolin converts to mullite at around 1600°C . The associated volume shrinkages are 20% and 15% respectively. The mullite content of each is about 85–90% and 75–80% respectively while the remainder is glass.

Both sintered and fused mullite are available on the market; but because of their higher cost in the pure form, neither of them is used in large quantities in monolithic compositions. As described before, calcined clays comprising of larger quantities of bauxitic clays provide high ratios of mullite. Fused mullite is sometimes used to provide better thermal-shock resistance particularly in high alumina castables. Table 9 shows the chemical compositions of sintered and fused mullite.

Silica

Silica happens to be the major component of the earth's crust. It constitutes 60% of the crust's composition, occurring both independently and as silicates in combination with other oxides. The silica used in monolithic refractories includes quartzite, silica sand, diatomaceous earth and fused silica glass. Silica sand was originally used for containment of molten iron and steel. Today, silica is often used as mortar and some kinds of special plastics such as tap-hole

clay. Fused silica is used in making castables and pumpables primarily for use in the coke oven. Fused silica containing low-cement castable shapes are now also used in repairing coke ovens. Pumpable fused-silica compositions have better physical and thermal properties than silica bricks. They have better strength, lower thermal expansion and better load-bearing capacity than silica bricks.

Table 9. Chemical composition of sintered (calcined kyanite) and fused mullite.

Composition, %	Sintered (Calcined kyanite)	Fused
Al ₂ O ₃	60.1	72.2
SiO ₂	37.8	27.0
Fe ₂ O ₃	0.92	0.13
TiO ₂	0.98	0.01
CaO	0.01	0.15
MgO	0.43	0.05
K ₂ O	0.15	0.04
Na ₂ O	0.27	0.38

Finely ground silica (below 325 mesh) is used in various compositions, particularly coatings. The use of these fine silicas is somewhat restricted because of the associated health hazards (fine silica can cause silicosis after prolonged inhalation of the dust without appropriate protective safety gear).

Vermiculite

Vermiculite³ is a group of mica like hydrated ferro-magnesium aluminum silicate minerals. Most commercial vermiculite has been geologically formed by the alteration of biotite or phlogopite mica. Vermiculite has the ability to expand (exfoliate) 16–20 times when heated quickly at elevated temperatures. Exfoliation of vermiculite concentrate particles is done in vertical furnaces with brief residence time when the particles fall through a zone of 900°C. Although the major uses of vermiculite are in areas other than refractories, it provides a good source for low-temperature insulating castables and

gun mixes. Typical chemical analyses of vermiculite are given in Table 10.

Table 10. Chemical composition of vermiculite.

Composition, %	South Africa	Kenya
Al ₂ O ₃	11.1	14.0
SiO ₂	42.9	42.0
Fe ₂ O ₃	6.30	4.70
TiO ₂	0.85	1.40
CaO	1.30	0.95
MgO	26.5	28.1
K ₂ O	4.25	0.70
Na ₂ O	0.22	0.30
L.O.I.	6.33	8.00

Perlite

Perlite is a metastable amorphous aluminum silicate that expands when subjected to rapid heating which drives off the water, increases the volume and significantly lowers the density. Perlite is expanded by dropping the crude through a vertical furnace where the expansion occurs. Expanded perlite is extremely low in density with a bulk density around 0.3 g/cm³. Perlite is a very common raw material used in making insulating castables and gun mixes. Insulating refractories made with perlite are resistant to temperatures up to 1400°C. A typical average chemical analysis would be 71–75% SiO₂, 12.5–18% Al₂O₃, 4–5% K₂O, 1–4% Na₂O and CaO, and 2–5% water.

Silicon carbide

Commercial silicon carbide is produced in electric furnaces from a mixture of coke and silica sand (>99.4% SiO₂) which sometimes also contains sawdust and salt or another binder. An electric current passed between permanent electrodes located at the two ends of the furnace, through a graphite core produces a temperature higher than

2200°C–2400°C. At this temperature, crystals of silicon carbide form from the sand-coke charge. Beta-SiC (cubic) forms at 1400–1800°C and alpha (hexagonal) forms at temperatures greater than 1800°C. If used, the saw dust burns out keeping the mass porous, and the salt assists in the removal of impurities through the formation of volatile chlorides. A furnace run takes 2–20 days depending on the transformer and furnace size. Another method used to form silicon carbide pieces is vapor deposition of silicon onto a heated graphite or carbon surface.

The intrinsic characteristics include a molecular weight of 40.1, specific gravity of 3.2, dissociation temperature of around 2500°C, high hardness, high thermal conductivity, low thermal expansion and resistance to neutral to acid slag compositions.

Commercial grades vary in composition in the range of 90%–99.5% SiC. Light green SiC is 99.8% pure, and as the purity decreases to 99%, the color changes to dark green and at 98.5% to black. Material that is >99.5% in purity is used for most abrasive and refractory applications. High-purity green SiC is used in high-performance ceramics and as resistance-heating rods.

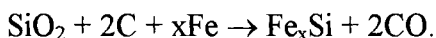
In monolithic refractories, a wide range of purity of SiC is used depending on the application. The most common application of SiC is in the blast-furnace cast house where low purity (90%) SiC is used. Higher purity SiC (97–98%) is used in ramming, gunning or plastics compositions used for thermal power stations since one of the main functions of the refractory is to conduct heat. In castable or pumpable refractories, the major problem encountered with SiC is the presence of metallic impurities that cause the evolution of gases during the application. Hence, it is a normal procedure to test SiC for metallic impurities before use in the castable or pumpable compositions.

Silica fume

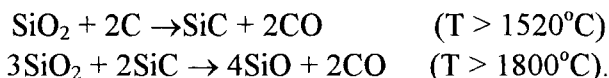
Silica fume is a by-product of ferrosilicon and silicon metal production. Silicon and ferro-silicon are produced in large electrical

smelting furnaces by the reduction of silica above 2000°C. The raw materials consist of quartz and carbon (such as coal, coke and wood chips). For ferrosilicon production an iron source is also added.

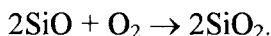
The desired chemical reaction for ferro-silicon production may be written as⁴



However, the chemistry of the process is much more complex than above and involves a number of side reactions. Two of the more important reactions that are thought to take place are listed below:



In other words, silicon carbide (SiC) and the unstable silicon monoxide (SiO) gas play an important intermediate role in the production process. Some of the SiO gas tends to escape through the surface, and is subsequently oxidized by air:



This is known as silica fume or microsilica. Typically 10–20% of the quartz added ends up as volatilized silica, i.e., silica fume.

On a macroscopic level, silica fume is a fine powder of a color, ranging from off-white to dark gray that indicates the carbon content. The carbon present may found in several different forms, e.g., coke or coal, silicon carbide, some tar and finally carbon black (possibly from the cracking of hydrocarbons in the volatiles of the raw materials).

Silica-fume particles are spherical in shape and have a mean particle diameter of 0.15 μm that ranges from 0.02 to 0.45 μm and has a surface area of 15–25 m^2/g . Typical bulk chemical compositions of silica fume are given in Table 11.

Table 11. Bulk chemical composition of silica fume.

Element/Compound	From silicon metal production. %	From 75% ferro-silicon production, %
SiO ₂	94 – 98	85 – 95
C	0.2 – 1.5	0.8 – 2.5
K	0.2 – 0.7	0.5 – 3.5
Na	0.1 – 0.3	0.2 – 1.5
Mg	0.1 – 0.4	0.5 – 2.5
Ca	0.05 – 0.3	0.1 – 0.5
Al	0.05 – 0.2	0.1 – 1.0
Fe	0.01 – 0.3	0.1 – 2.5
Ti	0.00 – 0.01	0.1 – 0.03
P	0.01 – 0.1	0.02 – 0.1
S	0.1 – 0.2	0.05 – 0.5

The bulk density of silica fume is 150–250 kg/m³ in the regular form. It is also available in densified form with a bulk density of 500–700 kg/m³. The densified form has the advantage of costing less to transport and needing less storage area. But this form may also cause problems in application since the densified agglomerates are hard to break down into the individual particles during the mixing operation and, hence, may lose the expected rheological properties.

Due to the enormous growth in demand in the market during the last decade for silica fume, some silica fume is produced as the primary product instead of a secondary one. This silica fume is off-white in color, and is much purer and more consistent. Obviously, the cost is significantly higher than that of common silica fume. Due to the absence of surface impurities, this kind of silica fume shows good rheological properties particularly in the formulation of self-flow castables.

Magnesite

Although magnesite is a mineral (MgCO₃), in the refractories industries the term magnesite is identified with MgO (magnesia).

Magnesite is produced either by calcining the mineral magnesite (MgCO_3) after appropriate beneficiation or by synthetically extracting it from sea water or brine. Naturally occurring magnesite is often associated with dolomite, talc, chlorite, serpentine, mica, pyrite and magnetite. Most deposits are mined selectively and the ores are often beneficiated. The beneficiation method varies according to the location but usually includes crushing, screening, washing, magnetite separation, heavy media separation and froth floatation.

The most important process in the production of magnesia from sea water and brines is the precipitation of magnesium hydroxide from solutions of magnesium salts by the addition of a strong base (calcined limestone or calcined dolomite). The magnesium hydroxide precipitate is then washed, thickened, filtered and calcined to produce magnesia. In another commercial process, concentrated magnesium chloride (MgCl_2) brine is sprayed into a thermal reactor where hot gases convert it to magnesia and hydrogen chloride. The magnesia is slurried forming magnesium hydroxide that is washed, filtered and calcined to produce magnesia.

For refractory applications, dead-burnt ($>1400^\circ\text{C}$) magnesia is used. Naturally occurring dead burnt magnesite is often associated with higher quantities of silica and iron, while the synthetic magnesites are made by controlled chemistry with desired quantities of silica and calcia to attain higher density. Fused magnesia is produced at temperatures $>2750^\circ\text{C}$ by fusing magnesia in an electric arc furnace. The fused magnesite has much larger crystal size and is superior to dead-burnt magnesite in strength, abrasion resistance and chemical reactivity.

In monolithic refractories the greatest use of magnesia is in the gunning maintenance of basic oxygen furnaces and electric furnaces. In recent years, the use of magnesite in the tundish working lining has become a regular practice and its consumption is increasing with time. The magnesite used in the tundish lining does not need to be of high quality since it is mixed with silicates and clay minerals to obtain the

desired properties, and it can tolerate much higher amounts of impurities compared to other applications. Table 12 shows the chemical analyses and densities of some available magnesites.

Table 12. Chemical and physical characteristics of some dead burnt and sintered magnesites.

Source	Natural	Natural	Sea water	Brine	Brine
Density, g/cc	3.25–3.30	3.30–3.35	3.45–3.47	3.45–3.48	3.28–3.33
Chemical analysis, %					
MgO	85–88	90–91	98–99	95–96	97–98
SiO ₂	5.0–5.7	1.0–1.5	0.2–0.3	0.5–0.6	0.5–0.6
Fe ₂ O ₃	3.0–4.0	5.0–6.0	0.2–0.3	0.2–0.4	0.1–0.2
Al ₂ O ₃	1.0–2.0	1.0–1.5	0.2–0.4	0.2–0.4	0.1–0.2
CaO	2.0–3.0	1.5–2.0	1.0–1.5	2.2–2.4	1.5–2.0

Olivine

Olivine³ was named after its olive-green color. It is a generic term used to indicate a group of orthosilicate minerals in a solid solution series, with forsterite (2MgO.SiO₂) and fayalite (2FeO.SiO₂) as end points. Olivine is a major constituent of earth-forming rocks. Therefore, it is found in varying quantities of alteration products, such as serpentine (3MgO.2SiO₂.2H₂O). Dunite is an olivine-containing rock with about 90% olivine. Olivine has natural properties that support its use for a variety of applications, viz., melting point of 1800°C, low heat conductivity, good insulating values (60–80% less than magnesite), high refractoriness level (1760°C), no hydration, i.e., no calcination is required before use, inert nature, Mohr's hardness of 6.5–7.0, specific gravity of 3.27–3.37 and bulk density of 1.5–2.0 g/cm³. Moreover, it is environmentally friendly (contains no free silica), high chemical and mineralogical stability (due to strong forsterite mineralogical binding) and high resistance to metal penetration (to both basic and acidic iron oxide-rich slags, alkaline oxides, sulfates carbonates and halides).

Because olivine is inexpensive, it competes favorably against higher-priced raw materials of similar chemistry. Olivine competes with magnesite as a refractory mineral as well as in castables and tundish liners. In refractory incinerators, olivine has technical advantage over other refractory materials including resistance to slag attack, temperature and spalling. Table 13 shows the typical chemical composition of refractory-grade olivines.

Table 13. Typical chemical compositions of refractory-grade olivines.

	Aaheim, Norway	Raubergvika, Norway	Raubergvika, Norway	Hamilton, Wash., USA
MgO	48–50	45–48	45–48	45–49
SiO ₂	41–43	41–43	39–42	39–42
Fe ₂ O ₃	6.8–7.4	7.6–8.0	6–8	6–8
Al ₂ O ₃	<1.0	<1.5	<1.0	<1.0
CaO	<0.5	<1.0	<1.0	<1.0
L.O.I.	0.5–1.2	0.8–1.2	0.8–1.4	0.5–1.0

Graphite

Natural graphite³ is a form of carbon that occurs in nature. Graphite is generally described as grayish black in color with a lustrous black sheen. It crystallizes in the hexagonal system with rhombohedral symmetry. Natural graphite is usually described in three generic terms: amorphous, flake and high crystalline. Amorphous (which is a misnomer) graphite generally occurs in areas similar to coal deposits and the carbon content varies between 75–90%. The base raw material for amorphous graphite is generally that of ordinary coal as determined by chemical analysis. Amorphous graphite is primarily found in Mexico, Korea, China and Austria.

Natural flake graphite is a term also used to describe a naturally occurring graphitic mineral that has been homogeneously distributed through its host mineral. The flaky crystalline structure is easily identified when compared to amorphous graphite. Natural flake graphite is different from amorphous graphite; its properties are more

directional because of its high crystallinity. Natural flake graphite has a graphitization of 99.3%. The base material for high crystalline graphite is believed to consist of crude oil deposits that have been converted to a solid mass of graphite with time, temperature and pressure. High crystalline graphite occurs only in Sri Lanka and is usually used as a standard for comparison with all other graphites via X-ray diffraction.

The other forms of graphites are artificial forms that are produced from calcined petroleum coke heat treated to $>2800^{\circ}\text{C}$. These materials are 99.3% graphite with usual carbon contents of 99.9%. The other artificial graphite is produced by the processing of graphite electrodes and has a graphite content of 85–95% and a carbon content of 98–99.5%. The physical and chemical characteristics of graphite are given in Table 14.

Amorphous and artificial graphites are used more often in castables and pumpables since crystalline and flake graphite tend to impede the flow. All other monolithic compositions use different varieties of graphites depending on the use and economy.

Table 14. Chemical and physical characteristics of graphites.

	Amorphous	Flake	High crystalline	Flake artificial
Carbon, %	81.0	90.0	96.7	97.0
Sulfur, %	0.1	0.1	0.7	0.07
True density, g/cc	2.31	2.29	2.26	2.24
Morphology	Granular	Flake	Plates needles	Flake

Pitch

Both coal-tar pitch and petroleum pitch are used in monolithic refractories. Although coal-tar pitch has higher residual carbon than

petroleum pitch, both of them are effective in providing the carbon component to the refractory. The retained carbon from either coal tar or petroleum pitch is amorphous in nature. These are used in both pulverized and pelletized forms depending on the formulation. The advantage of using pitch over other forms of carbon, e.g., graphite, is that pitch melts at low temperatures and tends to coat the grains, thus providing a good protective layer which resists slag corrosion.

Coal tar

Coal tar has been used primarily in blast-furnace tap-hole plastics. Its characteristic behavior has been advantageous for the specific properties needed for this application. Coal tar allows the plastic to retain its workability over a long period of time. The coal tar used for this application usually has quite stringent specifications regarding the volatile contents at different temperatures, the residual pitch content, water content, carbon disulfide solubility and residual carbon content. The specifications vary according to the limits set by the manufacturers.

Calcium aluminate cement

Calcium aluminate (CA) or high-alumina cements continue to be the most important hydraulically setting cements used for bonding refractory castables (concrete) and gunning mixes. They develop high strength within 6 to 24 hours after placement. The early strength development enables high-temperature processing units to be put back on stream with short turnaround time, thus providing a good cost/performance ratio. Compared to other conventional refractories, castables have the following advantages:

- quick installation and low construction costs;
- reduced and simplified furnace maintenance;

- much lower thermal conductivity than for fired bricks permitting thinner linings;
- good thermal shock resistance;
- monolithic (no joints);
- prefired shapes need not be stocked;
- ready availability; and
- greater flexibility in design.

Limited use of CA cements as refractory concrete binders began in the 1920s in the U.S. after first being marketed in France in 1918. Initially, most refractory concretes were mixed on the job site, similar to portland concrete. In the late 1940s and early 1950s, specially formulated dry concrete aggregate mixes (castables) became more available. Until the early 1950s, commercially available HAC contained large amounts of iron oxide and SiO_2 as impurities. The presence of these oxides limited the use of these cements to relatively low-temperature applications. Higher-purity calcium aluminate cements were introduced during the 1950s which expanded the use of refractory castables to include higher-temperature applications.

The predominant method of manufacture of HAC is by the sinter or clinker process. Relatively pure limestone is used as the lime source for producing all CA cements. Calcined alumina is used to manufacture high-purity CA cements, whereas low-iron, low-silica bauxite is the preferred source of alumina for intermediate- and low-purity HAC. Commercial HACs shown in Table 15 are listed according to purity and the alumina content.

Low-purity cements are manufactured from bauxites and contain about 18% Fe_2O_3 and 9% SiO_2 . These impurities limit refractory-concrete service to around 1425°C . Low-iron (2–4% Fe_2O_3) and low- SiO_2 (5–7%) bauxites are sintered with limestone to manufacture the intermediate-purity cements. These types of cements are commonly used castable-concretes having a service limit of $<1650^\circ\text{C}$. Impurities

totaling <2% in the high-purity cements permit use to service limits of 1870°C, depending on the refractoriness of the aggregate in the mix.

Preliminary crushing and drying is required for the bauxite before being ground with crushed limestone to a fineness of 200 mesh. However, calcined Bayer Al_2O_3 (100–325 mesh) requires no processing prior to concurrent grinding with the crushed limestone. The product is sintered at 1315°C to 1425°C, cooled and then ground to cement fineness. They are mixed with additives, as desired viz., calcined Al_2O_3 to obtain the desired Al_2O_3 content, gypsum or other minerals to control the set, as well as fillers and plasticizers to improve workability.

Table 15. Chemical composition of calcium aluminate cement binders.

Chemical components	Low purity	Medium purity	High purity
Al_2O_3	39 – 50	55 – 66	70 – 90
Fe_2O_3	7 – 16	1 – 3	0.0 – 0.4
CaO	35 – 42	26 – 36	9 – 28
SiO_2	4.5 – 9.0	3.5 – 6.0	0.0 – 0.3

Table 16 shows the mineralogical phase assemblages in CA cements according to their relative reaction rate with water. They form the hydrated cement phases responsible for developing strength after curing the concrete in a humid environment. Monocalcium aluminate (CA) is the main active ingredient in all HAC. It hydrates sufficiently fast to satisfy the short turnaround demands of the industry. Minor cementitious constituents such as, C_{12}A_7 and CA_2 contribute to development of the strength of high-alumina cement. The presence of C_{12}A_7 to any significant amount causes fast setting in low-cement castables. In formulation of low-cement castables, the high-alumina cement should contain as little C_{12}A_7 as possible. The mineralogical phase* assemblages in CA cements are given in Table 16.

Table 16. Typical mineralogical phase* assemblages in CA cements.

Relative hydration rate	Low purity	Medium purity	High purity
Fast	CA $C_{12}A_7$ CA_2 C	CA $C_{12}A_7$ CA_2 C	CA $C_{12}A_7$ CA_2
Slow	C_2S C_4AF C_2AS	C_2S C_4AF C_2AS	
Non-hydrating	CT A	CT A	A

*C=CaO; A=Al₂O₃; S=SiO₂; F=Fe₂O₃; T=TiO₂

High-alumina cement castables require only 24 h to develop 70-80% of its full strength compared to 28 days for normal portland cement concretes. The dried and fired strengths of refractory concretes depend on the use of: 1) the correct amount of tempering water required for specific placements to assure installation of a sound lining with minimum structural defects; 2) proper curing conditions to ensure development of the desired hydrated phases to maximize the structural integrity; and 3) a controlled uniform heating schedule. These requirements have a sequential effect on the service life and the performance of a refractory-concrete lining bonded with HAC, i.e., the degradation in strength resulting from structural defects caused by improper placement is amplified by poor curing practices as well as nonuniform and excessive heating. The net result can be a premature shutdown resulting from catastrophic failure of the thermo-mechanical lining.

The processing variables that must be controlled during installation and initial heat up of HAC-bonded castable concretes are the material, equipment, water temperature and content, mixing intensity and time, temperature and humidity, surface characteristics of the working face and heating rate and schedule. Temperature is the

most important variable during the installation of refractory concrete linings since it influences the sensitivity of these materials to all processing steps. It has a significant effect on the extent to which cement bonding phases develop and on the sensitivity of the initial heating to explosive steam-spalling.

The profound effect of casting and curing temperature of HAC-bonded concretes are shown in Table 17.

Table 17. Effect of temperature on hydration-reaction products of Ca-cement and their properties.

Curing temp., °C	Hydration products of CA phase	Physical properties	
		Density, g/cc	Solid volume change from CAH_{10}
<21	$\text{CAH}_{10} + \text{AH}_3$ (gel)	1.72	
21 – 35	$\text{C}_2\text{AH}_8 + \text{AH}_3$ (gel)	1.95, 2.42	–37
>45	$\text{C}_3\text{AH}_6 + \text{AH}_3$ (crystalline)	2.52, 2.42	–53

As shown in Table 13, HAC-bonded cement concretes can run into severe problems if the installation and curing temperature is <21°C. Both CAH_{10} and AH_3 (gel) are formed around this temperature. During heating, the gel prohibits the removal of steam due to the low permeability. Also during heating, the CAH_{10} changes into C_2AH_8 with significant volume change as indicated above. A combination of these two factors causes problems during drying which may result in explosive spalling and cracking of the installed lining.

If the installation and curing of the HAC-bonded concrete lining is accomplished between 21 and 35°C, the lining can be heated at a rate depending on the thickness of the lining as indicated in Table 18.

Calcium aluminate cements are used for both castables and gunning mixes. Low- and medium-purity cements can be used for either casting or gunning. High-alumina cement clinkers (fast setting)

are generally used for gunning. But for castables, various additives are either supplied by the manufacturers or users to conform to the conditions of casting — working time, setting time and attainment of specific strength characteristics after a certain time during the heating period.

Table18. Heating rate of firing refractory concrete specimens.

Cross section of specimens, mm	Maximum heating rate °C/h
114 by 76	38
114 by 65	38
51 by 51	150
25 by 25	370

Alumina-magnesia spinel

During the past decade, alumina-magnesia spinel has been largely used as a high-temperature refractory material. The technical advantage of alumina-magnesia spinel for both shaped and monolithic refractories are:

- high resistance to thermo-mechanical stresses;
- low thermal expansion;
- high resistance to changes in the environment;
- low content of secondary oxide phases, providing good refractoriness; and
- high purity in the material providing impurity-free refractory.

Alumina-magnesia spinel is available with various levels of magnesia content below and above the stoichiometric level of 28.2% MgO. The spinels with MgO content above the stoichiometric content (28.2%) are generally used in brick making. These, however, are not recommended for monolithic refractory formulations because of two possible problems. First, the extra MgO may have a tendency to

hydrate and form minor cracks during heating and firing. Secondly, the extra MgO may form spinel at high temperatures that will create unwanted expansion. Currently, alumina-magnesia spinels are available from 10% MgO to 33% MgO. The commonly available alumina-magnesia spinel compositions are given in Table 19.

Table 19. Alumina-magnesia spinel compositions.

Properties	1	2	3	4	5
Chemical Analysis, %					
Al ₂ O ₃	66.0	70.4	74.3	23.0	90.0
MgO	33.0	28.5	25.0	76.0	9.0
Fe ₂ O ₃	<0.1	0.23	<0.1	<0.1	<0.1
CaO	0.4	0.1	0.28	0.3	0.25
SiO ₂	0.09	0.22	0.25	0.06	0.05
Bulk Density, kg/m ³	3270	3400	3300	3250	3300
Apparent Porosity, %	2.0	3.9	7.5	2.0	2.5
Phases present, XRD					
Major	Spinel	Spinel	Spinel	Spinel	Spinel
Minor	Periclase	Periclase	Cor-undum	None	Cor-undum

Alumina-magnesia spinel is made by sintering or fusing Bayer alumina and magnesia in an electric furnace. These spinels are extremely pure and completely silica-free but usually cost more. Spinels are also formed by fusing or sintering bauxite and magnesia containing small quantities of silica. The composition of the spinel depends on the silica content in the bauxite.

Colloidal silica

Colloidal silica is an aqueous colloidal dispersion of silica particles. It is opalescent liquid with a little sliminess to touch. Because of its colloidal nature, colloidal silica has high specific surface area which accounts for its novel properties and wide variety of uses.

The particles in colloidal silica are discrete uniform spheres of silica that have no internal surface area or detectable crystallinity. They are dispersed in an alkaline medium which reacts with the silica surface to produce a negative charge due to which the particles repel one another and result in a stable product. The stabilizing counter ions could be of sodium or ammonium. In many applications the desired effects of colloidal silica are realized after the sol has been converted to a dry solid, usually a gelatin. Colloidal silica can be gelled by removing water, changing pH or by adding a salt or water-miscible organic solvent. During drying, the hydroxyl groups on the surface of the particles condense by splitting out water to form siloxane (Si-O-Si) bonds resulting in coalescence and inter-bonding.

Colloidal silica is available in various particle sizes and concentrations (as shown in Table 20). Usually the larger sizes give more stable formulations and can be used in higher concentrations.

The use of specific colloidal silica depends on the application. In general, 8 nm particles at 30% concentration and 15 nm particles at 40% concentration are used in refractory formulations.

Table 20. Different types of colloidal silica.

Particle size, nm	Surface area, m ² /g	% Silica (as SiO ₂)	pH @ 25°C	Specific Gravity	Viscosity cP
4	750	15	10.5	1.10	<10
5	600	15	9.0	1.09	<10
8	375	30	10.0	1.21	<10
13	230	30	10.2	1.20	<10
15	200	40	9.7	1.29	15
20	150	40	9.0	1.29	15
60	50	50	8.5	1.39	15
75	40	40	8.4	1.29	10

Colloidal silica has been widely used in investment-casting refractories for a long time. It has more recently been used in coatings, castable/pumpables, ramming mixes and gunning mixes. Colloidal silicas are normally stable at elevated ambient temperatures, but they gel below freezing point in an irreversible manner. Hence, anti-freezing agents such as ethylene glycol should be added to store colloidal silica below its freezing point. Colloidal silica with added antifreeze may affect the properties of refractories.

Phosphoric acid and phosphates

Phosphoric acid reacts with metal oxides to form salts that act as bonding agents. Various forms of the phosphates are also used as bonding agents. Although the use of a phosphate as a refractory binder was first reported in 1941, it was not until 1950 that Kingery⁶ reported wide use of phosphates in refractories.

In phosphoric acid, H_3PO_4 , $(\text{PO}_4)^{3-}$ is considered the acid radical, but phosphoric acids and phosphates⁵ also include other radicals, such as, $(\text{P}_2\text{O}_7)^{4-}$, $(\text{P}_n\text{O}_{3n+1})^{(n+2)-}$, and $(\text{PO}_3)^-$. The ion $(\text{PO}_4)^{3-}$ is referred to as orthophosphoric acid radical and the others as pyrophosphoric acid radical, polyphosphoric acid radical and metaphosphoric acid radical respectively. Their corresponding salts are known as orthophosphates, pyrophosphates, and metaphosphates.

The most commonly used are orthophosphoric acid, known usually as phosphoric acid, and its salts. Phosphoric acid is available on the market in various concentrations. Table 21 shows the properties at different concentrations.

Tribasic phosphoric acid dissociates in three stages in aqueous solutions as:

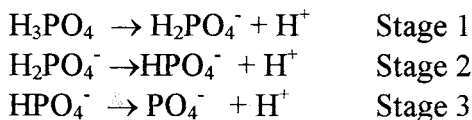
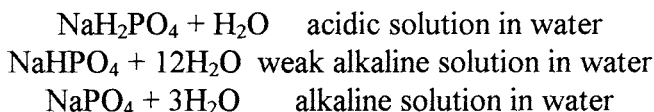


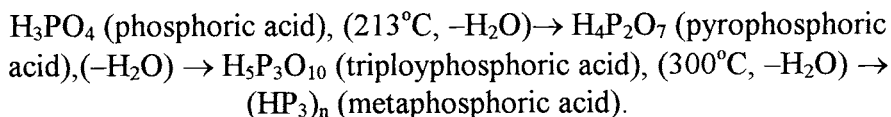
Table 21. Properties of commercial phosphoric acids.

Phosphoric acid, H_3PO_4 %	75	80	85	105
% P_2O_5	54.5	58.0	61.2	76.3
Density, g/cc @ 20°C	1.57	1.64	1.68	1.94
Viscosity, cP @ 25°C	12	17	23	440
Melting point, °C	-17.5	+4.5	+21.2	+16

The corresponding possible sodium salts are

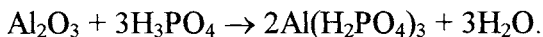


Since phosphoric acid is not very strong, the strong alkali metal sodium quickly changes towards the alkaline after the first dissociation of hydrogen ion. Phosphoric acid dehydrates as it is heated and the dehydration products are as follows:



Various oxides and hydroxides react with phosphoric acid at room temperature whereas the reaction rate and setting time depend on the oxides. Of the different oxides, MgO and $\text{Mg}(\text{OH})_2$ react violently with phosphoric acid but take a long time to set. Calcined (900°C) MgO reacts with heat evolution and sets in a short time. Of all the oxides reacting with phosphoric acid, Al_2O_3 and $\text{Al}(\text{OH})_3$ are used most frequently in refractories. Phosphoric acid and aluminum hydroxide react readily, but alumina and phosphoric acid react slowly at room temperature. With increase of temperature, phosphoric acid

undergoes dehydration and reacts with alumina to form monoaluminum phosphate, $\text{Al}(\text{H}_2\text{PO}_4)_3$, according to the reaction



The reaction starts above 127°C and continues until 427°C . With further increase in temperature (between 732°C and 1327°C), orthoaluminum phosphate, AlPO_4 , is formed and completely converted to the cristobalite form above 1327°C .

At room temperature there is no reaction between SiO_2 and H_3PO_4 forming no bonding properties, whereas at elevated temperatures $\text{SiO}_2 \cdot \text{P}_2\text{O}_5$ and $2\text{SiO}_2 \cdot \text{P}_2\text{O}_5$ are formed. A liquid phase is formed (between 1100°C and 1300°C). Hence, $\text{SiO}_2 \cdot \text{P}_2\text{O}_5$ compounds are not suitable for refractory products.

Magnesia and magnesium hydroxide react rapidly with phosphoric acid to form various phases of magnesium phosphate as shown below:

Hayesite	$\text{MgHPO}_4 \cdot 2\text{H}_2\text{O}$
Newberyite	$\text{MgHPO}_4 \cdot 3\text{H}_2\text{O}$
Phosphorresslerite	$\text{MgHPO}_4 \cdot 7\text{H}_2\text{O}$

Although, the form Newberyite has the highest melting point (1327°C), it is still not refractory enough for most applications.

Aluminum phosphate

The most useful among the phosphates for application in refractories are the aluminum phosphate in forms shown in Table 22.

The important phosphate compound useful for refractory applications as a ceramic binder is monoaluminum phosphate (MAP), $\text{Al}(\text{H}_2\text{PO}_4)_3$ known for its solubility in water, bonding strength and stability. Monoaluminum phosphate is manufactured by reacting phosphoric acid with aluminum hydroxide at $100\text{--}200^\circ\text{C}$ and is mostly

available in liquid form in a water solution. The dry form, made in a spray drier, is extremely hygroscopic and soon dissolves again.

Table 22. Various forms of phosphates.

Acid radical	Type	Chemical formula
$(\text{PO}_4)^{3-}$	Monoaluminum phosphate	$\text{Al}(\text{H}_2\text{PO}_4)_3$
	Dialuminum phosphate	$\text{Al}_2(\text{HPO}_4)_3$
	Aluminum orthophosphate	AlPO_4
	Aluminum sesquiphosphate	$\text{Al}(\text{H}_3\text{PO}_4)_2$
$(\text{P}_2\text{O}_7)^{4-}$	Aluminum pyrophosphate	$\text{Al}_8\text{H}_{12}(\text{P}_2\text{O}_7)_9$
$(\text{P}_3\text{O}_{10})^{5-}$	Aluminum tripolyphosphate	$\text{AlH}_2\text{P}_3\text{O}_{10}$
$(\text{PO}_3)^-$	Aluminum metaphosphate	$\text{Al}(\text{PO}_3)_3$

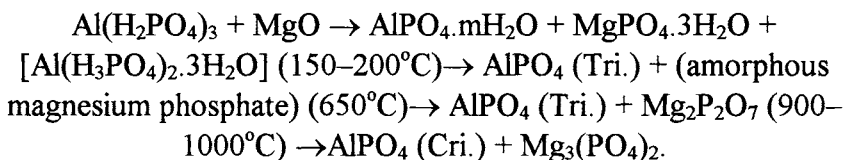
The common commercially available form of monoaluminum phosphate (MAP) is 50% solution in water. The solution is acidic with a pH of 1.4. Monoaluminum phosphate does not harden rapidly at room temperature. After heating to 100°C, it loses part of its water and hardens, but its bonding strength decreases because it reabsorbs moisture when exposed to air. To obtain a non-hygroscopic stable phase, it has to be heated above 350°C.

In the presence of Al_2O_3 , aluminum metaphosphate, $\text{Al}(\text{PO}_3)_3$ formed at 400°–500°C changes at 700°–1000°C to aluminum orthophosphate (AlPO_4) which resembles silica in structure. The form corresponding to that of quartz is berlinite (AlPO_4), which has other polymorphs of the cristobalite and tridymite forms.

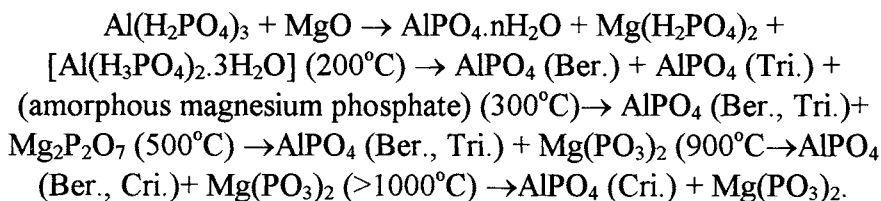
A plastic refractory with monoaluminum phosphate bonding tends to show shorter storage life than other plastic bonding systems. This could be due to small amounts of impurities in the aggregates or the mineral constituents of the clay. The short storage life is also thought to be caused by the nonstoichiometric composition of typical monoaluminum phosphates since it is difficult to produce according to its theoretical composition.

As a bonding agent, monoaluminum phosphate does not rapidly harden at room temperature. The most commonly used setting agent is magnesia (MgO). The reaction between MgO and monoaluminum phosphate is exothermic, developing a considerable amount of heat. The setting rate and time depend on the ambient temperature, degree of crystallinity and grain size of magnesia, all of which must be taken into consideration when manufacturing the refractory. The reactions between monoaluminum phosphate and magnesia can be depicted as follows:⁷

(a) System forming $\text{MgHPO}_4 \cdot 3\text{H}_2\text{O}$ (Newberite) as a result of the reaction in the composition of $\text{P}/(\text{Al} + \text{Mg}) = 1$ at room temperature.



(b) System forming $\text{Mg}(\text{H}_2\text{PO}_4)_2$ as a result of reaction in the composition of $\text{P}/(\text{Al} + \text{Mg}) = 1.5$ at room temperature.



As these reactions show, MgO added as a setting agent forms low-melting point compounds like $\text{Mg}_3(\text{PO}_4)_2$ (melting point 1357°C) and $\text{Mg}(\text{PO}_3)_2$ (melting point 1165°C). Furthermore, MgO breaks the three dimensional bond of aluminum phosphate. Therefore, above 1300°C , MgO decreases the hot strength of phosphate bonded refractories.

Sodium Phosphates

Other phosphates used for refractories include various kinds of sodium phosphate that are mainly used for coagulation and deflocculation and as bonding agents for basic gunning refractories. The classification of sodium phosphates available on the market is shown in Table 23 which shows the anhydrate forms of the phosphates.

In small quantities added to a powder-water system like a clay slip, sodium phosphate causes deflocculation, reducing the apparent viscosity of the slip. This allows attaining the requisite fluidity with smaller quantities of water. Deflocculation occurs when the state of the charged ions on the surface of the particles change with the addition of an electrolyte. Thus, the pH of the aqueous solution of sodium phosphate acting as an electrolyte has an important effect on deflocculation.

Sodium phosphates, on the other hand, cause coagulation by reacting with alkaline earth metal compounds like CaO and MgO. The application of sodium phosphates to magnesia-based gunning refractories is based on this property.

Table 23. Types and characteristics of various available sodium phosphates.

Name	Chemical formula	Composition, %	
		P ₂ O ₅	Na ₂ O
Monosodium phosphate	NaH ₂ PO ₄	25.8	25.8
Di-sodium phosphate	Na ₂ HPO ₄	43.7	43.7
Sodium orthophosphate	Na ₃ PO ₄	56.7	56.7
Sodium pyrophosphate	Na ₄ P ₂ O ₇	46.6	46.6
Acidic sodium pyrophosphate	Na ₄ H ₂ P ₂ O ₇	28.0	28.0
Sodium tripolyphosphate	Na ₅ P ₃ O ₁₀	57.9	42.1
Sodium tetrakisphosphate	Na ₆ P ₄ O ₁₃	60.4	39.6
Sodium hexametaphosphate	(NaPO ₃) ₆	60.6	39.4
Sodium ultraphosphate	Na ₂ P ₄ O ₁₁	82.1	17.9

Sodium silicates

The most prevalent and commonly used silicates are sodium silicates of the general formula, $\text{Na}_2\text{O} \cdot n\text{SiO}_2 \cdot m\text{H}_2\text{O}$. Potassium and lithium silicates are also available but are rarely used in refractory application. The anhydrides are usually transparent like glass and dissolve in water. They are sometimes described as water glass.

The molar ratio of $\text{SiO}_2/\text{Na}_2\text{O}$, which varies in the range of 0.5 – 4.0 in industrial products, determines the characteristics of a sodium silicate. Sodium silicates in the molar ratio of 2.0–3.35 are commonly used in refractories. The molar ratio of crystalline sodium silicates is 1 or less, whereas, sodium silicates with molar ratio of more than 1 are amorphous. The types of sodium silicates available on the market are shown in Table 24.⁵

In the dry process, sodium silicate is manufactured by fusing silica sand and sodium carbonate at around 1200°C . The product is cooled and the glass obtained is dissolved in water. In the wet process, water glass is manufactured from powdered silica sand or amorphous silica and an aqueous solution of caustic soda. Dry sodium silicate is an intermediate product between sodium silicate anhydrate and liquid sodium silicate (water glass). It can also be obtained by dehydrating sodium silicate and pulverizing it.

Table 24. Types of sodium silicate.

Name	Chemical formula	Mol. ratio ($\text{SiO}_2/\text{Na}_2\text{O}$)
Sodium orthosilicate	$2\text{Na}_2\text{O} \cdot \text{SiO}_2 \cdot n\text{H}_2\text{O}$	0.5
Sodium pyrosilicate	$3\text{Na}_2\text{O} \cdot 2\text{SiO}_2 \cdot n\text{H}_2\text{O}$	0.67
Sodium metasilicate	$\text{Na}_2\text{O} \cdot \text{SiO}_2 \cdot n\text{H}_2\text{O}$	1
Liquid	$\text{Na}_2\text{O} \cdot 2\text{SiO}_2 \cdot \text{aq.}$	2
Liquid	$2\text{Na}_2\text{O} \cdot 5\text{SiO}_2 \cdot \text{aq.}$	2.5
Liquid	$\text{Na}_2\text{O} \cdot 3\text{SiO}_2 \cdot \text{aq.}$	1.5
Liquid	$\text{Na}_2\text{O} \cdot 4\text{SiO}_2 \cdot \text{aq.}$	4

The molecular weight corresponding to a $\text{SiO}_2/\text{Na}_2\text{O}$ molar ratio of 3.3 ranges from 325 to 1500 but these are not considered to be macromolecules. Although there is a rapid increase in viscosity with concentration, it is not due to polymerization but due to the sudden rise in the intermolecular attraction forming a network structure. The higher the $\text{SiO}_2/\text{Na}_2\text{O}$ molar ratio, the more the likelihood of this phenomenon.

The viscosity of an aqueous solution of sodium silicate is influenced both by the molar ratio and concentration, and remarkably varies with temperature. The viscosity is also affected considerably by the addition of water-soluble organic substances like alcohol or starch.

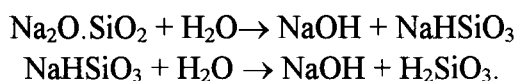
An aqueous solution of sodium silicate forms a film when in contact with air. This film prevents the vaporization of moisture-forming sodium carbonate (Na_2CO_3 and or $\text{Na}_2\text{CO}_3 \cdot 2.5\text{H}_2\text{O}$) which increases with decreasing molar ratio. Consequently, there is a tendency for sodium silicate containing refractories to bloat when heated since the moisture cannot escape because of the formation of the film.

Sodium silicate with a molar ratio of one, starts crystallizing at 550°C , continues to crystallize up to 700°C to anhydrous metasilicate ($\text{Na}_2\text{O} \cdot \text{SiO}_2$) and becomes porous. With increasing in temperature between 1000°C and 1100°C , the crystalline phase disappears with a drop in porosity. With the molar ratio of 2, sodium silicate crystallizes at 600°C with increasing amounts of the crystalline phase up to 800°C . The crystalline phase vanishes at 900°C with vitrification and corresponding decrease in porosity. Sodium silicate with a molar ratio of three starts to crystallize between 600°C and 650°C and continues up to 700°C . Complete vitrification occurs at 900°C with a decrease in porosity.

With the addition of a setting agent, hydrogel silica forms with increased permeability and bloating as vaporization decreases. Due to the formation of a liquid phase upon heating, sodium silicate does not indicate any definite melting point, but starts to soften above certain

temperatures with a gradual decrease in viscosity. This relationship depends on the molar ratio.

Sodium silicate hydrolyses in aqueous solutions and exhibits alkalinity as indicated by the following equations:



Smaller molar ratio of sodium silicates shows more pronounced hydrolysis and the pH drops with decreasing molar ratio. Sodium silicates with higher molar ratios provide a buffer action.

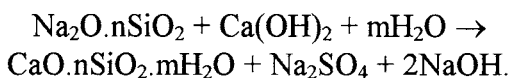
Sodium silicate also has a deflocculating property and is often used with casting slip in molding pottery. There it forms sodium hydroxide and silica gel developing a protective glutinous film on the surface of particles and changes the viscosity of the cast slip.

Aqueous solutions of sodium silicates react with acids or salts allowing changes in their viscosity, precipitation or coagulation to take place. With addition of acid, the alkali of the solution is neutralized and the viscosity is increased due to polymerization between hydroxysilicate radicals. A typical reaction of sodium silicate with sulfuric acid that is applied for production of hydrosilicate gel follows:



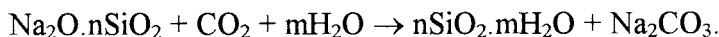
Alkali salts (such as sodium chloride), when added to a sodium silicate solution, increases the viscosity and forms a gel that may be the result of either an increase in the intermolecular force due to neutralization of the charge or dehydration. But addition of a dibasic acid salt or alkali carbonates does not readily form a gel.

Sodium silicate reacts with alkali earth metal hydroxides or their salts and coagulates. The process by which the sodium silicate reacts with different acids and salts and hardens is shown below:



The setting agents of a sodium silicate-bonded refractory are selected according to the applications of the refractory. These setting agents include sodium silicofluoride, aluminum polychloride, sodium phosphate, aluminum polyphosphate, magnesium polyphosphate, ammonium pentaborate and glyoxal.

Since sodium silicofluoride does not readily dissolve in water, the setting reaction is a solid-liquid reaction between sodium silicofluoride particles and sodium silicate solution. Silica hydrogel ($\text{SiO}_2 \cdot n\text{H}_2\text{O}$) and sodium fluoride (NaF) are formed when sodium silicofluoride reacts with Na_2O in sodium silicate. The viscosity change observed at the beginning of the reaction varies with the molar ratio ($\text{SiO}_2/\text{Na}_2\text{O}$) of sodium silicate. The higher the molar ratio, the earlier the viscosity increases. The setting time becomes shorter, on the other hand, at low concentrations and molar ratios ($\text{SiO}_2/\text{Na}_2\text{O}$). The synergesis is highest when the concentration of sodium silicate is 25–30% since it is influenced by the internal pressure in the gel and the resistance to squeezing out a liquid. The internal pressure is low at low concentrations, whereas the resistance to squeezing a liquid increases at high concentrations due to the thinning of the capillaries in the gel with high internal pressure. As expected, the setting reaction is accelerated with increasing temperature. Carbon dioxide (CO_2) gas also hardens sodium silicate-bonded refractories according to the reaction which explains the formation of the hardened surface (Na_2CO_3) of a sodium silicate bonded refractory:



(The sections of phosphoric acid and phosphates and sodium silicate are partially reproduced by permission of Plibrico Japan Company Ltd.)

References

1. S. Banerjee and G. L. Ramsey, "High Fired Refractories for Continuous Casting of Steel," *Cer. Eng. Sci. Proc.*, **9**, NO. 1-2, pp. 67-73, 1988.
2. S. Banerjee and G. L. Ramsey, "High-Fired Refractories for Continuous Casting Tundishes," *Adv. Refr. Met.. Ind. Proc.*, **4**, pp. 305-312, 1987.
3. Annual Minerals Review, *Amer. Cer. Soc.*, 1995.
4. M. Dastol, "Production and Properties of Elkem Microsilica," *Elkem Microsilica Tech., 1st. Seminar*, Sept. 1984.
5. A. Nishikawa, Editor, "*Technology of Monolithic refractories*," Jan. 1984, Japan.
6. W. D. Kingery, "Fundamental Studies of Phosphate Bonding in Refractories: I, Literature Review, II, Cold Setting Properties and III, Phosphate Adsorption by Clay and Bond Migration," *J. Amer. Cer. Soc.* **33**, No. 8, pp. 239-250, 1950.
7. Ando, Shinada and Hiraoka, *Cer. Soc. Japan*, **82**, (12), 1974

CHAPTER II: CASTABLE REFRACTORIES

Introduction

Portland cement concrete has been used for a long time in the building industry. It is universally accepted and used for construction purposes of domestic and industrial premises where the ambient temperature does not exceed more than 400°C. When used above this temperature it expands, cracks, and loses its strength. Hence it is not suitable for industries where the use temperature exceeds 400°C.

The manufacture and commercial availability of calcium aluminate cements provide a good hydraulic bonding system for higher temperature applications. The gradual improvement in purity and quality of the calcium aluminate cements helped improve refractory castable materials for more critical applications. Although conventional castables provided applications in several areas in industries using high temperatures, they still lacked in versatility because of deficiencies in properties.

The use of castables increased with the development of low and ultralow cement castables in areas where bricks or other monolithic refractories were prevalent. Subsequently, cement-free pumpables and self-flow castables further increased the application of castables. To date, the application of low-cement castables as "shotcrete" technology, which combines the benefits of gunning and castables, has

taken the monolithic technology to another level. Refractory consumers benefit significantly by using the new technology from labor and cost reduction and an increase in productivity. The present section will deal with the progressive development of castable technology, its uses and areas of application.

Conventional castables

Calcium Aluminate Cement:

The phase diagram of the $\text{CaO-Al}_2\text{O}_3\text{-SiO}_2$ ($\text{C} = \text{CaO}$, $\text{A} = \text{Al}_2\text{O}_3$, $\text{S} = \text{SiO}_2$) system shows that $\text{CaO-Al}_2\text{O}_3$ compounds are better for high temperature applications than CaO-SiO_2 . The first calcium aluminate cement was introduced in France in 1918 and in the United States in 1924. These cements had low purity, containing fair amounts of SiO_2 and Fe_2O_3 which made them unsuitable for high temperature use. They were made from limestone and bauxite.

During the sintering process, the hydratable forms of calcium aluminates, such as C_{12}A_7 , CA , and CA_2 are formed. The silicate-containing phases such as $\beta(\text{beta})\text{-C}_2\text{S}$ and C_2AS are also formed. The compound C_{12}A_7 is formed in small quantities and hydrates very rapidly. If the $\text{CaO/Al}_2\text{O}_3$ ratio is increased, the amount of C_{12}A_7 increases. The presence of C_{12}A_7 in larger quantities makes the cement very quick-setting. The compound CA is the most important one present in high alumina cements (HAC) because it is primarily responsible for their particular cementitious behavior. It is slow-setting initially but it subsequently hardens rapidly.

Pure CA_2 reacts very slowly with water at ordinary temperatures but once the hydration starts, the rate increases with the growing concentration of the hydration products. In the presence of CA , pure CA_2 hydrates at a reasonable speed.¹ The compound C_2S is hydraulic but hydrates very slowly, thus reducing the likelihood of contribution to concrete strength at early stages. C_2AS (gehlinite) appears as a

regular constituent in these HAC but hydrates at a very slow rate. The presence of higher amounts of gehlinites in HAC causes a serious reduction in the rapid hardening properties of HAC. The iron oxide forms both calcium ferrite (CF) and dicalcium ferrite (C_2F) and may be present in the HAC. Both CA and $C_{12}A_7$ take ferric oxide into solid solution.

The low-purity HACs are formed at around 1430°C , thus preventing their use as high temperature concretes. However, high-purity calcium aluminate cements were introduced during the 1950s. The combination of high-purity limestone and Bayers alumina² followed by high-temperature sintering led to the desired properties of the HAC. It has been surmised that the HAC should have the following characteristics for successful use in refractory castables:

1. Castables made with the cement should maintain the fusion point of the mixture at least 40°C above the maximum temperature of use.
2. The cement should impart satisfactory working properties for placement, setting and subsequent installation of the concrete.
3. The cement should develop sufficient strength within a 24-hour period to allow demolding and application of heat.
4. Castables should develop and retain sufficient strength throughout the entire temperature range used in order to remain structurally sound during service.
5. The cement must retain a dimensional stability after placement, curing, initial heat treatment and subsequent reheating.
6. The castables made with the cement must have resistance to spalling during the initial heat treatment and at high temperatures.

At present, several types of HAC are available to satisfy the requirements of various applications. The highest-purity material should have less than 1% of the impurities MgO , Na_2O , SiO_2 and Fe_2O_3 . These are available as a pure clinker with 27–28% CaO or co-

milled with extra alumina with 18–19% CaO content. Typical analyses of the high-, medium- and low-purity high-alumina cements are shown in Table 1.

Table 1. Chemical composition of high alumina cements.

Oxide	High purity		Medium purity	Low purity
	%	%	%	%
$\text{Al}_2\text{O}_3 + \text{TiO}_2$	70–71	80–81	57–58	46–47
Fe_2O_3	<0.1	<0.1	1.0–1.5	6.0–6.5
CaO	27–28	18–19	34–35	35–36
SiO_2	<0.1	<0.1	5.0–6.0	8.0–9.0

Hydration characteristics of HAC

The binding properties and strength development of HAC arise from the hydration of different phases of calcium aluminates. This rate of hydration will differ with different phases of calcium aluminates. The rate of hydration diminishes as the C/A ratio of decreases, i.e., C_{12}A_7 is the most reactive and CA_3 is the least reactive. The temperature to which the cement is subjected during curing has severe effects on the hydration reaction products of concretes with CA cements.³ The predominant hydrate formed at low temperatures (<21°C) is CAH_{10} . Along with AH_3 (gel), C_2AH_8 is formed in the temperature range 21–35°C. At temperatures above 45°C, C_3AH_6 is formed along with crystalline AH_3 as shown in Table 2.

The temperature dependence of the hydration products of CA cements is one of the most critical parameters in the high alumina refractory concrete technology. The effect of temperature on the hydration of a tabular alumina-based castable (with 20% CA cement) during curing and dehydration during the heating process has been shown in Table 3.³

As indicated in Table 4, several problems may arise when the curing temperature is low (<21°C). Apart from the formation of the

high-hydration phase of CAH_{10} , the gel phase of AH_3 is also formed. The permeability of the concrete decreases due to the gel phase and CAH_{10} crystallizes into hexagonal prisms. Upon heating, the hexagonal prisms of CAH_{10} are transformed into hexagonal C_2AH_8 plates followed by conversion to the somewhat spherical, cubic C_3AH_6 crystals.³ The trapezohedral-type crystals are about 10 times the size of the residual CAH_{10} crystals.

Table 2. Effect of temperature on hydration-reaction products of CA cements.

Hydration products of CA phase	Curing Temperature, °C	Physical Properties	
		Density, g/cm ³	Solid volume change from CAH_{10} , %
CAH_{10}	<21	1.72	
C_2AH_8 and AH_3 (gel)	21–35	1.95, 2.42	–37
C_3AH_6 and AH_3 (Crystalline	>45	2.52, 2.42	–53

Table 3. Phases of high alumina castable at different temperatures.

Temperature, °C	Tab. alumina + 20% CA–25 24h cure/90% humidity
0 – 20	AH_3 gel, CAH_{10} , CA (unhydrated)
20 – 40	AH_3 , C_2AH_8 , C_3AH_X (X=8-12), C_3AH_6
40 – 60	AH_3 , C_2AH_8 , C_3AH_X (X=8-12), C_3AH_6
60 – 100	AH_3 , C_3AH_X (X=8-12), C_3AH_6
100 – 350	AH_3 , C_3AH_X (X=8-12), C_3AH_6
600 – 1300	A, CA, CA_2
1400 – 1650	CA_6

The water requirement for formation of different hydrates during curing are given in Table 4.

This massive transformation causes two primary problems. There is a great possibility of spalling during the heating period and an

accompanying loss of strength due to the reduced specific surface and voids occurring around the large cubic crystals. These voids occur as a result of the nominal 50% volume change upon conversion of CAH_{10} to the denser C_3AH_6 and AH_3 . The gross restructuring and high shrinkage of the bonding matrix and conversion of the low density CAH_{10} phase formed at $<21^\circ\text{C}$ explain the loss of strength in drying HAC castables cured at low temperatures.

Table 4. The Hydrates of alumina cement and their water of crystallization.⁴

	Hydrates	Water of crystallization, (wt.%)
1	CAH_{10}	53.3
2	C_2AH_8	40.3
3	C_3AH_6	28.6
4	AH_3	34.6

High-alumina castable compositions

In formulating a castable composition, several factors need to be considered. First, the end-use application dictates the overall chemistry of the composition and the alumina content of the castable, and thus determines which raw material should be used as the major aggregate in the castable. The aggregates may be derived from calcined flint clay, calcined fireclay, andalucite, calcined semi-bauxitic fireclay, calcined bauxite, brown and white fused alumina and tabular alumina. Sometimes the aggregates are used by themselves and sometimes they are used in combination with others. The application temperature, the environment in contact with the castables, both gaseous and liquid metal, and/or slag govern the nature of the aggregate selection.

The physical properties of castables depend on the packing of ingredients in the castable. Density, porosity, strength characteristics and thermal stability are dictated by the compactness of the castable. Regardless of the choice of aggregates for a particular use, the grain-

size distribution must be carefully determined to obtain the maximum compaction. The water requirement of the castable also has a direct correlation with selection of the grain sizing in the aggregates and the matrix. The methods of securing compactness in concrete mixes were demonstrated by Furnas⁵ and Anderegg.⁶

Furnas designed two systems. One involves intermittent or gap sizing based on mixing two, three or four sizes of materials. The other system involves continuously graded size distributions. Selection of either the gap-sizing or continuous-sizing depends on the availability of the sizing for the different ingredients in the mix. The main objective is to obtain the maximum packing and minimum water requirement for a given composition.

The water requirement of a castable depends on the porosity of the aggregates. High-alumina aggregates like tabular alumina and brown fused alumina require minimum water whereas calcined flint clay needs the maximum amount of water. The selection of the purity of HAC depends on the end use of the castable, particularly the temperature of application. The quantity of cement is also determined by the end use. Normally, a cement content of 15–35% is used. The higher the cement content, the higher the casting water requirement that leads to a decrease in density and increase in porosity. Hence, the amount of cement needed is compromised with the physical properties desired for particular use.

Effects of additives on high-alumina castables

A detailed study has been done by MacZura et al.⁷ on the effect of different additives on the water requirement, flowability, working time, and strength characteristics. They studied the effect of liquid isopropyl alcohol, sodium gluconate, sodium gluconate with filler (Plastiment), sodium tetraborate, plaster of Paris, lithium chloride, sodium aluminate, sodium chloride, calcium chloride, calcium hydroxide, calcium carbonate, magnesium carbonate and supplementary additives like methocel, and polyvinyl alcohol. The

castable was based on tabular alumina with 15% HAC (80% alumina). Liquid isopropyl alcohol, 1–2%, cement weight basis, was found to be the most functional retarder with water reduction, good plasticity and flow property with minimum strength reduction. Sodium gluconate also reduces water but is more critical in strength impairment. A combination of 0.25% each of sodium bicarbonate and citric acid proved to be effective as additive in retarding free lime in unstable calcium aluminate cements.

Lankard and Hackman⁸ reported that admixtures providing safe water reduction and/or set reduction with low-purity cement castables did not have any significant effect on a high-purity cement castable when used in amounts of up to 0.05% on total-weight basis. Plaster of Paris seemed to be a good accelerator in amounts of 1–3 wt.% of cement basis while small additions of lithium compounds (below about 0.1% cement weight basis) have been found to be good accelerators for strength development.

Various water-reducing agents have been used by many authors during the past 20 years to enhance properties of the castables. The common water-reducing agents used are sodium or calcium lignosulphonate, gluconic acid, sodium citrate as well as superplasticizers like melamine formaldehyde sulphonate and naphthalene formaldehyde sulfonate. The drying and heating of the reduced water castables needs more attention.

Drying and heating of high alumina castables

Drying and heating of HAC castables are two of the important criteria for obtaining the desired properties. Castables normally require 25–30% more water than the neat cement-hydration water requirement to hydrate the cement and to bring the castable to a shaping consistency. As the drying process begins, the castable loses both the mechanically introduced water and the water of hydration. As indicated earlier, the water of hydration varies according to the curing temperature. Lower curing temperatures (<21° C) tend to form CAH_{10} with a water of

hydration level almost twice that of C_3AH_6 . Thus for the same amount of water added, the free water and hydrates depend on the curing temperature.

The rate of drying has a significant effect on the castable refractory. Too rapid a rate of heating may cause an explosion from the quick evolution of steam by raising the vapor pressure inside the mass. There is a distinct relationship between the curing temperature and the explosion of castable refractories. Figure 1 shows the relationship between curing temperature and the explosion limit.⁴ As the plot shows, specimens cured at low temperatures ($<15^\circ\text{C}$) explode at all temperatures. Above 20°C , the explosion temperature suddenly rises. Although the explosion-limit curve varies by the type of the castable composition, the relationship between curing and explosion-limit temperature is consistent. If the permeability of the castable is increased, the explosion-temperature limit also increases.

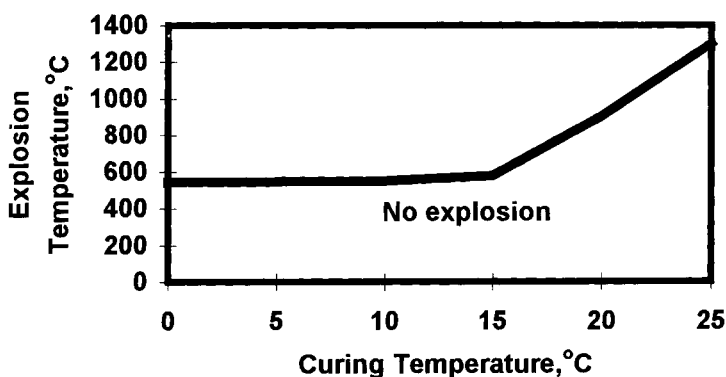


Fig. 1. Relationship between curing temperature and explosion limit.

1. In refractory castables, most of the water content — both mechanical and the water of hydration — is driven out between 100°C and 550°C . During the initial heating, the mechanically

combined water leaves the system, followed by the higher hydrates. The water of hydration of C_3AH_6 is the last to leave the system. In order to get successful results from installation of refractory castables, the following conditions should be maintained.

2. Care must be taken during installation to maintain the temperature of the dry material and the mixing water above 20°C .
3. After installation, the curing temperature should not be allowed to drop below 21°C , preferably maintained in the temperature range $30\text{--}40^{\circ}\text{C}$.
4. The rate of heating should follow the schedule:
 - a. Raise the temperature from ambient to 110°C @ 15°C/h .
 - b. Hold at 110°C for one hour per 25 mm thickness.
 - c. Raise from 110°C to 300°C @ 15°C/h .
 - d. Hold at 300°C for one hour per 25 mm thickness.
 - e. Raise from 300°C to 550°C @ 25°C/h .
 - f. Hold at 550°C for one hour per 15 mm thickness.
 - g. Raise from 550°C to the application temperature @ 75°C/h .

It is important to adhere closely to this schedule to avoid cracking or spalling of the installed lining. The attainment of the desired castable properties depends to a certain extent on proper choice of drying and heating conditions.

Properties of conventional high-alumina castables (HAC)

Properties of HAC castables are governed by the type of aggregates and matrix as well as the quantity of cement used. These variables dictate the amount of water needed for the castable. If low-quality porous aggregates are used, more water will be needed to fill up the pores of the aggregates. The result will be a castable with higher porosity, lower density and lower strength. If the cement content is higher, more water is needed increasing the cold strength in the

process. The open porosity of a castable usually varies between 15–20% after drying at 110°C. Although the strength of these castables is good after drying, it gradually drops off with heating due to the dehydration of the cement bond.

The hydraulic bond starts losing water at temperatures between 250–350°C forming calcium aluminate phases. The loss of strength continues up to 700°C. The strength does not increase until the ceramic bonding phase begins above 800°C. At higher temperatures, the hot strength of HACs reaches the values equal to or higher than comparable fired bricks. However, silica-containing castables lose their strength due to the formation of liquids around 1400°C. Typical cold- and hot-strength characteristics of silica containing high alumina castable are shown in Fig. 2.

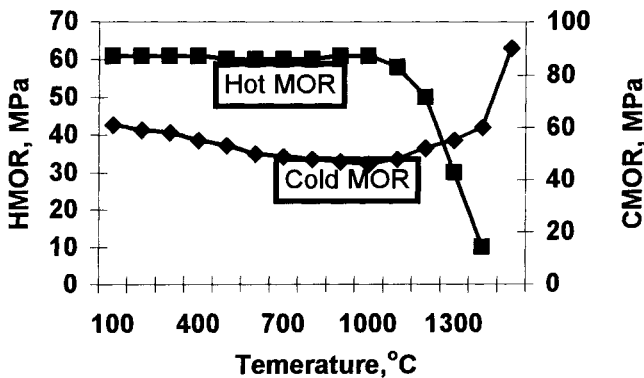


Fig. 2. Cold and hot modulus of rupture of HAC with temperature.

The amount of water needed for a conventional castable is determined by the ball in hand method or by the flow-table method (ASTM – C 860). The ball-in-hand method is subjective while the use of the flow-table method is less dependent on the operator. Castables are formed into 76 mm cubes to determine the porosity, density and compressive strength and 51 x 51 x 228 mm bars are made to measure

modulus of rupture. To determine the hot modulus of rupture, bars 38 x 38 x 205 mm in size are made. Specimens are allowed to cure in a humid atmosphere at a temperature varying between 22°C and 45°C for at least 18 h but usually 24 h. These specimens are demolded and their dimensions are measured. They are then put in a drier set at a temperature of 110°C and dried for 24 h. Physical properties are then determined according to ASTM standards. Typical properties of HAC-based castables are given in Tables 5 and 6.

Table 5. Chemical analysis of high alumina cement-base castables.

%Al ₂ O ₃ Class	45	50	55	65	95
Chemical Analysis, %					
Al ₂ O ₃	43.8	48.0	55.8	65.0	95.8
SiO ₂	37.3	40.0	38.6	30.0	0.2
Fe ₂ O ₃	2.5	1.4	1.0	1.1	0.1
TiO ₂	2.1	1.9	1.5	1.6	-
CaO	10.2	7.1	3.9	3.8	3.6

Low cement castables (LCC)

Significant improvement in the application of castables started after the commercial availability of high-purity high-alumina cements. Particularly, high-alumina (such as tabular alumina) aggregate-based castables have been widely accepted in critical high-temperature applications. MacZura et al⁹ demonstrated that tabular alumina castables with 80% alumina cement provide exceptional hot strength (16 MPa @ 1510°C) and excellent resistance to sag deformation. But conventional castables have not been suitable for variety of applications even when high-purity cements are used, for three reasons:

1. Higher quantities of cement are used whenever more strength is needed in order to raise the cold strength of the castable.

- However, since the greater amount of cement needs more water, the bulk density decreases and porosity increases.
2. In alumina-silicate castables, the increased amount of cement introduces more flux, which decreases refractoriness and high temperature strength.
 3. Conventional castables suffer the loss of strength at intermediate temperature ranges of 300–900°C during the dehydration of the cement. It results in an increase in porosity and decrease in density until the temperature is reached where ceramic bonding takes place (>900°C).

Table 6. Physical properties of high alumina cement-base castables.

%Al ₂ O ₃ Class	45	50	55	65	95
CMOR, MPa					
After heating to					
110°C	6.9	11.0	7.2	7.9	8.2
815°C	3.5	9.0	4.1	6.2	7.6
1370°C	14.5	17.2	13.8	6.6	12.3
CCS, MPa					
After heating to					
110°C	34.5	55.2	34.8	34.4	45.0
815°C	28.0	44.8	24.1	31.0	34.8
1370°C	41.0	62.0	40.9	49.0	75.0
HMOR, MPa at					
815°C	3.1	11.0	2.0	6.6	8.3
1370°C	1.1	1.7	1.4	4.6	8.3
PLC, %					
815°C	0.1	-0.3	-0.2	-0.2	-0.2
1370°C	-0.3	0.2	+0.1	+0.2	-0.3

The use of castables has been limited to a few industrial applications through the mid-1970s because of the above factors. Although the initial concept of low-cement castables was presented¹⁰ as early as 1962, it was not adopted by the industry because purer varieties of calcium aluminate cement were not available. In the early

invention volatilized silica was used in conjunction with calcium aluminate cement with a $\text{CaO}/\text{Al}_2\text{O}_3$ ratio of 1:1 to 1:4 in castables containing aggregates such as chrome ore and calcined alumina. These formulations produced improved strength and abrasion resistance in castable materials for use between 250 and 1100°C.

The initial low-moisture and low-cement castable (LCC) compositions as adopted by the industry in later years, were first introduced in 1974.¹¹ A dense refractory body could be formed by the use of only 5–8% calcium aluminate cement along with pulverulent materials selected from the group consisting of clay, kaolin, micronized silica, micronized alumina, micronized magnesia, micronized chromite and micronized forsterite. This mix also contained a dispersing agent selected from alkali metal phosphate, alkali metal carbonate and alkali metal humate. These castables did not exhibit any loss of strength in the intermediate-temperature range. The use of 3% clay helped reducing the water content of the castable from 10% to 6%. The porosity decreased from 27% to 16.9% while the density increased from 2.60 to 2.85 g/cm.³ This was one of the first inventions of low-moisture, low-cement castables. The increased density and lower porosity caused the refractory castable to be less permeable and considerable problems arose as a consequence in drying the castable. Unusually long periods of drying were needed to avoid cracking and spalling of the refractory body.

The largest number of papers on the developmental work of low moisture and low-cement castables were presented at the International Symposium on New Developments in Monolithic Refractories held during April 30–May 1, 1984, in Pittsburgh, PA.

At the 1984 symposium the manufacture of vibratable castables with thixotropic behavior¹² resulting from the use of bonding agents were reported. Low-moisture castables of cold-setting compositions without cement and bodies containing carbon and silicon carbide were described. A silicate compound (SILUBIT FB 10) that is basically a comilled silica fume and kyanite with a small amount of sodium

phosphate was used to replace a part of the cement. Results were shown with 8% high-alumina cement in combination with 4% of the silicate compound, using fused alumina as the aggregate. The mixing water was reduced considerably from 7% to 5.3%. For cement-free castables, an aluminosilicate compound with a minor proportion of a phosphate (LITHOPIX AS 85) was used instead of the cement in a fused alumina-aggregate base. The use of 7% of this additive reduced the mixing water to 5% with good vibration-cast property.

Another composition was mentioned with tabular-alumina aggregates including silicon carbide (15%) and graphite (2%) where the aluminosilicate compound was used in conjunction with an electrolyte (ZUSOSET TH 3). The use of 4% of this additive produced a good vibratable castable with only 4% mixing water. There was significant improvement in the strength values of these castables compared to those of high-cement conventional castables with 20% cement both at room temperature and at high temperatures. The densities were higher and porosities were lower. The different property characteristics are shown in Figs. 3 and 4.

A series of papers¹³⁻¹⁵ were presented on the development and application of LCC. "High-technology castables" (meaning LCC using 0–5% HAC with a lime content of 0–1.5%) reported water requirements in the range of 4–6%. This resulted in denser particle packing, higher density and lower porosity. When compared with other monolithic refractories and brick compositions with similar alumina contents these castables demonstrated the distinct superiority in all physical properties including erosion, abrasion, mechanical and thermal spalling. These castables have been successfully used in steel ladles, tundish back-up lining, and trough hoods.

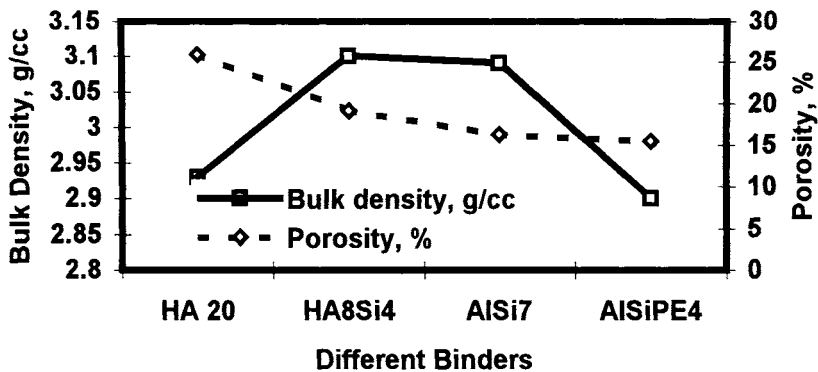


Fig. 3. Density and porosity of low-moisture castables with different binders.

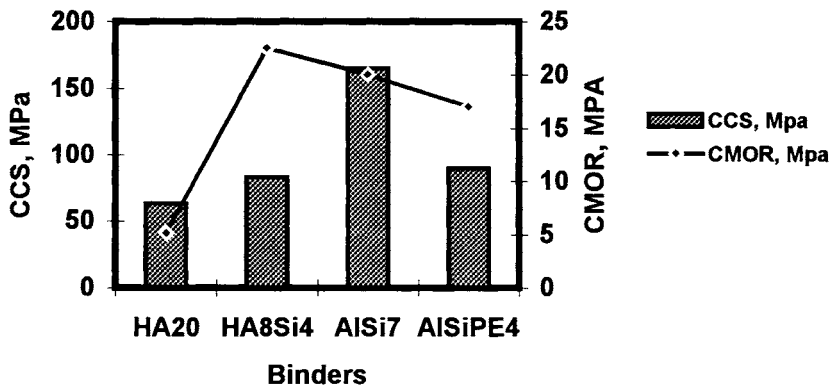


Fig. 4. Cold modulus of rupture and cold crushing strength of LCC with different binders after firing at different temperatures.

"High-performance castables" (LCC) were reported with a range of alumina contents using different aggregates (such as calcined kaolin, bauxite and fused or tabular alumina). These castables were developed for severe applications including torpedo ladle throats, continuous-casting steel tundishes and reheat-furnace hearths. They were also applied in palletizing kilns, direct-reduction kilns, nose rings,

planetary cooler bends, lifter blocks and retaining rings in cement kilns, oil refineries, ferrosilicon furnaces, kiln cars and aluminum ladles.

The development of new types of monolithic refractories (also LCC) using different types of additives — deflocculants, superfine particles and other active powders in calcium aluminate cement containing castables were also reported. The effect of different polyphosphates on the zeta potential of calcium aluminate cement was demonstrated. It mentioned that the use of appropriate amounts of deflocculants prevented flocculation among aluminous cement grains and increased the zeta potential of oxide grains in castables, thus lowering the mixing water. The deflocculants also had this effect on the concentration of Ca^{2+} in the liquid phase of aluminous cement-deflocculant-water system.

Deflocculants impart a significant effect on the superfine powders and other fine aggregates. A clay-bonded castable utilizing the deflocculants was reported to have properties better than conventional cement-containing castables. Since the addition of superfine powders and deflocculants produce dense castables, the permeability decreases and causes problems in drying of castables. The use of Al (metal) powder helps produce clay-bonded castables capable of quicker setting, rapid dry-out and improved strength.

A detailed study¹⁶ on the properties and applications of LCC indicated the importance of the direct relationship between water addition and these properties. The importance of adding water to LCC and the corresponding deterioration of properties was quantitatively demonstrated. Figures 5 a), b) and c) show the dependence of water addition on the properties.

Apart from mixing water, the importance of the curing temperature (to avoid explosive spalling) and the use of appropriate vibrating equipment was mentioned. The LCC was defined as a pseudothixotropic material in nature. It was theorized that during the dehydration of the cement, recrystallization of the CA phases occurred

on the submicrometer particles and not in void spaces, thus resulting in strength increase during the dehydration process. At higher temperatures, the formation of mullite strengthened the cement. The cold modulus of rupture of LCCs with different alumina contents after firing at different temperatures is shown in Fig. 6.

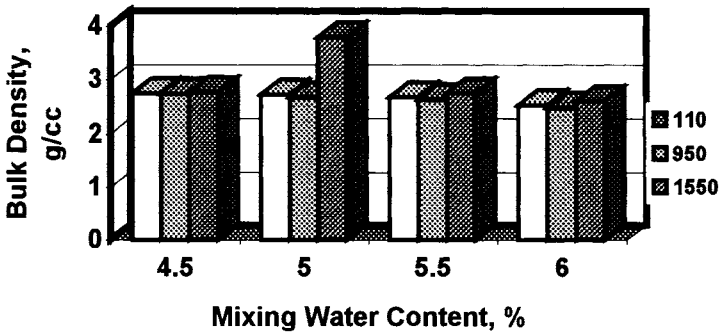


Fig. 5a. Variation of bulk density with mixing water after firing at different temperatures.

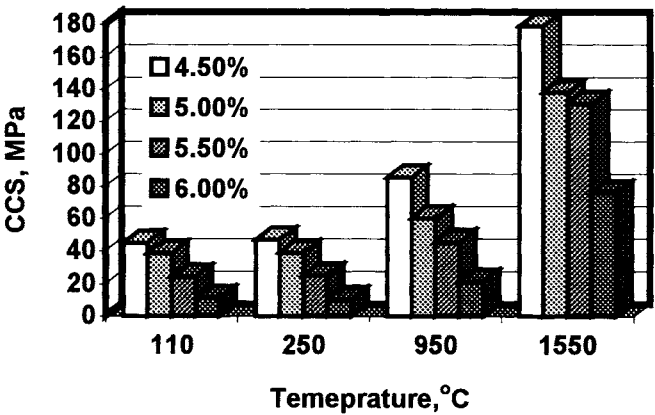


Fig. 5b. Variation of cold-crushing strength with mixing water after firing at different temperatures.

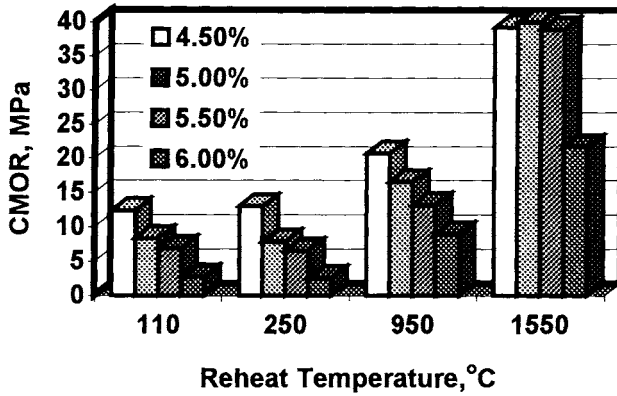


Fig. 5c. Variation of cold modulus of rupture with mixing water after firing at different temperatures.

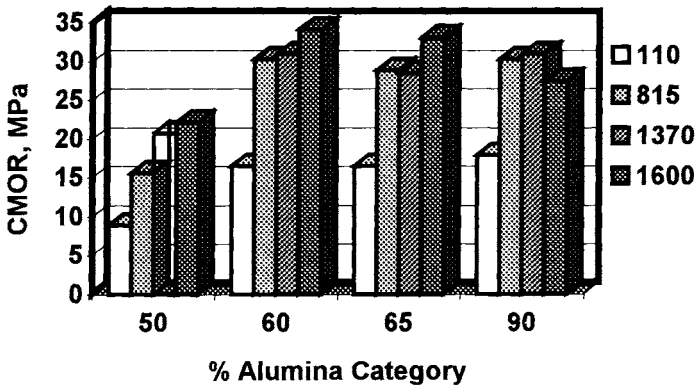


Fig. 6. Cold modulus of rupture of LCC with different alumina content after firing at different temperatures.

A series of papers¹³⁻¹⁵ were presented on the development and application of LCC. "High-technology castables" (meaning LCC using 0–5% HAC with a lime content of 0–1.5%) reported water requirements in the range of 4–6%. This resulted in denser particle

packing, higher density and lower porosity. When compared with other monolithic refractories and brick compositions with similar alumina contents these castables demonstrated the distinct superiority in all physical properties including erosion, abrasion, mechanical and thermal spalling. These castables have been successfully used in steel ladles, tundish back-up lining, and trough hoods.

The HMOR of different alumina-containing LCC at 1370°C are shown in Fig. 7. It is interesting to note that the HMOR of 75% alumina castable is higher than that of the 90% alumina composition. The 75% alumina castables had mullite both in the grains and in the matrix. This provided a much better bonding than that between the corundum aggregates and the matrix in the 90% composition.

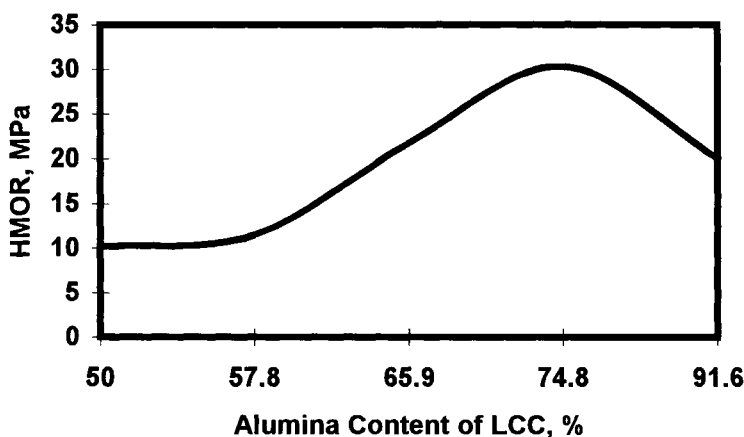


Fig. 7. HMOR of LCC with varying alumina contents.

Figure 8 shows the creep deformation of 60% and 65% alumina low-cement castables. It was evident that the creep deformation of the 65% alumina castable was lower since the 60% contained more siliceous phase.

The application of these castables in the blast-furnace cast house (trough, iron and slag runners), steel making, coke ovens, reheat

furnaces, foundries, petroleum and aluminum industries was also mentioned in the report.

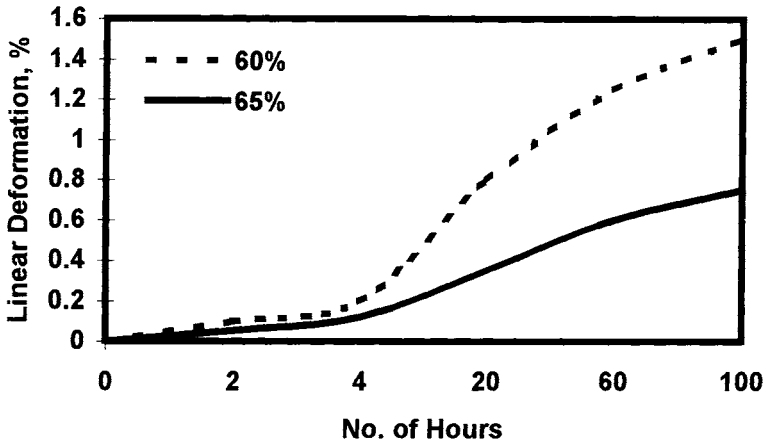


Fig. 8. Creep deformation of 60% and 65% alumina LCC at 1425°C under 0.17 MPa load with time.

Further work¹⁷ was described in the development of LCC with significantly improved hot strength achieved by replacing superfine silica particles with ultrafine chromium oxide. Since the use of superfine silica particles forms low-melting phases, (such as anorthite) the hot strength falls drastically at 1500°C. The use of chromium oxide in place of silica powders, forms a high-melting solid solution between Al_2O_3 and Cr_2O_3 . Figure 9 shows the variation of hot modulus of rupture with temperature of microsilica and fine chromium oxide-containing low-cement castable with fused alumina base. The studies also describe work on ultralow-cement castables (ULCC — with CaO contents of 0.3% and their improved high temperature properties) compared to LCC primarily because of the lower CaO content.

A study of LCC using silica fume, and/or fine superground calcined alumina was also described¹⁸ using 80% and 90% (new) calcium aluminate cements in an aggregate distribution derived from Furnas. It showed that the sag deformation at 1760°C was far less with fine alumina than with silica fume. The 90% cement castable was even better than the 80% cement castable with alumina added and without any silica fume.

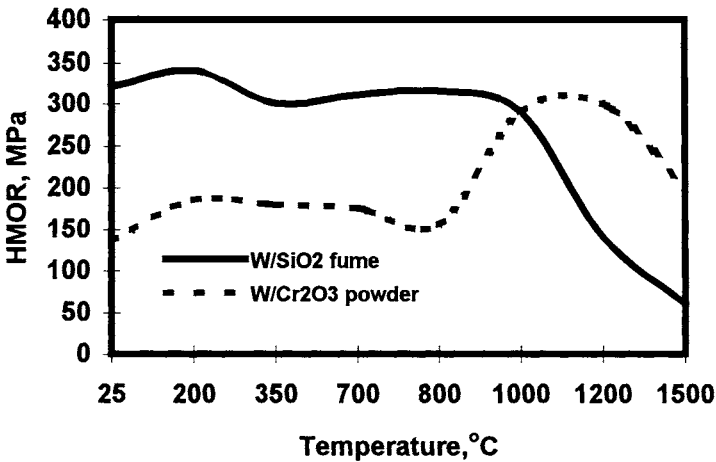


Fig. 9. Variation of HMOR with temperature of a fused alumina-base castable containing microsilica and fine chromium oxide.

The mineralogical analyses¹⁹ showed the hydration characteristics about the effect of microsilica in LCC which clearly reflected the physical properties and mineralogical composition of refractory concretes. It showed that partial replacement of cement by microsilica improved the properties of the castables significantly without decreasing the refractoriness (load-bearing capacity). The improved strength, as postulated, may have been due to the lower degree of hydration and conversion during curing and drying. The CA and CA₂ in the microsilica-containing castable were only partially hydrated at all

temperatures. Figure 10 shows the intensities of the x-ray diffraction pattern of the mineral phases of an LCC. This was one of the major findings that explains (as mentioned in the previous studies) the continuous increase in LCC strength properties with increasing temperature. This is a reaction unlike conventional castables. The increased strength was also attributed to the distribution of microsilica particles in the voids between the coarse cement and aggregate particles.

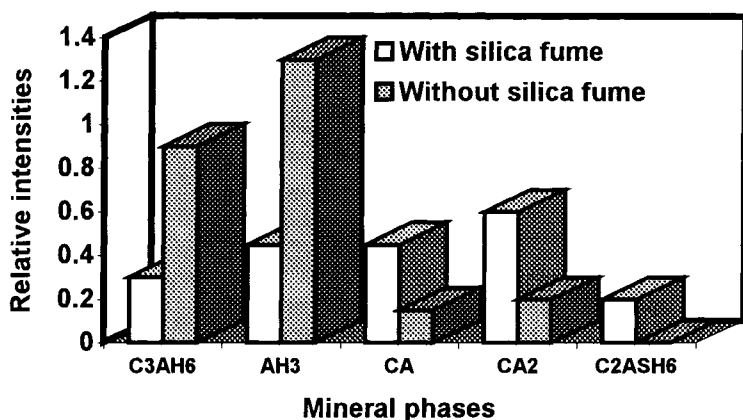


Fig. 10. X-ray diffraction intensities of an LCC showing different mineral phases with and without fume silica.

Silica fume for low-cement castables

Ultra-fine particles such as silica fume (microsilica) in the matrix are used to fill void spaces between the aggregate particles with solid particles. This is the basis of low-cement castables. Besides the chemistry of the silica fume, density has a profound effect on the setting characteristics of the castable. To produce low-cement castable products with consistent flowability and setting time, the density of the silica fume should stay within a narrow range. Higher quality silica fumes have been commercially available in the past few

years. Composition of four silica fumes differing in quality are given in Table 7. These silica fumes reduce the water content of the castables (with different cement contents) as shown in Fig. 11 and the corresponding improvement in properties²⁰ (Fig. 12).

Table 7. Quality of silica fumes chemical analysis.

Silica fume type	D	E	F	G
Oxides	%	%	%	%
SiO ₂	95	95	98	98
Al ₂ O ₃	0.4	0.3	0.4	0.3
Fe ₂ O ₃	0.08	0.08	0.05	0.07
K ₂ O	0.5	0.5	0.2	0.3
Na ₂ O	0.05	0.09	0.02	0.02
CaO	0.2	0.3	0.2	0.2
MgO	0.2	0.2	0.1	0.1

Silica fume is also available in regular and densified forms. Although the densified form tends to use less volume and reduces the water content, it has been proven not to be very effective in regular castable production. The densified particles form agglomerates that do not break down into the smaller size easily. Consequently, the primary benefit of the silica fume is not fully realized. A direct correlation has been found²¹ between the soluble soda levels of silica fume and the setting time. The higher the soda content, the shorter is the setting time.

The use of silica fume alone and in combination with finely ground reactive alumina has been studied in detail.²¹ This study was conducted with ratios of silica fume to ultrafine reactive alumina as 100:0, 70:30, 30:70 and 0:100. The ratio was found to have a profound effect on the placement, as well as dried and hot properties.

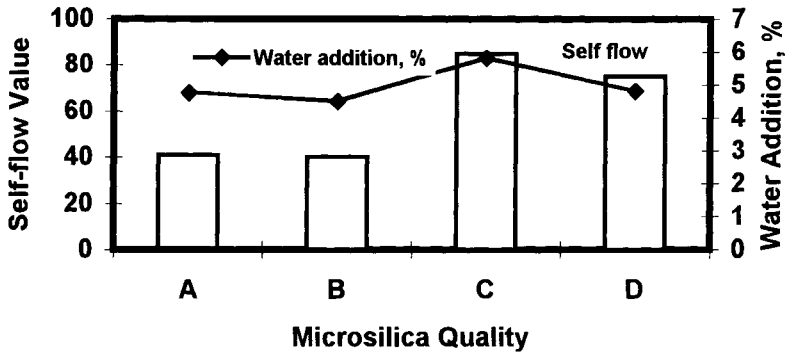


Fig. 11. Variation of flow and mixing water for various qualities of microsilica.

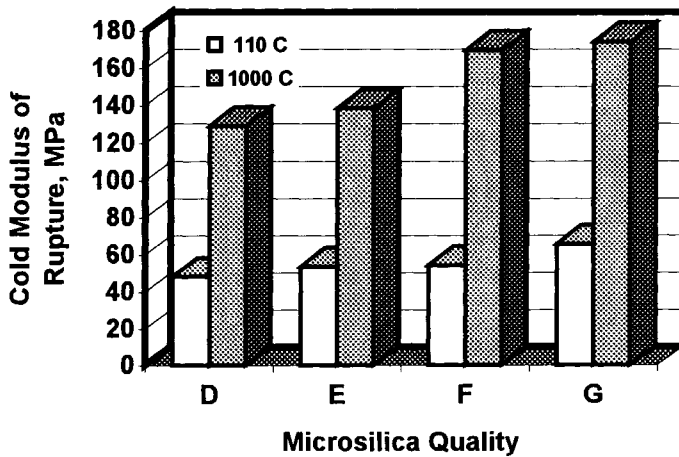


Fig. 12. Variation of cold modulus of rupture with microsilica quality.

As the ratio of silica fume to alumina decreases, the water requirement for placement increases and the physical properties deteriorate correspondingly. The determining factor for the strength is the ratio of silica fume to calcined alumina. At ratios below 70:30 (silica fume : calcined alumina), the strength decreases markedly. The

reason is twofold. First, the increase in the porosity has an inverse relationship to the flexural strength. The second reason is the pozzolanic nature of the silica fume forming hydrates such as C-S-H and C-A-S-H (C = CaO, A = Al₂O₃ and S = SiO₂) that provide strength.

The most significant effect of the ratio of silica fume to calcined alumina is reflected in the hot properties. The HMOR of samples prefired and tested at 1400°C is the highest for the 100% silica fume, decreasing rapidly to 0 for a 30:70 ratio of silica fume to calcined alumina. But with calcined alumina as the only ultrafine material (without silica fume), HMOR shows similar values at 1400°C as of 70:30 silica fume : calcined alumina ratio. These results confirm that the low-cement castable containing 70:30 (silica fume to calcined alumina) shows the optimum placement characteristics as well as dried and hot properties. The high hot strength of 100% silica-fume castables at 1400°C is attributed to the formation of mullite and highly viscous liquid. It has also been reported¹⁷ that a low-cement castable (with silica fume as the only ultrafine material) exhibited significantly higher HMOR at 1500°C where the ratio of silica fume : calcium aluminate cement is increased from 1:1 to 1.5:1.

Properties of low-cement castables

The main characteristic of low-cement castables is a low water requirement that results in higher density and low porosity. Another noticeable characteristic is a strength that increases with temperature. Unlike conventional castables that lose strength in the intermediate range during the dehydration of the cement, low-cement castables actually become stronger during the process. Some typical chemical and physical properties of low-cement castables are shown in Tables 8 and 9 respectively.

It has been shown^{22,23} that the properties of precast shapes made from low-cement castables are far superior when the shapes are fired at temperatures above 1350°C compared to those fired around 500°C.

The above fact has been established by measuring hot-load deformations at 1593°C under a load of 172 KPa for 1.5 h where the high fired shapes have been found to be more stable. This observation can be explained by the fact that the formation of mullite from fine alumina and silica goes through a vitreous phase. The phase transformation is already completed in high-fired shapes, whereas, for low-fired pieces the shapes go through vitreous transformation at the temperatures while under load, thus giving rise to deformation.

Table 8. Chemical composition of low cement castables.

Alumina Class	60	65	70	85	90
Chemical analysis, %					
Al ₂ O ₃	60.4	5.5	68.7	85.3	91.5
SiO ₂	35.5	31.6	28.5	7.9	7.3
Fe ₂ O ₃	1.0	1.0	0.8	1.0	0.1
TiO ₂	2.0	1.2	1.0	3.0	0.1
CaO	1.4	1.4	1.4	1.8	1.4

In this investigation, 70% alumina castable was found to be the most stable because the matrix and the aggregates had perfect mutual bonding and were both mullite. It was remarkable to see that the 95% alumina castable sheared at an angle around 1450°C and did not reach the top temperature. This was the result of a lack of bonding between the tabular alumina aggregates and the mullite matrix. Hence, the castable sheared as soon as the vitreous phase of mullite started to form. This fact corroborates with microstructural analysis of the 70% and 95% alumina castables by optical microscope (Figs. 13a and 13b).

Ultralow-cement castables (ULCC)

The development of low-cement castables has been a significant improvement over conventional castables. It expanded the horizon of applications for monolithic castables. But one major problem still remained. With the presence of highly reactive silica, reaction with the

CA cement forms low-melting phases such as anorthite and gehlinit. In low-cement castables, the CaO content usually varies 0.9–2.5%. This CaO content is still high enough to affect the high-temperature properties of the low-cement castables. The significant strength reduction at 1400°C and above is caused by the formation of viscous liquids from anorthite and gehlinit which increases the pore size of the castable body and contributes to the loss of strength.

Table 9. Physical properties of low cement castables.

Alumina Class	60	65	70	85	90
BD, kg/m ³					
After heating to					
110°C	2480	2560	2640	2930	2995
815°C	2475	2555	2635	2925	2990
1370°C	2470	1550	2630	2925	2990
CMOR, MPa					
After heating to					
110°C	16.5	16.5	16.2	15.2	17.9
815°C	30.3	29.6	28.9	26.0	30.3
1370°C	31.0	31.0	28.6	24.6	31.4
CCS, MPa					
After heating to					
110°C	66.9	66.9	40.6	68.9	111.7
815°C	76.5	106.2	67.6	93.6	126.2
1370°C	92.0	119.3	69.6	75.2	129.6
HMOR, MPa, at					
1093°C	29.2	28.5	27.0	30.3	20.0
1370°C	4.0	6.5	9.3	6.5	15.2
1430°C	1.8	3.2	4.5	-	-

The loss of high-temperature strength can be avoided by the addition of superfine chromium oxide²⁴ or super-ground calcined alumina in place of microsilica (Fig. 9). But the addition of these superfine materials becomes cost prohibitive. Also the addition of chromium oxide is environmentally unacceptable in many countries.

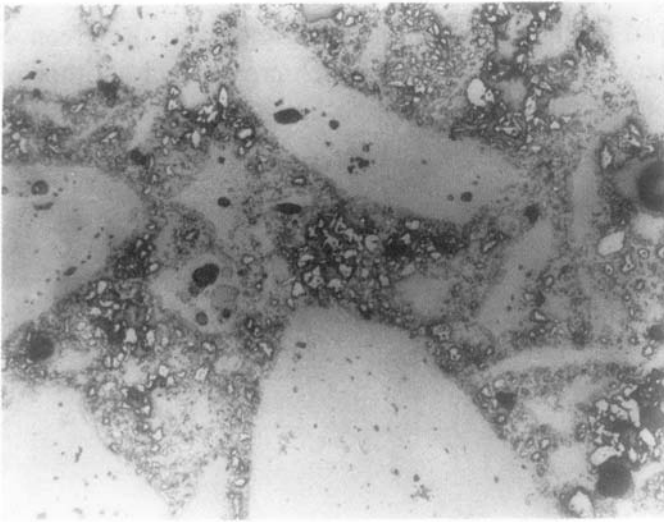


Fig. 13a. Microstructures of 70% Al_2O_3 low-cement castables after hot-load test (1593°C/172 kPa/1.5h).

To improve the high-temperature strengths, attempts have been made to reduce the CaO content further to 0.25–0.8%. In tabular alumina-based ultralow cement castables (ULCC), the addition of reactive alumina to the castable along with microsilica gives a strong mullitic reaction²⁵ from approximately 1300°C. The mullite formation occurs via a liquid phase and produces a network of needles which strengthens the structure at high temperatures. The CaO content of the castable does, however, influence the mullite formation. Mullite is attacked by CaO which inhibits its formation of the former so that a low-cement castable loses most of its strength at 1400°C. Reducing the cement content (CaO – 0.25%) does theoretically lessen the liquid formation but the hot strength does not improve until a minimum of 6% microsilica is present. At this point, the hot strength increases to 12 MPa at 1400°C.²⁰

Lower cement content considerably reduced the green and hot strengths of castables. The increased silica-fume content not only

improved the hot strength but also improved the green strength with its pozzolanic characteristics which created yet another problem, viz., a prolonged setting time that became unacceptably long (several days) for demolding and further handling of the castable.

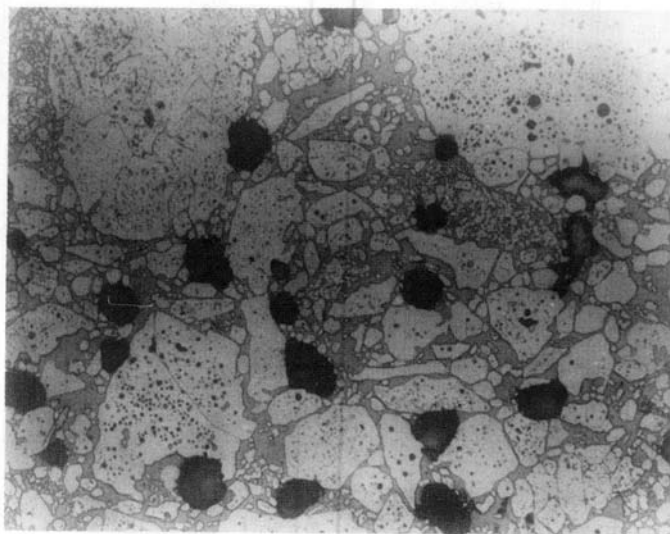


Fig. 13b. Microstructures of 90% Al_2O_3 low-cement castables after hot-load test ($1593^\circ\text{C}/172\text{ kPa}/1.5\text{h}$).

The addition of lime or small amounts of portland cement shortens the setting time but, at the same time, affects the flow characteristics of the castable. The addition of white portland cement (CaO –65% and SiO_2 –24%) has been studied by Myhre.²⁶ Results showed that the flow decreases rapidly with small additions but increases with larger additions. In a bauxite-based ULC castable with low cement content (CaO –0.25%), an addition of 0.2–0.25% white cement seemed to be optimum. It offered good flow characteristics and reduced the setting time to 4.5 h.

Alumina-silicate compounds were added to the above ULCC to find the effect of flow, setting time as well as cold and hot strengths.

Fine ($<75 \mu\text{m}$) andalucite, kyanite, and mullite were added by 2 wt.% substituting calcined alumina. All the doped castables exhibited a low strength at 1200°C but increased significantly at 1300°C . Out of the different alumina-silicates, kyanite showed the highest strengths at 1200, 1300 and 1400°C . Andalucite also performs well between 1200 and 1300°C but loses strength considerably at 1400°C (Fig. 14). Of the alumina-silicates, mullite addition had the least effect on strength, producing the lowest strength of all at 1400°C . With mullite as dopant, the mullite tended to grow as rounded particles instead of the desired needle-like shape. The purpose was to have mullite crystals available (andalucite and kyanite converts to mullite around $1300\text{--}1350^\circ\text{C}$) on which new mullite crystals could grow. This growth forms the needle-shaped crystals enhancing the hot strength.

The transformation is also associated with volume increase. There is a volume expansion of 6–8% from andalucite and 20–22% from kyanite upon conversion to mullite. The positive effect on strength of the alumina-silicates may be caused by a combination of two mechanisms. First, the transformation, which may begin below 1300°C in the presence of impurities, acts as a seed for further mullite formation. Second, the volume increase of andalucite and kyanite occurring in the liquid phase will force the liquid to fill in the pores. This movement of the liquid phase may also be beneficial to the solution/precipitation process from which mullite is supposed to develop.²⁷ It shows that kyanite gives the highest strength because of higher volume expansion than andalucite and also that mullite causes the least improvement in bonding because of no volume change.

Effects of additives on LCC

In high alumina cements, introduction of additives²⁸ raised the zeta potential of calcium aluminate cement to a certain concentration. After this point, the zeta potential dropped with further increase in concentration. The materials studied were calcium and sodium lignosulfonates, sodium hexametaphosphate and tetrasodium

pyrophosphate and the phosphates in between. Among the phosphates, the longer the phosphate chain, the more effective was the dispersing agent. It has been concluded that the commonly used additives — such as water reducing agents for high alumina cements — function by being adsorbed on the surfaces of the cement particles. The adsorption may be physical and some chemisorption may also take place.

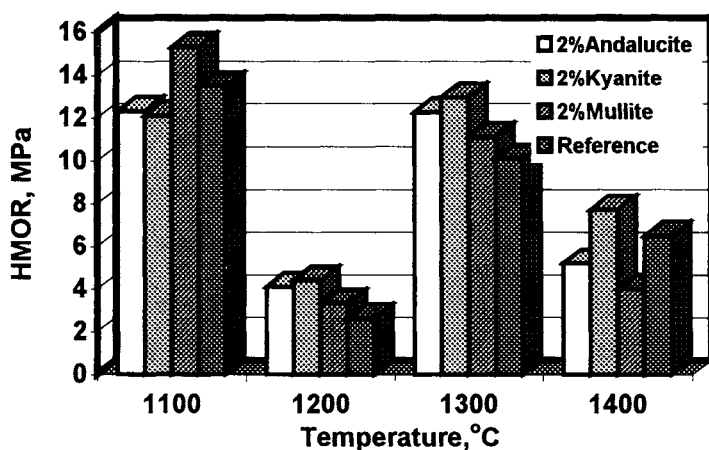


Fig. 14. HMOR of ULCC using aluminosilicate minerals.

The effects of calcined alumina and deflocculants were investigated²⁹ in a high-alumina LCC. The comparative result of regular-ground calcined alumina and reactive alumina were tested with the deflocculants. The deflocculants used in the investigation were Calgon (sodium hexametaphosphate), Empiphos (sodium tripolyphosphate), Darvan 7S (sodium polymethacrylate) and Darvan 811D (sodium polyacrylate). In all cases the reactive alumina resulted in better flow with all the deflocculants at all levels. The water content was kept constant at 4.5% for the tests.

Out of the deflocculants, Calgon led to the best flow at all levels. The most optimum addition was 0.05% (total weight basis) particularly with the reactive alumina. Neither Darvan 811D nor Darvan 7S provided workable flow below 0.1% and their flow values were lower than Calgon across the board. As expected, the physical and thermal properties correlated with the flow properties. The one exception was Empiphos, which showed higher porosity and lower density compared to the rest of the castables. Both Calgon and Empiphos exhibited similar hot strengths at 1200°C but Calgon showed the highest strength at 1400°C. As expected, the reactive alumina compositions showed better hot strength for all the deflocculants than those of calcined alumina.

The effects of oxides and salt additions using different kinds of commercially available calcium aluminate cements were looked at in a comprehensive study.³⁰ A single lot of cement was used to determine the effect of various salt additions at different concentration levels. The addition of LiCl has been found to have the most accelerating effect on the set time. Although the initial set times were only reduced from 50 to 15 min by increasing LiCl addition from 0 to 0.9%, the final set was reduced significantly from 158 to 20 min by the same addition.

The effect of other salt additions with magnesium chloride, barium chloride, lithium carbonate, zirconyl chloride, ammonium chloride, magnesium hydroxide, sodium hexametaphosphate, potassium chloride, sodium chloride, zirconyl nitrate, sodium citrate, and chromium chloride on the set time were plotted according to the following equation:

$$\ln t \rightarrow m X^{1/2} + B$$

where t = set time
 X = salt addition, percent
 B = constant.

The plot of $\ln T$ against the square root of the additive concentrations was linear (Fig. 15) for all the salt additions. The accelerating effect of the different salt additions varied as lithium chloride >>> magnesium chloride > barium chloride > lithium carbonate > calcium chloride > zirconyl chloride > calcium hydroxide > ammonium chloride > magnesium hydroxide > barium chloride. The retarding effect of the different salt additives were in the order as sodium hexametaphosphate < potassium chloride < sodium chloride < zirconyl nitrate < sodium citrate < chromium chloride. The ultrafine particles of alumina (A-17) and silica (silica fume) seem to have a retarding effect on the set times. In this characteristic plot, setting accelerators produced negative slopes and retarders positive slopes.

The deflocculating mechanism may vary³¹ between an inorganic- and organic-type dispersant. The most widely used dispersants are sodium hexametaphosphate (inorganic) and sodium polyacrylate (organic). Laboratory studies indicate that phosphates might adhere to the microsilica surface, more or less cleaning impurities from the surface, whereas, high molecular-weight organic molecules might actually curl up between the particles, keeping them apart. The effect of the dispersants is only observable after the salts are dissolved in water. The dissolution produces ions that find oppositely charged particles in the cement. The microsilica, keeping them apart, produce movement of the particles which, under the vibrational energy, causes the mix to flow.

Cement-free castables

The development of low- and ultralow-cement castables provided a significant step forward for monolithic refractories. The new castables provided longer life, easier and quicker installation, and reduced the refractory cost to the users. However, addition of microsilica to calcium aluminate cement producing lower-melting phases prohibits the refractories from high-temperature applications. The combination

of silica with the CA cement produces anorthite and gehlinites, which have low melting phases in the temperature range 1380–1450°C.

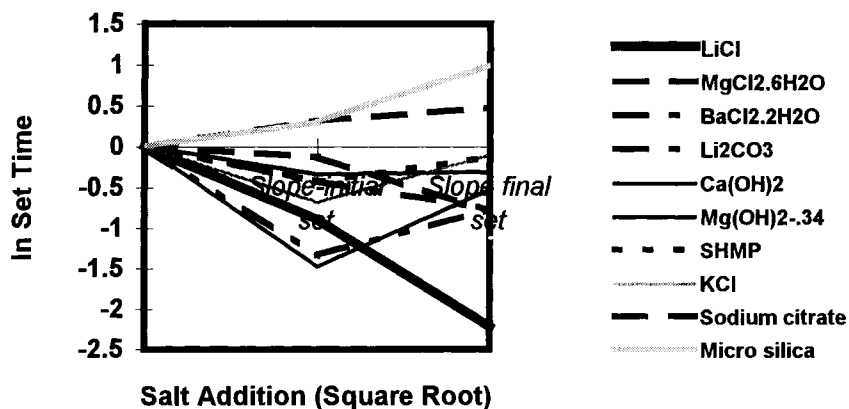


Fig. 15. Variation of setting time with different additives.

The development of a hydratable alumina binder³² provided a means to improve high-temperature properties for the low-moisture castables. The hydratable alumina particles are basically amorphous (x-ray patterns show only a few diffused bands). The particles are extremely fine with a surface area of more than 200 m²/g. Since the CaO content of the hydratable alumina is <0.1%, this no-cement binder can be used in conjunction with microsilica without affecting the high-temperature properties. Like calcium aluminate cement, the hydratable alumina develops strength through hydration. But unlike CA cement, the cast refractory does not need to be in humid conditions to develop strength. The hydratable phases are bayerite and boehmite (gel). Formation of bayerite (only at temperatures below 25–30°C) is more desirable. This is achieved by the addition of alkali metal salts.

The development of this binder centered on the use of tabular alumina as the aggregate system, using reactive alumina (A-3000 FL

and A-1000 SG) to provide better flow properties and hot strength. Potential binder contents are recommended in the range 3–7%. The hydratable alumina containing castables behave like calcium aluminate cement castables, requiring similar quantities of water and the same type of mixers. Generally, dense castables using the hydratable alumina require additional time to wet after water addition is made (two minutes dry mixing followed by five minutes of wet mixing). During mixing, the compound will climb and stick to the mixer just prior to the wetting out. Manufacturers should avoid the typical inclination to add more water to accelerate wetting and flow beyond optimum levels of casting. After mixing, the working time ranges from 40 to 100 min. Longer air curing does not improve the strength to any appreciable extent.

Vibration levels and times used for vibratable LCC containing high alumina cement exhibiting thixotropic casting behavior should also be applied to hydratable alumina-containing castables. To avoid crazing or dusting, cast parts should be covered with plastic. Some drying in the mold may be needed to develop suitable strength prior to demolding for firing. Drying at 110°C increases the strength significantly.

To avoid explosive spall and achieve an optimum strength development, the dewatering process must be understood. The thermogravimetric analysis of the hydratable alumina is given in (Fig. 16). The most rapid water loss peaks at 140°C and slows at 300°C. Hydratable alumina-containing castables with microsilica exhibit very low permeability at these temperatures. Although small amounts of water vapor still evolve at temperatures from 250°C to 650°C, it will not contribute to explosive spall. Hence it is recommended that precast shapes be heated above 650°C where intolerance to steam is warranted. The addition of organic fibers that melt or shrink at temperatures around 150°C should be used to vent the matrix and minimize spalling tendencies.

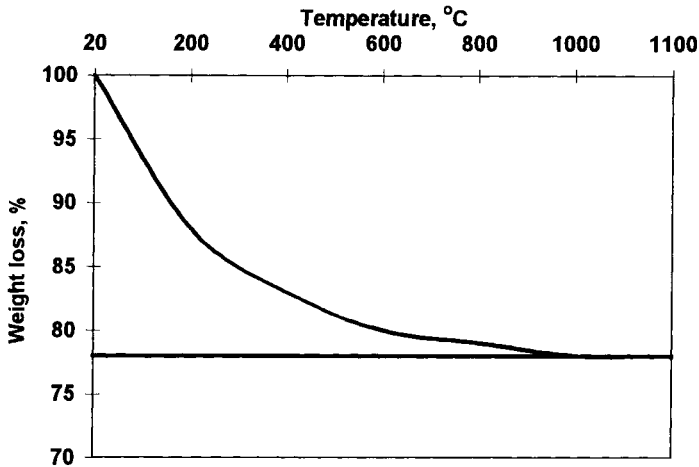


Fig. 16. Thermogravimetric analysis of hydratable alumina.

The hydraulic bonding of the alumina binder produces a strength development curve similar to calcium aluminate cements. The addition of microsilica improves the strength after firing at 815°C. Hydratable alumina-containing castables with silica fume develops acceptable bond strength after firing above 1100°C, whereas the temperature needs to be above 1400°C to develop acceptable bond strength in reactive alumina. Hot strengths are also improved by the addition of microsilica. Hydratable alumina binder can be used in either pure alumina systems without any microsilica or with microsilica in aluminosilicate systems. The composition and typical properties obtained from a tabular alumina-based composition with hydratable alumina are shown in Tables 10 and 11.

A comparative study has been done³³ between a high-alumina cement and a hydratable alumina binder on a bauxite base vibratable castable. Two compositions were made with the hydratable alumina

binder — one with microsilica and the other with reactive alumina for the sake of comparison. All the castables were cast with constant water content of 5.6%. Table 12 shows the chemical compositions and Table 13 shows the physical properties of the castables.

Table 10. Composition of hydratable alumina-based castables.

	Mix 1	Mix 2	Mix 3
Tab. Alumina -6 mesh	75	16	16
Tab. Alumina -325 "	10	10	10
Cal. Alumina A-3000FL	5	5	5
Cal. Alumina A-1000SG	5	—	—
Silica Fume	—	5	5
HAB (Alphabond 100)	5	5	—
HAB (Alphabond 200)	—	—	5
Casting water	5.4	5.4	5.4

Table 11. Physical Properties of hydratable alumina-based castables.

	Mix 1	Mix 2	Mix 3
CMOR, MPa after 110°C	14.3	11.6	10.1
CMOR, MPa after 815°C	3.5	13.1	10.8
CMOR, MPa after 1370°C	31.5	44.3	34.5
HMOR, MPa at 1500°C	3.5	6.8	6.1

The advantages of using hydrated alumina binders include the reduction of microsilica as well as better green strength while retaining other refractory properties. Microsilica inhibits the hydraulic bond formation in the castables using the hydrated alumina bonds, but improves the flow and workability of casting mixes. HAB 1 with 1% microsilica resulted in the best combination of flow and workability, green strength, and refractory properties. In contrast, in the castables containing HAB 2 with 5% microsilica addition, the hydraulic bond is

no longer inhibited. The result is better green strength and improved flow and hot strength.

Table 12. Bauxite-based castable formulation.

Material	LCC	NCC	NCC-1	NCC-5
Cal. Bauxite, 3.35-0 mm	80	80	80	80
Cal. Alumina	10	10	10	10
Reactive alumina	—	5	4	0
CA cement (71% alumina)	5	—	1	5
Alumina binder	—	5	5	5
Microsilica	5	—	—	—
Empiphos, STPP	0.1	0.1	0.1	0.1
Water, %	5.6	5.6	5.6	5.6

Table 13. Summary of physical properties.

Castable	Micro-SiO ₂ , %	CCS MPa	HMOR MPa @	HMOR MPa, @	HMOR MPa @	HMOR MPa, @
		110°C	1200°C	1500°C	1200°C	1500°C
Al. Bond1		60.6	20.0	87.0	2.1	3.9
Al. Bond 1	1	85.0	85.7	106.0	8.3	4.0
Al. Bond 1	5	50.0	135.0	105.0	28.7	4.5
Al. Bond 2	—	33.1	15.5	96.5	2.0	4.0
Al. Bond 2	1	36.0	75.2	95.0	7.3	3.5
Al. Bond 2	5	55.0	129.8	125.0	28.5	4.0

Proper setting, curing and firing methods determine the success of the products made with hydratable alumina. In handling large precast shapes, the mold should be placed inside the furnace before setting or the shape should be left in the mold long enough to develop sufficient strength for handling. The strength development as a function of temperature in a high alumina composition is shown in Fig. 17.³⁴

The data in Fig. 17 was gathered from a laboratory specimen bar after curing for 24 h. The strength for a particular castable composition varies according to the ambient conditions, size of part, and mold design.

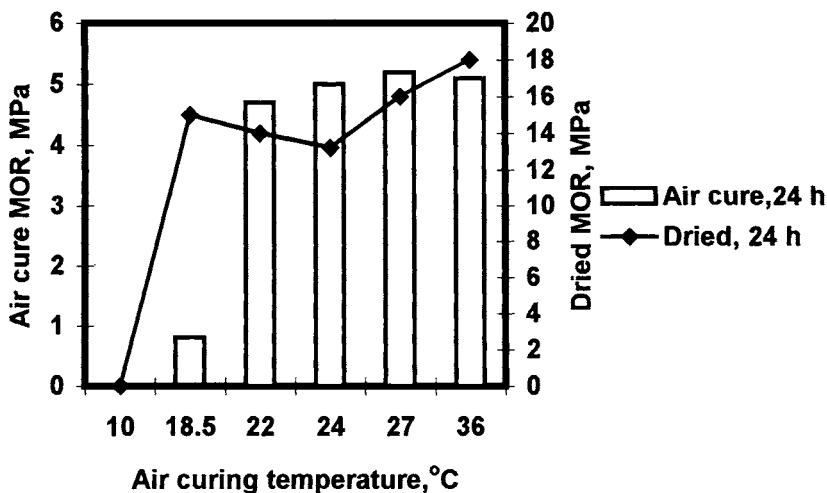


Fig. 17 Strength development of dried versus air-curing temperature of high alumina/HAB composition.

One of the greatest concerns about using HAB in castables has been the drying and firing of in-place installed castables. Compositions with microsilica in particular have very low permeability especially at low curing temperatures. The addition of organic fibers and small amounts of aluminum powder have been added to ease the release of steam. For large shapes and thick installed linings, extreme precaution must be taken to prevent explosive spalling. This is still one of the major disadvantages of using HAB in castables.

Castables containing HAB have been successfully used for injection lances in steel-ladle furnaces. They last longer than calcium

aluminate cement containing compositions and show less erosion in the slag environment. These castables are also successfully used in the delta section of electric-furnace roofs, showing longer life and less susceptibility to slag erosion with occasional splashes from the electric furnace. Of course, extreme care needs to be taken in the drying out of the delta section since these are usually very thick in cross section and are prone to explosive spalling.

In a recent study³⁵ it was observed that both cement-free and LCC, cement-containing castables have good resistance to high temperature creep but when small amounts of silica fume was added, the cement-containing castable showed significant creep deformation whereas there was no creep deformation observed with cement-free binders. This observation agrees with the behavior of the low-melting phase formed in the $\text{CaO-Al}_2\text{O}_3\text{-SiO}_2$ system responsible for the deterioration of the creep resistance.

Alumina-magnesia spinel LCC

It has been only in the past few years that high-alumina castables³⁸ containing alumina-magnesia spinel appeared on the market. Magnesia-alumina spinel is the only high-temperature stable phase in the magnesia-alumina system. $\text{MgO} \cdot \text{Al}_2\text{O}_3$ has high refractoriness with a melting point of 2135°C . It is available in both sintered and fused forms. The stoichiometric composition of magnesia-alumina spinel is MgO -28.3% and Al_2O_3 -71.7% by weight. It is available either in the magnesia-rich form (up to 32 wt. % MgO) or in the alumina-rich form (up to 95 wt. % alumina). The commercially available spinels are usually of high purity (<1.0% impurity). The impurities in the spinel are concentrated in the grain boundaries (Fig. 18).

The spinels used for castable compositions are usually of the stoichiometric composition or in the high alumina compositions since the magnesia-rich compositions tend to create problems of hydration of the magnesia. The magnesia-rich spinels are commonly used in brick making. Alumina-magnesia spinel castables contain about 20–

25% spinel usually in the fine grain sizing. Typical properties of an alumina-magnesia spinel castable are shown in Table 14 and the hot modulus of rupture is shown in Table 15 and Fig. 19.

Table 14. Properties of alumina-spinel low-cement castable.

Chemical Analysis, %	Prefired to, °C	BD, kg/m ³ / Por, %	MOR, MPa	CCS, MPa	PLC, %
Al ₂ O ₃ -92.8	110	3040/12.9	9.5	29.6	0.0
MgO-5.8	955	2995/20.0	7.9	28.8	0.1
Fe ₂ O ₃ -<0.1	1370	2995/18.7	30.0	76.2	0.09
CaO-1.2	1550	2995/18.2	30.5	85.2	-0.06

Table 15. HMOR of alumina-spinel low-cement castables after prefiring to different temperatures.

HMOR, MPa	At 1370°C	At 1480°C	At 1540°C	
Prefired to, °C				
110	13.6	12.3	9.3	
1400	14.8	13.9	9.7	
1540	21.3	17.5	16.3	

The alumina magnesia spinel castable is a pure system. It is highly intolerant to the presence of free silica. The presence of silica affects the high-temperature properties significantly due to the formation of the low-melting phase cordierite.

Figure 20 shows the thermal expansion characteristics of an alumina-spinal castable demonstrating its high stability. The thermal expansion curve shows that sintering begins at 1300°C when it starts to shrink. After shrinking up to a temperature of 1500°C, it begins to expand again. Mineralogical analysis of the specimen fired at 1600°C indicates that the major phase is spinel. Calcium hexa-aluminate and corundum are the minor phases. Calcium aluminate cement reacts aggressively with the free alumina to form significant amounts of

calcium hexa-aluminate leading to the densification of the castable structure.

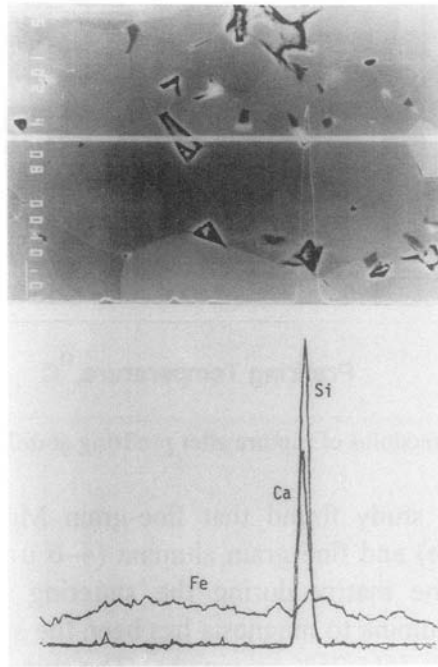


Fig. 18. Electron-microprobe-analysis composition image and the line-analysis profile of impurities of synthetic spinel. (Courtesy of Onoda Cement Co., Japan).

A recent study³⁹ indicates the performance of an alumina-spinel castable in steel ladle can be improved by the addition of fine grain magnesia and alumina which results in the formation of spinel *in-situ* instead of the addition of spinel grains. Peeling and cracking have been observed at the hot face of steel ladles at work temperatures and longer holding in between heats which decreases the service life of the ladle.

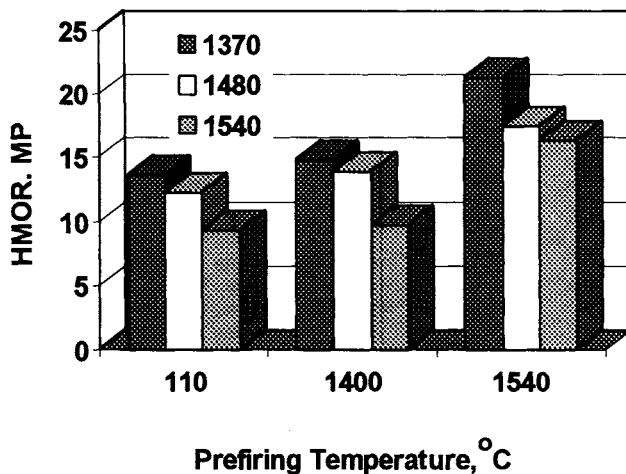
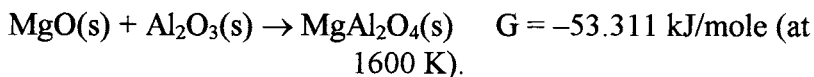


Fig. 19. Hot modulus of rupture after prefiring at different temperatures.

The above study found that fine-grain $\text{Mg}(\text{OH})_2$ (used as the magnesia source) and fine grain alumina ($4\text{--}6\ \mu\text{m}$) was ideal to form spinel within the matrix during the sintering process. The ideal proportion of alumina to magnesia has been the stoichiometric ratio of $\text{Al}_2\text{O}_3:\text{MgO} = 1, 75\%:25\%$ by weight. The spinel formation is caused by the chemical reaction



The hot modulus of rupture, refractoriness under load and spinel formation have all been optimized at 25% of MgO. The castable behaved adversely when spinel grains were added in the mix along with alumina and magnesia. The reaction process began at 1300°C and continued through sintering at 1500°C . The introduction of this composition (with 75% Al_2O_3 and 25% MgO) resulted in less cracking and peeling while the service life increased by 20%.

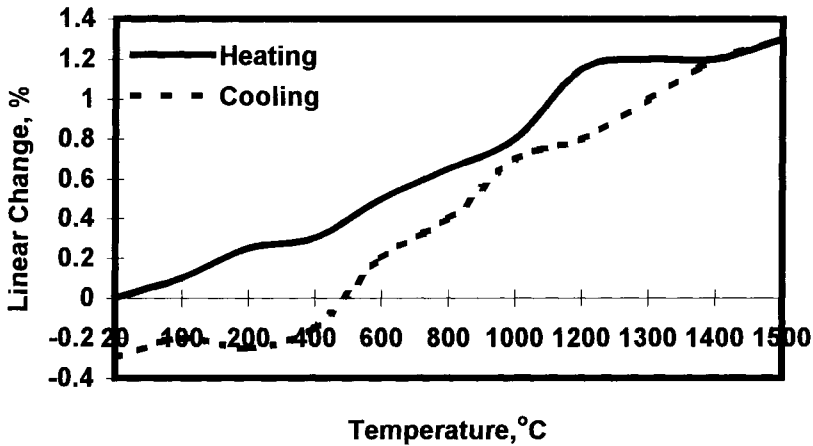


Fig. 20. Thermal expansion curve of alumina-spinel castable.

Basic low-cement castables

Cement-free MgO castables have been developed⁴⁰ using microsilica as the bonding phase. These castables have good casting characteristics and high strength at low temperatures. At higher temperatures, however, the strengths are considerably decreased. To increase the hot strength, the reaction between MgO and SiO₂ producing forsterite has been used. Earlier attempts in using AlF₃ and Al₂O₃ were not successful because refractoriness was lowered due to the formation of cordierite. Silicon nitride and Si metal have been found to be suitable bond-phase modifiers and improved the hot strengths considerably.

Users have observed that a finer distribution of particles needs less water but high temperature strength is better with coarser distribution. Calgon and Venisperse CB were found to be equally effective as the dispersing agent but hot strengths with Calgon were lower than those of Venisperse CB (probably due to the presence of Na in Calgon). Hot modulus of rupture at 1400°C using the two

dispersants with different amounts of Si metal and silicon nitride additions are shown in Figs. 21 and 22.

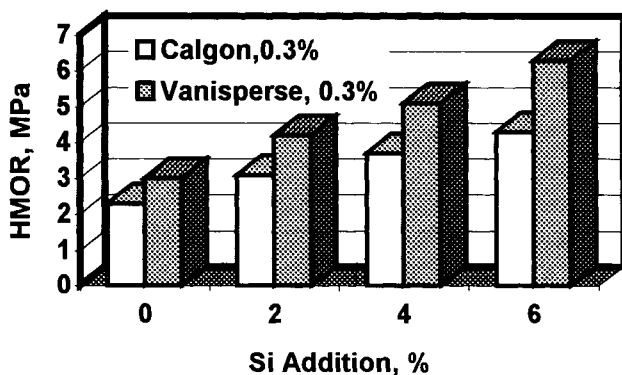


Fig. 21. HMOR at 1400°C (20 h hold) with Si addition using Calgon and Vanisperse CB (0.3%).

Samples prepared with 4% and 6% Si and also with 4% and 6% silicon nitride additions usually developed a black core inside the bars. The formation of the black core was attributed to the unoxidized portion of Si metal and the decomposition of silicon nitride into metallic Si and nitrogen. The hot modulus of rupture was, however, higher with samples having the black cores. Heating the specimens at 1250°C for 12 h prevented the formation of black cores.

Although the HMOR showed a minimum value at 1000°C, it progressively increased at higher temperatures particularly with higher silicon nitride addition. Figure 23 shows hot modulus of rupture at different temperatures and silicon nitride contents (all specimens were pre-fired at 1250°C for 12 h).

Magnesia-containing castables have been reported⁴¹ to be useful in preventing slaking of magnesia in contact with water. They also provide resistance to basic slags in steel-ladle furnaces. Magnesia clinkers were produced by adding Al_2O_3 , SiO_2 , TiO_2 and ZrO_2 to a

98% MgO base material. The amount of the additive remained at the level of 2–8%. The slaking resistance of the resulting clinker was determined by forming samples with various sizes of the clinker into a castable with 6% water. An autoclave was used to subject the prepared specimens to 110°C at 0.14 MPa for 24 h. The deterioration of the specimens was determined qualitatively by visual examination. Slag-penetration resistance was determined in a rotary slag furnace at 1650°C for 3 h using a slag composed of Al_2O_3 – 20, SiO_2 – 15, Fe_2O_3 – 10, CaO – 45, MgO – 5 and MnO_2 – 5%.

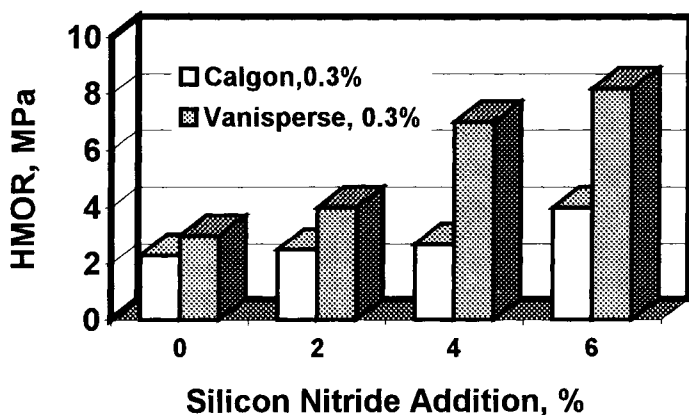


Fig. 22. HMOR at 1400°C (20 h hold) with SiN addition using Calgon and Vanisperse CB (0.3%).

The addition of 2–4% Al_2O_3 provided the best resistance to slaking; additions of higher amounts led to cracking and disintegration. Since addition of Al_2O_3 , SiO_2 , and ZrO_2 , SiO_2 is known to reduce slag penetration, the effect of the addition of these materials was studied for slag-penetration resistance and volume stability. Fused silica was used as the SiO_2 source, calcined alumina as the source of Al_2O_3 and natural zircon as the source of ZrO_2 . The amount of the

additives was 10 wt.% and the sizes were medium (<1 mm) and fine (<0.63 mm). It was concluded from the slag test that medium- and fine-grain fused silica and fine-grain alumina abated the problem of slag penetration effectively because they dissolved in the slag and increased the viscosity of the slag.

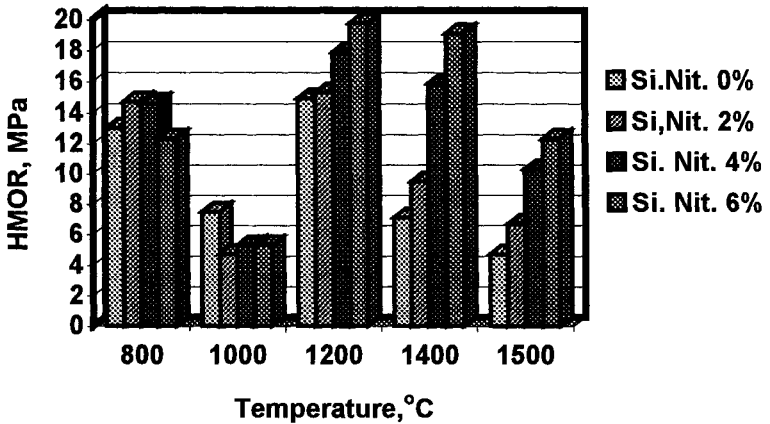


Fig. 23. Hot modulus of rupture at different temperatures and silicon nitride contents (specimens prefired at 1250°C for 12 h).

Large shapes based on magnesia castables have been made⁴² for applications such as electric-furnace runners and basic oxygen furnace tap-hole blocks. Initial castables were made by adding white fused alumina to magnesia-based castable (98% MgO with a CaO:SiO₂ ratio of 1:1) using phosphate as the binder. The hot strength of the castable was lower and addition of alumina made it worse. The next generation of castables was developed using calcium aluminate cement and without adding any low-melting additives. Results have been reported with the addition of 0–50% alumina in the castable with 5–5.3% water added for casting. These castables showed much better corrosion resistance to basic slags although they were not as good as all-magnesia specimens. The addition of alumina also improved

structural spalling and thermal shock resistance. The hot modulus of rupture improved with alumina addition but started to go down with alumina contents above 25% as shown in Fig. 24.

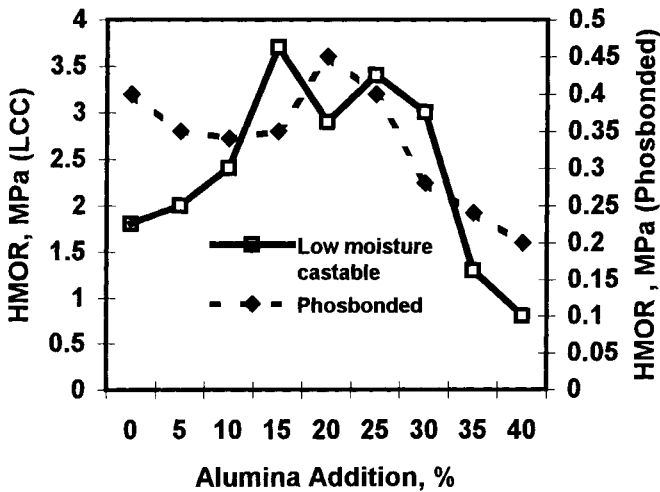


Fig. 24. Variation of HMOR at 1370°C with alumina addition.

Drying aid for castable refractories

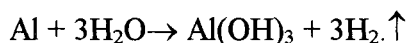
Drying aids are important for calcium aluminate cement-containing castables. Significant care must be taken in drying CA cement-containing castables. Although curing temperatures have significant effect on the formation of the different phases, the drying process needs to be carried out slowly even when the curing temperatures are in the range of phase formation less susceptible to explosive spalling. During hydration of the CA cement, $\text{Al}(\text{OH})_3$ gel is formed, thereby reducing the permeability of the castable and causes it to spall or explode.

Attempts have been made to reduce the spalling problem by increasing the permeability with channel-forming agents such as synthetic fibers⁴³ and gas-forming agents like Al powder. The

synthetic fibers are based on polyethylene or polypropylene that can provide fibers of small diameters ($15\mu\text{m}$) and customized to 5–6 mm lengths. These polymers have unique characteristics suitable for the intended use. The polymers shrink to 50% when heated to 60–66°C. This shrinkage opens channels and helps drying at temperatures well below the boiling point of water. The polymers absorb only 0.1 wt. % water, so their use does not increase the water requirements of the castables.

For most applications, fiber additions of 0.2–1.6 vol. % have effectively improved castable refractories' resistance to steam spalling. The fiber addition also minimally affects the density/porosity and strengths of the castable refractories. Even in metal contact-area uses, addition of fiber has been found to be beneficial. When applied in molten aluminum contact refractories, fiber-containing refractories helped in better non-wettability characteristics compared to similar refractories without any fiber addition. Addition of organic polymer fiber is now being widely used in monolithic refractories as an aid to successful drying.

Addition of metal powders (particularly aluminum) has been used with organic fibers as an aid in drying dense castables. Aluminum powder reacts with water at ambient temperature as follows:



The hydrogen gas formed escapes from the castables producing small channels. These channels then help steam leave the castable during the drying process. The $\text{Al}(\text{OH})_3$, on the other hand, reacts with microsilica at low temperatures producing aluminosilicates.

But the use of aluminum powder has some inherent problems. Evolution of hydrogen and its exposure to oxygen in air causes it to produce flames (visible blue color). It can cause serious hazards if the hydrogen gas produced is trapped somewhere in the forming molds, which can cause an explosive reaction in contact with air. Aluminum

powder should only be used where there is plenty of ventilation and minimal chance of evolved hydrogen gas entrapment.

Application of low- and ultralow-cement castables

Since the development of LCC and ULCC, a significant number of applications have evolved where once only brick could be used. Now LCC are used because of convenience and superior performance in areas where other monolithic refractories had been the traditional choice.

The uses of LCC and ULCC are numerous. These are currently used in all types of industries from primary and secondary metal production to cement, chemical and hydrocarbon waste incinerators, thermal power plants and other industries.

Iron and steel making

Coke ovens — Fused silica has long been used for costly coke-oven door plugs that suffered from coke penetration causing spalling. The use of 50% alumina LCC has been highly successful in this application. Not only does it cost far less than fused silica compositions but it also prevents the penetration of coke and, hence, lasts longer. Fused silica-based LCC is also used as preformed shapes for repairing flue walls.

Blast furnace cast house — The most important application of ultralow-cement castables has been in alumina-silicon carbide-graphite compositions for trough, slag and iron runners in blast-furnace cast house. The CaO content varies between 0.2% and 0.5%. It has been highly successful in an *in-situ* application where the most important factor is the resistance to blast-furnace slag corrosion and erosion. Optimization of the compositional ingredients is highly critical in the proper performance of the castable. The grain sizing of the components, the optimum amounts and quality of silicon carbide and appropriate selection of the quality and quantity of graphite play

important roles in the performance of the castable. These castables, primarily based on fused alumina grains, are dense (14–17% porosity) and susceptible to explosive spalling and cracking.

Various attempts have been made to ease the drying out of the castable avoiding spalling and cracking by adding metallics, organic fibers and a combination of both. The addition of metallic powder (mainly aluminum) (properly vented), and addition of organic fibers that shrink and melt during the drying process have been used extensively. Although organic-fiber addition helps in the drying process, it does not completely eliminate the problem. For this reason, other methods have been introduced to avoid the dry-out problems (i.e. using venting tubes along the trough⁶⁶ for letting out the evolved steam).

Tilting runner — Low and ultralow cement castables have been used in tilting runners with good results. The design of the tilting runner needs to be properly adjusted, particularly near the neck, to avoid cracking. The impact area can be patched repeatedly to obtain longer life of the refractory.

Steel ladle — LCC have been used in steel ladles for many years in Japan and Europe but with little success in the United States. A long dry-out process, resulting in delayed turnaround time, has been the major impediment to using the LCC in steel ladles, particularly in the U.S. Impact pads and well blocks with or without porous plugs are made from precast LCC extensively throughout the world. In modern ladle furnaces LCC are used for making gas-purging lances with or without zoning with different compositions.

Continuous-casting tundish — Low cement castables have been used in continuous-casting tundishes in back-up lining for the past several years. In some special steel making, low-cement castables are used as the working lining as well. Depending on the number of heats per sequence in the tundish, the back-up lining life can vary from 300 to 2,000 heats per installation with some intermittent repairs.

Precast shapes from low-cement castables have used as flow-control devices, such as impact pads, dams, weirs and baffles. Tundish pocket blocks and covers are also made from LCC.

Non-ferrous metals

Aluminum — In the metal industry, the second most common use of LCC is in aluminum production, particularly in the secondary process. LCC has been used in the bottom and sidewalls in direct contact with molten metal. With certain chemical additives, low cement castables can be made non-wetting and non-penetrating to the molten aluminum in direct molten-metal contact areas. LCC is also used in troughs, primary and secondary transfer ladles, and filter boxes.

Copper — Various compositions of LCC including alumina spinel castables have been used in different parts of copper brazing furnaces with good success.

Industrial applications

Hydrocarbon processing industries — There has been significant use of LCC in petrochemical and related industries since its development. Except for cyclones, LCC is used in almost every area of the industry either as a monolith or precast shape. Great advantage and longer life have been achieved, especially in the risers where primary cracking of the oil takes place. The ease of installation, quick turnaround time and longer life have helped the industry to a great extent.

Waste incinerators — Various LCC compositions have been used in waste incinerators depending on the type and design. Compositions range from high silicon carbide to high alumina to medium alumina depending on the area and the type of exposure.

Cement kilns — One highly successful application of LCC has been in the drying and cooling sections of rotary cement kilns.

Insulating castables

Since the prohibition of asbestos for insulation purposes, ceramic fiber wools, blankets and boards have been extensively used for industrial and domestic insulation purposes. Apart from the higher cost, there are specific areas where the use of ceramic fibers is not possible. In these cases, insulating castables have been used as pouring castables or gunning castables.

Insulating castables play more important role in industrial applications than in the production of iron and steel. Hydrocarbon industries (petrochemical refineries) are greatly concerned about conservation of energy while at the same time striving to preserve the strict requirements of the process parameters.

Insulating castables are needed for a wide range of temperatures. The composition and formulation depend upon the specific needs of each application. A variety of insulating raw materials are now used -- insulating brick grogs with various temperature ranges, vermiculite, haydite, perlite, Verilite[®], and bubble alumina. For low temperature insulation, vermiculite or haydite and low-purity calcium aluminate cements are used along with some clays. The highest hot-face temperature of these castables may be up to 1000°C. Perlite can be used in combination with medium-purity CA cement for temperatures up to 1400°C at the hot face. Verilite[®] along with a high-purity cement can be used up to a hot face temperature of 1550°C. Bubble alumina, a high cost material, is only used with high-purity CA cement for temperatures above 1600°C. Since the development of LCC, small amounts of fumed silica are usually added to provide better cohesion of the mix. These castables are often installed by a pump since the insulating castables light weight and do not require high strength.

A recently developed pumpable insulating castable⁶⁷ has been found to be superior to the previously existing insulating castables. Instead of calcium aluminate cements, colloidal silica is used as a

binder and provides significant advantage in use over a wide range of temperatures and ease of application.

Characteristics of insulating castables

Thermal conductivity, a measure of insulating properties, is directly related to the density of the castable. The density is controlled by the amount and type of lightweight aggregates used in the formulation. The choice of lightweight aggregates depends primarily on the hot-face temperature where insulating castables are to be exposed. In the majority of cases, except in the hydrocarbon industries, the insulating castables should withstand the hot-face temperature and be thermally stable.

The hydrocarbon industries demand more than just the insulating characteristic of the castables. They also need better dimensional stability, better strength characteristics and higher abrasion resistance. In the past, hydrocarbon industries used a dual lining — one was an insulating layer and the other a heavier and stronger abrasion-resistant working layer. The use of the double layers often created problems causing premature shutdown of the process.

Since the development of the low-cement castable technology, dual-layer systems have been eliminated with single layer refractories that provide both the needed insulation as well as the strength and abrasion-resistance properties. The development of these castables was even more important since the catalytic cracker units have been replaced by risers, where the cracking of the oil occurs. Table 16 illustrates the refractory requirements of petrochemical refineries (such as Exxon).

Refractory castables for aluminum industry

Molten aluminum behaves quite differently from molten iron toward alumina and alumina-silicate refractories. When refractories are

exposed to molten aluminum at around 815°C for a period of 72 h, aluminum reacts with the refractory at the point of contact and the molten aluminum gradually gets into the refractory.

Table 16. Refractory properties for petrochemical industries, measured at room temperature after firing to 815°C/10h.

Material (Type)	Dry BD, g/cm ³	CCS, kg/cm ²	Erosion, cc	PLC %	T.C. W/m.°C
Castable (1)	Cast 2.40–2.72	560	10	—	—
Plastics (2)	Cast 2.40–2.72	560	8	—	—
Insulating (3A)	Cast 1.92–2.24 Gunned 2.08–2.24	350	14	–0.3	Cast 1.08 Gunned 1.08
(3B)	1900 max	210	20	–0.3	0.79
Insulating (4)	Cast 1.12–1.31 Gunned 1.15–1.44	105	—	–0.3	0.45
Insulating (5)	Cast 1.12–1.44 Gunned 1.32–1.52	35	—	–0.4	Cast 0.4 Gunned 0.45
Insulating (6)	Cast 0.96–1.12 Gunned 0.96–1.28	49	—	–0.8	Cast 0.3 Gunned 0.33
Insulating (7)	Cast 0.96–1.12 Gunned 1.04–1.200	21	—	–0.8	Cast 0.26 Gunned 0.35

The penetration of aluminum does not depend upon the porosity or the composition of the refractory. Aluminum has been found to penetrate a 95% alumina LCC with a high density and extremely low porosity. Of course, the different alloys of aluminum behave differently and penetrate to different extents depths the refractory but almost all varieties of aluminum alloys show qualitatively the same type of behavior.

The penetration of aluminum inside a refractory is quite detrimental to the refractory life since the penetrated aluminum forms corundum by oxidation. The latter gradually grows and disintegrates the refractory as the outside growth enlarges. In course of time, the furnace capacity is significantly reduced. The mechanism of the particular behavior of molten aluminum toward alumina and alumina-silicate refractories is not clearly known. But over the years, different additives have been developed that are quite effective in avoiding the problem. The different additives have not only been able to deter the penetration of the molten aluminum but also prevent the molten aluminum from wetting the refractory surface. This behavior has been found to be true not only with dense refractories but also with insulating refractories.

Earlier, a glass frit additive was reported to be effective in retarding the reaction and penetration of aluminum into the refractory. Later on different refractory manufacturers started to use their own proprietary additives and have been quite successful in preventing the penetration and eventual degradation of the refractory.

A laboratory test that determines the effectiveness of the additive is ALCOA's aluminum cup test. Initially, this was designed for testing refractory bricks. A cup is sawed on the larger face of a standard brick 20 cm long, 10 cm wide and a depth of 5 cm at the center (as is shown in Fig. 24). With the development of better castable refractories, a cup is built in during the casting of the refractory. The refractory with the cup built in it, is heated to 815°C for ten hours and then cooled.

The test refractory and aluminum are then put inside the furnace and heated until the aluminum is completely melted. The molten aluminum is poured into the cup in the refractory and the refractory is put back into the furnace and heated for 72 h at 815°C. Occasionally observations are made to determine whether anything detrimental has happened to the refractory. After the heating period, the refractory piece is removed from the furnace and the molten aluminum poured out. The refractory is allowed to cool, and cut through the middle. A cross section of the refractory is examined for any visible penetration. With well designed additives, the aluminum not only does not penetrate the refractory but it also does not wet the surface and peels off from the refractory.

Over the years, refractories often passed the cup test but did not perform well or last as long as expected in the field. Hence, modifications of the test and newer test procedures have been designed to predict the expected life in the furnace. Tests were run in actual furnaces on a reduced scale with refractory blocks made up of test pieces. The furnace was maintained at 982°C to accelerate the reactivity. The test was run for three months while continuously changing the dross to simulate the actual conditions. After the run was completed, individual test blocks were examined visually and then by x-ray diffraction and spectrometers to find out the mechanism and extent of the effect of the metal on the refractories. The test process eliminated a good number of refractories that were previously approved by the cup test. Although the test has been expensive and time consuming, the results proved to be beneficial in the proper selection of acceptable refractories.

The new test resulted in one unique observation. In the refractory test block that used a predominantly mullitic grain, the molten aluminum penetrated the grain and practically shattered it into smaller pieces. This could have resulted from the residual thermal stress in the grains while cooling from the furnace, leading to disintegration of the grain upon contact with molten aluminum.

Apart from the resistance to penetration by molten aluminum, other physical and thermal properties like HMOR at 815°C, and thermal expansion need to be satisfied for the refractories to be approved for use in the furnace.

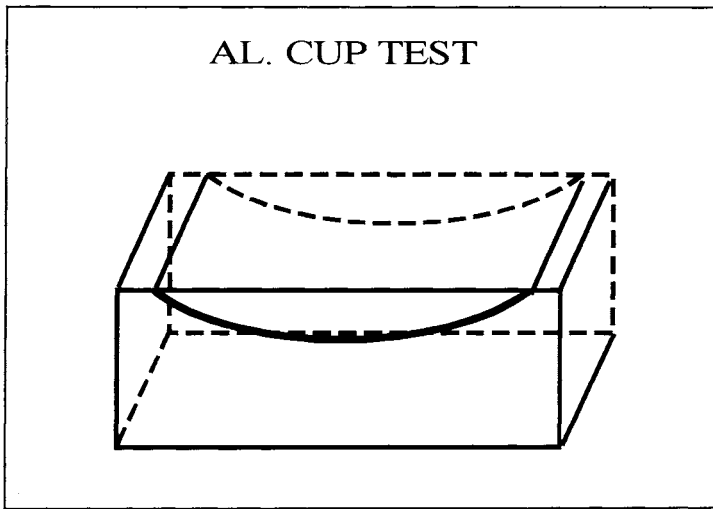


Fig. 25. Design of the cup used for aluminum cup test.

In recent years, many monolithic refractories — primarily castables — have been used in aluminum melting, holding and casting furnaces. They are used in the form of castable monoliths of precast blocks, belly bands and above furnace walls, troughs, transfer ladles and in dross furnaces for the recovery of aluminum from the dross.

References

1. T. D. Robson, "High Alumina Cements and Concretes," *John Wiley & Sons*, 1962.

2. W. H. Gitzen, L. D. Hart and G. MacZura, "Properties of Some Calcium Aluminate Cement Compositions," *J. Am. Cer. Soc.*, **40**, (5) pp. 158–167, 1957.
3. G. MacZura, L. D. Hart, R. P. Heilich and J. E. Kopanda, "Refractory Cements," *Cer. Proc.* Feb. 1983.
4. A. Nishikawa, Editor, "*Technology of Monolithic Refractories*," Pub. Plibrico Company Japan Ltd. Jan. 1984, Japan.
5. C. C. Furnas, "Grading Aggregates: I," *Ind. Eng. Chem.*, **23** (9) pp. 1052–58, 1931.
6. F. O. Anderegg, "Grading Aggregates: II," *Ind. Eng. Chem.*, **23** (9), pp. 1058–64, 1931.
7. G. MacZura, F. J. Rohr, D. V. Ruhs, P. T. Rothenbuehler, V. Gnauck and M. Van Den Berg, "Effect of Additives on High Purity Calcium Aluminate Cement (CAC) Castable Properties," *Paper presented at ALAFAR XVIII Cong.*, Oct. 24–26, 1988.
8. D. R. Lankard and L. E. Hackman, "Use of Admixtures in Refractory Concrete," *Amer. Cer. Bull.*, **62**, (9), 1091, 1983.
9. G. MacZura, J. Kapanda, F. J. Rohr and P. T. Rothenbuehler, "*Paper presented at Amer. Cer. Soc. Fall Meet.*, Oct. 4–6, 1984.
10. A. L. Renkey, "Refractory Castable," *U.S. Patent* 3,060,043, Oct. 23, 1962.
11. L. Prost, L. T. De Millery and A. Pauillac, "Hydraulically Setting Refractory Compositions," *U.S. Patent* 3,802,894, Apr. 9, 1974.
12. R. Stieling, H. J. Kunkel and U. Martin, "Vibrated Castables with a Thixotropic Behavior," *Adv. Cer.* **13**, pp. 211–218.
13. E. P. Weaver, R. W. Tally and A. J. Engel, "High Technology Castables," *ibid*, pp. 219–229.
14. C. Richmond and C. E. Chaille, "High Performance Castables for Severe Applications," *ibid*, pp. 230–244.
15. Y. Naruse, S. Fujimoto, S. Kiwaki and M. Mishima, "Progress of Additives in Monolithic Refractories," *ibid*, pp. 245–256.
16. S. Banerjee, D. W. Knowlton and R. V. Kilgore, "Low Cement Castables -- Properties and Applications," *ibid*, pp. 257–273.

17. G. MacZura, J.E.Kopanda, F. J. Rohr and P. T. Rothenbuehler, "Calcium Aluminate Cements for Emerging Castable Technology," *ibid.*, pp. 285–301.
18. B. Sandberg, B. Monsen, A. Seltveit and S. Bentsen, "Effect of Microsilica on Physical Properties and Mineralogical Composition of Refractory Concretes," *ibid.* pp. 201–210.
19. C. M. George, J. P. Bayoux and J. P. Letourneux, "Theory and Practice of Silica Fume Interactions," Lafarge Calcium Aluminates, *Private Communication, Intra-Office Rep.*, 1990.
20. B. Myhre, "The Effect of Microsilica Addition on Flow and Strength of Alumina Castables," *Paper presented. Amer. Cer. Soc. Meet.* April 18–22, 1993.
21. P. White, N. C. Fletcher and T. D. Reeves, "Use of Fume Silica and Other Ultrafine Particles in Low Cement Castables," *Proc. UNITECR'91*, Aachen, Germany, pp. 181–185.
22. S. Banerjee and G. L. Ramsey, "High Fired Refractories for Continuous Casting of Steel," *Cer. Eng. & Sci. Proc.*, **9**, NO. 1–2, pp. 67–73, 1988.
23. S. Banerjee and G. L. Ramsey, "High Fired Refractories for Continuous Casting Tundishes," *Adv. Refr. & Met.. Ind. Proc.*, **4**, pp. 305–312, 1987.
24. B. Clavaud, J. P. Kiehl and J. P. Radal, "New Developments in Monolithic Refractories," *Adv. Cer.* **13**, pp. 274–283.
25. A. Seltveit, G. S. Dhupia and W. Kronert, "Microstructural Aspects of Microsilica-Blended High Alumina Castables," *Cer. Eng. Sc. Proc.*, **7**, No. 1–2, pp. 243–260, 1986.
26. B. Myhre, "Development of a Bauxite-Based Utralow Cement Castable," *Elkem Materials Report* April 1991.
27. U. Schumacher, "Untersuchungen an Zementarmen und Ultazementarmen Korundfeuerbetonen," *Dr. Ing. Thesis*, Aachen, Germany.

28. H. C. Bassalo and W. E. Worrall, "The Effect of Additives on the Physical Properties of High Alumina Cements, Part I: Adsorption and Zeta Potential," *UNITECR '89 Proc.*, pp. 1183–1193.
29. K. Hughes, J. M. Roy and K. J. Willis, "The Influence of Calcined Alumina and Deflocculant on Low Cement Castable Properties," *Proc. IREFCON '96*, 1, pp. 91–95, Feb. 1996.
30. J. L. Mendoza, G. Lewis and R. E. Moore, "Effects of Dopants and Additives on the Setting Behavior of Refractory Concretes," *Paper presented at Amer. Cer. Soc. Meet.*, May 8, 1985.
31. B. Myhre, "The Effect of Particle Size Distribution on Flow of Refractory Castables," *Paper presented at Amer. Cer. Soc. Meet.*, April 1995.
32. Hydratable Alumina Binders, *Prod. Bull., Alcoa Industrial Chemicals Division*, Nov. 1994.
33. F. Azizian and M. A. Azam, "Development of Alumina-Based Binders for No-Cement Castables," *Paper presented at Amer. Cer. Soc. Meet.*, April 14–17, 1996.
34. M. W. Vance and K. J. Moody, "Steel plant Refractories Containing Hydratable Alumina Binders," *Paper presented at ISS, 53rd. Elec. Con. Meet.*, Nov. 12–15, 1995.
35. M. Strothmann, H. Huber, A. Rendtel, V. C. Pandolfelli, J. A. Rodrigues, A. P. Margarido, R. T. Noronha and C. Pagliosa, "High Temperature Performance of Cement-Free and Low Cement Alumina Castables," *Proc. UNITECR'95*, pp. 343–350, Nov. 1995.
36. M. C. Langenohl and G. O. Hughes, *U. S. Patent # 5,512,325* Apr. 30, 1996.
37. W. G. Allen, "Current Developments in Equipment for Refractory Placement," *Presented at 32 Ann. Symp. Refr. Amer. Cer. Soc. St. Louis Sec.*, Mar. 22, 1996.
38. S. Banerjee, "Thermal and Corrosion Properties of Monolithic Refractories for Iron and Steel Applications," — "Microstructure

- and Properties of Refractories," *Transtec Pub.* edited by J. P. Singh and S. Banerjee, pp. 1–20, 1993.
39. J. Mori, Y. Toritani, and S. Tanaka, "Development of Alumina-Magnesia Castable for Steel ladle," *UNITECR '95 Proc.*, pp. 171–178, Nov. 1995.
40. B. Myhre, "Cement-free Castables in the System MgO-SiO_2 ; The Effect of Bond Phase Modifiers on Strength," *Private communication*.
41. A. Watanabe, H. Takahashi, S. Takanaga and M. Uchida, "Magnesia Containing Basic Castable," *UNITECER '95 Proc.*, pp. 181–188, Nov. 1995.
42. F. W. Henry and J. W. Stendera, "Production of Basic Castables and Basic Precast Shapes for Steel Applications," *ibid.* pp. 1–8, 1995.
43. T. R. Cleeb and J. A. Caprio, "Properties and Service Experience of Organic, Fiber-Containing Monoliths," *Adv. Cer.*, **13**, pp. 149–160, 1985.
44. K. N. Singh and R. S. Hoffer, "Successful Implementation of Metal-free Low-Moisture Castables in Blast Furnace Troughs and Runners," *UNITECR '89 Proc.*, **1**, pp. 193–211, 1989.
45. S. Banerjee, et al, US Patent # 5,422,323, Jun. 6, 1995.

CHAPTER III: PUMPABLE CASTABLES

Introduction

The development of low and ultralow cement castables (LCC and ULCC) in recent years has provided significant advantages in monolithic refractories. The new LCC and ULCC are now used in variety of applications where conventional castables could not be used before. But there are some basic problems inherent in the application and performance that are restrictive to their further use. The following points address these problem areas in the application of LCC and ULCC:

- **Controlled water addition** — Both LCC and ULCC are very sensitive to water addition. Laboratory Investigations and field experience have shown that when more water is added than its optimum level the properties are significantly affected. Every 0.5% increase in water for casting can affect the properties by as much 15–20%. Increased water addition defeats the basic principle of this technology. Often installers lack the proper understanding of this technology and are responsible for many of the problems in performance.
- **Drying sensitivity** — Cement containing castables develop strength from the hydration of the cement. Depending on the

temperature during casting and curing, the cement can form various hydrated phases (mentioned in Chapter II). Even with the ideal temperature during casting and curing, the rate of drying has to be slow because of the phase changes during dehydration. It has been found out that the permeability of LCC and ULCC is very low up to 1000°C. Several attempts have been made to facilitate the drying of castables, e.g., addition of plastic fibers and metallics (Al, Si etc). However, even with these additions, the problem exists although to a lesser degree. In addition, there are inherent problems, such as auto explosion associated with the addition of metallic powders.

- **Need of vibration during installation** — Since the LCC and ULCC are pseudothixotropic in nature, vibrational energy is needed for flow and compaction. Application of vibration can sometimes cause problems from too little and too much vibration. The properties of the installed lining can differ in performance depending on the nature of compaction from vibrational installation. Too little vibration will cause a lack of consolidation and compaction whereas too much vibration will cause segregation. Neither is desirable for optimum performance.
- **Thermal shock resistance** — The low and ultralow cement castables exhibit poor thermal shock resistance due to the increased density and lower porosity.
- **Thermal conductivity** — Increased density and lower porosity in LCC and ULCC also lead to a higher thermal conductivity compared to that of other monolithic refractories and brick with comparable chemical composition.
- **Refractoriness** — The refractoriness of LCC and ULCC is limited to 1450°–1500°C. The formation of low-temperature phases formed above these temperatures limits the use of LCC and ULCC at higher temperatures.

These points clearly indicate that there are basic problems with the calcium-aluminate cement and hydratable-alumina cement (that also exhibit some of the shortcomings listed above) castables. It is obvious that some improved bonding system is needed to minimize or eliminate some of the problems associated with LCC and ULCC. This need led to the development of a separate bonding system that was compatible with the refractory materials while, at the same time, more versatile in its application. The use of colloidal silica as a binder not only reduced (and in most cases eliminated) the drawbacks of LCC and ULCC (also hydratable aluminas) but also created a new class of castable/pumpable refractories.

Gel-bond castables/pumpables

During the late 1980s, a new bonding system was developed providing another step forward for the monolithic refractories. The use of colloidal silica has long been accepted in the investment casting process, but its use in refractories had not been perceived as a viable alternative. The principle is based on a sol-gel bonding system, which has long been used in the production of high performance ceramics. The advantages of the sol-gel based ceramic body are its uniformity, superior properties and its low temperature sintering characteristics. Instead of having the sol-gel as the main body, the sol-gel process is used as a binder in refractories for the aggregates and refractory components. With proper selection of controlled gelling agents, the colloid is gelled around the refractory particles which, after the drying process, forms a gel skeleton and provide dried strength to the refractory. Since colloidal silica is a stable colloid, available easily and economically at high concentrations, it is used as the main source of the gel-bond refractories.

In alumina, alumina-silicate and silica refractories, the availability of the colloidal silica bonding made it the favored bonding agent. With the use of calcium aluminate cement for bonding the refractoriness has been limited due to the introduction of CaO.

However, colloidal silica as the bonding agent does not introduce any component to reduce the refractoriness. In fact, the use of colloidal silica provides several advantages over other monolithic refractories, particularly the cement-containing castables. The colloidal silica bonding (gel-bond) has been applied to various castable compositions across the board resulting in the following advantages:

- **Less mixing time during installation** — The mixing time for LCC and ULCC is usually between five and 10 minutes. The minor additives needed for deflocculation, acceleration or retardation need to go into solution to be active in the mix., thus needing a longer mixing time. Gel-bond compositions, on the other hand, do not have those types of additives, so the mixing time is significantly less. Gel-bond compositions need only 30–50% of the mixing time required by LCC and ULCC compositions (two to three minutes compared to five to 10 minutes).
- **No vibration during installation** — The gel-bond refractory compositions are very cohesive and self-flowing. As such, no vibrational energy is necessary, eliminating some of the parameters that can result in inconsistent performance.
- **Shorter drying time during installation** — In gel-bond compositions, no water is used. Instead, colloidal silica is the only liquid phase used for mixing. Colloidal silica is more viscous than water (40% solids and a viscosity of 13–15 cP) and, as a result, the liquid phase keeps the solid particles slightly more separated than mixing with water. The permeability of the gel-bond materials is significantly higher, which helps drying of the installed material. The drying time is about 30–50% that of LCC and ULCC compositions.
- **Use of lighter consumable form for installation** — Since the gel-bond composition does not need vibration, lightweight consumable/reusable forms can be used, making the composition versatile in a wide range of applications.

- **Reduced drying flaws** — Due to the nature of the bonding system, the gel-bond compositions have higher permeability at low temperatures during the removal of steam and can be dried much more easily and quickly. Thus, they are free of most drying defects such as cracks and spalls.
- **Better refractoriness** — In alumina and alumina-silicate compositions, the nanometer-sized colloidal silica particles are highly reactive and form mullite with the available alumina particles. Since mullitization is the high temperature bond, the refractoriness is much higher because no low-melting phases (such as anorthite or gehlinites) are formed.
- **Better chemical resistance** — The gel-bond materials form the high refractory mullite and do not form any low melting phases. The result is a gel-bond composition that resists chemical attacks better than LCC and ULCC compositions.
- **Better oxidation resistance** — In compositions containing carbonaceous materials, the gel-bond materials provide much better oxidation resistance since the colloidal silica provides a protective coating around the carbonaceous phase and thus inhibits the materials from reacting with oxygen.
- **Longer lining life** — With the combination of better chemical and thermal shock resistance and excellent oxidation resistance of the gel-bond composition, the overall campaign life is much longer since the unaffected parts of the lining do not deteriorate with time.
- **Longer shelf life** — The shelf life of the gel-bond pumpable compositions is much longer since there is no hydratable phase like cement as found in LCC and ULCC compositions.

Characteristics of gel-bond castables/pumpables

Sol-gel technology has been used effectively in ceramics for the past few decades because the ceramic body is highly homogeneous, can be formed in thin layers and sintered at a relatively low temperature. The

gel-bond technology differs from sol-gel materials since it serves as a bonding system rather than the main body of the ceramics. The water-based colloid (colloidal silica) in the gel-bond technology surrounds the refractory particles and, after gelling, forms a network encapsulating the refractory particles. After removal of water, the gel leaves a skeleton that serves as a bond that is highly reactive and forms mullite with fine alumina at relatively low temperature. The colloid has higher viscosity than water and, hence, keeps the particles farther apart. This provides the gel-bond compositions higher permeability at low temperatures with quicker and easier dry-out characteristics. The fine coating surrounding the particles in carbon containing compositions also provides better oxidation resistance. The viscous colloid creates fine channels between the particles that provide better thermal shock resistance as well. Since the colloidal silica is compatible with silica and aluminosilicate compositions, the gel-bond refractories have been successfully developed for a wide variety of compositions ranging from high alumina to alumina-silicate with or without silicon carbide and graphite, as well as all silica compositions. This compatibility provides a wide range of applications. Some of the characteristic chemical and physical properties of these compositions are given in Tables 1 and 2.

Table 1. Chemical compositions of Gel-Bond Refractories.

Refractory Classification	50	60	70	85	70/SiC
Chemical Composition, (Calcined basis), %					
Al ₂ O ₃	51.0	58.5	67.5	86.5	72.5
SiO ₂	46.5	39.7	30.5	9.5	7.5
Fe ₂ O ₃	1.5	1.8	1.1	0.7	1.0
TiO ₂	1.0	1.0	1.1	2.8	1.5
SiC	—	—	—	—	17.5

The gel-bond compositions have better thermal properties than LCC and ULCC. This is evident from a comparison of the thermal

expansion curves of the gel-bond castable and LCC — both with 70% alumina (Fig. 1) and a high alumina-silicon carbide-carbon containing gel-bond castable and ULCC (Fig. 2). These comparisons clearly indicate that the gel-bond castables are more thermally stable than the cement containing castables. This fact is demonstrated in their applications as less and fewer cracks have been observed in the continued application of gel-bond castables.

Table 2. Physical properties of gel-bond refractories.

Refractory Classification	50	60	70	85	70/SiC
BD, kg/m ³ After firing to					
110°C	2310	2480	2610	2925	3000
815°C	2310	2450	2610	2925	3000
1370°C	2320	2430	2620	2925	2990
Porosity, % After firing to					
110°C	17.5	15.6	14.5	16.5	13.5
815°C	17.5	16.3	14.5	17.5	16.5
1370°C	16.5	16.4	14.5	16.5	15.5
CMOR, MPa After firing to					
110°C	5.5	4.4	5.5	5.5	3.7
815°C	7.5	6.6	9.5	9.5	5.7
1370°C	12.8	12.8	17.5	13.5	7.3
HMOR, MPa @ 1370°C	11.5	12.8	13.5	12.5	6.7*
*Under reduced condition					

As mentioned earlier, one of the main advantages of gel-bond compositions is their self-flow characteristics. This enables them to be installed by a pump. The wet mix coming out of the pipes flow freely on the surface and finds its own level. No vibration is needed to make the material move and thus provides a great advantage in installations.

Figure 3 shows the wet mixed material coming out of the pipe and Fig. 4 shows that the material is leveling automatically without any external force.

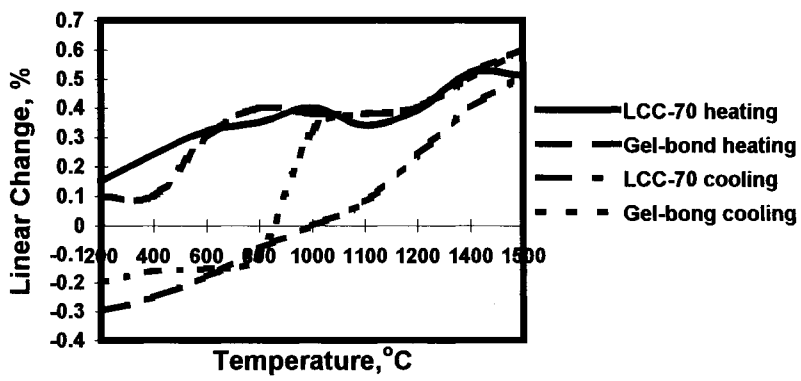


Fig. 1. Comparative thermal expansion of a 70% alumina LCC and gel-bond castable.

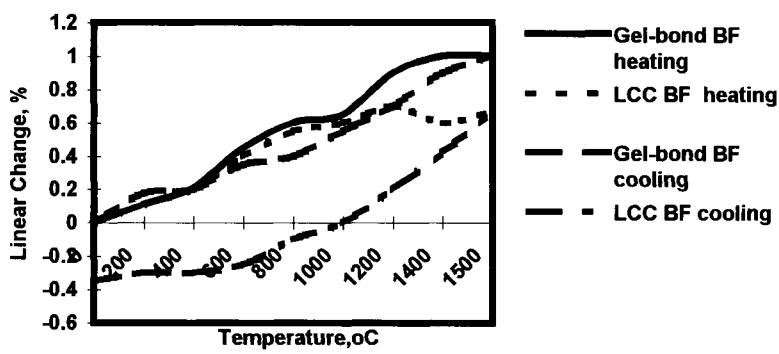


Fig. 2. Comparative thermal expansion of a 70% alumina LCC and gel-bond castable both containing silicon carbide and graphite.

Since the major application of ULCC has been in the blast furnace cast house trough, iron and slag runners, studies were needed to compare gel-bond compositions with ULCC. Several of the high alumina-silicon carbide-graphite gel-bond compositions have been investigated¹ to find the optimum composition to have maximum resistance to slag/metal corrosion and erosion and while being at least comparable to ULCC. The slag/metal corrosion/erosion has been investigated for four different compositions with varying amounts of silicon carbide and alumina in an induction furnace. Wear has been measured directly from the remaining portion of the specimen. Cross sections of the maximum wear area of the specimens have been further examined by scanning electron microscope or slag penetration levels. The concentration of CaO and MgO inside the specimens indicate the level of penetration (the details with the pictures are shown in Chapter X). These indicate the relative extent of slag penetration in the refractories. Slag penetration in the refractories causes densification that eventually promotes spalling of the refractory. Figure 5 shows the relative cost performance index of the four different compositions of the gel-bond material leading to the best use of the composition.



Fig. 3. Wet mixed gel-bond composition coming from the pump out of the pipe.

There is a significant difference between the thermal properties of the gel-bond compositions and those of LCC. Although the porosity of LCC is much lower than the gel-bond compositions, the thermal shock resistance is remarkably lower. This may be due to the fact that the colloid has a higher viscosity than water and, hence, keeps the particles at a greater distance than water base mixes creating a built-in micro-crack system that arrests crack propagation. Also, since the gel-bond compositions' high temperature bonding phase is mullite, the hot Modulus of rupture is far better than that of LCC compositions — particularly above 1400°C as shown in Table 3.

Another gel-bond composition has been developed using pure silica as the aggregate for use in place of silica bricks. This product can either be used as a castable or pumpable depending on the application. Table 4 shows the properties of the silica based gel-bond composition compared with two different silica bricks. Figure 6 shows the thermal expansion compared to silica brick and also clearly indicates that the gel-bond composition is more stable than silica brick.



Fig. 4. Wet mixed gel-bond composition coming from the pump out of the pipe finds its own level without any external force.

Table 3. Comparative thermal properties of LCC and gel-bond pumpables.

	60% Al ₂ O ₃ LCC	70% Al ₂ O ₃ LCC	50% Al ₂ O ₃ Gel-bond	70% Al ₂ O ₃ Gel-bond
Porosity, % after 1400°C	11.2	9.2	16	14
HMOR, MPa @ 1400°C	4.5	14	16	17
@ 1500° C	1.5	6.0	9.7	11.5
Thermal shock, % MOR loss, ASTM C-1171	74	90	20	16

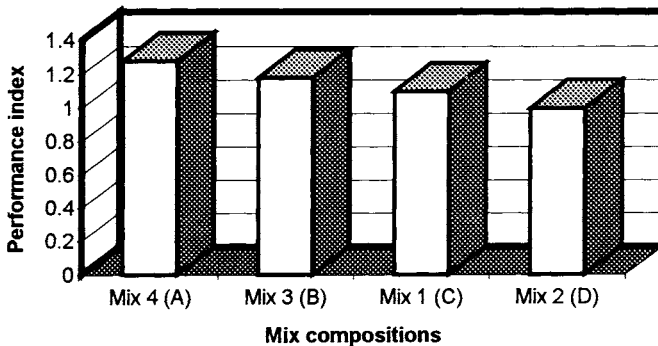


Fig. 5. Cost performance index indicating the gradual improvement in performance.

Thermal properties such as thermal shock resistance, hot modulus of rupture and refractoriness are the most distinguishable characteristics of the gel-bond compositions from the LCC and ULCC. Moreover, the gel-bond compositions are cohesive in nature so they can flow under gravity, making them self-flowable in nature. This special characteristic led the gel-bond compositions to be installed by pumping, thus ushering in a new era in refractory technology. Several papers have been written

on this topic²⁻⁶ and patents⁷⁻¹⁰ have been issued. The ease of installation by pumping the gel-bond compositions led to a new era in developing low and ultralow cement castables suitable for pumping.

The characteristic properties that distinguish gel-bond compositions from the LCC and ULCC are thermal properties — thermal shock resistance, hot modulus of rupture and refractoriness. Moreover, the gel-bond compositions are cohesive in nature so they can flow under gravity, making them self-flowable in nature. This special characteristic led them to be installed by pumping, thus ushering in a new era in refractory technology. Several papers have been written on this topic²⁻⁶ and patents⁷⁻¹⁰ have been issued. The ease of installation by pumping the gel-bond compositions led to a new era in developing low and ultralow cement castables suitable for pumping.

Table 4. Comparative properties of gel-bond composition and silica brick.

	Gel-bond	Silica brick #1	Silica brick #2
Bulk density, g/cc			
After firing to 315°C	1.83	1.70	1.79
1178°C	1.87	(as received)	(as received)
Porosity, %			
After firing to 315°C	15.7	23.7	22.0
1178°C	16.0	(as received)	(as received)
CCS, MPa			
After firing to 315°C	26.9	17.9	22.7
1178°C	31.4	(as received)	(as received)
Abrasion loss, cc ASTM C-704 @ 1178°C	22	47 (as received)	31 (as received)

The gel-bond compositions can be pumped with the grain size distribution devised by Furnas¹¹ and Anderreg¹² for close packing. LCC

and ULCC form into high density/low porosity monoliths by vibration using the gap-size or continuous distribution (Furnas). Compositions of LCC and ULCC suitable for vibration could not be pumped since they need much higher quantities of water leading to segregation.

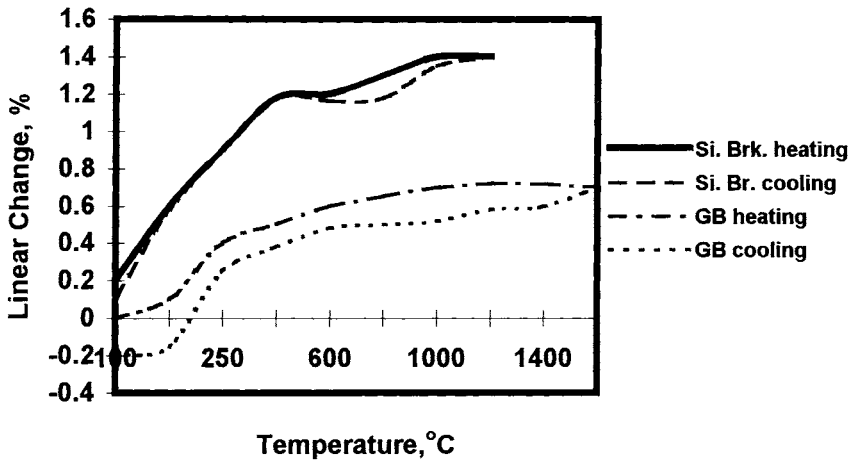


Fig. 6. Thermal expansion curve for silica brick and silica-based gel-bond composition.

Gel-bond pumpables also led to another area of application that has been of significant importance in prolonging the life of blast furnaces used in iron making before they must be shut down for major repairs. The pumpability of gel-bond castables allowed the composition to be pumped at high elevations needed for grouting.

Injectable grouts

Blast furnaces are built to produce iron on a continuous basis for several years. The refractories inside the blast furnace are mostly alumina-silicate brick down to the bosh area. Below this point, carbon or silicon carbide bricks are used. During the course of use bricks wear and sometimes get loose creating a void in the lining. This not only

weakens the integrity of the lining but also creates hot spots on the shell. This can cause distortion of the shell and pose a severe problem in the steel structure. To avoid further deterioration and elimination of hot spots, refractory masses are injected into the furnace.

The term “grout” is used to inject a refractory material through a small pipe into the blast furnace lining by transporting the wet mixed material by a pump. Grouting is used when the steel shell shows sporadic wear patterns as evidenced by hot spots. When hot spots become visible on the steel shell, the blast furnace is temporarily shut down for repair by injecting refractory at and around the hot spot. Usually, a hole (35 – 50 mm) is drilled at the hot spot and a refractory mass is injected into the blast furnace to replace the worn out refractory.

The grouting material is mixed near ground level and transported by a pump at the place of repair through flexible pipes. The material should be capable of being transported vertically for a long distance without causing any problem to the pipes. The grouting material is intended to fill in the worn area and stop when it heats the burden inside. It should remain intact when the furnace is heated and the burden goes down. It must also have good thermal and mechanical stability and erosion resistance so the material remains in good condition for a reasonable period of time.

For a long time, grouting materials were made by using sodium silicate as the binding agent with different alumina contents. Although the application and physical characteristics were quite satisfactory, the grouting refractories did not perform on a consistent basis. Sodium silicate, by nature, forms a crust at the surface when heated prohibiting the removal of moisture, and causes swelling and cracking. The swelling causes the grouting material to be laminated with a weakness in its integrity and causes wear when it comes in contact with the burden.

The use of colloidal silica as the bonding agent helps eliminate the problems that existed with the sodium silicate bonding. Typical properties of a gel-bond grout composition are given in Table 5.

Table 5. Properties of a Gel-bond Grout

Physical properties	Bulk density, kg/m ³	MOR, MPa	CCS, MPa	Linear change, %	HMOR, MPa
Prefired to, °C					
110	2410	21	5.5	0.0	
955	2410	22	7.5	-0.12	
1500	2380	22	15.5	0.4-0.5	
@ 1370 °C					4.5
@ 1425 °C					3.5

Applications of gel-bond pumpable compositions

Gel-bond compositions have the convenience of application either by casting or by pumping with similar binder contents. Also, a wider variation of grain sizing can be used without affecting the properties. The convenience of a pumping application prompted the use of gel-bond compositions in various industries. Since large amounts of coarse aggregates can be used in the compositions, the gel-bond materials can be used for corrosion-erosion resistant applications more effectively.

Gel-bond compositions are widely used with pure alumina and alumina-silicate compositions as well as in combination with silicon carbide and graphite. Gel-bond compositions with 50–85% alumina are used in reheat furnace hearths and subhearths, walking beam furnaces, torpedo and transfer ladles, steel ladle barrels, tundish back-up linings, soaking pit walls and waste incinerators. The alumina-silicon carbide-graphite composition has been used extensively in blast furnace cast house in trough, iron and slag runners, in the mouth of torpedo ladles where desulfurization is carried out, and in hot metal mixers. Fused silica based gel-bond materials are ideal for repairing

glass tank furnaces, coke oven door plugs and flue wall repair materials. High alumina (85–90%) gel-bond material has been used in high temperature shaft kilns, burner blocks for high temperature use and blast furnace tuyers.

Pumpable (self-flow) castables

A refractory castable may be characterized as a coarse structure consisting of larger aggregates bonded together by a finer, viscous bond phase (matrix) that develops the strength, usually by the hydration of a cement. The aggregate and matrix may be of similar or vastly different chemical and physical characteristics as dictated by the specific type of refractory. Almost infinite combinations may be generated when different aggregates, bonding matrices and particle size distributions are considered. The application and economic conditions often dictate the choice of aggregates, bond phase, their proportions and particle size distribution for a particular refractory castable. The relative proportions of the aggregates and bond phases, as well as particle size distributions and, to some extent, the chemistry and mineralogy affect the flow properties and the physical and thermal properties of the refractory castable.

Particle packing

Castables with low porosity have been developed¹³ on the basis of the particle packing principle. High compaction in mortar and concrete was developed after Furnas and Anderegg in the early 1930s. Furnas developed a mathematical relationship with two different types of grading — intermittent or gap grading based on mixing two, three or four sizes and the other based on a continuous size distribution. The gap sized distribution gives poor workability¹⁴ while the continuous distribution gives improved workability, lower water requirement and good compactness resulting in low shrinkage and high strength.

To obtain a particle size distribution close to Furnas' distribution, it is necessary to separate the system into a number of screened

fractions. In Furnas' distribution, the particle sizes should be in the range of 0.2 to 4760 μm (4 mesh) for obtaining maximum density by optimum particle packing. This distribution is based on the assumption that particle packing of narrow-sized fractions of component materials is constant at 50% voids. Particle packing curves are generally displayed on a volume basis, but can be regarded as weight for one component system or for particles with similar density.

Simultaneously Andreassen¹⁵ worked on the size distribution for particle packing, but with a different approach. Although the approach is more theoretical, it partly represents an empirical theory of particle packing.

The equation is represented as

$$\text{CPFT} = (d/D)^q \cdot 100$$

Modified Andreassen:

$$\text{CPFT} = [(d^q - d_m^q / D^q - d_m^q)] \cdot 100$$

where

CPFT is Cumulative Percent Finer Than (Volume),

d is particle size,

D is the maximum particle size,

d_m is the minimum particle size of distribution and

q is a quotient.

Andreassen concluded that the exponent should be between one-third and one-half to obtain optimum packing. If the exponent increases, it means an increase of the coarse material; if it decreases, the amount of the fine material is increased. If $\log(\text{CPFT})$ is plotted against $\log(d)$, then q becomes the slope of the curve. Therefore, it is common to represent the particle size distribution in log-log plot.

Dinger and Funk¹⁶⁻²⁰ made a comprehensive treatment of the packing theories. According to computer simulations, the exponent of Andreassen's equation should be 0.37 to get optimum particle packing. Dinger and Funk concluded²⁰ that an exponent greater than the optimum (i.e. 0.37) pack in a scheme approaches a minimum non-zero porosity. Regardless of the width of the distribution, the porosity will never approach zero. The limiting porosity is related to the value of the exponent. Distributions with exponents less than 0.37 approach zero porosity as the width of the distribution increases. With increasing finer particles in the distribution, the porosity will decrease and approach zero.

Hence, it is meaningless to use exponents greater than 0.37 if the objective is to create a maximum packing density. Furthermore, for castable systems, a mobility of particles is needed to place the castable. Mobility may be obtained by using more water, but water should be minimized because it fosters porosity as it evaporates. The desired degree of fluidity can be obtained by keeping the coarser particles apart by a suspension of fine particles. The viscosity of this suspension is determined by the amount of fines and superfines as well as the state of flocculation within the suspension.

The important factor in making dense packing castables is the widening of the particle size distribution. Larger amounts of maximum particle size may reduce the total porosity of a system. However, other factors, such as flowability and cohesiveness, usually determine the maximum particle size. The obvious option is to expand the particle size distribution toward smaller and smaller particles.

The availability of superfine powders like microsilica that are inert to water provides a unique method of keeping the particle size distribution toward the smallest section of the composition. The microsilica, which is small enough to replace water molecules in the open spaces, has a surface area of approximately 20 m²/g and may contribute more than 50% of the total surface area in certain mixes.

Microsilica is available at various purity levels. The purity of microsilica used LCC and ULCC is not highly critical. Although purer varieties may provide only marginally better properties, the purity of the microsilica does play an important role in designing the self-flow castables. The surface impurities, particularly the carbonaceous compounds (polymeric forms), may create characteristics that are an impediment to the self-flowing properties.

The effect of microsilica in a castable is at least twofold. Once properly dispersed, the main function is to reduce the cement content in the refractory castable. The use of the superfine microsilica is based on the assumption that, in a castable with a classical particle size distribution, the density is limited by intergranular voids that are filled with excess water during installation. These voids may be filled with ultrafine particles of microsilica replacing water. The remaining micropores are filled by a gel made of hydrated cement. The use of these principles led to the low-water, high-density LCC.

The role of a filler is by no means the only function of microsilica in refractory castables. It has been demonstrated²¹ that in cement pastes with microsilica, a certain fraction of the microsilica reacts with the cement and water to form calcium alumina-silicate hydrates (CASH) phases. This is in addition to the calcium aluminate hydrate (CAH) and alumina hydrate (AH) phases normally found in hydrated cements. The CASH phases are of zeolithic nature and the amount of these hydrated products is dependent on the quality (purity) of the microsilica. The addition of microsilica was found to alter the pore size distribution toward finer pores, an effect that is probably more physical than chemical in nature. During heating, the CASH phases convert to CAS_2 and probably to cristobalite or quartz.

The particle size distribution is one of the most important variables in casting properties. It is desirable to expand the particle size distribution toward smaller and smaller sizes. Because of its simplicity, the Andreassen model has been mostly used in improving the flow property of castables.

It has been established from the Andreassen equation that the q value of a self-flowing castable must be less than 0.37. A fused alumina castable composition,¹³ having self-flow properties with 1.5% high alumina cement and about 10% microsilica, is shown in Fig. 7 with a q value of 0.3. The log-log plot of the cumulative % (volume) value against the particle size of the castable gas has been plotted with the Andreassen value indicated.

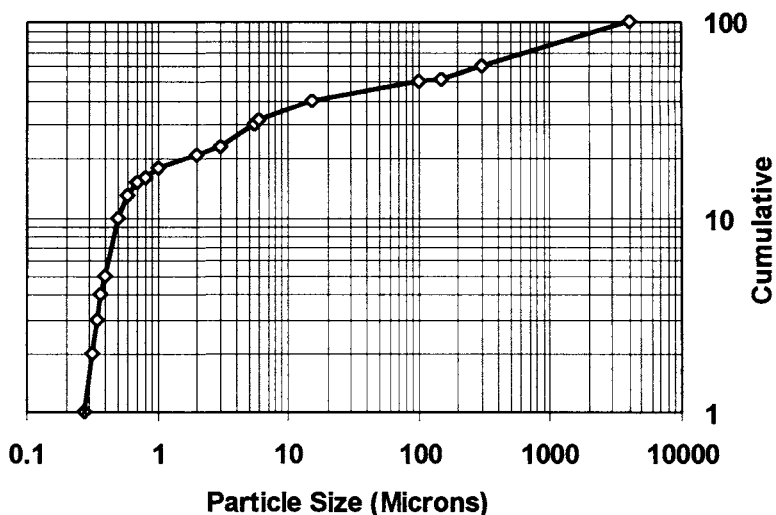


Fig. 7. Plot of cumulative % retained versus particle size (log-log) in a tabular alumina castable with 1.5% HAC and 10% microsilica.

The strength varies with the q value of the particle size distribution. The best strengths have been obtained from particle size distribution with a q value of 0.25. The bulk density is also found to maximize at a q value of 0.25.

In comparison, the particle size distribution of a high cement castable¹³ suitable for vibration casting is shown in Fig. 8. The castable, which shows good flow properties under vibration and high

strength, has a q value of 0.29. Since a higher q value indicates lower fines content, the coarser particles of such castables will be more closely packed and could interfere with each other. This type of interference could obstruct the flow unless vibrational energy is introduced during casting. On the other hand, a castable with a lower q value, i.e., having higher fines and superfines content, will result in increased distance between the coarser aggregates. Higher amounts of fines also make the castable less susceptible to separation with more water addition.

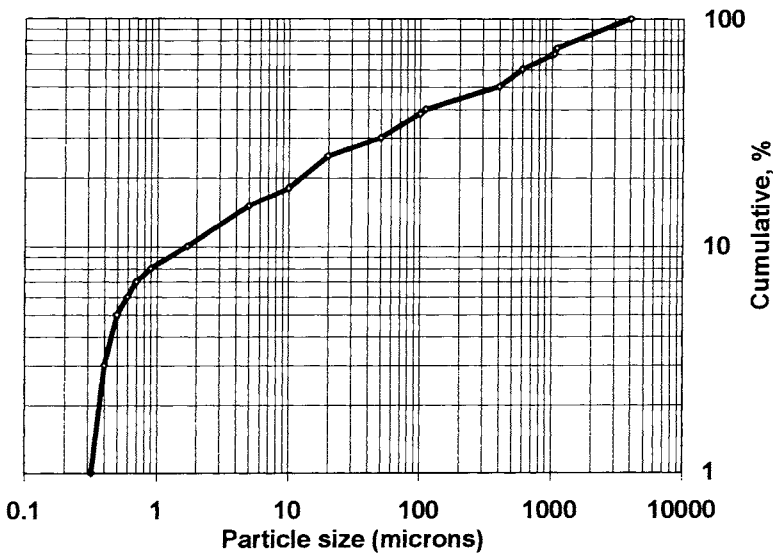


Fig. 8. Particle-size distribution of a high-cement vibratable composition.

Figure 9 shows a high cement castable¹³ based on the same raw materials with self-flowing characteristics at the same water content as the previous one will flow under vibration.

The self-flowing, ultralow cement castable shown in Fig. 10 is characterized by good flow without vibration and low water addition

(about 4%). This castable has high refractoriness and high hot strength (HMOR of 30 MPa at 1400°C). The particle size distribution is gap-sized (with the gap between 60 and 15 μm). The coarse end of the distribution has a q value of 0.21 and the fine end has a steeper distribution of a q value of 0.37 providing minimum interference between coarse particles. The size distribution of the fine fraction (which should have ultra-fine particles to a sufficient extent) is also important to prevent the castable from becoming dilatant.

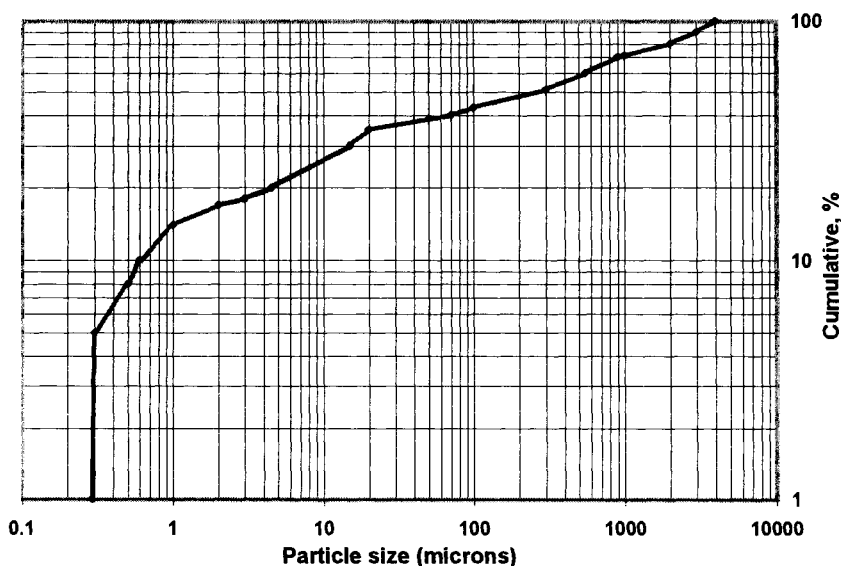


Fig. 9. Particle-size distribution of a self-flowing, high cement castable.

The presence of a sufficient amount of ultrafine submicrometer size particles, usually provided by microsilica, is important. A successful use of high amounts of microsilica is dependent on microsilica quality. When impurities are present in the microsilica, the mix is sticky and does not flow properly. Hence, it is advisable to use purer varieties of microsilica for self-flowing castables.

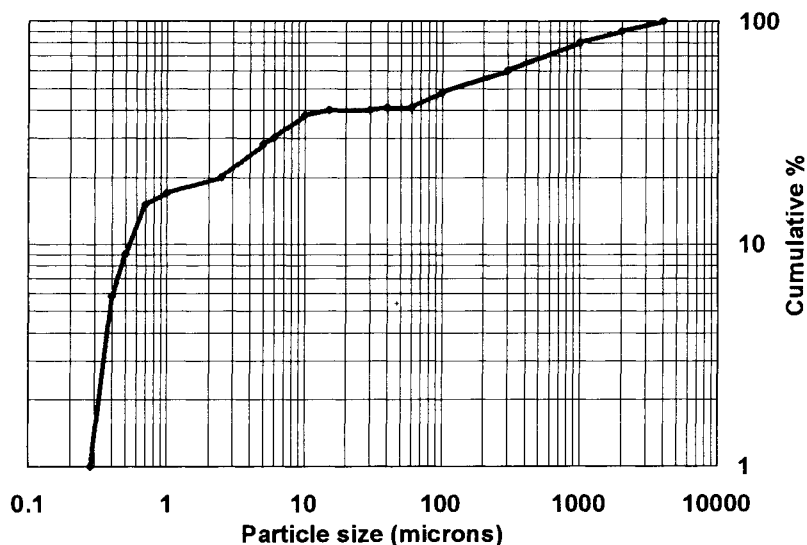


Fig. 10. Particle-size distribution of self-flowing, gap-sized ultralow cement castable.

Recent investigations²² have shown that there is a difference in the rheological and refractory properties of reactive alumina and microsilica even with similar particle size. Figure 11 shows a plot for reactive alumina and microsilica used as superfines in a fused alumina based self-flow castable. The plot shows the amount of water needed to obtain a specific amount of flow (100% increase in diameter) as a function of the superfine addition.

It is evident from Fig. 11 that an increased addition of the superfines resulted in decreased water demand up to a certain addition level. The water reducing effect is more pronounced with the addition of microsilica than with the addition of reactive alumina. The figure also shows that the water reducing effect stops when the addition of superfines exceed a certain level. From the above figure it is seen that

6 vol. % of alumina and 9–10 vol. % of microsilica are the limiting amounts for the water reducing effect. Addition of more superfines only expands the system while the flow remains constant. The above plot also shows that the porosity from water addition is 16% with alumina whereas it is only 9–10% with microsilica. Calculating in weight percent, for alumina the limit of addition is 6 wt % alumina with 5% water whereas for microsilica, it is 6–7 wt % with 3% water.

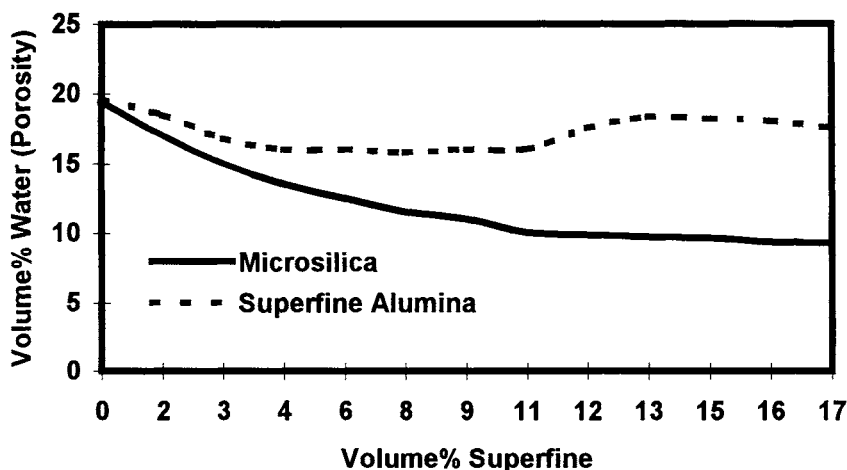


Fig. 11. Water addition to obtain the same flow (100% increase in diameter) as a function of superfine addition.

Self-flow castables have been prepared for HMOR studies by using a total of 8% superfines and making the castables from all microsilica to all reactive alumina compositions. Figure 12 shows the HMOR values at 1400°C.

It is obvious from this study that the composition using all microsilica has noticeably higher HMOR than that using all reactive alumina. The combination of microsilica and reactive alumina in different proportions has lower HMOR compared to the individual systems.

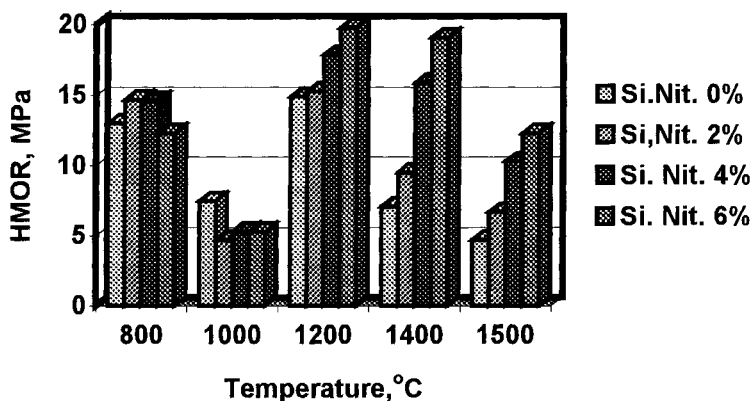


Fig. 12. HMOR of fused alumina-based castable with different microsilica/reactive alumina ratio at 1400°C with time.

A self-flowing castable has been developed²³ for application in the cement industry as free-flowing castable without the use of vibrators. It provides long life and improved working conditions. The study was conducted by adjusting the particle size distribution of a low cement castable with the addition of ultrafine particles and determining the level of the dispersing agent.

The influence of superfine alumina (low Na₂O content, 0.17 to 0.25%) was studied by adding different particle sizes ranging from 0.5 to 5 µm individually and in combination. The mix composition chosen is shown in Table 6.

The nature of the superfine alumina seemed to have a great influence on the flow properties, water addition and mechanical strength of the castable. A combination of the finest and the finest to the coarsest alumina in the ratio of 4:1 showed the best flow with the least amount of water.

The investigation of the dispersant content showed that ideal flow property was obtained with 0.1% addition. Higher amounts decreased

the working time of the castable. Comparative data have been presented between the low cement and self-flow castable with sintered alumina and bauxite as per base compositions mentioned above, in Table 7.

Table 6. Mix composition of a castable for cement industry

Component	Wt. %
Sintered Alumina	81.0
Superfine Alumina	10.0
Micro Silica	4.0
Calcium Aluminate Cement	5.0
Additives	0.1

Table 7. Typical Properties of Low Cement Conventional and Self-flowing Castables.

	Conv. LCC	SFC	Conv. LCC	SFC
Casting water, %	5.0	5.0	6.5	6.5
Chemical Composition, %				
Al ₂ O ₃	91	90	83	82
CaO	1.2	1.2	1.5	1.5
Bulk density, g/cm ³ ,				
110°C	3.1	3.08	2.85	2.85
1400°C	2.95	2.95	2.78	2.77
Linear change, %,				
110°C	-0.03	-0.03	-0.03	-0.03
1400°C	-0.20	-0.15	-0.10	+0.10
CCS, kg/cm ² ,				
110°C	850	860	680	720
1400°C	1250	1200	1050	1020
Application	Burner parts	Burner parts	Cooler nose	Cooler nose

Along with the particle size distribution, the distance between particles is another factor considered to be important in the self-flow castable technology.

Interparticle separation distance

Interparticle separation distance (IPS)²⁴ — the distance between any two adjacent particles in the mix (assuming all the particles as separate entities), has recently been considered to be a factor that determines the flowability. Assuming no agglomerates exist, the average IPS in a suspension can be calculated from the volume, surface area, porosity, and volume fraction of the solids in the wet mix. When the pore volume is completely filled, further water addition separates the particles. The IPS is double the thickness of the separation of the water spread over the surface area of the body. It is doubled because each adjacent particle has its own water film.

The equation to calculate the IPS is:

$$\text{IPS} = 2/\text{VSA}(1/\text{V}_s - 1/(1 - \text{P}_{\text{of}}))$$

where

IPS = interparticle separation distance in m,

VSA = volume surface area, calculated from the major specific surface area,

V_s = volume fraction of solids in the mix,

P_{of} = pore fraction of the body when all particles are in contact after drying.

Control of IPS is necessary to control the rheology of the body. Obviously, if the moisture content is less than or equal to the porosity of the dry body, the mix will not flow because the particles are touching each other. The body will behave like a dry, solid mass. Only when the moisture content sufficiently exceeds the porosity, separating the particles to a minimum of about 50 nm apart, can any reasonable fluidity be expected. Therefore, the porosity of the composition is controlled by the particle size distribution. Ceramic bodies with different particle size distribution may have the same

porosity but significantly different surface areas, so the addition of the same amount of water will produce the same IPS but different porosity. The amount of additional water needed to impart fluidity depends upon the particle size distribution of each of several ingredients in the body and the blend.

The above investigation concludes that the interparticle separation distance and not the particle efficiency, is the most important criterion for designing a very highly loaded, self-flowing refractory castable mix.

Gun/spray (shotcreting) LCC

The use of wet gunning shotcreting systems has been well known and common in portland cement based concrete where sand, gravel and portland cement are mixed with water into a pumpable consistency. The concrete is then placed through a swing valve concrete pump and a cement set accelerator. Air is added at the nozzle to blow the material onto a surface where it hardens rapidly enough not to slump off the applied surface. The hardening is achieved by the addition of chemical accelerators that quickly raise the viscosity of the concrete to a level high enough to prevent slumping. Usually, in practice, 3–6 wt % accelerator is added for each 100 wt % of the dry concrete mix to attain the desired stiffness.

Since portland cement and calcium aluminate cements are quite different in their chemistry, hydration characteristics and rheological behavior, the accelerators used for one do not necessarily work for the others. Although low and ultralow cement castables show excellent performance when installed by vibration, their properties decline significantly while they are being used for gunning. Castable installation by gunning provides significant advantages, e.g., no forming or vibration is needed.

Within the last few years, new application technology has been developed whereby calcium aluminate cement containing refractory castables can be applied like portland cement concrete shotcretes.

This system mixes the low cement castable with a measured quantity of water. A pump transports the mixture and applies it through a nozzle. An accelerator is added as in portland cement shotcrete where the air comes for spraying at the nozzle.

Two systems work in the shotcreting of LCC. First, the addition of appropriate deflocculating and dispersing chemicals makes the castable flowable at the minimum water level. This is an important criterion since the refractory castable properties depend on the amount of water added for mixing just like vibratable LCC. Physical properties such as density, porosity and strength characteristics are directly related to the water addition to the castable. The accelerator is then added at the nozzle and capable of sticking to the surface without slumping. Although the properties obtained from shotcreting LCC are slightly inferior to LCC installed by vibration, they provide good properties as well as ease and convenience of installation. The slight deterioration in properties is due to a certain amount of air entrapped while applying the spraying air. This method has been highly effective in different applications and its uses are increasingly gaining market shares throughout refractory user industries.

In the literature,²⁵ various additives are known to accelerate the hydration of calcium aluminate cement and can freeze the material sprayed on the surface. Some of these additives are more effective than others and hence the selection of the accelerator will be dependent on the composition and nature of the mix. Hydrated lime,²⁶ calcium chloride and dipotassium phosphate have been cited²⁷ as the accelerators in various alumina containing castables.

Comparative properties — such as installation rate, porosity, cold modulus of rupture of the gun/spray materials — of a 60% alumina castable are compared with LCC and conventional gunning in Figs. 13, 14 and 15 respectively.

The chemical analysis and the physical properties are given in Tables 8 and 9.

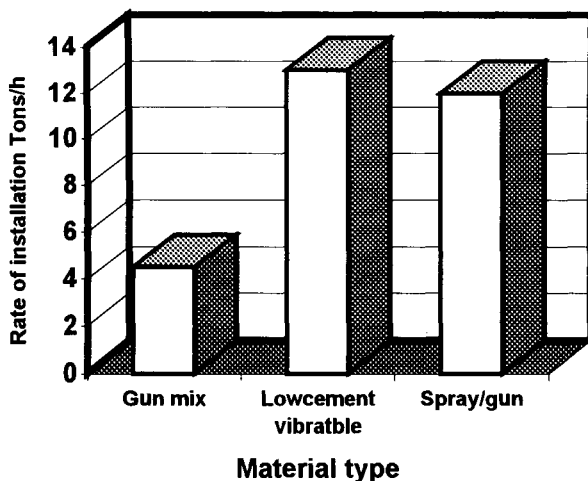


Fig. 13. Comparative rates of installation of Gunning, LCC and Spray/Gunning materials.

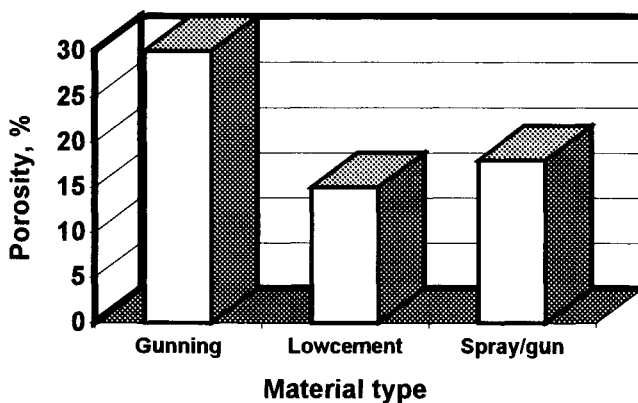


Fig. 14. Comparative porosity of Gunning, LCC and Spray/Gunning materials.

Although somewhat lower than vibratable castables, the physical properties clearly show the advantages over gunning refractories with similar compositions. Attempts have been made to improve the

properties further, particularly with ultralow cement castables. Table 10 shows such an improvement in the 80% alumina ultralow cement spray/gunning composition along with the vibratable version of the ULCC.

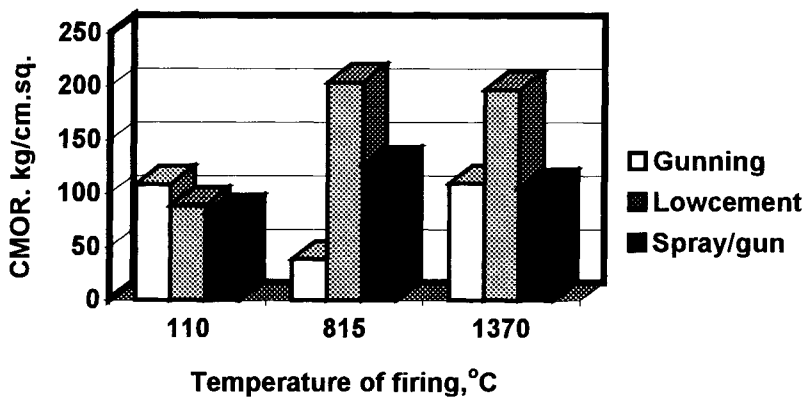


Fig. 15. Cold modulus of rupture of Gunning, LCC and Spray/Gunning materials after firing at different temperatures.

Table 8. Chemical composition of spray/gunning castables.

Alumina Class	50	60	65	80	90	95
Al ₂ O ₃	48.8	58.8	63.5	80.7	91.5	93.9
SiO ₂	46.4	36.1	31.7	13.0	4.3	4.3
TiO ₂	1.4	1.7	1.4	2.7	1.2	—
Fe ₂ O ₃	0.8	0.9	0.8	1.0	0.5	0.1
CaO	2.1	2.1	2.1	2.0	2.1	1.4

With the recent popularity of self-flow castables, the shotcreting of low cement castables will be more advantageous. Since self-flow castables have similar (sometimes better) properties compared to low cement vibration castables, the shotcreting process provides more convenience for users. Shotcreting of castables provides the positive

aspects of gunning while eliminating dusting and rebound. In addition, shotcreting provides consistency and improved properties.

Table 9. Physical Properties of Spray/Gunning Castables.

Alumina Class	50	60	65	80	90	95
Apparent Porosity, % After firing to 815°C	18.0	15.2	18.5	19.0	19.7	17.0
CMOR, kg/cm ² , After firing to						
110°C	85	82	107	89	112	109
815°C	92	127	129	118	118	173
1100°C	118	116	143	127	80	109
1370°C	74	106	120	81	63	106
1600°C	130	198	145	130	75	74
CCS, kg/cm ² After firing to						
110°C	130	165	338	366	350	278
815°C	151	211	278	458	472	401
1100°C	278	207	304	371	294	525
1370°C	236	218	421	405	359	535
1600°C	267	324	670	658	621	74
PLC, % After firing to						
1370°C	0.6	0.6	0.4	1.3	0.1	0.1
1600°C	0.1	-0.1	-0.5	-1.3	0.8	0.8

Castables are usually mixed in batch-type mixers with capacities ranging from 500 kg to 1500 kg. Unlike continuous mixers which can create problems from drops in water pressure and segregation in the mix, the batch type provides the control needed for precise water addition. The wet-mixed material is then dumped into a twin cylinder reciprocating engine, where one cylinder draws the refractory from the hopper on the return stroke and the other pushes it on the forward stroke into the swing valve leading to the line. The pistons are driven

by hydraulic cylinders and a hydraulic pump that can be diesel, electric or air powered.

Table 10. Chemical Composition and Physical Properties of Ultralow Cement Vibratable and Spray-Gunning Composition.

	Vibratable	Spray-gunning
Al ₂ O ₃	82.6	77.2
SiO ₂	12.9	18.0
TiO ₂	2.0	2.4
Fe ₂ O ₃	1.0	1.1
CaO	0.8	0.7
Apparent Porosity, %, After firing to 815°C	14.6	13.6
CMOR, kg/cm ² , After firing to		
110°C	123	111
815°C	236	171
1100°C	239	268
1370°C	222	169
CCS, kg/cm ² , After firing to		
110°C	521	338
815°C	1090	451
1100°C	1100	592
1370°C	1230	528
P.L.C., % After firing at		
110°C	—	−0.1
815°C	−0.1	−0.1
1100°C	−0.3	−0.3
1370°C	−0.1	+0.9

The following factors have been cited²⁸ as directly affecting the pressure required for the pump:

- Mix design;
- Swing tube reduction;
- Pipeline reduction;

- Pipeline system (steel pipe vs. rubber hose);
- Nozzle;
- Head pressure; and
- Pumping output rate.

With the recent development of self-flow castables, the mix design is well controlled for good flow of the material through the pump, thus putting less pressure on the mechanism. The most important factor in successful pumping is the amount of reduction from the swing tube and reductions before entering the pipe and hose system. Reductions are directly related to the amount of cross-sectional surface area. Because flexible hoses offer more resistance to material movement, the use of steel pipes wherever possible reduces the pressure on the pump. Nozzles can be either straight or spiral since neither design causes any extra pressure on the pump provided the diameter of the nozzle is the same as that of the flexible pipe.

Extra pressure is exerted on the pump if the mix has to be moved vertically rather than horizontally or downwards. If the designed output rate is high, a higher capacity pump is desirable to avoid running at full capacity. For a successful shotcreting installation, the following points need to be considered:

- Properly designed refractory mix composition and associated mixing requirements;
- Appropriate choice of the pump, cylinder size and subsequent reduction in pipe size;
- Properly designed surface and anchor system to ensure bonding of the refractory and proper depth of lining; and
- Familiarity with the conditions and safety requirements of the high pressure pumping equipment.

References

1. S. Banerjee, K. N. Singh and C. T. Youmans, "Use of Pumpable Castable in Bethlehem Steel Blast Furnace at Sparrows Point," *Steel World Qurtly.*, Oct. 1994.
2. S. Banerjee, M. W. Anderson, J. P. Singh and R. B. Poeppel, "Reaction of Iron and Steel Slags with Refractories," *UNITECR '93 Proc.* 3, pp.1481–1492.
3. S. Banerjee, "No-Cement Castables or Pumpables," *Proc. Indian Refr. Congress*, 1993, pp. 575–586.
4. S. Banerjee, "State of the Art of Blast Furnace Cast House Refractories," *Steel World Qurtly.*, Sept. 1993, pp. 53–58.
5. S. Banerjee, "A New Era of Blast Furnace Cast House Refractories," *Proc. AISE Mtg.*, Sept. 1993.
6. S. Banerjee, "A Leap into the Nineties with Innovation in Refractory Technology," *Trans. Indian Cer. Soc.*, **50**, 3, pp. 49–60, 1991.
7. S. Banerjee, et al., *U.S. Patent* # 5,147,830, Sept. 15, 1992.
8. S. Banerjee, et al., *Australian Patent* # 634,580, Feb. 25, 1993.
9. S. Banerjee, et al., *Europe. Patent* # 0,425,086, May 2, 1991.
10. S. Banerjee, et al., *Canada. Patent* # 2,027,378, Oct. 31, 1995.
11. C. C. Furnas, "Grading Aggregates: I," *Ind. Eng. Chem.*, 23 (9) pp. 1052–58, 1931.
12. F. O. Anderegg, "Grading Aggregates: II," *Ind. Eng. Chem.*, 23 (9), pp. 1058–64, 1931.
13. B. Myhre, "The Effect of Particle Size Distribution on Flow of Refractory Castables," *Proc. on Refract.*, St. Louis Sec., March 1994.
14. G. MacZura, V. Gnauck and P. T. Rothenbuehler, *Proc. 1st. Int. Conf. Ref., Tech. Assn. Refr.*, Japan, Nov. 15–18, pp. 560575, 1983.
15. Andreassen and J. Anderson, *Kolloid Z*, 50, pp. 217–228, 1930.
16. D. R. Dinger and J. E. Funk, *Interceram*, **41**,(1), pp. 10–14, 1992.

17. *Ibid.*, (2), pp. 95–97.
18. *Ibid.*, (3), pp. 176–179.
19. *Ibid.*, (5), pp. 332–334.
20. *Ibid.*, 42, (3), pp. 150–152, 1993.
21. A. Mohmel and W. Gessner, "Untersuchungen zur Bindungsgenese in Zementarmen Feuerbetonen (LCC)," *Proc. XXXVI Int. Coll. Refr.*, Aachen, 1993.
22. H. S. Kang, C. H. Cha and S. Y. Yang, "Development of Self-Flowing Castable," *UNITECR '95 Proc.* pp. 264–271, Nov. 1995.
23. B. Myhre and A. Hundere, "On the Influence of Superfines in High Alumina Castables," *Private communication*.
24. D. R. Dinger and J. E. Funk, "Particle Size control for High Solids Castable Refractories," *Proc. 13th. Ann. Symp. Refr., Amer. Cer. Soc.* St. Louis Sec. 1994.
25. G. MacZura, F. J. Rohr, D. V. Ruhs, P. T. Rothenbuehler, V. Gnauck and M. Van Den Berg, "Effect of Additives on High Purity Calcium Aluminate Cement (CAC) Castable Properties," *Presented at ALAFAR XVIII Con.*, Oct. 24–26, 1988.
26. E. Allison, U. S. Patent # 5,628,940 May 1997.
27. M. C. Langenohl and G. O. Hughes, *U. S. Patent # 5,512,325* Apr. 30, 1996.
28. W. G. Allen, "Current Developments in Equipment for Refractory Placement," *Proc. 32nd. Ann. Symp. Refr., Amer. Cer. Soc.*, St. Louis, 1997.

CHAPTER IV: PLASTIC REFRACTORIES

Introduction

Until the early part of the twentieth century, shaped firebricks were the only refractories used for high temperature use. However, the use of such refractories posed certain restrictions and problems.¹ Firstly, the mortars used for the joint material introduced weak links and were the main points of failure. Secondly, as the furnace construction design became progressively complex with firebricks, skilled brick layers were needed which increased the cost of construction. Thirdly, the removal process of bricks from the damaged area for repair caused the remaining brick structure to become weak and needed more unnecessary repair incurring more time and cost.

To avoid such problems of using exclusively firebricks, the first solution was put forward by W. A. L. Shaefer. He invented the first pliable refractory brick that could be put in place and molded to the desired shape. It was the first instance of the use of plastic refractory. As the name implies, the refractory behaved like a plastic that could be shaped as desired. Initially, the standard installation was done by a mason with a 1.0 to 1.5 kg hammer and a helper who fed material to him. Later, in the 1940's the pneumatic or air hammer was introduced

for installing refractory plastics. It eliminated the metal hand hammer increasing rates and improving lining integrity.

Composition

Plastic refractories usually contain two kinds of refractory raw materials. The main aggregates used must be stable at high temperatures like prefired materials and high-temperature-stable raw materials such as chrome ores and calcined flint clay. In recent years, other materials like tabular alumina, calcined bauxite, andalucite, fused or sintered mullite and silicon carbide aggregates are being used for various applications. A second ingredient added to provide the pliability or plasticity of the refractory body, is plastic fire clay, usually containing montmorillonite as the main or part of the clay mineral.

Evaluation

The main index by which a plastic refractory is evaluated is its workability or plasticity. It is usually an empirical value obtained by standard methods prescribed by ASTM or DIN. Plastic refractories usually lose their workability with time that is termed as aging. It may be a result of evaporation of the moisture, polymerization reaction of some ingredients in the mix, or the mutual reaction between some of the ingredients themselves. Unlike many other refractory materials, plastic refractories always have a time limit for their effective usage before they become too hard to use. To avoid losing the plasticity before the predicted time, plastic refractories should be stored away from areas exposed to direct sunlight or high atmospheric temperatures.

Plastic refractories freeze when the temperature falls below 0°C and need to be winterized. Although there is a slight loss of strength after thawing of frozen plastic refractory, it can still be used in most cases after thawing. The freeing can be avoided by using organic antifreeze such as ethylene glycol along with water.

Bonding systems

Several types of bonding systems are available for plastic refractories. They are:

1. Clay bonding — they have been used in the earlier types of plastic refractories and are quite inexpensive. They can suffer high shrinkage particularly, during the drying and dehydration period, but part of this shrinkage can be compensated by appropriate selection of raw materials in the composition.
2. Clay and sulfate bonding — this type of bonding is still in use in plastic refractories and is inexpensive. However, they provide additional low temperature strength and shrinkage stability.
3. Phosphate bonding — it is the most prevalent type of plastic refractories in the market today. Phosphate bonded refractories can range from low alumina-silicate to very high alumina, and may contain chrome oxide, silicon carbide, carbon or graphite in various proportions. Care must be taken to retard the chemical reaction of phosphoric acid with the ingredients in the refractory because it may drastically reduce the workability. Most common retarding agents are the monohydroxy, polyhydroxy and polycarboxy organic acids, such as, oxalic, citric and tartaric acid.
4. Organic bonding — other types of bonding by organic materials like molasses, various synthetic resins and coal tars are used in this class of bonding. They usually contain carbon or graphite as one of the ingredients. The use of organic binders often leave a carbon structure in the matrix that introduces special properties to the refractory.

Types of plastic refractories

Heat-setting super-duty fireclay plastics² form a solid monolithic surface that is highly resistant to thermal shock and many acid slags. They have excellent workability and low shrinkage properties, making

them the ideal choice for many applications. Cold-setting plastics with similar features are also available in the market.

Super-duty heat-setting plastics with graphite exhibit excellent resistance to wetting and corrosion by molten metal and slag. Plastics with 50% alumina are resistant to spalling due to thermal shock and many types of acid slags. Heat-setting 60% alumina plastics offer higher refractoriness than super-duty plastics with increased strength and volume stability through their temperature range. Air-setting high alumina plastics in the 80% alumina class are primarily used over 60% alumina plastics where improved refractoriness is desired. They offer good resistance to fluxing oxides and slags up to their service temperature.

Phosphate-bonded high-alumina plastics ranging in alumina contents from 50 to 90%+ are widely used in many applications as primary lining materials and for patching existing refractory linings. These products typically have high density and strength, combined with excellent volume stability throughout the temperature range. Plastics ranging from 70 to 85% alumina content are often used in applications where resistance to slags and metal wash are required whereas, 85% alumina refractory is widely used for high abrasion resistance. Plastics in the 90% alumina range are based on high-purity alumina and hence have high strengths at high temperatures.

Phosphate-bonded alumina chrome plastics have significantly high strength at high temperatures. These compositions based on high-purity alumina aggregates and chromic oxide, form an alumina-chromia solid solution bond with purple to pink color at temperature above 1400°C providing good resistance to high iron oxide slags and coal slags. Other phosphate-bonded alumina-chrome plastics with lower alumina contents are also available for specific service conditions. These include mullite based alumina-chrome plastics that have good slag resistance to acid and neutral slags. The presence of mullite grains allows for earlier reaction, making this type of refractory ideal for slagging applications between 1370°C and 1590°C.

Silicon carbide-based phosphate-bonded plastics with aluminum phosphate bond have high thermal conductivity, and non-wetting properties to many acid slags and non-ferrous metals.

The principle of phosphate bonding is based on the reaction of phosphoric acid or acid phosphates with reactive alumina. The reaction between phosphoric acid and alumina goes through different phases of aluminum phosphate as the temperature increases. The progressive phase changes with temperature are shown below:

1. Room Temperature: $\text{H}_3\text{PO}_4 + \text{Al}_2\text{O}_3 \rightarrow 2\text{Al}(\text{H}_2\text{PO}_4)_3$
(Aluminum acid phosphate; water soluble)
2. Around 260°C: $2\text{Al}(\text{H}_2\text{PO}_4)_3 + \text{Al}_2\text{O}_3 \rightarrow \text{Al}_2(\text{H}_2\text{P}_2\text{O}_7)_3 + 3\text{H}_2\text{O}$
(polyphosphate; water soluble)
3. Around 500°C: $\text{Al}_2(\text{H}_2\text{P}_2\text{O}_7)_3 + 3\text{H}_2\text{O} \rightarrow \text{Al}(\text{PO}_3)_3$
(beta metaphosphate)
4. From 500°C to 1200°C: $\text{Al}(\text{PO}_3)_3 \rightarrow \text{Al}(\text{PO}_3)_3, \text{AlPO}_4$
(Beta-800°C and Alpha-1200°C Orthophosphate)
5. Above 1370°C: $2\text{AlPO}_4 \rightarrow \text{Al}_2\text{O}_3 + \text{P}_2\text{O}_5$
(P_2O_5 gradually becomes volatile that is time temperature dependent).

Properties

Properties of plastic refractories are somewhat unique in nature. For attaining the pliable behavior of the refractory body some type of plastic fire clay must be used in the composition. These plastic fire clays are characterized by high shrinkage during their drying period. These clays have plate like structures (montmorillonitic clay) which absorb moisture between layers and tend to swell somewhat that gives them the pliable behavior. During the drying period the water dries out allowing the body to shrink somewhat. All plastic refractories shrink around 1% during the drying period. At high temperatures, above 1400°C, the shrinkage is somewhat compensated by the formation of mullite. Although the plastic refractories show shrinkage during drying, they are quite stable at service temperatures.

All plastic refractories need to be dried very carefully with predetermined heating schedule otherwise cracking or spalling may occur. Phosphate -bonded plastic refractories tend to form a hard crust on the surface if the drying rate is too fast, especially below 350°C. Phosphate bonded plastic refractories must be dried to a temperature of at least 350°C before they can be used, particularly before exposing to steam. Phases formed at temperatures below 350°C are water soluble and the refractory loses its strength and disintegrates if exposed to steam environment.

The cold and hot modulus of rupture (CMOR and HMOR) measured for a typical plastic refractory shown in the Fig. 1. The cold MOR is low at low and intermediate temperatures, but increases above 1300°C due to sintering. Between 1100°C and 1200°C HMOR is higher than CMOR, but is the highest around 1000°C which is pronounced in the silica-alumina system. HMOR does not drop below CMOR until the temperature reaches 1200°C.

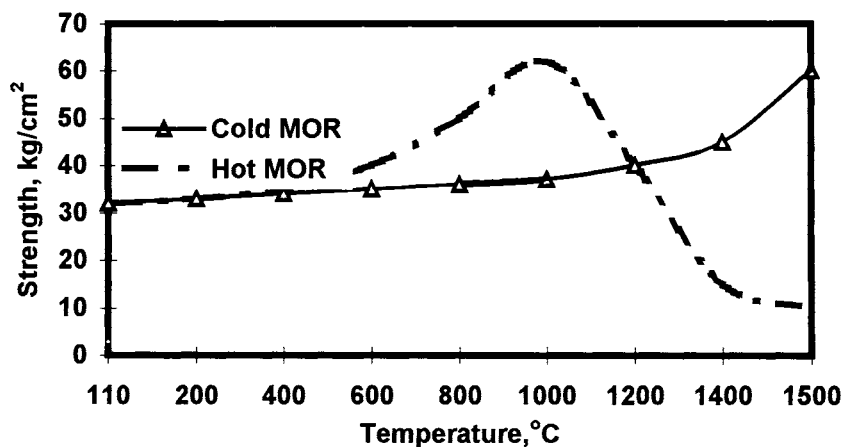


Fig. 1: Hot and cold modulus of rupture of a plastic refractory.

Depending on the composition, plastic refractories can be designed for abrasion and corrosion resistance. A phosphate bonded high alumina plastic is the standard for abrasion resistance and has

been used in petrochemical industries for a long time. Corrosion resistance to molten metal and slag can be achieved by the addition of silicon carbide and graphite in the composition. Chemical and physical properties of some typical plastic refractories are given in Tables 1 and 2.

Table 1. Chemical analysis of some typical plastic refractories.

Category, %Al ₂ O ₃ Class	40	50	60	70	90	80-SiC
Bond and Setting	Air & heat	Chem.	Air & heat	Chem.	Chem.	Chem.
Chemical Analysis, (Calc.),%						
Al ₂ O ₃	41.6	53.6	58.2	69.3	90.5	7.1
SiO ₂	50.9	38.5	35.4	24.7	4.9	7.9
Fe ₂ O ₃	1.4	1.1	1.4	0.9	0.5	0.1
TiO ₂	1.5	1.2	2.2	1.3	0.9	—
P ₂ O ₅	—	2.8	—	2.7	2.7	2.7
SiC		—	—	—	—	80.6

Krietz³ et al. examined the fracture behavior of plastic refractories with various types of bonding for plastic refractories, such as, clay, clay and sulfate, and phosphate-bonded ones containing various types of alumina. From their study they concluded that

1. There is a radical difference in the behavior in clay and sulfate bonded plastics vs. phosphate bonded plastics. Clay and sulfate bonded plastics show a dramatic increase in work of fracture (WOF) between 800°C to 1100°C, indicating considerable resistance to crack growth in this temperature region. It is further demonstrated by the ribbon thermal-shock test and also

by their usage in the steel mill such as soaking pit covers. The WOF for the phosphate bonded plastics declines in this region.

Table 2. Physical properties of some typical plastic refractories.

Category, %Al ₂ O ₃ Class	40	50	60	70	90	80-SiC
CMOR, MPa After firing to						
110°C	2.41	7.21	2.45	9.65	11.7	6.21
815°C	2.1	9.60	2.43	7.23	13.8	6.90
1370°C	4.82	13.1	3.75	12.5	20.7	13.7
PLC, % After firing to						
110°C	-0.6	-0.6	-0.7	-0.4	-0.6	-0.4
815°C	-0.7	-0.5	-0.8	-0.4	-0.6	-0.5
1370°C	+0.5	-0.3	+0.3	+0.2	-1.1	-0.4
HMOR, MPa at						
815°C	2.76	9.65	2.10	12.41	12.40	6.21
1370°C	1.2	6.90	3.80	10.34	8.28	8.95

- Below 800°C, clay and sulfate bonded plastics have lower HMOR than phosphate bonded plastics, but the differences in the WOF are less significant.

Although the work was not totally conclusive, it certainly explains the behavior of the plastics in actual usage.

Application

As mentioned earlier, plastic refractories are installed by pneumatic rammers with a convex aluminum head. To eliminate voids, joints are usually staggered for succeeding courses. Ramming should always be done with the ramming handle perpendicular to the hot face. Since the installation of plastics is totally manual, the performance depends on the quality of installation of the entire material. Plastics can be

installed around metal or ceramic anchors and specific care should be taken so that no gaps remain around the anchors.

For certain installations, such as the cyclones of petroleum refineries, the plastic refractory is usually installed by hand with rubber mallets. The installed thickness is about 35–40 mm and the installation is done in hexmesh or around S-anchors. After installation, the surface of the refractory is trimmed to provide a smooth surface. If ceramic anchors are used the surface should be trimmed to the face of the refractory anchors. For metal anchors, there should be at least 30–40 mm of refractory above the tip of the anchors, excepting hexmesh or S-anchors.

By nature of the formulation, plastic refractories are difficult to dry out. Hence to facilitate the drying process, holes are pierced through the plastic body with rods with 3–5 mm diameter at intervals of 20–30 cm up to two-thirds of the thickness of the installed refractory.

Plastic gunning refractories

In 1980, a technique was developed whereby gunning of plastic refractories became possible without the addition of water at the nozzle. Since then, widespread field experience on this technique has been gained around the world.⁴

This new application method provided certain distinct advantages over conventional plastic and gunning refractories, namely, (1) high material replacement rates, (2) reduced man-hours for installation, (3) uniform installation without lamination, (4) eliminating for water addition at the nozzle avoiding human variance and (5) good physical properties.

The plastic refractory is often tailored to the specific application where the water content, workability and the overall composition must be controlled at the manufacturing plant. The plastic material is then modified for gunning by reducing the material to small pieces by a shredder-granulator. The shredded material then goes through a feed

conveyer to an extensively modified rotary gun. The material is then applied in place directly from the gun without any water addition. Chemical and physical properties of some plastic gunning refractories are given in Tables 3 and 4.

This system not only eliminates the human factor involved in gunning and ramming installation of plastic refractories, but provides an easier method where the addition of water to the refractory becomes a problem because of ambient temperature conditions and availability of appropriate water source. There is another advantage for the method over conventional plastic application method. Since conventional plastic refractory cannot be used for gunning, there are certain positive aspects of the gunning plastic over conventional ones. The shrinkage for gunning plastic is less, the creep resistance is better and the application scope is much better than the conventional plastic refractories.

Table 3: Chemical analysis of plastic gunning materials.

Category Class, Alumina	50	60	70	80	85	90
Bond & Setting Characteristics	Air/heat	Air/heat	Chem	Air/heat	Chem	Chem
Chemical Analysis (Calc.),%						
Al ₂ O ₃	44.0	58.5	69.0	78.0	84.5	89.0
SiO ₂	49.0	24.5	24.0	17.0	8.5	6.0
Fe ₂ O ₃	2.1	1.4	1.0	1.5	1.2	0.5
TiO ₂	1.4	0.5	1.6	2.2	1.0	0.6
P ₂ O ₅	—	—	2.7	—	2.7	2.7

In 1985, a gunning machine emerged in the market with the capability of transporting large quantities of refractory plastics which made the plastic gunning practice quite popular. However, due to their sticky nature, refractory plastics, are difficult to gun. In 1986, it

was found that an addition of small quantities of certain organic materials⁵ enhanced gunning characteristics greatly that allowed rapid installation of all qualities of refractory plastics.

This method differs from the previous plastic gunning system. In this process the plastic is made as a granular material in the plant with initial plasticity level as that of a regular plastic refractory. The granular material, fortified with the organic additives, is transported in cardboard boxes to the place of installation. During installation, a small quantity of water is added at the nozzle for gunning. These materials available in a wide variety of compositions showed smooth flowability through the hose, minimum rebound, high degree of adhesiveness, rapid installation, and uniform lining. In service, they showed better resistance to spalling, volume stability, and thermal shock resistance. Properties of some of the gunning plastics are shown below in Table 5:

Table 4: Physical properties of plastic gunning materials.

Category class, Alumina	50	60	70	80	85	90
CMOR, MPa						
After firing to						
110°C	3.5	4.1	8.2	3.8	8.9	10.3
815°C	3.1	3.8	9.0	2.8	10.3	13.8
1370°C	6.2	5.2	13.8	4.8	13.8	12.9
HMOR, MPa at						
815°C	4.1	4.1	13.1	4.8	13.8	13.8
1370°C	1.0	1.4	5.5	2.8	5.5	8.3
PLC, %						
After firing to						
110°C	-0.5	-0.4	-0.5	-0.5	-0.5	-0.4
815°C	-0.7	-0.4	-0.6	-0.6	-0.4	-0.4
1370°C	+0.6	+0.1	+0.1	+0.6	+0.2	+0.3

Table 5: Properties of Plastic Gunning Materials.

Classification, % Al ₂ O ₃	85	50	80
Bonding type	Phosphate bond	Air set	Phosphate bond - hot gunning
Bulk Density, g/cc, as made	3.04	2.42	2.93
CMOR, MPa, After firing to			
110°C	8.15	1.52	8.32
815°C	7.57	1.55	15.53
1093°C	14.5	3.78	12.46

Drying and heating

The drying and heat-up of installed refractory should be started as soon as possible after installation is finished since such is often the desire of the users. If not, the surface of the plastic refractory should be covered by curing compounds to prevent moisture from escaping which gives rise to shrinkage cracks. If shrinkage cracks appear they can be covered by curing compounds or with a slurry of heat-setting mortar to prevent them from getting larger. The drying and heat-up schedules for heat- and cold-setting plastics are different from phosphate-bonded plastics. In general, the following schedules can be followed for safe drying and heat-up for the different bonding types of plastics:

Drying and heat-up schedule for heat- and cold-setting plastics:

1. Heat from ambient temperature to 250°C @ 25°C per hour.
2. Hold at 250°C for one hour per 25 cm of thickness.
3. Raise to 550°C @ 40°C per hour.
4. Hold at 550°C for one hour per 25 cm of thickness.
5. Raise to operating temperature @ 60°C per hour.

Drying and heat-up schedule for phosphate-bonded plastics:

1. Heat from ambient temperature to 125°C @ 25°C per hour.

2. Hold at 125°C for one hour per 25 cm of thickness.
3. Raise to 250°C @ 30°C per hour.
4. Hold at 250°C for one hour per 25 cm of thickness.
5. Raise to 550°C @ 30°C per hour.
6. Hold at 550°C for one hour per 25 cm of thickness.
7. Raise to operating temperature @ 60°C per hour.

If at any time steam appears to be evolving from the refractory surface, the temperature should be kept constant at that temperature until the evolution is completed.

From a study of the effect of curing and drying of phosphate-bonded plastics⁶, it was observed that there is a significant migration of the phosphate bond to the surface. Repeated abrasion tests of the phosphate plastics revealed that there was not much difference in material losses, even though there was evidence of the presence of significant amounts of aluminum phosphates on the surface. Differences were noticed both in abrasion loss and strength when specimens were cured according to ASTM C-1054, air-dried in the mold for 24 h followed by 66°C for 24 h and 24 hr at 105°C compared to other specimens air dried out of the mold for 6 days and at 105°C for 24 h. Most improvement was noticed in the last set of specimens.

From electron microprobe point analysis, the surface and interior bond compositions of both 50% and 90% alumina plastics were found to be significantly different. Almost twice the phosphate content was present on the surface than in the interior part. Significant bond migrations were found in all samples, but the extent of bond migration imparted by the curing and drying methods was less pronounced for high-alumina plastics. X-ray diffraction (XRD) studies of the bond phases of both 50% and 90% alumina plastics indicated that significant quantities of both aluminum metaphosphate (AlPO_3)₃ and aluminum orthophosphate (AlPO_4) were present in the 50% alumina plastic fired to 815°C, whereas only aluminum orthophosphate was present in the 90% alumina plastics fired to 815°C. This observation is in agreement

with other findings that orthophosphate formation is favored in plastics where higher amounts of alumina is present.

It can be inferred from this study that longer air curing of phosphate bonded plastic provides better abrasion resistance and strength which is effected by bond migration. This study further indicates that the drying schedule of phosphate-bonded plastics has to be well controlled to prevent any imperfection of the surface due to cracking or spalling.

Tap hole plastics (mud)

One major unique type of plastic refractories is in continuous use all over the world. They are used for plugging the tap hole of blast furnaces. The major objective of this material is not only to plug the tap hole of the blast furnace but also to be drilled to allow the tap for the subsequent cast. In the earlier stages the plastic was made from silica, fireclay and some kind of carbon (coke) with an aqueous binder. As blast furnaces increased in size, inside pressure and productivity, the demand for improved tap hole clay became obvious.

At present, all blast furnace tap hole plastics contain an organic binder like coal tar or synthetic resin (phenol-formaldehyde type or similar) and are combined with carbon and graphite to provide certain specific properties.

Following each cast at the blast furnace, the tap hole must be closed quickly. The tap hole plastic is injected through a hydraulic mud-gun that pushes the refractory plastics into the tap hole and closes it to prevent further molten material (iron/slag) from coming out. In recent years, a soaking bar practice has been introduced. The soaking bar method consists in introducing a rod about 25 mm thick through the tap hole following the plugging of the hole. While opening the hole, the rod is pulled out and the molten iron comes out in a controlled small stream through the hole. The specifications for these products particularly for use with the soaking bar method, are

very rigid. To produce a successful tap-hole plastic the following conditions need to be satisfied:

1. Depending on the size and operational conditions of a specific furnace, the plasticity of the material must be properly designed. The plasticity of the material has to be satisfied and constantly monitored by the user since the performance criterion is critical.
2. For the operation with soaking bar, the consistency of the material should be such that the rod can be introduced without difficulty and also be withdrawn smoothly during the following cast.
3. It is desirable that the tap-hole plastic maintain a good retained length in the hole (indicating good resistance to corrosion) to protect the integrity of the tap-hole refractory.
4. During casting the tap-hole passage should resist corrosion/erosion by the iron/slag to maintain a uniform size and not open up with the progress of the casting.

The functional role and properties of the tap hole are discussed in detail by Delabre et al.⁷ The functional role of the tap hole plastic are:

1. After injection, it assures the obstruction of the blast furnace tap hole as well as the hearth tightness.
2. Before casting, it provides the passage through drilling to assure a regular steady output of the molten iron and slag.
3. Provides the hearth refractory protection in the tapping hole zone.

The tap hole plastic should have the following properties:

1. Pliability in a close range at the injection temperature, so that the material can be pushed through the mudgun with the available pressure.
2. After plugging of the hole, it should have mechanical and chemical resistance to protect the tap hole from breakout.

3. It should provide relative ease of drilling the hole before casting.
4. It should resist rapid erosion of the hole to provide a steady narrow stream of the iron and slag outlet.

The above properties depend on chemical and mineralogical compositions, proportions and the grain-size distribution of its different components. Refractory aggregates consist of silica, alumina, calcined bauxite, calcined fireclay, and sintered mullite. Carbon containing components are composed of carbon black, anthracite, graphite and calcined coke. Additives may consist of silicon carbide, silicon nitride, ferro-silicon nitride and metal powders. The binder is usually coal tar or synthetic resin. Phenol formaldehyde is used most commonly whereas furan resins are also used.

The rheological behavior of the plastic refractory is primarily dependent on the binder characteristics. Coal tar, most commonly used, is required to be within close specified range regarding its distillation characteristics at certain temperatures. The specific quantity of the residue after distillation, carbon disulfide solubility, moisture content and above all a close range of viscosity values at a certain temperature are all critical. The specification for the coal tar is extremely important to assure proper function of the tap-hole plastic in use. The characteristics of the resin and the hardener are also important where a resin is used as the binder. The viscosity of the liquid resin, free phenol content as well as the nature and quantity of the hardener also contribute to the rheological behavior of the refractory.

Apart from the binder characteristics, the rheology of the tap-hole clay depends on the ratio of the binder to the refractory materials, particularly the grain size distribution, shape and porosity. For both coal tar or resin binder, the tap-hole plastic goes through an aging process. It is caused by adsorption and absorption of the binder in the grains and transformation of the binder itself by the evaporation of the most volatile products as well as by oxidation and polymerization. As

mentioned before the tap hole plastic has a definite shelf life that is strongly dependent on the storage climate.

Customarily, various additives are used in the tap-hole plastic to achieve special properties such as extrusion resistance, permanent dimension, hot strength and corrosion resistance. These additives are: silicon carbide, ferro-silicon nitride, metallic aluminum and silicon, fine and ultrafine alumina and silica. As expected, the effects of metallic aluminum and silicon are the most pronounced and those of fine and ultrafine alumina and silica are the least pronounced.

Recently, a pollution-free taphole mix has been reported.⁸ The majority of taphole mixes, used for blast furnace taphole mud applications, are kneaded with coal tar and/or liquid resin as the binder to adjust mix plasticity and to achieve good binding property. However, these binders contain harmful materials that pose health hazards and are regulated by environmental laws in many countries. The problem with coal tar is the emission of benzo(a)pyrene which is known to be a carcinogen whereas liquid resin contains formaldehyde and various parts of phenol that are also health hazards. The investigation was done by the use of fractional oil of coal tar having similar features to coal tar but containing significantly lower amounts of benzo(a)pyrene. Since the boiling point of benzo(a)pyrene is 392°C, it is theoretically possible that coal tar fractional oil, fractionated at a temperature lower than 392°C contains no benzo(a)pyrene. For this project, two kinds of fractional oils, such as heavy oil and anthracene oil are used since they have higher carbon contents and are expected to have higher bonding strength. Table 5 below shows different fractions of coal tar.

Table 5: Different fractions of coal tar.

	Light oil	Medium oil	Heavy oil	Anthracene oil
Fractional Temperature, °C	<160	160–240	240–270	270–370
Specific gravity	0.88	0.98	1.05	1.09

From plasticity tests (Marshall Value at 50°C) using standard mixes it was found that the mix using heavy oil dramatically stiffened after two months, and hence further experiments were restricted to anthracene oil only. It is also necessary that the taphole mix should have enough strength when receiving heat from the molten pig iron so that it does not run just after the taphole is filled up with the mudgun. It has been observed that mixes with direct replacement by anthracene oil alone do not have enough strength compared with the coal tar mixes. Hence mixes with anthracene oil have been supplemented with carbon, metallic and boron additives. One such combination produced physical properties similar to the coal tar containing mix and contained benzo(a)pyrene well under the regulatory amount (less than 50 ppm). Properties of this mix compared to coal tar containing mix are given in Table 6.

Table 6: Comparison of physical properties between tar bonded taphole mix with newly developed taphole mix (base mix containing bauxite, tabular alumina, silicon carbide, graphite and clay).

Physical Properties after Sintering*	Conventional Tar-bonded Mix	Pollution-free Newly Developed Mix (Anthracene Oil-Based)
CMOR, MPa	3.5	3.6
CCS, MPa	17.4	16.5
Apparent Porosity, %	24.7	26.3
Bulk Specific Gravity, g/cm ³	2.46	2.41
Wear Resistance (Rotary Erosion and Corrosion Test Method)**		
Wear Depth, nm	14.8	14.5
Index	100	98
Benzo(a)pyrene, ppm	1182	10

* Physical properties after sintering test method: The test specimen is formed to a shape of 50 x 50 x 130 mm under 100

kg/cm² pressure and fired in a reducing atmosphere at 1450°C for three hours.

- ** Wear resistance test method: The test specimens are formed under 100 kg/cm² pressure and baked at 350°C. The test specimens are tested in Rotary Erosion and Corrosion Test Machine using blast furnace pig iron and slag.

From the above investigation, it may be concluded that pollution-free taphole mix can be made by using the anthracene fraction of coal tar using carbon, metallic and boron additives to attain properties comparable to a conventional coal-tar based taphole product.

Use of plastic refractories

The types of plastics used vary significantly with applications. Although some of the refractory plastics have been replaced by other monolithic refractories such as LCC, ULCC and no-cement pumpables in recent years, significant amounts of refractory plastics are still used in various industries. Heat-setting plastics are particularly suitable for reheat furnaces, soaking pits, rotary kiln hoods, incinerators, aluminum furnace upper sidewalls, iron ladles, and various foundry applications. Air-setting plastics are used in electric furnace roofs and reheat furnace hearths.

Depending on the chemical nature of the refractory plastic, phosphate-bonded refractories have various applications. High alumina phosphate-bonded refractories are used as primary linings and for patching material for applications such as cyclones of petroleum refineries, iron, steel and foundry ladles. Phosphate-bonded alumina-chrome refractories are used for industrial incinerators, thermal power station boilers, electric furnace deltas and runners, and applications where refractoriness, corrosion resistance and thermal stability are required. Mullite-chrome based phosphate-bonded refractories are used in foundry troughs, ladles and induction furnaces, and electric furnace deltas and runners. Silicon carbide based aluminum

CHAPTER V: RAMMING MIXES

Introduction

Ramming mixes came onto the market after pliable refractories (plastics) had been used for installation and repair. The plastics could be molded into desired forms and shapes. The major difference between plastic refractories and ramming compositions is in their physical nature. The plastic refractories are semi-solid pliable in nature whereas the ramming mixes are either granular semi-wet or dry consistency.

During the 1950s through middle of 1980s, ramming mixes were quite popular and used extensively in a variety of places such as blast furnace cast house trough, iron, slag and tilting runners. They are also used in cupola bottoms, launderers, breast walls and slag boxes, alloy steel melting induction furnaces, electric furnace deltas, spouts and runners. Precast shapes for troughs, runners, skimmers and slag notches are also made from ramming mixes. But the popularity of ramming mixes diminished after low- and ultralow-cement compositions came on the market. Not only did these compositions have superior properties but they were also capable of veneering to previously-used lining surfaces. Ramming mixes could not offer this improved refractory usage and cost effectiveness.

Characteristics

Ramming mixes are usually refractory compositions installed by ramming. Most ramming compositions are semi-dry that knead together when the ramming pressure is applied, usually by pneumatic rammers in recent years. Certain classes of refractories are also rammed in dry form primarily with basic compositions.

Ramming mixes consist of refractory aggregates and a semi-plastic bonding phase. The semiplastic nature is created by the use of clay that depends on the nature of the specific application. The mixes may be based on a wide range of base materials e.g., silica, high alumina, corundum, mullite, calcined bauxite, dead burned magnesite, chrome ore, zircon and other refractory materials.

The major advantages of ramming mixes are their abilities to be — 1) installed rapidly, 2) installed at a relatively high temperature, and 3) to be dried out much quicker than castables or plastics. Depending on the bonding system, the rammed monolith usually produces a lining with relatively lower porosity and high density.

The grain size distribution for ramming mixes is different from castables. Since there is insufficient liquid phase in the system, the relative grain packing must be achieved by mechanical force. This calls for more of the intermediate size distribution in the mix. Unlike castables, where the binding phase is the cement or a similar binder, ramming mixes can be formulated with a variety of types of binders depending on the application.

Binder systems

The binder system can vary widely depending on the application. Generally, ramming mixes are supplied in semi-wet form such that the mass compacts upon ramming. There is a wide variety of materials now used as binders for ramming mixes — both organic and inorganic compounds. The common binders are:

- Inorganic:** Clays, phosphoric acid and phosphates, sulphates, silicates, chromates and chromic acid, colloidal alumina and silica; boric acid and borates for dry ramming mixes;
- Organic:** Carboxymethyl celluloses, lignosulphonates, molasses, resins (phenol-formaldehyde, furfuryl alcohol and furan), coal tar and pitch.

A wide variety of raw material aggregates are being used for ramming mixes. The use of particular aggregates will depend on the end use. The different varieties of raw materials used are listed below:

High alumina compositions —

Tabular alumina, brown fused alumina, calcined bauxite, mullite, and calcined fireclay (Depending on the ultimate use, these aggregates can be mixed with silicon carbide and graphitic materials.)

Graphitic compositions —

Anthracite, graphite, carbon black and coal tar or petroleum pitch

Others —

Silicon carbide, various spinels and zircon

Basic compositions —

Dead burnt magnesite, dolomite, chrome ore, magnesia-chrome grains and chromic oxide

Alumina-silicon carbide ramming mixes are used for copper and brass industries. In lining induction furnaces they are particularly useful because of their excellent resistance to copper penetration. Mullite grain-based high purity ramming mixes are used in electric furnace roofs, induction furnaces, burner blocks and ports.

Because of their excellent resistance to shrinkage and thermal spalling at high temperatures, high alumina (> 80%) ramming mixes

are used in electric-furnace roofs, burner rings and ladle linings for ferrous and non-ferrous industries.

Properties and uses

Phosphate bonded high purity alumina-chrome ramming mixes offer high strength at high temperatures and excellent resistance to acid and neutral slags including coal ash. They are used in working linings where high iron oxide slags in the acid to neutral range are prevalent. Typical chemical analysis and physical properties of some of these mixes are shown in Tables 1 and 2 respectively.

Table 1. Chemical analysis of some alumina-chrome-phos-bonded ramming mixes.

Alumina Class	90	85	80	60
Chemical Composition, %				
Al ₂ O ₃	87.5	84.3	80.0	62.0
SiO ₂	0.2	0.5	7.4	24.7
Fe ₂ O ₃	0.1	0.15	0.25	0.9
TiO ₂	0.1	1.9	0.22	1.4
Cr ₂ O ₃	9.8	9.7	9.0	7.8
P ₂ O ₅	2.1	2.1	2.7	2.7

Alumina-silicon carbide-graphite ramming mixes are used in foundry troughs and electric furnaces spout patching. Typical chemical analyses and physical properties of these are shown in Tables 3 and 4 respectively.

For application under the slag line and in the metal line, high graphitic compositions have been effective because of their non-wetting characteristics. Chemical analyses and physical properties of some typical high graphitic compositions are given in Table 5 and 6 respectively.

Dry ramming mixes based on high purity dead burnt magnesite, along with low temperature sintering aids (e.g., borates and boric

acid), are widely used in electric-furnace bottoms. High-purity magnesite ramming materials are used for the lining of coreless induction-furnace melting ferrous alloys. Magnesia-chrome fused grain based ramming mixes having exceptional density and strength throughout the temperature range are used in degasser linings and under other extreme service conditions.

Table 2. Physical properties of some alumina-chrome-phos-bonded ramming mixes mentioned above.

Alumina Class	90	85	80	60
Bulk Density, kg/m ³				
110°C	2995	3060	2580	2590
815°C	2805	3010	2575	2565
1370°C	2835	3010	2575	2565
CCS, MPa				
110°C	31	28	26	24
815°C	52	55	47	44
1370°C	70	46	58	40
CMOR, MPa				
110°C	14	13	13	12
815°C	22	21	19	18
1370°C	41	28	26	24
PLC, %				
110°C	-0.2	+0.6	-0.5	-0.5
815°C	+0.1	+0.6	-0.5	-0.5
1370°C	+0.8	+0.6	-0.5	-0.4

High alumina (brown fused alumina and calcined bauxite), silicon carbide and graphite containing ramming mixes with various kinds of bonding systems have long been used in blast furnace troughs, slag and iron runners. They offer excellent results because of their good slag resistance and non-wetting characteristics. Glass applications use zircon grain based air-setting high density ramming mixes.

Characteristics of bonding systems

Once the appropriate aggregate system is identified for a specific application, the performance of the ramming mix will be primarily governed by the bonding system until the high temperature ceramic bonding starts to operate. Air-setting and heat-setting bonding based on silicates are widely used. Care has to be taken during the drying and heating of these ramming mixes since they tend to form a layer of crust on the surface which prevents steam from escaping and may cause surface defects. Also, the high temperature uses are somewhat limited because of the presence of alkali metals.

Table 3. Chemical analysis of typical alumina-silicon carbide-graphite ramming materials.

Alumina Class	65	60	60	40	30
Chemical Composition, %					
Al ₂ O ₃	64.0	62.0	58.0	42.0	32.0
SiO ₂	10.0	10.0	10.0	18.0	17.0
Fe ₂ O ₃	1.3	1.3	1.2	1.3	1.9
TiO ₂	1.7	1.8	1.8	1.9	2.3
SiC	13.5	12.0	11.0	13.0	10.0
C	3.0	5.0	6.0	5.0	13.0

The use of chromates and chromic acid has proven to be excellent with respect to their high temperature strength, thermal stability and slag resistance to a variety of compositions. But because of the hazardous nature of the hexavalent chromium (an identified carcinogen), the use of this type of bonding system has been extremely limited in many places.

Table 4. Physical properties of typical alumina-silicon carbide-graphite ramming materials.

BD, kg/m ³					
110°C	2775	2675	2480	2465	2280
815°C	2765	2665	2400	2415	2250
1370°C	2760	2655	2495	2410	2170
CCS, MPa					
110°C	18.5	12.5	10.0	19.0	10.0
815°C	35.0	34.5	10.5	32.5	10.9
1370°C	47.5	55.0	14.0	49.0	12.0
CMOR, MPa					
110°C	6.5	5.5	4.0	5.5	4.0
815°C	7.5	9.0	3.0	8.0	3.6
1370°C	9.5	10.0	4.0	10.0	4.0
PLC, %					
110°C	+0.2	-0.25	-0.25	-0.35	-0.45
815°C	+0.2	+0.2	-0.2	-0.3	-0.45
1370°C	+0.25	+0.25	+0.25	+0.2	-0.35

Table 5. Chemical analysis of some high graphitic ramming compositions.

Classification	60	50	15	15	10
Chemical Composition, %					
Al ₂ O ₃	12.5	20.1	48.8	59.6	73.9
SiO ₂	24.0	27.7	31.4	22.1	9.7
Fe ₂ O ₃	0.5	1.6	1.2	0.6	1.2
TiO ₂	0.7	0.7	1.5	0.8	2.2
C	61.5	48.2	16.4	15.9	12.8

Ramming mixes have long been used in trough, as well as iron and slag runners in the blast furnace cast house. In the early stages, ramming mixes comprising sand, carbon and clay (and later on with

coal tar pitch) were used. But due to its short life cycle, the material needed to be changed quite frequently interrupting the casting of the iron from the blast furnace. Also the workplace hazards have been high because of the higher ambient and cast-house temperatures. Since then, high alumina-silicon carbide-graphite containing ramming compositions have been used with much better performance and needing to be changed only after several days of casting.

Table 6. Physical properties of some high graphitic ramming compositions.

Classification	60	50	15	15	10
.BD, kg/m³					
110°C	1360	1618	2210	2210	2498
815°C	1185	1298	2050	2066	2275
1370°C	1250	1313	1840	2080	2338
CCS, MPa					
110°C	5.0	3.5	8.3	4.1	3.4
815°C	6.0	4.2	11.4	15.2	16.3
1370°C	6.0	4.8	8.3	31.0	20.0
CMOR, MPa					
110°C	2.0	2.1	5.2	2.7	2.7
815°C	2.7	2.1	4.1	8.3	4.8
1370°C	3.1	2.6	4.9	17.3	13.8
PLC, %					
110°C	+0.1	+0.1	-0.5	-0.1	+0.1
815°C	0.0	+0.1	-0.5	-0.1	+0.1
1370°C	-0.2	+0.2	+0.8	-0.2	+0.6

These ramming mix compositions invariably contain organic binders. Organic bonding systems (such as lignosulphonates, molasses, various types of synthetic resins, coal tar and petroleum pitch) all suffer from the lack of high temperature strength after the burn out of the carbonaceous materials in cases where the areas of application are prone to oxidation. Oxidation of the carbonaceous bonding makes them more vulnerable to slag attack and it also decreases the strength.

Phenol-formaldehyde resin which produces toxic fumes is the most successful organic binder used in the ramming mixes for blast furnace cast-house applications. In recent years, much emphasis has been placed on safety and hazard at the work place. Emission of hazardous pollutants in the blast furnace cast house area has come under scrutiny. Although most of the cast house troughs and runners are covered to minimize work place emissions, workers are exposed to potential hazards during installation and tear out of refractories. The phenolic fumes evolved during installation cause problems to workers on direct contact with the skin and mucous membranes, and are particularly objectionable at higher temperatures.

A ramming material with a unique inorganic bonding system has been developed¹⁻⁶ to provide a good workplace environment along with better performance, particularly in the blast furnace cast house areas. The composition does not contain any organic binder and there are no emissions at any temperature level during installation and use as well as tear out.

Tables 7 and 8 show the comparative chemical and physical properties of the resin bonded and the colloidal silica-bonded ramming compositions both used for blast-furnace cast-house trough, iron and slag runners.

The non-toxic ram mix has several advantages over the resin-bonded ram mix. First, as illustrated in Table 8, the density and porosity of the nontoxic mix remain unchanged throughout the temperature range. The reheat strengths increase with temperature unlike the resin-bonded mix that loses strength and density with a corresponding increase in porosity. Although the resin-bonded mix performed well in initial campaigns in the blast furnace trough lining, the unused material in the back needed to be removed due to oxidation of the carbonaceous materials. The unused material, although unattacked by the slag or iron, becomes fragile and powdery. This should be removed to avoid a significant increase in the probability of breakout.

Table 7. Chemical analysis of resin and colloidal silica-bonded ramming mixes.

Chemical Composition, (Calcined basis) %	Resin-bonded	Non-toxic mix
Al_2O_3	62.4	73.9
SiO_2	9.0	6.6
Fe_2O_3	1.0	0.5
TiO_2	2.0	1.8
SiC	25.5	17.0

The non-toxic ram mix, on the other hand, does not get oxidized in the unused portion of the lining because of a protective coating on the carbonaceous components. This is also reflected in its higher hot strengths throughout the temperature range.

Installation of ramming mix

Before the advent of air rammers, ramming mixes were installed by tamping with heavy flat-bottomed iron heads and sometimes shovels. The mechanical force applied to the material does not transmit a great distance due to its attenuation by intergranular friction. Because of this, ramming mixes are installed in several layers, usually ramming to a depth of 100–150 mm at a time. After finishing the ramming of one layer, the surface is scratched to foster bonding between the layers. In spite of the care and precaution taken, rammed linings are invariably found to have laminations differing in degrees.

The techniques of ramming differ depending on the target material. For ramming blast furnace troughs, it is the normal practice to ram the bottom first, place the form and then ram, and gradually the sidewalls are rammed to the height by gradually building the sidewall. But for ramming small runners, e.g., electric furnace runners, the mold is placed inverted and the sidewalls are rammed first gradually building up to the bottom at last. Ramming is always done by vertical build up;

otherwise compaction will be minimized and the performance will be diminished.

Table 8. Comparative physical properties of resin-bonded and non-toxic ramming mixes.

	Resin-bonded	Non-toxic mix
BD, kg/m ³ /Porosity, %		
110°C	2883/9.0*	3043/15.1
815°C	2755/16.2	3027/15.2
1370°C	2800/16.5	3027/13.9
CCS, MPa		
110°C	58.6*	19.6
815°C	24.8	45.5
1370°C	44.8	84.1
CMOR, MPa		
110°C	22.7*	5.2
815°C	11.7	11.6
1370°C	14.8	19.6
PLC, %		
110°C	+0.15*	0.0
815°C	-0.3	-0.13
1370°C	-0.1	+0.3
HMOR, MPa, @		
1093°C	3.8	13.8
1370°C	1.7	3.1
(Under N ₂ atmosphere)		

*175°C

The consistency of the material has to be within an acceptable range to produce the best rammed lining. If the material is too dry, it will not compact well. The result is higher porosity and less density, making the lining more vulnerable to quick deterioration. If the mix is too wet, the rammed lining will tend to slide and result in less compaction. The extra liquid present in the mix will cause higher porosity and lessen strength, also resulting in poor performance. To

get the best performance from the ramming material, the workability of the mix needs to be in the range of the acceptable values predetermined by the material, application and installation conditions.

Magnesia containing dry ramming mixes are usually used to line induction furnaces and electric furnace bottoms. These materials are rammed dry in place in layers, scratching the surface after each rammed layer until the installation is complete. The surface scratching provides better bonding between layers. These ramming mixes contain low temperature sintering aids that develop the initial strength and are specifically used for contact with highly basic slags.

Ramming mixes based on periclase-graphite have been tried⁷ for two steel ladles as a complete lining with lives of 81 and 103 heats. However, no mention was made as to whether the ladles were used for metallurgical purposes. Postmortem examinations showed that the material had good thermal stability with few cracks developing. It also showed good resistance to steel penetration and slag corrosion.

Application

Ramming mixes are widely used particularly in iron and steel making, foundries, non-ferrous metals, and other industrial applications. The greatest application has been in the area of blast furnace cast house in the main trough, iron and slag runners. Although the use of ramming mixes in cast houses has been significantly minimized since the development of LCC and ULCC, they are still used in a few plants in USA and many other parts of the world. Some plants in the USA use ramming mixes in iron and slag runners. Resin-bonded high silicon carbide based ramming mixes are used in thermal power plants. MgO-based dry ramming mix is widely used at the bottom of electric arc furnaces. Iron foundries use alumina-silica ramming mixes in parts of cupola furnaces particularly at and near the bottom. The ease of installation and rapid heat up provides convenience in the operation.

References

1. S. Banerjee, and S. Shah, "Blast Furnace Casthouse Refractories," *ISS Trans.*, pp. 89–93, March 1989.
2. S. Banerjee, and S. Shah "Non-Toxic Blast Furnace Cast House Refractories," *I&SM Proc.*, Vol. 48, pp. 17–20, 1988.
3. S. Banerjee, *U.S. Patent* # 5,064,787, Nov. 12, 1991.
4. S. Banerjee, et al., *Australian Patent* # 634,580, Feb. 25, 1993.
5. S. Banerjee, *Europe. Patent* # 0,417,898, Mar. 20, 1991.
6. S. Banerjee, *Canada. Patent* # 2,029,826, Oct. 10, 1995.

CHAPTER VI: GUNNING MIXES

Introduction

In using monolithic refractories, initial linings are usually installed by castables, pumpables, ramming mixes or plastics that takes considerable amount of time. Gunning refractories, on the other hand, can be installed quickly and are mostly used as a repair material (except some special uses like thermal power plant cyclone areas and some areas in the hydrocarbon processing industries). The practice of using gunning mix is normally for an interim repair material to continue operation until a major repair or replacement can be done. Although the properties of gunning refractories are not as good as other monolithic refractories, the installation is quick and reduces downtime.

Gunning mixes are granular refractory mixtures designed for application with air replacement guns. A variety of guns are used to spray the mixes at high velocity and pressure to form homogeneous, compact linings, essentially free from laminations and cracks. In certain industrial furnaces, the turn-around time and installation costs are the major factors when choosing a refractory lining. In other cases, repair can be done in a short time with little or no downtime. In either of these circumstances, application of gunning mixes becomes the obvious choice. The installation of gunning mixes involve the

transportation of the dry or semi-dry material pneumatically by compressed air. Water is added at the nozzle. Successful installation of gunning mixes often depends on the uniformity of mixing of the dry ingredients with the added water.

In conventional gunning, the quality of the installed lining depends on the experience of the nozzleman (the worker at the nozzle). The worker must adjust the appropriate flow of water so that the mix is neither too dry (creating dust and rebounding without sticking to the wall) nor too wet (running off the installed surface). The pattern of the gunned lining (devoid of laminations) often dictates the performance. For thick linings, the installation is done in layers to avoid slumping. Rebound losses and dusting can be minimized by predampening the dry materials with a predetermined amount of water.

Many castables, ramming mixes and specially designed plastics can also be applied by pneumatic gunning. Magnesites and hot gun mixes are not predampened but placed in a batch pressure gun. Gunning mixes should be designed to wet well and as wide a water range as possible, and provide an excellent coverage in a variety of applications. Gunning mixes can either be lightweight for insulation purposes or dense compositions for more severe conditions and can be used with different binders depending on the application.

Binder systems

A variety of bonding systems are used in gunning mixes and the choice depends upon the application parameters; they can be either air-setting or heat-setting type. Common bonding agents are clays, silicates (dry), colloidal silica, dry phosphates (all kinds), calcium aluminate cements, and organic compounds (molasses, lignosulphonates and synthetic resins). The aggregates and other components of the mix are determined by the end application. They include silica, calcined fireclay, other aluminosilicates (mullite and calcined bauxite), high alumina, dead-burnt magnesites and dolomite and chromes. Aggregates such as vermiculite, haydite, perlite, insulating fireclay

grog, cenospheres from fly ash, pyrophyllite and bubble alumina are used for lightweight insulating products, depending on the end use temperature, environment and desired properties.

Calcium aluminate cement bonding

Calcium aluminate cements of various purity are used in gun mixes depending on the application since the purity is defined by the amount of iron present in the cement. Fireclay gun mixes include multi-purpose high-fired fireclay and standard calcium aluminate cement compositions for easy installation and low rebound. Such mixes are used in boilers, incinerators, process heaters, stacks, breechings and a variety of other service areas. Fireclay gun mixes with high purity calcium aluminate cements are designed for more severe conditions experienced in incinerators, preheaters in iron ore furnace, wind boxes, fluid bed catalytic cracking process transfer lines, and slide valves. Different versions are designed for high alkali applications such as cement kiln preheaters. Gun mixes are also designed with medium purity calcium aluminate cements to provide resistance to CO disintegration and hot load deformation as required in blast furnace stack linings.

High purity alumina mixes provide exceptional refractoriness, volume stability and a service temperature of up to 1650°C. Gun mixes that use high purity high fired alumina aggregates (such as tabular alumina combined with high purity calcium aluminate cement) can withstand severe environments with a maximum service temperature of 1820°C. Primary application areas of these gunning mixes include ammonia reformers, combustion chambers and catalytic or naphtha reformer cyclones. Gunning mixes can also be designed based on high alumina aggregates for hot gunning applications particularly for slag and corrosion resistance that can withstand a temperature up to 1650°C.

The chemical composition and physical properties of some calcium aluminate cement bonded gunning mixes are given in Tables 1 and 2 respectively.

Table 1. Chemical composition of calcium aluminate cement containing gun mixes.

% Alumina class	50	60	70	90
Chemical Composition, (Calcined basis), %				
Al ₂ O ₃	52.1	59.5	69.8	90.8
SiO ₂	41.5	32.4	23.7	3.8
Fe ₂ O ₃	0.9	1.0	1.1	0.3
TiO ₂	1.1	1.7	2.1	0.4
CaO	3.9	5.0	2.9	4.2

Gunning mixes based on silicon carbide using either calcium aluminate cement or phosphate bonding are generally used in coal fired thermal power stations that need high abrasion resistance combined with high thermal conductivity. The cement bond has the advantage of room temperature set while the bond is insoluble in water at room temperature. The phosphate bond has better adhesion to metal studs found in waste-to-energy units but it takes a higher temperature to form a strong water-insoluble bond.

Resin - bonding

Gunning mixes containing silicon carbide, graphite and antioxidant metals such as Si, using phenol-formaldehyde resin, have been used for a number of years for temporary repair of blast furnace trough iron and slag runners. These mixes were in use in the industry particularly when resin bonded ramming mixes were applied in the blast furnace

cast house runners. Since the use of low cement castables and the frequency of gunning have been reduced, major repairs are normally done by recasting low cement castables.

Colloidal silica bonding

Using colloidal silica as a binder in compositions containing brown fused alumina, silicon carbide and graphite gunning mixes¹⁻³ have been highly effective in the blast furnace cast house application, particularly in the trough. The gun mix could be applied either as a granular material or as a pelletized product to facilitate flow from the receiving hopper.

Table 2. Physical properties of cement containing gun mixes.

% Alumina class	45	50	60	70	90
BD, kg/m ³					
110°C	1843	2085	2090	2485	2565
815°C	1790	2040	2043	2445	2530
1370°C	1850	2090	2097	2493	2572
CCS, MPa					
110°C	2.0	27.8	34.3	23.3	49.7
815°C	2.0	24.3	22.3	16.5	30.9
1370°C	5.8	35.7	23.7	20.6	47.8
CMOR, MPa					
110°C	0.7	9.6	10.6	4.1	11.9
815°C	0.8	4.8	3.8	3.1	6.3
1370°C	3.4	11.3	10.7	3.4	20.0
PLC, %					
110°C	-0.2	-	-	-	-
815°C	-0.2	-0.5	-0.3	-0.2	-0.2
1370°C	-0.9	+0.4	+0.1	-0.5	-1.0
HMOR, MPa, At					
1093°C	1.0	4.8	4.5	5.2	7.8
1370°C	0.4	1.0	1.0	1.1	1.9

The gunning mixes specifically used for the maintenance of blast furnace cast house trough containing phenol-formaldehyde resin and colloidal silica as the binder are worth looking on a comparative basis, since the performance of the colloidal silica binder based gunning mix is far superior to the resin binder based gunning mix. The chemical and physical properties are given Tables 3 and 4.

Table 3. Comparative chemical analysis of resin and silica bonded gunning mixes.

Chemical Composition, (Calcined basis), %	Resin-bonded	Silica bonded
Al_2O_3	68 – 70	63 – 65
SiO_2	8.0 – 9.0	12 – 14
Fe_2O_3	0.4 – 0.6	0.3 – 0.5
TiO_2	1.5 – 2.0	1.5 – 2.0
SiC	19 – 21	19 – 21

Other bonding agents

Gunning mixes based on dead-burnt magnesite using silicates, phosphates and chromates are used in the maintenance of the working linings of basic oxygen furnaces, electric furnaces and argon oxygen decarburisers. Chromates are gradually being eliminated from the bonding system because of their health hazard problems.

A pelletized magnesia-graphite gunning mix⁴ using phenolic resin as a binder has been developed which is particularly useful for gunning the slag line and slag-metal interface of steel ladle furnaces. The pelletizable gun mix consists of a mixture of high purity, high density and fused magnesite along with different varieties of graphite. It also contains anti-oxidant metallic powders such as aluminum, silicon, magnesium, magnesium (or a combination of these metallic powders). Phenol-formaldehyde resin is added both as a powder and liquid form.

Pelletization is done preferably in a rotating pan with an angle and speed suitable for controlling the size of the pellets. After pelletization is complete, small amount of fine powder is added to keep the pellets separate and prevent them from agglomerating into larger pellets. The pelletized gun mix has been found to very effective in prolonging the life of a ladle. Experience has shown that the ladle life can be increased by as much as 33% from its original life by using this gun mix when the ladle is about to be taken out of service.

Table 4. Comparative physical properties of resin and silica bonded gunning mixes.

	Resin-bonded	Silica bonded
BD, kg/m ³ /Porosity, %		
110°C	2125*/35.4	2435/–
815°C	2245/28.8	2275/31.1
1370°C	2530/24.5	2355/26.9
CCS, MPa		
110°C	31.4*	8.8
815°C	55.2 (coke breeze)	8.3
1370°C	58.0 (coke breeze)	14.8
CMOR, MPa		
110°C	–	3.1
815°C	7.6 (coke breeze)	2.4
1370°C	17.3 (coke breeze)	5.2
PLC, %		
110°C	–0.15*	–
815°C	+0.3	–0.13
1370°C	+0.3	+0.3
HMOR, MPa, @		
1093°C	6.8**	4.5
1370°C	5.9**	2.2
**(Under N ₂ atmosphere)		

*175°C

Alumina-spinel gun mix

Alumina-magnesia spinel castable and gunning mixes have been quite popular in recent years because of their high refractoriness, thermal stability and corrosion/erosion resistance. Gun mixes have been used with alumina-magnesia spinel in the matrix or having reactive alumina and magnesia as separate ingredients in the matrix along with high-purity calcium aluminate cements. These products have been found to be effective in maintaining steel ladles for long periods of time. The use of alumina-magnesia spinel composition in the matrix not only reduces cost and increases productivity but it also provides high purity in the steel having no silica in the composition. A typical chemical analysis and the physical properties are shown in Tables 5 and 6 respectively.

Table 5. Chemical analysis of an alumina-magnesia spinel gun mix.

Oxides	%
Al ₂ O ₃	94–95
CaO	2.5–2.7
MgO	2.5–3.0

Table 6. Physical properties of a alumina-magnesia spinel gun mix.

Prefired to °C	Bulk density, kg/cm ³	Porosity, %	CCS, MPa	PLC, %
110	2600–2620	25–27	12.0–13.0	–
955	2550–2570	28–29	9.0–10.0	–
1500	2550–2570	28–29	56.0–58.0	–

Characteristics of Gunning Mixes

Gunning mixes have long been used by transferring dry materials pneumatically through flexible pipes and adding water at the nozzle. The primary disadvantage of gunning mixes has been associated with

dusting since all materials are not completely wetted. The result of this inevitable dusting of the dry materials is an unclean atmosphere. Another disadvantage is rebounding — a process in which the material being partially wetted does not stick to the application surface and falls to the bottom. To eliminate these problems in gunning application, the dry mix is predampened with a specified quantity of water in a mixer before being pneumatically transferred to the nozzle. The pre-wetting significantly eliminates the dusting and reduces rebounding. If the lining is too thick, pre-wetting can create slumping problems. The total thickness is usually built up at one time to eliminate lamination. For gunning vertical surfaces, it should be started at the bottom, gradually filling upward; otherwise rebounds can be trapped at the bottom and result in poor performance.

Recently, both dusting and rebounding have been minimized by modification of gunning nozzle designs. To increase the residence time, spirals are designed inside the nozzle so that the material has a spiralling movement and can be in contact with water longer improving wet mixing. Also, non-cement gunning mixes have been pelletized to eliminate dusting and promote better flow at the hopper.

Sometimes a Mini gun⁶ is used to repair areas of brick or monolithic refractory damaged by spalling, to gun-coating materials to protect brick linings, and to build and repair small areas. A Mini gun is a portable device capable of applying about 5 kg of mixed castable to a surface and able to apply the same materials as larger gunning machines. The amount of water needed for mixing with the refractory should be adjusted to a consistency somewhat wetter than mortars used for trowelling.

Applications

Blast Furnace

The maintenance of blast furnace linings by gunning has been practiced for a long time. Particularly, when the other parts of the furnace are

still good, application of gunning is a temporary solution for continued use of the furnace. One of the criteria for blast furnace gunning, both upper and lower stacks, is its resistance to CO disintegration since the atmosphere inside the furnace is usually rich in CO. Blast furnace gunning repairs are usually temporary, since the gunned materials often spall off partially, during the heating and the properties usually are not as good as the operators would like. The materials usually are medium impurity CA cement bonded and the Al_2O_3 contents vary between 45 to 65% depending on the area of gunning. In recent years the gunning applications have been performed by robots with remote controls so that the repair process can be performed at higher than ambient temperatures.

Blast furnace cast house

Maintenance of blast furnace cast hose is of prime importance since the molten iron produced in the blast furnace cannot be used without the cast house being in proper condition. In recent years the refractory practice for the initial lining in the cast house has changed significantly from ramming to casting to pumping using various bonding agents. Maintenance of the trough and iron runners has been done with gun mixes. Still today most of the maintenance, particularly the interim repairs, is done by gunning. These gun mixes usually are similar to the base lining materials containing high Al_2O_3 , SiC and carbon. Although the frequency of gunning has been reduced from all maintenance by gunning to intermittent usage between partial casting/pumping, the practice is still quite common in almost all blast furnace cast houses.

Basic oxygen furnace (BOF)

BOF is usually lined with magnesia-chrome or MgO-C bricks and are maintained with magnesia-based gunning mixes. In recent years, the life of BOF has been improved significantly by the newly developed technique of slag splashing. Slag splashing is the process of covering

the refractory with slag by tilting the vessel and forcing the slag to move up the lining by compressed nitrogen gas. By this process the slag in the steel making process comes in contact with the slag and the refractory is protected. The life of the BOF is prolonged by slag splashing and controlled gunning of the lining until the cost of gunning exceeds that of the benefit in increased lining life.

Since the restriction in the uses of chromic acid or chromate bonded refractories, the primary bonding agents use either sodium silicate or sodium phosphate, depending on the steel makers choice. In recent years the application process has been improved by remote controls so that the vessels can be gunned while hot.

The composition of the basic gun mixes vary according to the specific needs of the steel maker. The MgO content may vary between 80 and 90% and the other ingredients are dictated by the binder and the matrix materials.

Electric arc furnaces (EAF)

During the past 15 years production of steel by the electric arc method has undergone a large increase. The convenience of steel making using scrap metal helped a large number of companies — so called min-mills. The EAF process involves a different type of refractory wear. The electric arc produced during the process creates localized severe temperatures that affects the life of the refractory. EAF is also maintained by gunning with basic mixes that are somewhat different from the BOF application. But the binder systems usually are common to both applications.

Tundish

Since the beginning of continuous casting, steel making has become more consistent, economical and less labor intensive. In the initial stages of continuous casting, tundish has been considered just as a transfer vessel that will carry the steel from ladle to the casting mold.

But in recent years, tundish has gained significant importance contributing to the quality and cleanliness of steel.

In the earlier stages, the tundish used to be lined with brick only which contributed to problems. The brick joints become weak causing weakness to the lining which in turn contribute to its failure. Also, since the remnant steel skull in the tundish is discarded by dumping upside down, the possibility of sticking due to the penetration of steel in the brick joints creates severe problems. Moreover, since the lining life of the tundish with brick has been considerably short tundishes have needed frequent lining. The time needed to line the tundish with brick is quite long and has needed skilled masons and large number of tundishes.

The next phase of tundish lining has been to use the brick as the back-up lining and magnesia based gunning mix (25 to 40 mm thick) as the working lining. The purpose of the use of the working lining is to protect the back-up lining, not to interfere with the chemistry of the steel and to ease the de-skulling after casting. This method improved the process considerably. In the earlier stages, the working lining consisted of high magnesia with silicates or phosphates as bonding agents. The general characteristics of the gunning mixes are shown in Table 7.

Table 7. Chemical and physical characteristics of tundish gun mixes.

Chemical composition, %	
MgO	80–90
SiO ₂	6–8
Al ₂ O ₃	1–2
Physical characteristics	
Bulk density, kg/m ³	1950–2050
Cold crushing strength, MPa after firing at 1500° C	20–35
Permanent linear change, % after firing at 1500° C	–(4.0 –5.5)

The gun mixes and spray materials are based on the use of varying amounts of magnesite and blending with olivine to arrive at the desired MgO content. This makes good sense both chemically as well as economically. Also, since the magnesite is blended with clay as well as other binders like sodium silicate and phosphates, lower quality magnesites can be used which makes the material more economical. In addition, various other materials are being used to reduce the density of the working lining. The commonly used materials may be shredded papers, paper fibers, synthetic fibers, straws, saw dust plastic fibers and other various similar materials that could contribute to lowering the density and are decomposed at high temperatures without interfering with the steel chemistry.

The general characteristics of the spray-gunning mixes are shown in Table 8.

Table 8. Chemical and physical characteristics of tundish spray-gunning mixes.

Chemical composition, %	
MgO	50–90
SiO ₂	6–40
Al ₂ O ₃	1–2
Physical characteristics:	
Bulk density, kg/m ³	1250–1950
Cold crushing strength, MPa after firing at 1500° C	10–25
Permanent linear change after firing at 1500° C	–(4.0 to –5.5)%

The spray material has certain advantages over the gunning compositions. Since the material is mixed before the application, there is no dusting and no rebound loss. Also, the spray materials are lighter and hence less material is needed for the application. Initially, the applications were carried out manually. But in recent years robotic application of the gunning or spraying mixes has been quite common.

This practice has resulted in more consistent linings and reduced labor requirement.

References

1. S. Banerjee, *US Patent* # 5,147,834, Sept. 15, 1992.
2. S. Banerjee, *Australian Patent* # Jan. 21, 1993.
3. S. Banerjee, *Canadian Patent* # 2,022,542, Oct. 27, 1995.
4. S. Banerjee et al., *US Patent* # 5,418,198, May 23, 1995.
5. A. Nishikawa, Editor, "*Technology of Monolithic Refractories*," Plibrico Company of Japan, Jan. 1984, Japan.

CHAPTER VII: MORTARS

Introduction

A refractory mortar is defined as a material that is used either by trowelling, dipping or pouring between brick joints or joining refractory shapes. The building of masonry units with refractory bricks consists of many relatively small units where the strength of the masonry depends on the strength of the individual brick, the manner in which they are laid together and the characteristics of the mortar material used in the joints. The function of the mortar is to fill up the joints between bricks and hold them together, protect them from attack by molten slags and other fluxes and to stabilize the integrity of the total structure. Hence the mortar material should be selected as carefully as the brick and should be physically and chemically compatible with the brick composition.

Classification

The bonding mortar¹ often must meet extremely exacting conditions while in service that may require an adjusted balance of properties. The mortar should be economical and convenient to use for either dipping or trowelling consistency. The mortar should be thermally stable and should not shrink excessively upon drying or heating and should become vesicular at the maximum service temperature by

overfiring. The thermal expansion of the mortar should be approximately the same as that of the brick or the that of the shape it is used for joining; otherwise, temperature changes will affect the bond brick or shape and cause coating to crack or peel. Strong joints are desired for certain applications for the brick or shape and the mortar forming a high strength bond. The refractoriness of the bond should be sufficiently high so that the mortar will not melt or flow from the joints at high temperatures. However, for refractory structures that are subjected alternately to soaking heat and cooling cycles, mortars that do not form a strong bond are often desirable.

Refractory mortars are generally divided into two classes - heat-setting mortars and air-setting mortars. Heat-setting mortars develop a strong ceramic bond only when heated to the furnace temperatures. These mortars provide flexibility in expansion and contraction as furnace temperature rises and falls. Heat-setting mortars may also compensate for the high thermal expansion of certain types of bricks, and for differences in the thermal expansion between bricks of different types in the same furnace. Various compositions of heat setting mortars for laying bricks in applications where full or solid joints with minimum shrinkage are especially desirable, or where a strong bond along the entire wall thickens is an essential requirement, are available in the market.

Air-setting mortars take a rigid set when dried that are prepared from suitable refractory materials by proper selection and combination. Chemical binders impart air-setting properties and maintain the strength of the bond up to the temperature at which the ceramic bond takes effect. Air-setting mortars form mechanically strong bonds with high resistance to abrasion and erosion. The wall is bonded throughout from hot to the cold surface.

Characteristics of mortars

Mortars are usually characterized by their setting — whether air-setting or heat-setting that is dictated by their use. The overall

composition is selected from the refractory ingredients that are appropriate for the end use. Binders used for mortars can be either inorganic or organic. The most common inorganic binders are clays, silicates, and phosphates among which silicates form the largest group of mortars. Lignosulphonates and phenolic resins are the common organic binders. Mortars can be either wet or dry depending on the type and the binder. For wet mortars, one of the main requirements is that the solids should keep in a good suspended condition when used, otherwise any separation of the solid from the liquid will cause problems. Whereas, dry mortars should be mixed with water or appropriate liquid as recommended prior to use. Depending on the requirement, mortars are used either by trowelling or dipping or in some instances by pouring on to the joints that need to be bonded. All mortars should be capable of being cured and bond development while the furnace or the site of application is heated for preheat treatment. This will, of course be dictated by the selection of the mortar. There are several types of mortars available in the market. The selection of the mortar depends upon the type of the brick or the shape that the mortar is going to join. In general, the following mortars are common:

Silica mortar

Depending on the specific use, silica mortars are either based on fine silica sand or fused silica. For coke oven application, the preferred choice will be the mortar based on fused silica due to its better thermal stability. The silica mortars are based on sodium silicate binders.

Fireclay mortar

There can be different classes of fireclay mortars depending on the brick used that vary from low-duty to super-duty class. All the fireclay mortars are also based on sodium silicate binders. The air-setting mortars contain a mixture of high-fired fireclay and high-alumina calcines and plastic clays. They are recommended for laying

high alumina brick in the range of 50–70% alumina as well as insulating brick. These mortars could be available in either dry or wet form.

High alumina mortars

High alumina air-setting mortars can be used in applications up to 1760°C because of their excellent refractoriness and resistance to attack by corrosive slags. Heat-setting high alumina mortars also have high refractoriness, volume stability and resistance to attack by molten metal or slag. These mortars are available in either dipping or trowelling consistencies. Heat-setting mortars based on high purity tabular alumina calcines can be used up to a temperature of 1870°C. These mortars also have the exceptional stability and load bearing capability at high temperatures and also have high resistance to corrosion to volatile alkalis and slags in all types of furnaces. These are specifically used in laying bricks of the 90% alumina class.

Alumina-chrome mortars

Phosphate-bonded high alumina-chrome mortars generate high bond strengths and show excellent resistance to corrosion by ferrous and non-ferrous metals. These mortars are usually used while laying alumina-chrome brick, brush coating over refractory walls, or other applications where strong bonded joints and resistance to slag and metal penetration are desired.

Basic mortars

Mortars based on high purity magnesites are used with refractories containing high MgO contents and other types of basic bricks. These mortars are produced in dry form. Dry air-setting mortars based on chrome ore possess excellent resistance to a wide range of slags and fumes in chemical applications. These mortars are used for laying basic bricks but also can be used as a neutral layer between basic and acid brick.

Mortars that contain special additives are also available for specific requirements involving penetration and corrosion inhibition and are used to lay both basic and high-alumina bricks.

References

1. “*Modern Refractory Practice*,” 5th. Edition, published by Harbison Walker Refractories, 1992.

CHAPTER VIII: COATINGS

Introduction

Refractory coatings are defined as fine particulate materials suspended in a liquid medium (in most cases) that is applied on the desired surface either by spraying or brushing to provide a thin layer covering the exposed surface. The liquid medium usually provides the bonding leaving a solid continuous surface that is supposed to provide protection by forming an impervious layer for the intended purpose.

Characteristics

Refractory materials are used for coating on the working surface of the lining having a thin layer of thickness. Refractory coating materials may be used for different purposes depending on the requirement. They can be classified as follows:

1. As a parting material between two refractory layers so that the individual layers can react independently from thermal changes.
2. As a working layer in front of the lining to protect the main lining from direct exposure to the working conditions.
3. As a parting material between metal-metal contact or metal refractory contact.

4. As an impervious material on insulating blankets or modules so that the furnace gases cannot permeate through the insulation that can cause ineffective insulation as well as affect the supporting structure by corrosion from reactive gases.
5. As a sealant for gas leaks, such as coke oven roofs where the coatings help in preventing toxic gases from being released in the atmosphere.
6. As heat reflector providing heat emissivity where considerable heat can be saved by using the coatings when used inside the furnace up to a certain temperature limit.
7. As a protective layer for oxidation resistance of materials containing carbonaceous components.
8. As an inert, non-reactive, high temperature protection, lubrication and release.

Since the refractory coating formulations and applications are widely diverse it is prudent to discuss the different types of coatings on the basis of their applications.

As parting layers

As a parting layer between two similar monolithic linings, the application of monolithic refractories in various high temperature liquid and metal contact areas has been increasing at a rapid rate. In blast furnace trough applications, the working lining has long been monolithic refractories from tamping to ramming to low cement castables to pumpable compositions. The back-up lining has been fireclay super-duty brick in most common furnaces and carbon brick in bigger high performances. But with the increased efficiency of the blast furnace and the need to produce iron on an uninterrupted basis, the refractory practices need to be changed so that the backup lining can last several times more than the working lining. In several cases, the use of precast shapes from ultralow cement castables became quite common. But the installation of the precast shapes needs special

arrangements, e.g., having a plastic layer on the base so that the shapes can seat evenly on the base and the sides also need to be fairly uniform. Instead, the use of a monolithic lining like a castable or preferably a pumpable layer with slightly lower quality than the working lining is much more convenient to use. This can be done with wooden formers. Once the material sets, a coating is recommended to cover the surface in contact with the working lining to make them independent of each other. The coating along with the surface can be dried before putting the working lining. The separation by the coating will allow the two linings behave separately and they can move with the thermal changes independently. A coating of this type may have the following characteristics:

Chemical composition —

SiO_2	40 – 45%
Al_2O_3	10 – 14%
Appearance	White liquid
pH	10 – 10.4
Specific gravity	1.44 – 1.46
Viscosity	2500 – 3000 cPs

Similar applications may be done in steel ladles where the working as well as the back-up lining are both monolithic castables. The safety lining can be cast with a relative lighter but metal penetration resistant material while the working lining can be either high alumina or alumina-spinel composition. After the safety lining is cast a thin coating (primarily alumina based) may be applied that can be dried along with the safety lining which will provide a breaking plane preventing the progress of any cracks generated in the working lining that will stop major breakouts through the lining.

Protective coating -type 1

As a protective lining before the main working lining, this type of coating is widely used to protect the metal- or the refractory lining so that the material that needs to be contained can have parting lining and can be removed easily without affecting the main lining. Typical examples are slag-pot coating for the blast furnace or steel-slag pots. In some plants the slag is carried directly over the steel lining or some places a castable is used on top of the steel lining. In either case the use of a coating will protect the lining. Typical characteristics of such a coating are as follows:

Chemical composition —	
SiO ₂	40 – 45%
Appearance	White liquid
pH	10 – 10.4
Specific gravity	1.34 – 1.40
Viscosity	1800 – 2200 cPs

Refractory-metal, metal-metal parting

As a parting material between metal-refractory contact metal-metal contact — This type of coatings have been used extensively in ingot casting to protect the ingot stool base by replacing part of the base by a precast refractory base that is being coated as parting layer so that the ingot will be easy to be moved from the base without sticking to the stool base. A typical characteristics of such a coating will be as follows:

Chemical composition —	
SiO ₂	58 – 62%
Appearance	White liquid
pH	10 – 10.2
Specific gravity	1.50 – 1.55
Viscosity	2700 – 2900 cPs

For metal-metal contact, such as while pouring ingots into mold, the coating needs to be such that it does not contaminate the liquid steel with any detrimental ingredients. The coating of this type is a totally organic base with fine graphite powder that will only work on the liquid steel and will burn out without leaving any residue. Typical characteristics of such a coating are as follows:

Appearance	Black liquid
pH	7.0 – 7.5
Specific gravity	1.15 – 1.25
Viscosity	400 – 500 cPs

Protective lining - type 2

As a protective layer for insulating ceramic blanket or module lining, this type of coating is used in furnaces where the insulating blankets or modules used experience evolution of corrosive gases that may permeate through the lining and corrode the steel shell. The selection of such a lining will be dictated by the type of gas evolved in the furnace operation. For acidic gases a silica-base coating such as mentioned in category 3 can be effectively used. For evolution of gases that are more alkaline in nature such as ammonia, typical characteristics may be as follows:

Chemical composition —	
Al ₂ O ₃	45 – 48%
Appearance	White liquid
pH	7.0 – 7.5
Specific gravity	1.45 – 1.48
Viscosity	2200 – 3000 cPs

Gas sealing type

As a sealant for gas leaks from furnace roofs or sides as in coke ovens — In recent years there has been a significant drive in reducing the

work place emissions in the atmosphere that are particularly objectionable and harmful to humans and the environment. The emissions from coke ovens and carbon baking furnaces are particularly harmful. Hence higher viscosity coatings are particularly useful in preventing such leaks. The coatings for such use may be similar to the one mentioned in category 3 with the high-silica composition with slight increase in viscosity and specific gravity, and are still sprayable.

Reflective coating

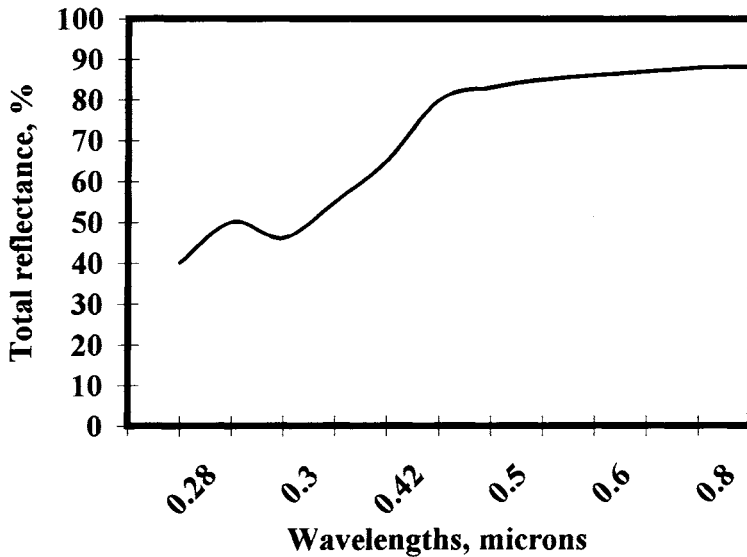
As a heat emissivity coating — Coatings of this type have been developed to reduce heat losses from high temperature furnaces working at temperatures up to 1400°C. This is called photon-diffusive coating. Most of these compositions are proprietary. A special variety of silica base comprising primarily of cristobalite and tridymite (60%) having very low indexes of refraction which does not capture and subject light to internal reflection. The photon reflectance of this material is shown in Fig. 1.

This mineral also has a natural micro-porosity with pores smaller than 0.17 µm. This has been used for reflecting heat inside the furnace and savings in heating cost up to 15 to 20% have been realized. The characteristics of the product are given below:

Chemical composition —	
SiO ₂	100%
Appearance	White liquid
pH	8.0 - 8.5
Carrying vehicle	Latex
Application	Brush, roller(as received), and spray by diluting 4:1 with water using airless sprayer.

There are several potential applications of this product. In steel industry in all varieties of reheating furnaces, in ceramic

industry in high heat, curing and glazing furnaces, and other industrial heat applications such as boiler interior, hoods and ducts, and in the glass industry in annealing,lehr and forge furnaces.



(Courtesy of Heat Shield Coatings, Florida, USA).

Fig. 1. Photon reflectance of silica based coating.

Oxidation protection

These coatings are for carbonaceous materials and are used in protecting electrodes of electric furnaces and graphite molds for casting. Because of their high temperature application the coating material needs to be highly refractory also. Hence, the use of all alumina material is recommended for these coatings. Typical characteristics of theses coatings are as follows:

Chemical composition —

Al_2O_3	50 – 55%
Appearance	White liquid
pH	7.0 – 7.5
Specific gravity	1.48 – 1.50
Viscosity	2400 – 3100 cPs

High temperature protection

As an inert, non-reactive, high temperature protection¹, lubrication and release, this is a high-performance coating used as for high temperature lubrication. It is used as an anti-sticking agent, for precision mold release, corrosion resistance and resistance to molten metals, weld spatter, braze alloys and molten salts. This type of coating is recommended where graphitic or other such coatings fail. It is ideal for super-plastic forming, forging and extrusion operations as well as providing a non-stick coating for glass and non-ferrous melting/casting industries. It has also the advantage of ready-to-use compositions with no mixing needed or, if desired, can be easily diluted with water. The characteristics of such coating are:

Appearance	White liquid
Dry composition	BN — 87.3%
	Al_2O_3 — 12.7%
Solids	25%
Specific gravity	0.96
Use temperature	1800°C (reducing, inert)
	1400°C (inert)
	1000°C (vacuum)

Other kinds of coatings are available in the market¹ that stop ceramics from degrading due to high-temperature reactions to above 1700°C. This type of coating has excellent chemical resistance, hardness abrasion resistance, and it prevents dusting and spalling, self

rigidizing and withstands cycle use. This type of water-based coating is protective to all ceramic bodies against most molten metals and corrosive environments above 1700°C. Typical characteristics are:

Appearance	Creamy white liquid
Solids	ZrO ₂
Use temperature	>1700°C (all atmospheres)

Certain characteristics need to be maintained in the manufacture of coatings. The coatings are finely suspended materials in a liquid medium. Hence the prime criterion of the coating should be to keep the particles in suspension at all times.

For most silica-based coating materials, colloidal silica provides a good medium for the coatings. In addition to the use of colloidal silica, some other organic compounds need always to be used as additives to defloculate and allow the particles to remain in suspension and not agglomerate or precipitate. Another criterion needs to be looked at is to stop any growth of microorganisms in the coating that would deteriorate the properties and cause failures. Hence, some fungicides need to be added particularly when the coatings are going to be stored in warm atmosphere over a period of time.

Reference

1. ZYP Coatings, Inc., Oak Ridge, TN, USA.

CHAPTER IX: DRY VIBRATABLES

Introduction

During the late 70s and early 80s, dry vibratable were developed and came into market for application in blast furnace cast house in the trough, iron and slag runners. They have already been used widely in foundries for different types of induction furnaces. Since they became quite popular in Japan, the introduction of the material in the USA was thought to be quite convenient. The major advantage of a dry vibratable is that it can be used within a short period of time after installation. Since there is no water or any other liquid present in the material, there is no reason for slow or prolonged heating process to avoid cracking or spalling associated with water or other liquid containing refractories. Also since the bonding is effected by heat, the hot surface sinters and hardens and a significant temperature gradient remains throughout the thickness providing a built-in insulating effect. The total thickness of the lining finally sinters as the heat gradually travels throughout the thickness particularly in installations with large thickness like blast furnace cast house applications.

Principle

The basic principle behind the dry vibratable is to use small quantities of low-melting components that form the initial bond around 400-

600°C and high-temperature ceramic bonding above 1200-1300°C. The low-temperature-bonding agents usually consist of boric acid, alkali borates, alkali silicates, alkali phosphates and metallics like Si and Al.

Boric acid/boric oxide/borate is used to allow for a "low temperature glass" to form within the matrix. The oxide components, such as iron, manganese, and calcium from the slag react with the glassy phase, lowering the refractoriness and resulting in wear of the refractory.

Sodium or potassium silicate is often used as a sintering aid to provide refractoriness at a low to intermediate temperature strength. Unfortunately, the silicate glass is affected by iron and manganese oxides in a way similar to that in a silica based refractory, forming low temperature fayalite or tephroite. With the dissolution of this sintering aid the refractory starts to erode.

Phosphoric acid/phosphates is also used as a low temperature bond forming agent. But here also both iron and manganese oxide react with the bond and decrease the refractoriness.

In combination with some oxidizing agents metallic powders like Si and Al are also used as a low- and intermediate-temperature sintering aid. Subsequently, they form oxides and provide good bonding medium.

Clay is often used to add strength to the matrix in the intermediate temperature range or to counteract the expansion of other refractory components. Again, the silicate component of the clay is susceptible to be attacked by iron and manganese oxides similar to other silica-based refractories. The overall effect is a general wear as the clay bond is washed out.

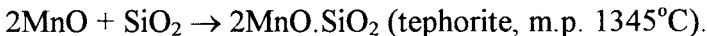
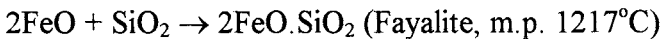
Formulating a dry vibratable for good performance depends a lot on its density. The grain size distribution is the main key in a dry vibratable for achieving a dense and strong product. Since there is no liquid in the composition of the dry vibratable, the grains should be in close proximity to have the most effective solid-solid reaction. The

early bond-forming components are low melting and hence may not be effective at high temperatures. Hence, the formulation should be such that the low-melting phases should react at high temperatures to form highly refractory phases.

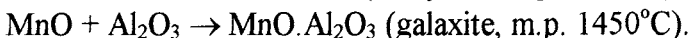
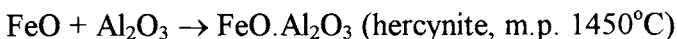
Applications

Foundry

Dry vibratables find their most and consistent use in iron-making operations in the foundry in coreless induction furnaces and channel induction furnaces. The use of a specific type of the dry vibratable is determined by the iron chemistry and slag practice. The nature of the slag in a coreless induction furnace usually varies from acidic to neutral as defined by the ratio of the sum of CaO and MgO to the sum of Al_2O_3 and SiO_2 . Besides these oxides the other corrosive components in the slag are FeO and MnO. The following reactions show the effect of both FeO and MnO on a silica-based dry vibratable refractory that is commonly used in such furnaces:



Alumina-based dry vibratable refractories are often used in channel furnaces for melting and holding iron and are frequently used also in coreless induction furnaces for melting iron.¹ In steel coreless melting, alumina-based dry vibratable is the choice. In high alumina (> 90%) refractory system, iron oxide can react with the alumina to form a spinel compound and manganese oxide behaves in a similar fashion forming a spinel as follows:



Although the initial formation of these compounds within the refractory hot face does not have any negative consequence, continuous exposure to these oxides promotes uncontrolled expansion of the saturated refractory which leads to hot-face spalling. In mullite-based dry vibratables, a reaction between the iron or the manganese oxides and the silica portion of the mullite may occur at much higher temperatures.

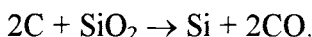
Magnesia-alumina spinel-based ($\text{MgO} \cdot \text{Al}_2\text{O}_3$) or spinel-forming refractories used in induction furnaces for ferrous melts will experience little reaction at the typical iron-melting temperatures. At steel-melting temperatures, however, reactions between iron and manganese oxides and spinel compound will occur more readily forming a modified spinel $[\text{Fe.Mn.Mg}]\text{O} \cdot \text{Al}_2\text{O}_3$. Hence, in alumina-rich spinel refractories either hercynite or galaxite may be formed.

Chrome-alumina-containing refractories are also affected by iron and manganese oxides forming a solid solution of $\text{Al}_2\text{O}_3 \cdot \text{Cr}_2\text{O}_3$. The iron or manganese oxides will replace one of the alumina to form $[(\text{Fe,Mn})\text{O} \cdot (\text{Al.Cr})_2\text{O}_3]$.

Magnesia-based dry vibratable refractories are typically used in inductors of channel furnaces for melting or holding iron. In steel melting, a magnesia base dry vibratable has been used for specifically high-temperature melting in coreless induction furnaces, especially, where high manganese steel grades are melted. The limiting factor for its use in larger coreless furnaces for steel is the unfavorable expansion that is attributed to the magnesia itself, leading to severe cracking due to thermal shock. When iron and manganese oxides are introduced into a magnesia system, both readily form a solid solution with the MgO . These can readily be substituted into a complex MgO solid solution compound. For example, in a forsterite ($\text{MgO} \cdot \text{SiO}_2$) compound, iron oxide is readily absorbed into the matrix to form $(\text{Fe.Mg})\text{O} \cdot \text{SiO}_2$. Similarly, in a spinel-forming MgO compound, the presence of FeO will cause the formation of $(\text{Fe.Mg})\text{O} \cdot \text{Al}_2\text{O}_3$ which is

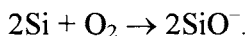
moderately stable but the overall refractoriness of the material is reduced by the presence of iron oxide.

Carbon (C) can reduce any silica present causing large gas voids to appear in a refractory.² This occurs as a result of the following reaction:



Silica is reduced to silicon and carbon monoxide gas is given off. Fine grain carbon added will penetrate the lining and cause gas pockets to form within the refractory. Coarser carbon will react with silica on the hot-face surface. It is important to add carbon with high solubility and purity.

A further reaction can take place when the reduced silicon comes in contact with air forming SiO (silicon monoxide) which is volatile and can leave the refractory creating more voids.



Blast furnace cast house

Although the dry vibratables are quite effective in induction furnaces of various types since the process is not very dynamic compared to the blast furnace cast-house application. The dry vibratables have been found to be quite effective in the iron area and at the bottom of blast-furnace trough. As a result, the bottom starts to build up and becomes hard which causes difficulty in tear out for complete trough installations.

The dry vibratables are usually installed by vibrating with a "Bosch" vibrator by installing in layers. During heating the hot face gradually becomes hard and since the lining has poor thermal conductivity, the back portion of the lining hardly gets enough heat for proper sintering and thus remains weak. This was the primary problem in the cast house. Also, the resistance to erosion and corrosion by the slag has not been competitive with other monolithic

refractories such as ramming mixes and low-cement castables. Hence, the application of dry vibratables has not been successful in blast furnace cast houses in direct contact with molten iron and slag.

The introduction of low and ultralow cement castables provided better erosion and corrosion resistance and longer lining life in the iron trough but gave rise to other severe problems. The comparatively higher thermal expansion created severe problems initially in certain cast house floors.³ It created frequent breakouts, severe warping of the steel shell and interrupted the cast house operation even though the castable had the potential for much longer life. The use of a dry vibratable against the insulating board next to the steel shell, was a viable solution to create a separating layer between the castable and the steel shell. An improved design of the trough^{3,4} by using thin precast shapes behind the castable to contain the dry vibratable and by putting the insulating board against the steel wall improved the process significantly and eliminated the frequent interruption in the cast-house operation. The dry vibratable was designed in such a way that there was enough sintering along the contact face with the precast shape but retained enough unsintered portion to provide the insulation against the insulating board and the steel shell.

The chemical compositions of some typical commercially available dry vibratable materials are given in Table 1.²

Table 1. Chemical composition of dry vibratables.

Typical base	Silica	Mullite	Spinel	Chrome-alumina
SiO ₂	99.1	4.0	0.6	1.8
Al ₂ O ₃	0.5	94.0	93.0	87.0
Fe ₂ O ₃	0.1	0.3	0.3	0.3
TiO ₂	0.1	1.6	1.7	3.3
Cr ₂ O ₃	—	—	—	7.5
Others	0.2	0.2	0.2	0.2

References

1. C. Williams, "Effects of Iron Slags on Refractories in Coreless Induction Furnaces," *AFS Trans.* pp. 433–436, 1994.
2. C. Williams, Y. H. Ko and T. M. Green, "Iron and Manganese Oxides: Culprits of Refractory Erosion," *Mod. Cast.* pp. 22–25, July 1996.
3. D. Fowles, J. B. Searles, W. R. Peay and R. G. Brenneman, "Single Taphole Blast Furnace Cast House Performance Optimizing Cost and Availability," *ibid.*, pp. 153–157.
4. L. Scheutt, J. P. Pawlak, L. Traina and R. G. Brenneman, "Gary Works No. 13 Blast Furnace — A New Removable Trough Design," *Iron Making Conf. Proc.* pp. 176–176, 1995

CHAPTER X: WEAR MECHANISMS

Introduction

Refractories are used at high temperatures where they are exposed to hot air and abrasive particles, and are in direct contact with molten metals, slag and sometimes both. The mode of wear of refractories depends on the specific industrial use because they create different environmental conditions. Certain wears are caused by one direct method of attack whereas others are caused by multiple factors. In most instances, the cause of wear is usually well identified and the refractory can be designed to provide proper resistance to the mode of wear. Wear mechanisms of monolithic refractories are characterized by:

- Abrasion
- Penetration
- Corrosion
- Erosion
- Corrosion/Erosion
- Spalling

The above classification of the wear mechanisms is defined by the specific modes whereby the refractories suffer loss and degradation

with time. In a variety of cases physical, mechanical and chemical actions are responsible for the wear either singularly or by combination of different actions.

Abrasion

The mechanical wearing of the surface of a refractory is defined as abrasion. It is caused by rolling, friction, impact, or high-speed hot gases with or without particles, from the charging of furnaces or by the dust particles in the furnace gases. Abrasive wear is most prevalent and is of major concern in petrochemical refineries. There the refractory is exposed to high speed catalyst particles during the cracking process and during the time when the spent catalyst is separated in cyclones. Wear due to abrasion is also experienced in other industries, such as, aluminum furnace charging areas, blast furnace walls, thermal power generating cyclones, waste incinerators and such other industries.

Since abrasion is a mechanical phenomenon, the maintenance of the mechanical strength of monolithic refractories is of prime importance. Compressive strength and modulus of rupture are important criteria in evaluating the abrasion resistance of a refractory material. The abrasion resistance of a monolithic refractory such as plastics, castables, or gunning materials, is determined by the standard ASTM C-704 method. In this method, the loss of the refractory abraded by a specific size of silicon carbide particles at a defined pressure for a certain period of time is measured. The loss of the mass is converted to volume loss of the material and refractories are classified according to their volume losses for use in specific areas of the process.

Studies have been undertaken to establish a relation and predict the abrasive loss with the compressive strength and the modulus of rupture. In petrochemical industries the abrasion-loss measurements are done after heating the monolithic refractory at 815°C for 10 h and then cooling to room temperature. The measurement is done at room

temperature. The size of the impinging particles and the velocity of the abrading particles have direct correlation with the abrasion loss.¹ Abrasion loss has been the highest when impingement is applied at right angles to the material provided that the impinging particles are not too much smaller than the aggregate size. For impinging particles smaller than the aggregate size, the matrix is abraded first and the aggregate is undercut and it falls off. Thus, if the particle is small in size, the quantity and strength of the matrix play an important role in abrasion and use of abrasion-resistant aggregate is not effective.

But for impinging particles larger than the average aggregate size, the matrix and the aggregates are simultaneously removed. Thus for impinging particles larger than the aggregate size, use of highly abrasion resistant materials such as SiC or Al₂O₃ (brown fused or tabular) is effective.

Penetration

Penetration is a phenomenon where either a gas, liquid or solid enters the refractory structure in inter-particular spaces creating new compounds or phases. Refractories used in several industrial environments suffer refractory loss due to penetration. When either the molten metal or the slag penetrates the refractory body, two mechanisms may affect the refractory. The penetrating medium may also react with the matrix and form new phases with different density and expansion characteristics affecting the refractory. Such phase changes often result in a spalling loss. Secondly, if the penetrating material does not react with the matrix or the aggregate, it may exert a pressure differential on the matrix and eventually the refractory may be lost due to spalling.

One of the most troubling penetration problems is observed in the aluminum industry where molten aluminum penetrates into the matrix of the refractory forming aluminum oxide which serves as a nucleating center for corundum formation. With time the corundum crystals grow forming hard nodules that ultimately grow diminishing the

furnace size. Since it is difficult to remove these formations by usual methods the furnace capacity is adversely affected. It has been found that the porosity or pore size of the refractory does not have any bearing on the penetration.² Although the mechanism is not fully understood, it is likely that the low viscosity of the molten aluminum helps in the penetration. If the contact angle between the molten aluminum and the refractory is increased by some additives, the penetration can be prevented.

Other penetration problems occur with slag in iron, steel and other metal industries. Penetration of slag densifies the matrix resulting in loss of refractory by spalling. Penetration of slag may be prevented by choosing the monolithic refractory composition properly so that there is minimum affinity of the slag for the refractory and wetting of the refractory by the slag is minimized.

Corrosion

Refractory corrosion may be caused by mechanisms such as dissolution in contact with a liquid, a vapor-liquid or solid phase reactions. It may also occur due to penetration of the vapor or liquid in the pores, forming an altered zone. Usually corrosion occurs due to some combination of these factors.

Dissolution of refractory in a liquid can be estimated from the equilibrium phase diagram of the compositions of the refractory and the liquid. The rate of dissolution is critical in the actual service. When the refractory comes in contact with molten slag and dissolves, a concentration gradient in the refractory composition occurs at the boundary region.³

The boundary layer, known as the diffusion layer exists along the refractory wall. Components from the refractory diffuse through the interfacial film and dissolve in the liquid followed by the resistance of the layer to diffusion. The rate of dissolution influenced by the interfacial film is known as the solution rate controlled diffusion. The

larger the concentration gradient the faster the dissolution rate and the thinner the diffusion layer, the more readily dissolves the refractory.

The viscosity of the slag has an effect on the thickness of the interfacial film. Generally, silica-rich slags have higher viscosity, whereas, basic slags have lower viscosity. Thus, a basic refractory easily absorbs the slag forming a thick layer of the altered zone and causes structural spalling. Also, larger quantities of FeO reduce the slag viscosity causing more corrosion of the refractory.

In the solid phase, the rate of reaction is controlled by the chemical reaction at the boundary between the different phases and by the diffusion of components through the layer of reaction products. However, corrosion in refractories rarely occurs due to solid-phase reaction. Mostly, refractories are corroded by a liquid phase or a reaction forming a liquid phase.

During service, refractory wear occurs faster when the wall is thick. The wear slows down with the decrease in wall thickness due to the cooling effect by increased heat loss and the service life of the lining is apparently lengthened after the wall reduces to an optimal thickness.

Slag reactions (corrosion/erosion/penetration)

Corrosion/erosion/penetration mainly occurs in iron, steel and other metal industries as well as thermal power stations and incinerators due to slag reaction. In iron and steel production, slag is a necessary evil. For the containment of iron and steel, the major reactions of corrosion/erosion of refractories take place in contact with molten slag. The slag compositions in iron and steel making are significantly different. The composition of the tundish slag varies greatly at various plants and are dictated primarily by the operational advantages and, to some extent, by the type of steel made. Consequently, every plant in the industry has somewhat different slag environment.

In iron making the most crucial area of slag reaction is in the main trough runner particularly in the impact area where significant

turbulence exists resulting in both chemical corrosion as well as mechanical erosion of the refractory material. Thus, it is not surprising that the life of a refractory material in iron making is dictated by the performance of the refractory material in the impact area of the trough. A significant amount of work has been done regarding the understanding and simulation of the slag corrosion/erosion reaction with the refractories over the years by various researchers. Since the most effective refractory composition has been found to be alumina-silicon carbide-carbon combination, the studies were primarily made using these materials. The studies cover ramming materials^{4,5} both clay bonded and resin/pitch bonded as well as sol-gel bonded⁶, low cement castable⁷⁻¹⁰ and cement-free pumpable materials.⁸ In all of these cases, it was found that the maximum wear occurred at the slag/iron/refractory interface rather than slag/air/refractory or slag/refractory interfaces, which is probably due to two reasons: (1) the slag at the slag/iron interface is at the highest temperature with the highest fluidity and hence the greatest reaction occurs at this point resulting in corrosion and (2) due to the partial oxidation of iron to ferrous oxide (FeO) at the slag/iron interface that has the high reactivity resulting in corrosive effects. Thus, when slag corrosion tests are performed with only slag, they do not represent the actual conditions as are experienced in the iron trough, whereas tests performed with combined slag and iron in a dynamic environment produce the best correlation with actual performance rather than tests with only slag.

In steel making, slag compositions in the steel ladle vary widely depending on the activities in the ladle, e.g., whether it is used as a ladle furnace or simply as a transfer vessel. The effect of the slag line also depends on the slag removal practice of a plant since the amount of slag carried over from the steel making process affects the performance of the refractory. Recently, the use of monolithic refractories in steel ladles especially in the barrel and the bottom, instead of brick, is becoming a common practice. This practice is

particularly prevalent in Europe and Japan. They are getting good service life from monolithic linings in the whole ladle where good slag removal practices are instituted, whereas, in most cases monolithic linings are used in the barrel and MgO–C brick in the slag lines.

Recently, the use of alumina magnesia spinel LCC has become more prevalent than high alumina castables. Many studies have been performed on the reaction between basic steel slag and alumina magnesia spinel castables.¹¹⁻¹³ Alumina magnesia spinel castables are used only in the barrel and at the bottom of the ladle. The use of such castables produces good results in plants where a slag-free clean steel is produced and the slag compositions are strictly controlled.

Tundish slag compositions can also vary significantly from plant to plant since they are dictated by operational advantages and also by the type of steel made at a particular plant. Only a limited number of publications are available in the area of tundish slag and MgO-based gunning refractory material.

Refractory-wear mechanism in Blast furnace trough

Corrosion/erosion effects of blast furnace slag and metal are best determined in the induction furnace where specimens (including a standard) are rotated in a pool of molten slag and iron for a period of 1 to 3 h, depending on the resistance of the specimens. Since the worst corrosion/erosion is observed at the metal/slag interface, the dynamic rotation of specimens in the slag/metal pool provides the condition closest to the actual operation.

Ever since the development of low-cement castables in the early eighties, blast furnace trough refractories have been dominated by alumina-silicon carbide-carbon containing low-cement castable refractories. Although, functionally, these refractories performed well in the blast furnace cast house, installation and dry out procedures are still being perfected. As indicated in Chapter III, the added mixing time, need for vibration for placement, long dry out and heating time, and lack of resistance to the oxidation of carbon hinder its versatile

use. Consequently, a new and improved bonding system was needed. The development of cement-free gel-bond material provides a solution to problems associated with ULCC compositions.

Studies of the slag/metal corrosion/erosion reactions show that the wear mechanism of the refractories are related not only to the corrosion/erosion but also to the penetration of the slag inside the refractory body which causes densification and gradual spalling of the refractory. Hence, it has been observed that some refractory compositions, which showed good wear resistance in tests do not necessarily perform well in actual application. The penetration of slag in the refractory body leads to poor performance. To solve the penetration problem the new gel-bond refractory was developed gradually by coordinating the slag-test results with actual performance data.¹⁴ Compositions of gel-bond castables are changed gradually as the performance of the refractory improved. The results of application are correlated with the laboratory-test results. The chemical analyses of the compositions are given in Table 1.

Table 1. Chemical analysis (calcined basis) of gel-bond castables for blast furnace trough.

Sample No.	1	2	3	4
Al ₂ O ₃	64.5	64.5	75.0	70.1
SiO ₂	8.8	8.9	6.7	8.8
Fe ₂ O ₃	0.2	0.2	0.5	0.8
TiO ₂	1.3	1.4	1.7	1.7
SiC	25.1	22.5	13.2	16.8

It is apparent that the silicon carbide contents need to be optimized to obtain better performance. Moreover, optimization of the raw material and their selection must be fine tuned so that the

refractory composition has both corrosion/erosion as well as penetration resistance to the slag. Physical properties of the above compositions are given in Table 2.

Table 2. Physical properties of gel-bond castables for blast furnace cast house.

Sample No.	1	2	3	4
BD, kg/m ³ , After firing to				
110°C	3012	3011	2980	3011
815°C	3028	3027	2947	2980
1370°C	3038	3043	2980	2980
Porosity, %, After firing to				
110°C	13.6	14.0	12.1	14.3
815°C	16.3	16.4	16.4	16.5
1370°C	15.3	14.6	115.1	15.0
CMOR, MPa, After firing to				
110°C	4.5	3.4	3.0	4.5
815°C	5.8	7.7	8.8	7.8
1370°C	7.4	7.5	11.0	9.8
*HMOR, MPa				
@ 1093 °C	7.6	8.6	10.5	12.1
@ 1370 °C	2.0	1.4	5.3	6.1
*Under reduced condition				

The slag corrosion test performed in the induction furnace is compared with the modified drip slag test under reducing atmosphere at 1600°C. Specimens run in the induction furnace with the above compositions are shown in Fig. 1 which clearly indicates that specimen #2 (marked 4 in the composition) has the minimum wear.

In the induction furnace test, areas lost by corrosion/erosion at the highest wear area were measured whereas, in the modified drip slag test, the areas lost by the slag corrosion were compared with the standard specimen (index = 1). The results are summarized in Table 3.

Specimens run in the induction furnace for the slag corrosion/erosion test were cut across the highest wear area to be examined with a scanning electron microscope (SEM). The structural

changes observed with SEM and energy-dispersive X-ray analysis (EDX) indicated penetration of the CaO and MgO in the body of the refractory. SEM micrographs with their corresponding EDX analyses along the sectional line are shown in Fig. 2 to 5. It proved that the gel-bond composition with the least wear and penetration of the slag performed the best in actual application.

Table 3. Slag corrosion results.

Sample No.	1	2	3	4
Induction furnace slag test (% Area lost at the highest wear)	57.5	54.9	46.4	41.5
Drip slag test (Standard index=1)	1.02	0.98	0.89	0.73

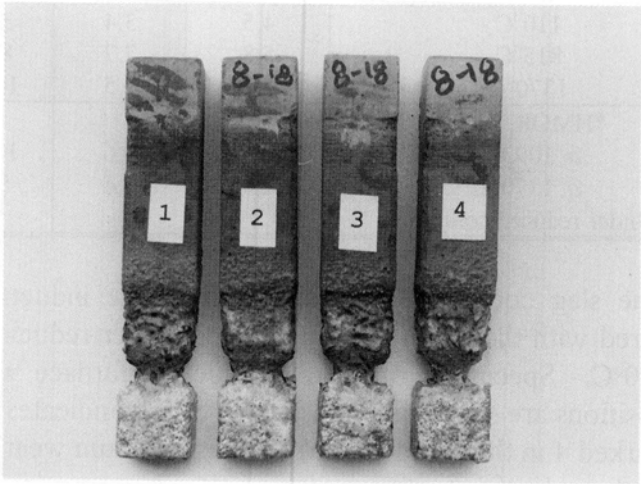


Fig. 1. Specimens run in the induction furnace showing relative wears.

From the above laboratory tests and the corresponding application data, it may be concluded that the increase in the silicon carbide content does not necessarily improve the slag resistance of the refractory. From similar investigations carried out with various silicon

carbide contents and different matrix characteristics. the following inferences can be made about the wear mechanism of the slag-refractory reactions.

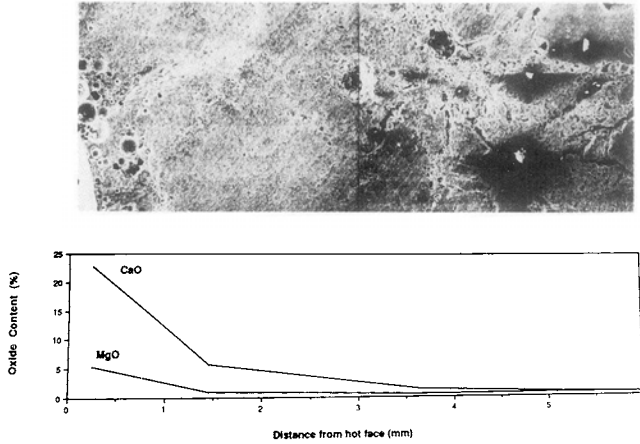


Fig. 2. SEM micrograph and EDX analysis of the highest wear area for CaO and MgO of specimen No. 1.

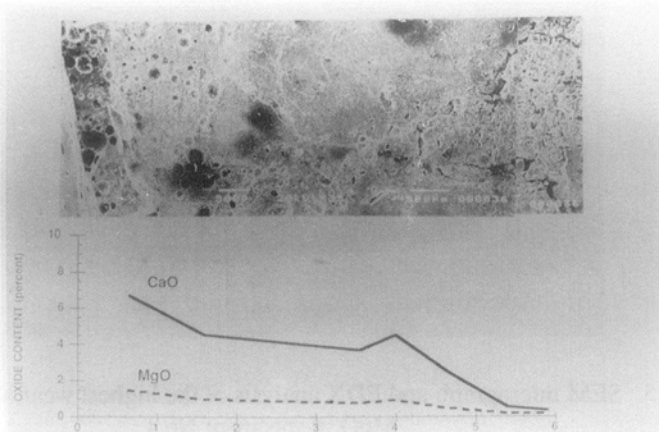


Fig. 3. SEM micrograph and EDX analysis of the highest wear area for CaO and MgO of specimen No.2

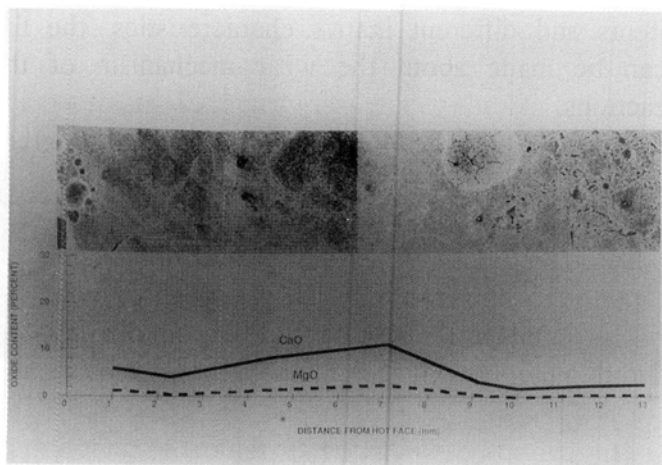


Fig. 4. SEM micrograph and EDX analysis of the highest wear area for CaO and MgO of specimen No.3.

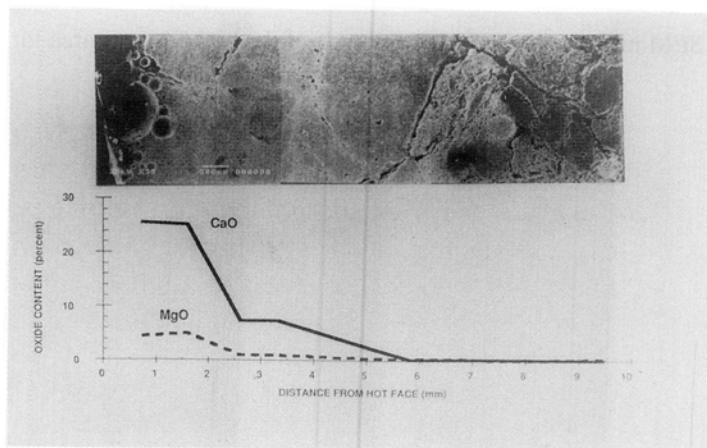


Fig. 5. SEM micrograph and EDX analysis of the highest wear area for CaO and MgO of specimen No.4.

Interestingly, slag corrosion/erosion/penetration was not found to be a factor of the silicon carbide content that has been believed to be

the most resistant to these wear agents. In spite of the lower SiC content, the mix composition No.4 showed the best wear resistance since the matrix composition was different from others. This means that the matrix composition and proper tuning of the finer phases play an important role in the wear mechanism. The correlation of the SiC content with the wear index both by the drip slag test as well as in the induction furnace is shown in Fig. 6.

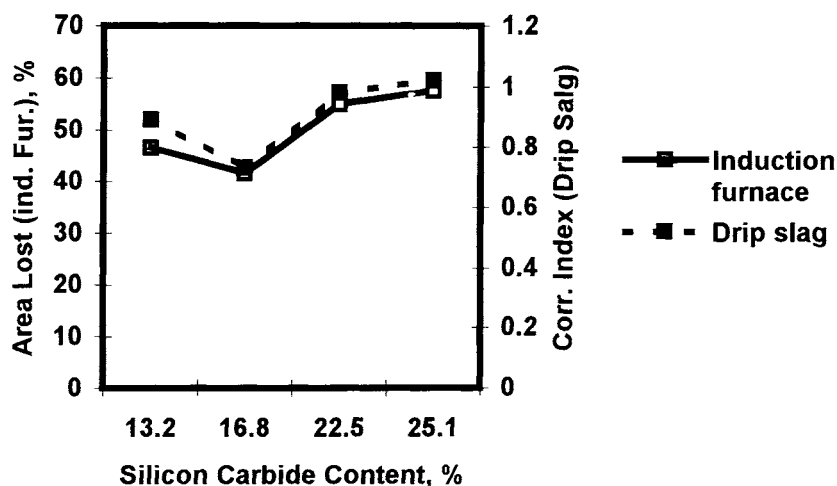
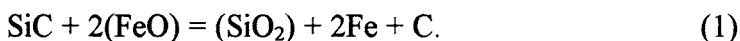


Fig. 6. Correlation of wear with SiC content.

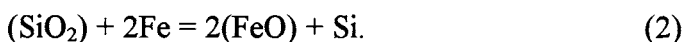
In the slag-refractory zone, the slag rich in SiO_2 and CaO , reacts with fine alumina grains to form anorthite and gehlinites that are liquid at the application temperature and hence start receding from the refractory matrix. After depleting the alumina in the matrix, the slag attacks the perimeter of the large alumina grains. The overall silicon carbide content in the refractory composition does not matter much, because once the matrix is attacked, the silicon carbide grains become loose and wash out in the slag.

At the slag/iron interface, the liquid iron gets oxidized to FeO that attacks the siliceous phase forming fayalite that is liquid at the temperature. Hence any free silica present on the matrix is prone to be attacked by this reaction. Also, FeO attacks the oxidized layer of the silicon carbide grain surface making it ineffective in slag corrosion protection. In general, the porosity and pore size of the ULCC and the gel-bond pumpables are low and the slag itself cannot be drawn into the pores. Penetration occurs from the dissolution of alumina and silica in the matrix that gradually proceeds through the main body of the refractory.

The following common phenomenon has been observed in all specimens run in the dynamic slag corrosion test in the induction furnace. A thin film of slag seemed to coat the specimen throughout the area under molten liquid including slag and iron. This might have happened either when the specimen was dipped in the molten iron through the molten slag or when the specimens were taken out through the slag layer at the top after the test was completed. The following reaction probably occurred in the slag film:



The presence of dissociated carbon and Fe particles was positively identified in the slag film.¹⁵ A reverse reaction may also occur as



Although the activity of FeO is assumed to be controlled by the C-O equilibrium in the metal, the oxidation of SiC by FeO may occur, FeO formed by reaction (2) may help reaction (1) to occur.

Another reaction may occur actively in the slag and at the slag-film interface as follows:



Reaction (3) occurs near the slag/metal interface, producing CO bubbles from the bottom that enhances corrosion, explaining the maximum corrosion occurring at the slag/metal interface.

Recent studies show that the corrosion of the trough lining along the slag-metal interface can be reduced by the addition of small amounts of alumina-magnesia spinel.¹⁶ The oxidation of iron to FeO at the slag-metal interface is responsible for the higher corrosion of the refractory compared to that at the slag-air interface. From the experience of alumina-magnesia spinel in slag corrosion resistance against FeO in steel ladles which forms a solid solution with MgO, the blast furnace trough refractory may have better corrosion resistance with the addition of alumina-magnesia spinel.

Slag corrosion studies were performed in an induction furnace using the specimens forming the wall of the furnace with compositions having alumina-silicon carbide-carbon and alumina-silicon carbide-spinel-carbon. The experiment was done at 1500–1550°C for three hours with blast furnace slag (with $\text{CaO/SiO}_2 = 1$) and pig iron. Subsequently, physical measurements were performed followed by microstructural evaluation of the corroded areas by SEM-EDX. Spinel containing specimens showed better corrosion resistance than those without spinel because the amount of liquid formed in the spinel containing specimen was smaller than without the spinel. Silicon carbide specimens showed better slag penetration resistance attributed to low wettability of SiC to molten slag. The oxidation of SiC of the spinel-containing specimens was larger than those without spinel, caused by the acceleration of the oxidation of SiC by MgO. The corrosion rate of practical trough became smaller by the application of the material with spinel. However, with larger amounts of fine SiC, the corrosion rate of the material was higher.

Slag reactions with steel-ladle refractories (alumina-magnesia spinel castable)

With the increase in metallurgical activities in steel ladles, the slag produced in the ladle becomes highly corrosive to the slag line refractories. In recent years, the most commonly used refractories in the slag line of steel ladles where significant metallurgical activity is performed, have been MgO-C bricks with varying carbon contents depending on the operational practice of the plant. After the development of alumina-magnesia spinel containing low cement castables, a significant amount of research has been performed to evaluate the effectiveness of this castable in the slag line of steel ladles. Japan has taken a lead in recent years, in studying the steel slag - alumina spinel castable reactions.

Slag corrosion and penetration mechanisms in castables were investigated^{17,18} with wide ranges of variation in alumina and spinel contents. In an induction-furnace-slag test it was observed that in the slag-penetrated region, the lattice constant of the spinel decreased with increasing alumina content and with increasing heating temperature. But it increased with increasing FeO content in the slag. The CA_6 content in the castable increased rapidly with increasing temperature whereas it decreased with increasing SiO_2 content. The use of spinel in the finer fraction is considered to be more effective to slag-penetration resistance than in the coarse fraction. The alumina-rich spinel seems to have much better corrosion resistance with slags containing high FeO content. A total of 5% MgO derived from an alumina rich spinel seems to be sufficient to provide good corrosion resistance.

Slag-penetration and corrosion-reaction mechanisms were studied.¹⁹ with different compositions of alumina-spinel monolithic refractories using spinels of various size fractions in a rotary slag furnace using both steel and basic slag furnace slag. It was concluded that near the hot face, fine spinel particles react with iron oxide in the slag to form a solid solution, increasing the slag viscosity and slowing

penetration into the refractory. When the liquid phase near the surface is cooled, the liquid composition changes with the precipitation of CA_2 and gehlinite which increases the silica content.

CaO in the slag and calcium aluminate cement react with alumina particles in the refractory to form additional CA_6 that reacts with CaO and SiO_2 in the slag at $1600\text{--}1650^\circ\text{C}$ to form a $\text{CaO-SiO}_2\text{-Al}_2\text{O}_3$ when the liquid is heated to 1600°C . It reacts with CA_6 which is soluble in the liquid and the liquid composition changes. CA_6 precipitates on cooling. The CA_6 resulting from the alumina particles is more accessible for reaction with the slag leading to deeper slag penetration for compositions containing more of fine alumina.

In another study,²⁰ various quantities of alumina-rich spinel of two different magnesia contents were used in both coarse and fine sizing and were cast with different cements to see whether the variations in the sizing of the spinel made any difference in slag penetration/corrosion characteristics. The optimum composition of the spinel has been found to be between 20 and 28% MgO spinel.

Basic tundish working lining and slag reaction

Since MgO based tundish working lining has been used extensively as the tundish working lining in the past few years, it has been necessary to look at the reactions occurring between the tundish slag and the lining to provide a guidance in formulating the working lining. A post mortem study was performed²¹ to understand the reaction between the slag and the working lining after the tundish had been in use for 10 heats and the specimen collected was from the middle of the lining to avoid any direct slag adherence. The tundish slag used had a composition as shown in Table 3.

Three different zones were identified in the specimen. A light zone consisted of calcium zirconate, spinel, calcium silicate and calcium magnesium silicate (monticellite). A dark zone contained mainly olivine, calcium magnesium silicate (monticellite) and a zirconia rich phase. The transition phase was seen to consist of mainly

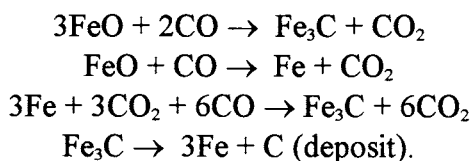
magnesium iron silicate, a zirconia rich phase and the spinel. It appeared that the virgin materials (magnesia and olivine) reacted with the slag to form monticellite, spinel and calcium zirconate. No periclase was observed in the specimen. A unique needle-shaped crystal was observed in abundance in the specimen which was identified as calcium zirconate.

Table 3. Chemical composition of tundish slag.

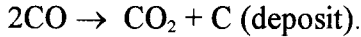
Constituents	%
Al ₂ O ₃	24.1
SiO ₂	37.0
Fe ₂ O ₃	6.3
CaO	2.6
MgO	3.5
MnO	3.3
Zr O ₂	14.0
Cr ₂ O ₃	4.7
P ₂ O ₅	0.5

Corrosion by vapor phase

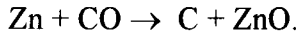
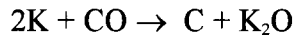
The most destructive gas-phase corrosion³ of refractories is caused by carbon monoxide. In the presence of metallic iron and free iron oxide (FeO) in the refractories, CO gas causes the following reactions at 400°-800°C, leading to carbon deposits or the formation of iron carbide (Fe₃C, cementite), destroying the refractory structure:



Formation and decomposition of Fe₃C alternates leads to the following reaction:



Some other reactions generate different forms of iron carbide and result in carbon deposit. CO-disintegration reactions are most prevalent in the refractory lining within blast furnaces due to the presence of large quantities of CO and the presence of alkali and zinc vapor which accelerates the reactions.



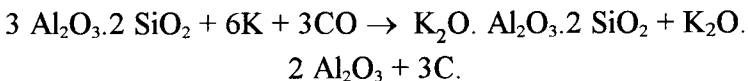
Carbon may also be deposited from hydrocarbons like methane gas (CH_4) through reactions occurring above 800°C as:



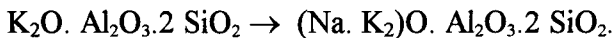
In the presence of glassy phases formed between iron oxide, SiO_2 and Al_2O_3 , carbon does not deposit readily. However, iron oxide becomes free in the presence of alkali vapors. Hence, monolithic refractories with low permeability and iron content are preferred for use in the presence of CO gas.

Refractory corrosion is highly aggravated with CO gas and alkali vapors in the presence of metallic iron and iron oxide (FeO), as in the case of blast furnace in the upper and lower stack regions. The alkali vapors not only accelerate the reaction with CO but also cause vapor-phase corrosion, weakening the texture of the refractory forming cracks and erosion.

Alkali compounds formed in the presence of CO are as follows:



In the presence of Na_2O



The formation of the different mineral phases depends not only on the average composition of the refractories, but also on the distribution of the minerals in the refractory or foreign components.

Spalling

Spalling³ is defined as the loss of refractory by extraneous forces not inherent in the refractory. Spalling can be caused by:

- Thermal stress resulting from sudden changes in the temperature
- Structural stress caused between the altered zone and the original zone in a refractory under service
- Stress caused by mechanical forces
- Internal vapor-pressure induced force

Thermal spalling

Usually, spalling occurs due to a combination of reasons - sometimes one dominating the others. The release of the stresses caused by one or more of the above reasons initially shows up as cracks that ultimately leads to spalling or loss of refractory. Spalling can be observed either in layers or in big portions of the refractory depending on the nature of the refractory and the cause of the spalling. Also spalling does not occur at a constant or predictable rate but usually the damage occurs unexpectedly.

When a metal is heated at the surface, the heat rapidly travels through the rest of the metallic body and the thermal gradient produced between the surface and the interior of the metal is less because of the high thermal conductivity of the metal and hence does not create high thermal stress. But when a refractory body is heated at

the surface large temperature gradients are produced between the surface and the interior creating considerable stress.

The temperature distribution in the material causes changes in the stress due to differences in the temperature, and the time-related variation in the gradient causes changes in the stresses produced within the material. Thus in a thick furnace wall or a massive block, a significant temperature gradient exists within the material; but if the cross section is small or the material has a higher thermal conductivity, then the thermal gradients will be small. Also, the coefficient of heat transfer changes if the temperature changes are due to the material being in contact with a liquid or if the temperature changes are due to air cooling.

The modulus of elasticity, coefficient of thermal expansion and thermal conductivity play important roles in the thermal stress generated in a body. Thus a material with a smaller thermal coefficient of expansion and higher thermal conductivity increases the resistance to cracking or eventual spalling off. Monolithic refractories generally have small moduli of elasticity and, in general, show excellent resistance to spalling.

The comparison between heating and cooling stresses show that compressive stresses develop on the surface while the material is heated causing tensile stress at the center. Since compressive strength is usually many times higher than the tensile strength, it is unlikely that compressive stresses will break the material. In practice, cracks develop at the surface more readily during cooling than during heating. However, if the material is heated too rapidly it will cause cracks due to tensile stress behind the surface, compressive stress will exceed the compressive strength of the refractory material. This will cause cracks to appear on the surface leading to peeling. Thus sudden heating or cooling may lead to unexpected damage and the refractory must not be subjected to thermal stresses exceeding the limits permitted by its properties or shape.

Structural spalling

Structural spalling occurs when an altered zone appears near the refractory surface during service since it tends to fall off due to the generation of internal stresses even with a small difference in temperature due to property changes with changes in the composition and mineralogy. The altered zone usually spalls as peeling or flaking of thin or thick layers. Overheating of the working face of a refractory promotes formation of the liquidus phase forming a vitreous or densified layer. Stresses arise due to shrinkage and temperature changes occur near the boundary between original and altered zones, causing cracking and failure.

Structural spalling in the altered zone is also caused by reaction or penetration from contact with slag. Thus, it is desirable to have low penetration of the slag in the refractory to avoid spalling and continued loss of refractory. The liquid phase formed must not be allowed to migrate into the unaltered refractory to avoid formation of the altered zones. Thus refractory designs should be aimed at factors like chemical resistance, porosity, pore-size distribution, and wettability of the slag in contact with the refractory.

Mechanical spalling

Mechanical spalling is the fracture phenomenon caused by concentrated stress due to nonuniform mechanical forces locally acting in a lining structure. This type of spalling is the fracture caused by the stresses due to differences in thermal expansions between refractories and also between metal fittings. Inappropriate design of a lining and improper allowance of expansion joints are usually responsible for mechanical spalling rather than the quality of the refractory material.

References

1. M. S. Crowley, *Amer. Cer. Soc Bull.* 48, (7), 1969.
2. S. Banerjee, D. Maas and S. Shah, "Refractories for Aluminum Industry", *Paper presented Pacific Coast Mtg.*, Nov. 1987.
3. A. Nishikawa, " *Technology of Monolithic Refractories*," Jan. 1984, published by Plibrico Company of Japan Ltd.
4. J. Wu and H. Young, "Erosion of Blast Furnace Trough by Molten Iron and Slag," *Amer. Cer. Bull.*, 62, No. 7, pp. 793–797, 1983.
5. A. S. Bowden and R. E. Williams, "A method of Evaluation Refractory Materials in Contact with Molten Metal and Slag," *Trans. Brit. Cer. Soc.*, No. 29, pp. 181–190, 1980.
6. S. Yoshino And Y. Hamazaki, "Wear Mechanism of Castable for Blast Furnace Troughs," *Taikabutsu*, 33, No. 283, pp. 28–31, 1985.
7. J. Yoshitomi, M. Hirada, T. Eguchi and K. Mukai, "Local Corrosion of Iron Trough Refractories in the Slag-Metal Interface," *Aachen Proc.*, pp. 85–89, 1988.
8. S. Bonsall and J. A. Koniuk, "Improved Laboratory Test Methods for Casthouse refractories," *Paper presented at the Amer. Cer. Soc. Fall Mtg.*, 1980.
9. S. Banerjee, J. P. Singh and R. B. Poeppel, "Comparison of Slag Tests for Blast Furnace Casthouse," *Paper presented at the Amer. Cer. Soc. Fall Mtg.*, 1980.
10. S. Banerjee and J. P. Singh, Editors, "Microstructure and Properties of Refractories," *Trans Tech Publ.*, 1993.
11. B. Nagai, O. Matsumoto, T. Isobe and T. Nishiumi, "Wear Mechanism of Castables for Steel Ladles," *Taikabutsu*, 42. No. 78, pp. 418–420, 1990.
12. Y. Sato, H. Hognchi and N. Hioroki, "Test Results of Alumina Spinel Castable for Steel ladle," *ibid*, 42, No. 7, pp. 376–382, 1990.

13. T. Yamamura, Y. Hamazaki, T. Kameshinga, T. Toyoda, M. Nishi and H. Kato, "Development of Alumina Spinel Castable Refractories for Steel Teeming Ladles," *ibid*, 42, No. 8, pp. 427–434, 1990.
14. S. Banerjee, K. N. Singh and C. T. Youmans, "Use of Pumpable Castable In Bethlehem Steel Blast Furnace at Sparrows Point, " *Steel World Qutrly.*, Oct. 1994.
15. J. Yoshitomi, M. Hirada, T. Eguchi and K. Mukai, "Local Corrosion of Iron Trough Refractories in the Slag-Metal Interface," *Aachen Proc.*, pp. 85–89, 1988.
16. M. Iida, Y. Toritani, S. Tanaka, "Effect of Spinel and Silicon Carbide on the Corrosion Resistance of Castable for Blast Furnace Trough," UNITECR'95 Proc. pp. 151–158, Nov. 1995.
17. S. Asano, E. Yorita, T. Yamamura and Y. Hamazaki, "Slag Penetration Mechanism on Alumin-Spinel Castable for Steel Ladle, " *Tiakbutsu*, 42, (9), pp. 518–519, 1990.
18. I. Tanaka, H. Shikano, I. Furusato, I. Takita and K. Furuta, "Effect of Spinel Raw Materials on Corrosion Resistance of Steel Ladle Castable," *ibid*, pp. 522–525.
19. Sato, "Slag Reaction with and Penetration of an Alumina-Spinel Monolithic Refractory," Sumitomo Metal Industries, Kashima Works, Priv. Com.
20. Wu and H. Yang, *Trans. Brit. Cer. Soc.*, pp. 58–63, 1988.

CHAPTER XI: MANUFACTURING

Introduction

Manufacture of monolithic refractories involves the mixing of the aggregates and matrix components that can vary over a wide range of the particle size. Usually, the aggregates range from as large as 8 mm to 75 μm and the matrix may be as small as 0.1 μm . The mixing of these ingredients together in a uniform manner is of utmost importance to achieve the expected performance. The particle size distribution of each size fraction indicated in the mix expected must be consistent to achieve repeatable performance. The overall size distribution in the mix as well as the uniformity of the particle size distribution will dictate the quality of the composition.

Monolithic refractories are either premixed materials ready to be installed or must be combined with the binder and placed at the installation site. Monolithic refractories are also used as precast shapes of various sizes and shapes and heat-treated as required so they can be used directly in operation.

Crushing and grinding

The equipment for crushing and grinding should be selected to yield the grain size required for efficient use of aggregates as monolithic refractories. A number of machines can be used for proper crushing

and grinding of the materials into coarse, medium and finer sizes. Lumps of large-size raw materials are primarily crushed with a jaw crusher, gyratory or cone crusher. The raw materials used for aggregates are crushed with an impact crusher that easily produces the required sizes. For finer grinding the ball mill, tube mill, conical ball mill, roller mill or vibrating ball mills are most efficient.

Grain size distribution

Whether the premixed refractory is to be installed on site or precast shapes are to be made, the grain size distribution plays an important role. In most cases of monolithic refractories, the density, porosity and the associated physical properties are directly related to the grain-size distribution of the product. In both low-cement vibratable or self-flow castables, the appropriate selection of the raw materials and their grain sizing is extremely important. According to ASTM, each specified closed sizing of the ingredients should not vary more than 5% on either side of the screen.

Mixing

Mixing of the ingredients homogeneously is another important factor in the manufacture of monolithic refractories. Dry mixing of the ingredients such as the aggregates, the binder, the matrix components and the minor additives should be done in a manner which not only is homogeneous but also eliminates any loss of ingredients added to the mix. In recent years, the formulation of monolithic refractories has consisted a of wide range of varying sizes from a few millimeters to less than a micrometer. Small amounts of different chemicals are added that are vital to the application and performance of the refractory. Both the mixing procedure and the mixing equipment are important so that the resulting product would be homogeneous and would retain all of the lighter and finer ingredients in the mix. Loss of the finer ingredients not only affects the properties of the product but also introduces inconsistencies between batches.

Dry mixing can be done in V-type, dry-pan, drum-type or enclosed-blade mixers. Muller type mixers are used in wet mixing for making ramming and plastics.. Kneaders, pug mill and de-airing pug mills are the mixers of choice for plastic refractories. Plastic refractories are usually mixed, kneaded and sliced into sizes such as 200 x 230 x 50mm thickness for convenient handling and installation.

Packaging

The grain size of monolithic refractories can vary from several millimeters to less than a micrometer and the density of the ingredients varies significantly. In cases where the variations in the grain size and densities are significant, it is advisable to put the total batch of mix into one bag instead of putting it into several smaller bags. The mixed batch can be put into separate smaller bags if the size and density differences are not significant. The packaging process should also be a closed system so that the finer and lighter ingredients do not escape by dusting.

Storage

Both the raw materials and the finished products should be stored in a dry place. Most of the monolithic refractories cannot use moist raw materials and hence care should be taken to protect raw materials from moisture. Raw materials that are susceptible to freezing should be stored in special storage areas where the temperature stays above freezing. If frozen they may change irreversibly and will be wasted. Products that are susceptible to freezing should also be stored in a warmer place to avoid damage.

Manufacturing of precast shapes

Since the development of low-cement castables more than a decade ago, the use of precast shapes has increased significantly throughout the user industries. The manufacturing of shapes of different sizes and

configurations needs special attention to provide defect-free products to users. The low-cement castables used for making shapes should be formulated in a way that satisfies the working time, setting time and need for ease of drying and firing.

The manufacture of shapes from low-cement castables is common for making baffles, dams and weirs, impact pads and pocket blocks for tundishes, ladle well blocks and impact pads, electric furnace deltas and runners. More recently, tundish back-up linings have also been made of large precast shapes from low-cement castables. Low-cement castables show a slight shrinkage during curing in the mold. Care should be taken, particularly while forming baffles with various sizes and shapes of holes, that the holes do not show any cracks. If any cracks develop during the curing process, they will grow larger during the drying and firing processes and the piece will end up as scrap.

The cores should be removed as soon as the material sets to touch. Once the shapes are made, proper care needs to be taken in handling for drying and firing. Since the cured pieces do not have high strength, they are susceptible to strains that may end up as cracks during drying and firing. High-fired (above 1350°C) shapes are usually less susceptible to failure in application than low-fired (500–600°C) ones.^{1,2}

Quality control and quality assurance

In modern manufacturing, it is of utmost importance that the manufacturer retains close control on manufacturing through process and quality control. This is the only way to provide users quality and consistency in all its refractory monolithic products. In the following sections, various areas of raw materials, process and product control are identified since this is a vital part of the total process for product performance on a consistent basis.

The manufacturer is encouraged to establish a formal data- and document-management system. At a minimum, the following data and documentation should be maintained:

- Supplier documentation on raw material;
- Raw-material inspection data;
- Manufacturing instructions and process-control data;
- Finished product-inspection data;
- External and internal laboratory test results;
- Customer work instructions, specifications, and drawings;
- Product-performance data as collected from the customer site including before and after conditions.

It is the responsibility of the manufacturer to maintain the documentation supplied and all changes to it as provided by the vendors. The manufacturer should establish a change-control procedure that includes procedures to identify obsolete data and documents. Inspection and performance data should be reviewed periodically to optimize the product performance..

Purchasing

Raw material suppliers should be qualified to assure their ability to supply quality material on a consistent basis. Cooperative work needs to be done with the vendors to provide them with current physical and chemical specifications to ensure successful maintenance of required specifications. Selection of the supplier should be based upon product quality, delivery, reliability, production capacity, cooperation, and price.

Procurement of all raw materials should be from vendors who are pre-qualified through Quality Audit of Candidate/Current Suppliers. The supplier should provide evidence that the materials are supplied with a consistent quality. This evidence must address limits given on the raw-material specifications. Periodic checks should be made to determine the correctness of the evidence.

A quality system of raw material suppliers assures the timely supply of correctly specified product. Although the incoming

inspection process may ensure the conformance to the specification, the use of reputable raw material suppliers, particularly for critical ingredients, is recommended. The use of good quality equipment, particularly blenders, mixers and pumps, as specified by the manufacturer is advantageous in the long run.

Subcontractors for installation services, if used, should be adequately evaluated prior to the award of any contract. Installation subcontractors should be capable of following instructions as provided. Whenever appropriate, references should be obtained before using installation subcontractors.

It is important to establish a non-conformance reporting process associated with purchasing. Quality audits of both current and prospective suppliers should be used. The audit should evaluate each supplier's quality capability through the use of a formal rating system. The audit score should then be used to determine the acceptability of the supplier and the degree of control which should be required on materials purchased from that supplier.

A similar formal rating system should be used to conduct internal audits of a company's own manufacturing plants.

Customer supplied products

Customers temporarily transfer their key facilities to the refractory manufacturer for installation and maintenance. During the installation and maintenance phases, the customer-provided facilities and documentation should be maintained properly. It is critical to evaluate them before and after conditions of the facilities. The manufacturer would be wise to establish a continuous performance-monitoring process as deemed necessary. Initial installation sites and installations for new customers should be kept under close observation. The use of photographic equipment, either still or video camera, should be used to record the setting process whenever possible. Such documentation should be maintained with textual documentation.

Product identification and traceability

Product identification and traceability must be maintained by the manufacturer throughout the life cycle of the product. Since formulations and processing form the heart of the product for monolithic refractories, the confidentiality of this information is critical. A codification of the finished products and installation kits should be devised that is easily understandable to the area workers.

Raw materials and their batches should be properly identified. All purchased materials should be identified so that inspected materials are traceable to the appropriate source. All materials should be clearly identified or separated according to the status in the inspection process (i.e. uninspected, approved or rejected). As appropriate, the manufacturer should be able to trace batches of raw materials used for each installation. Such information becomes valuable for cost reduction and performance improvement.

Process control

The conformance of raw materials to the specifications should be checked periodically for their quality and consistency according to the standard process control procedure used by the manufacturer. Manufacturing and installation processes described must be followed for every phase of operation. These procedures should be supplemented, where appropriate, with in-process inspection that determines satisfactory work completion and compliance to quality standards.

All employees should be informally checked every week to verify that they follow all current procedures correctly. A record of the checks should be kept as long as each employee works for the company.

Laboratory testing and inspection

The success of a monolithic refractory depends on both the laboratory testing and inspection during manufacturing and installation.

Laboratory testing

Materials made in the plant are to be tested in the laboratory on a regular schedule to assure that the properties of the materials are adequate. Acceptable service quality level of the in-house or external testing laboratories should be established before the test results can be accepted. The manufacturer must assess that they have the ability to conduct laboratory tests for different types of refractory products as described below.

Castables and pumpable refractories

Flowability:

Flowability of the materials should be tested for every batch since it determines the performance of the product. Every dry mix batch should be mixed with a predetermined amount of the binder in a Hobart mixer for four minutes. The Flowability is determined by the flow table by allowing the material to flow by itself for 30 seconds. to a predetermined value (60–80 on the flow scale) or under vibration for 30 seconds. to a preset value (110–130 on the flow scale) on a vibrating table with 3000–3600 vpm.

Setting time:

After the product meets the flowability value, it is examined for the setting time. For most castable and pumpable products, the setting time in the laboratory test should be two to four hours. Any batch setting outside of this time limit should be set aside for further evaluation.

Room temperature strength (ASTMC–133):

Periodically, the castable and pumpable materials should be tested for cold modulus of rupture (MOR) by casting specimens in plastic molds with dimensions 50 x 50 x 225 mm after drying

(110°C), heated to 1000°C and 1400°C for five hours. Cold crushing strength (CCS) is determined from the MOR specimens by cutting them into 50 x 50 x 50 mm cubes. These strength properties should be determined using a Universal machine.

Hot modulus of rupture ASTM - C 583):

The hot MOR at 1000°C and 1400°C of the products should be determined every three months to assure the consistency of the products by casting specimens into plastic molds with dimensions 38 x 38 x 150 mm.

For products that will be exposed to molten metal, slag or severe thermal changes, the following additional tests are to be performed on a regular basis to assure proper quality and consistency of the products.

Slag corrosion/erosion resistance:

For castable and pumpable products to be used in a blast furnace cast house, the information from a slag corrosion/erosion test is an important parameter. The most effective and informative method is to conduct a dynamic slag test in an induction furnace with molten slag and iron at 1510 to 1520°C. The specimen bars (prepared by casting the pumpable material into plastic molds with dimensions of 38 x 38 x 200 mm) are rotated at a predetermined speed from a supporting rod inside the molten slag and iron for a period of time (one to three hours). The test is always done with a standard specimen for comparison. The wear of the test bars is compared with the standard composition. Quantitative measurements (visual, physical and microstructural) are made and evaluated. This test provides an important guide to the performance of the product in service.

Thermal shock resistance (ASTM C-1171):

Since most refractories go through frequent temperature changes (particularly, for blast furnace cast house), it is imperative to

determine the thermal shock resistance of the products. The best method is to measure the modulus of rupture of the specimens before and after thermal cycling. Ten bars 50 x 50 x 225 mm in size are cast and are pre-fired to 1200°C. Five of these bars are subjected to alternative heating to 1200°C for 15 minutes and cooling by forced air for 15 minutes for five cycles. The MOR on these cycled pieces are determined. The percentage of loss of strength from the uncycled bar provides a measure of the thermal shock resistance.

Plastic and ramming refractories

Workability (ASTM C-181):

One of the main characteristics of plastic and ramming refractories is the capability of proper installation determined by workability. Hence, it is imperative that the plastic is checked for workability not only during manufacturing but also prior to shipment so that the data tally with the predicted values and will be capable of proper installation at the appropriate time.

Linear change (ASTM C-133)

Bars prepared either by ramming or pressing in molds with dimensions of 50 x 50 x 225 mm are dried for 18 h and heated to different temperatures up to the use temperature and held there for 10 h. The bars are then cooled and the %change in their dimension along the length are then measured

Receiving inspection and testing of raw material

Since monolithic refractories constitute a mixture of ingredients – mixed in dry, semi-wet or wet condition to form the refractory body, the following criteria are to be followed for inspecting raw materials:

1. The supplier of raw materials should provide the chemical analysis, size distribution and any other pertinent information

along with each shipment. Visual and physical contaminants should be identified promptly. All raw materials should be examined visually with proper sampling for contaminants or foreign materials. One of the most common foreign materials is tramp iron from the grinding medium. If tramp iron is a problem, magnetic separators should be installed to remove as much of it as possible.

2. One of the common contaminants in silicon carbide is silicon metal. The presence of metallic silicon and/or metallic iron causes the mix (with colloidal silica) to set very fast causing serious problems. Refractories used in CO environment such as inside the blast furnace, are susceptible to CO disintegration in the presence of either metallic iron or FeO. Products used in areas susceptible to CO exposure should be tested periodically for resistance to CO disintegration.
3. Particle size distribution is the most important parameter. Whether the raw material is processed in-house or by a vendor, it should be checked for sizing on a regular basis. Once the parameters are established, the sizing should not fall outside the permissible limits (as specified by the National Standards adopted by the country). All raw materials >325 mesh should be analyzed for particle size distribution by standard sieve analyses and materials <325 mesh should be analyzed by density. Raw materials like tabular alumina and raw kyanite – 325 mesh should be wet-screened for +325 mesh retains and should be within the specified limits. Submicron size materials such as silica fume should be tested by the bulk density.
4. All raw materials should be tested for chemical analysis on a regular basis. If it is a previously used material, the chemical composition should be checked every third or fourth shipment. If it is a new material, the chemical analysis should be done prior to accepting the shipment and should be examined to assure it tallies with the suppliers data. After that, the chemical

analysis should be repeated every third or fourth shipment. Any deviation from the accepted limits should be rejected.

5. Each shipment of liquid binders such as phosphoric acid, monoaluminum phosphate, liquid sodium silicates, colloidal silica, liquid resins and other liquid binders should be tested for their visual appearance, specific gravity, pH, solid content and viscosity as applicable.

In-process inspection and testing

Dry mix products

To predict proper installation and performance criteria, the dry mix (castables and pumpables) should be checked for the following :

1. The dry-screen analysis should be checked for sizes up to +200 mesh and wet screen analysis for sizes -200 mesh. The screen analyses should conform to preset limits set by the quality laboratory.
2. Each mix should be checked for flowability using a preset quantity of the binder such as water or colloidal silica specified by the laboratory. The flowability should be within the accepted limits (set by the laboratory) after up to 45 minutes of mixing.
3. Setting time of the wet-mixed material should be no less than two hours and no more than four hours for a hard set.

Wet and semi-wet products

Plastics and Ramming materials

Ramming materials are installed at the place of use, preferably by pneumatic rammers, to form a continuous solid monolithic refractory. It is important that the rammed material produces physical characteristics similar to those tested in the manufacturer's laboratory. In-process materials should be tested for higher workability that will be needed for in-place installation

since there is always a time lapse between the manufacturing and installation. This lapse is associated with a decrease in the workability due to liquid absorption of the refractory grains, liquid evaporation (either directly or diffusion through bags), or chemical reactions taking place between the liquid and some of the fine ingredients in the matrix.

In-process inspection of plant-equipment maintenance and calibration results should be scheduled according to a pre-established plan and inspection results must be documented.

Final inspection and testing

Final product must be tested to verify against the specified performance criteria. An installation inspection method, suitable to the organizational set up of the manufacturer, must be established. This method must include plans for:

- Site condition and customer requirements review;
- Inspection during installation and drying;
- Documentation of acceptance by the customer; and
- Post-installation inspection results.

Inspection and test records

The following records of inspection and testing must be documented and maintained for two years:

- Approval/rejection records;
- Equipment maintenance and calibration records; and
- Laboratory test records.

Control of inspection, testing, measuring, and test equipment

Inspection of materials upon receipt should be done according to the appropriate procedures established in this document and the existing manufacturer's procedure for purchased materials. Measuring and testing equipment should be calibrated according to the standards

provided by the equipment supplier or national standards. Inspection records should be kept for at least one year to provide a history of past supplier performance.

All scales, thermocouples, meters, and calipers should be calibrated on a scheduled basis either by an outside source or by correct calibration equipment within the company. All equipment should have a sticker stating the date of calibration, signature of the calibrator and due date for next calibration. Any equipment without a calibration sticker should not be used. The history of repair and calibration on major pieces of equipment should be kept for the lifetime of the equipment.

Inspection and testing status

Inspection and testing summary reports should be reviewed by the project management routinely. These reports should be made available, as necessary, for trouble shooting and material performance assessment.

References

1. S. Banerjee and G. L. Ramsey, "High Fired Refractories for Continuous Casting of Steel," *Cer. Eng. & Sci. Proc.*, **9**, NO. 1-2, pp. 67-73, 1988.
2. S. Banerjee and G. L. Ramsey, "High-Fired Refractories for Continuous Casting Tundishes," *Adv. Refr. Met. Ind. Proc.*, **4**, pp. 305-312, 1987.

CHAPTER XII: APPLICATION DESIGNS

Introduction

Appropriate design of the containment structure plays an enormous role in the life expectancy and performance of monolithic refractories. During the past 20 years, major changes in design for monolithic refractory applications had to be instituted to accommodate the productivity and life expectancy of improved refractory compositions and applications. In earlier days, designs of the containment structure were developed to cope with refractory compositions far less in performance of those produced in recent years.

Changes have been most prolific in two major areas. The blast furnace cast-house runners that accounted for the output of steel mills, have undergone major changes. Most blast-furnace casthouses have been redesigned during the past 25 years to take advantage of the improved and more dependable performance of vastly improved refractories. Profound changes have also been made in continuous casting tundishes. Refractories are now designed for placement inside the tundish in order to control flow to enable production of clean steel. This change has been more significant than the designing of the tundish itself. The purpose of the new designs inside the tundish is to make the steel as clean as possible as it leaves the tundish, the last vessel before the steel starts solidifying in the mold.

Steel ladles have often been used in recent years as a metallurgical furnace rather than a simple transfer vessel which simplifies the production of steel allowing a major part of the alloying and refining process to take place in the ladle. Since the function of the ladle has been more important, the erosion/corrosion of the refractory lining in the ladle has also been more severe. The refractory lining practice has been changed significantly although the preference of lining materials varies widely among different steel plants nationally as well as internationally. Of particular common interest has been the use of separately or in-situ cast bottoms especially in the impact areas.

Blast furnace trough design

The blast furnace is the key to keeping the process of an integrated steel mill operative. The function of an integrated steel plant is to convert the iron produced in the blast furnace into steel. The blast furnace has to produce and supply molten iron on a continuous basis as needed. After molten iron is produced in the blast furnace it comes out through a taphole on to the casthouse runners. The function of the main runner, or trough, is to separate the iron from slag. The barrier known as a skimmer allows the iron and slag to be separated by letting the slag leave from the top into the slag runner. The iron leaves through a passage in the skimmer at the bottom. Thus the productivity and efficiency of the blast furnace are highly dependent on the trough-refractory performance and its continuous operation. The trough runner refractory must be able to handle the iron produced in the blast furnace with minimum interruption. The performance of the trough refractory depends on the design of the cast house. A properly designed blast furnace cast house can greatly improve the life of the refractory.

As recently as 25 to 30 years ago, the trough designs did not allow iron to stay in the trough for any length of time. Appropriate refractories with high corrosion/erosion resistance enabling them to contain molten iron and slag for long periods of time were not

available. Troughs had a sharp incline allowing iron and slag to pass through rapidly. But with the development of improved refractories, trough designs were changed so that the iron could be held from a few days to several weeks at a time. Depending on the design, troughs are categorized as:

1. Non-pooling type;
2. Semi-pooling type; and
3. Pooling type.

The three different types of troughs are illustrated in sketches in Figs. 1, 2 and 3.

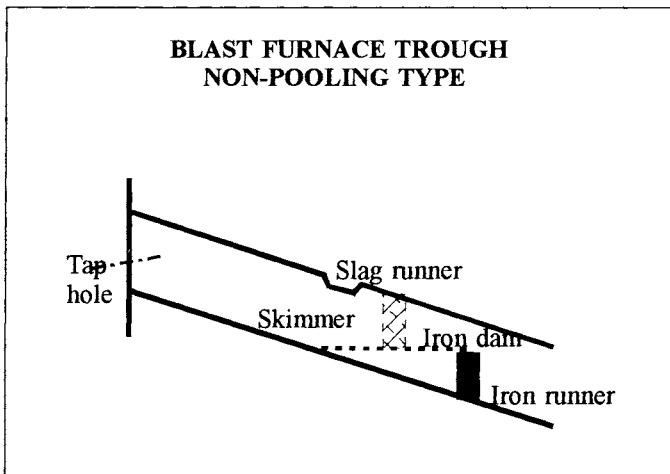


Fig. 1. Non-pooling type trough.

A **non-pooling** trough has a high descending angle (more than 20°) from the tap hole of the blast furnace to the skimmer into the iron well and dam. In this design, the iron coming out of the blast furnace rushes directly to the skimmer. The purpose of the skimmer is to separate the iron from the slag. However, due to the high velocity

and short residence time, the separation is inefficient and the iron carries some slag along with it. The slag in the iron causes more corrosion to the iron runner refractory and subsequently to the transfer or torpedo ladle. Also, the stream of iron coming out of the blast furnace always hits the refractory in the trough causing erosion due to direct impact. Moreover, the refractory in the trough is subjected to thermal shock following every cast.

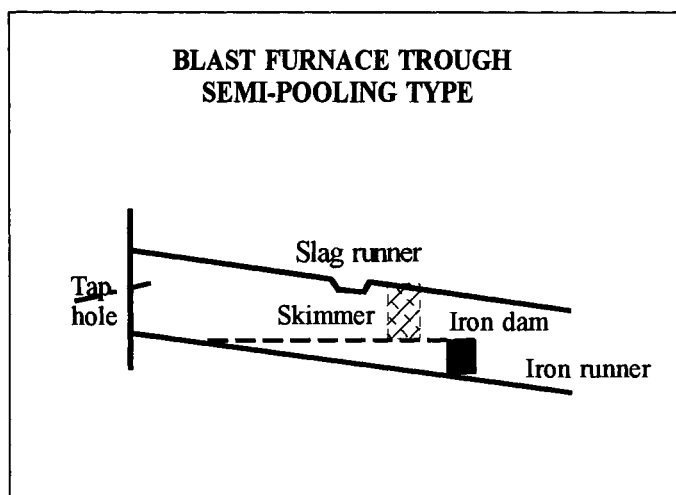


Fig. 2. Semi-pooling type trough.

A **semi-pooling** trough is designed with a descending angle that holds the molten iron half-way through. The rest of the design is non-pooling. In this design, the iron gets a better separation from the slag, but the trough refractory is still subjected to iron stream during the cast and the refractory is exposed to thermal shock between casts.

A **pooling** trough is designed with a descent that places the iron gate (situated beyond the iron well) at the same or slightly higher level than the entrance level of the trough from the blast furnace. The pooling trough design has three advantages.

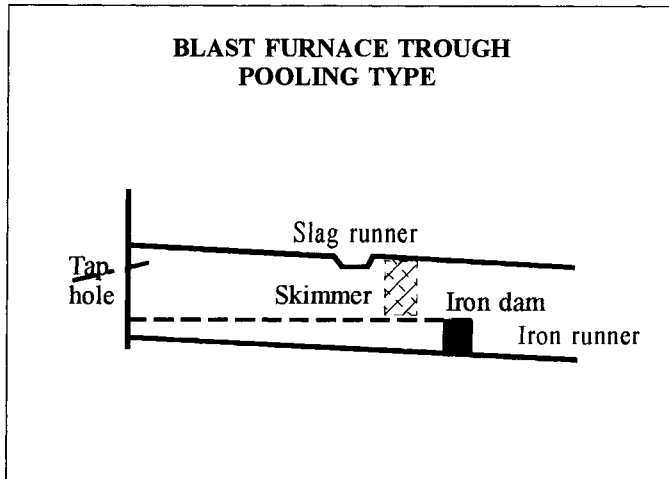


Fig. 3. Pooling type trough.

1. The residence time of the iron in the trough is much longer compared to other designs as a result of the low descending angle of the trough. The separation of iron from the slag is much more efficient in this design. Since the carryover slag is minimal, the iron runner and the transfer ladles experience less slag corrosion.
2. The iron stream from the blast furnace hits the pool of iron instead of the refractory thus increasing the life of the refractory. The life of the trough is also increased as a result of significantly less thermal shock from molten iron in the trough.
3. The lower descending angle reduces the velocity of the iron stream and, subsequently, the impact force which retards the erosion of the refractory and increases its service life.

Judging from the advantages of the pooling type trough, it is highly advisable to design or modify existing ones to a pooling-type

trough. Not only does it improve the life of the trough refractory but it also produces cleaner iron for the subsequent processing.

Over the years, refractory designs of the trough have evolved to improve the life of refractories. The tendency of the design engineers has been to cool the trough to remove the heat from the hot face. This technique worked when the trough refractories could not sustain continued high temperatures. The trough was cooled by water or forced air behind the steel shell of the trough. Carbon blocks were occasionally recommended for better heat transfer.¹

The concept of trough cooling has the following drawbacks:

1. Water cooling of troughs is potentially a serious safety hazard. If, at any time, the water comes in contact with the molten iron through cracks and ratholing of the back up lining, it can cause a severe explosion.
2. Cooling the trough creates a greater temperature gradient from the hot face of the working lining to the cold face of the back up lining which causes more thermal-shock damage, creating cracks and eventual spalling of the refractory.
3. The using of forced-air cooling in combination with carbon blocks creates another potential problem. The carbon back-up lining usually remains at such a high temperature that any cracks developing in the steel shell will oxidize the carbon block. This creates rat holes and increases the potential of trough breakout.

Trough lining

The refractory lining design of a blast-furnace trough has evolved through the years. A back-up lining and a working lining of almost equal thickness are now used to prevent the breakout of the trough lining. Figure 4 shows one method where the back-up lining is constructed of precast shapes keyed with each other with material similar to the chemical composition of the working lining. The working lining could be made from a low cement vibracast material or

a gel-bond pumpable material. The advantage of precast shapes, particularly in the impact zone, is the ease of replacement.

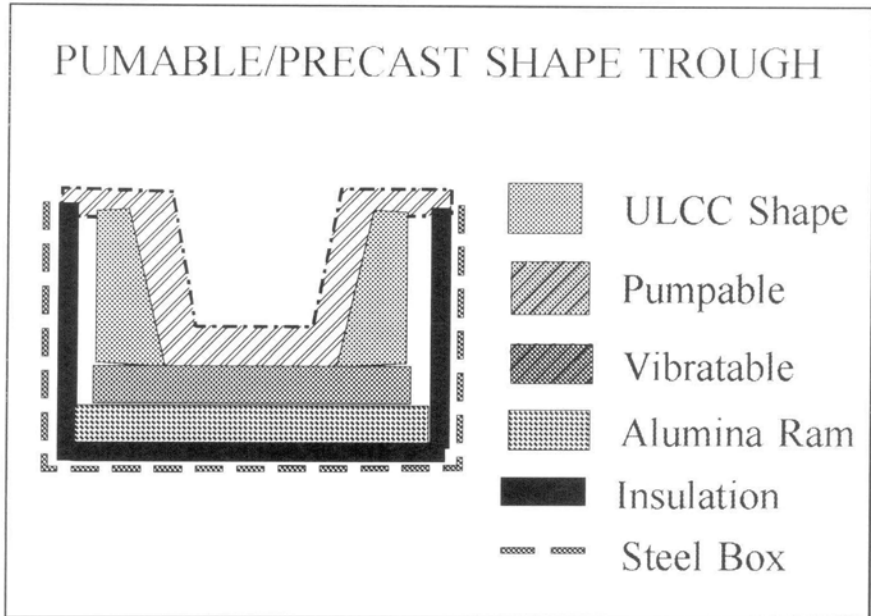


Fig. 4. Blast furnace trough refractory design with precast back up lining.

In troughs where installation of precast shapes causes logistic problems, the back-up lining can be installed either by vibrating or pumping with an appropriate material as shown in Fig. 5.

After drying out the back-up lining, the working lining should be installed on top of a parting compound placed on the back-up lining. The parting compound prevents the two linings from acting as a monolith where cracks could propagate from the working lining. The dry-out should be performed according to the heating schedule provided by the refractory supplier. Another important criterion in the design is the elimination of any expansion joint along the length of the

trough. The expansion joint should be located either at the furnace end or after the skimmer block.

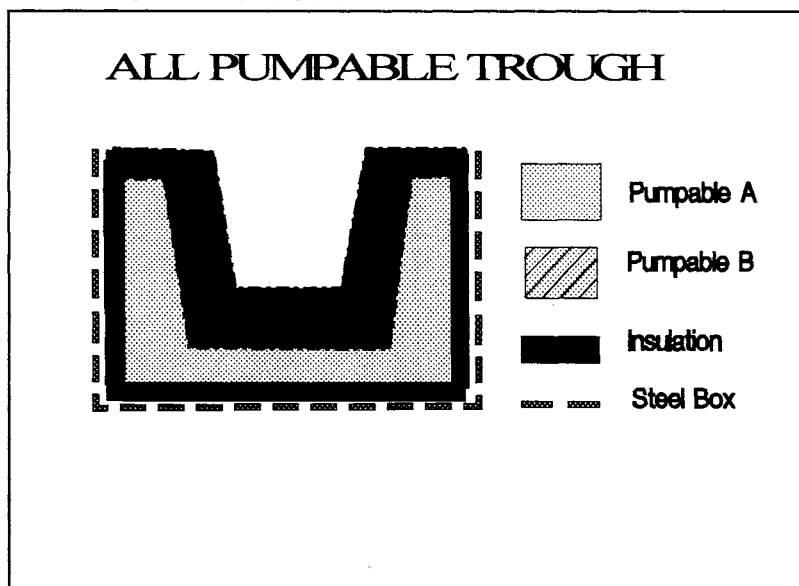


Fig. 5. Blast furnace trough refractory design with pumped or cast back-up lining.

Trough Designing with Water Modeling

The most extensive design work on blast furnace troughs was done using full scale water modeling.² The troughs were redesigned several times before producing good performance in actual use. The work was done with two single tap holes and one two-taphole furnace troughs. The full-scale water model consisted of two chambers — an outer chamber housing the candidate design and the inner chamber duplicating the actual design of the trough. One side of the model was made of clear acrylic plastic, the opposite side of steel. This design allowed convenient observation with reduced construction cost and greater ease of changing trough designs.

Both chambers were filled with water but the inner chamber, containing the iron trough, is sealed from the outer chamber. Water

was pumped in a closed circuit using a 2.3 m³/min capacity pump. The pump flow rate was controlled so the water velocity exiting the taphole matches the velocity of molten iron at the blast furnace. Only the molten iron was simulated with water; slag was not simulated since no suitable analog for slag could be found. The flow performance of each trough design was evaluated by visual observation of the flow patterns, measurement of the residence-time distribution (including proportion of the plug flow), back mixing and dead volume and point velocity measurements.

The residence time distribution was determined by injecting potassium permanganate, KMnO₄, rapidly into the water supply line ahead of the taphole. The output concentration of the KMnO₄ was monitored using a continuous colorimeter located at the overflow of the iron dam. The velocity profile was measured using an electromagnetic velocity meter. Typically, at selected trough cross sections, nine-point velocity measurements were recorded. The water velocity was only measured parallel to the trough walls and might not reflect the maximum velocity at the measured points. Initially, the changes in the design were aimed at reducing localized areas of high velocity and at increasing the hot metal residence time.

As a result of the water modeling experiments, many trough designs were converted from non-pooling or semipooling to pooling types that reduce the velocity at the walls and increased the quiet zone area considerably. Other changes associated with the change to pooling type trough were an increased length of the trough and adjustments to the Baker dam height providing a comfortable 200 mm pool of iron at the impact area. From a study of the wear pattern of several German blast-furnace troughs, it became clear that troughs with side wall angles greater than 20° and less than 6° experience greater wear. Hence, the side walls were designed to have an angle of 10° and exhibited less wear.

The recommended design derived by studying water modeling improved the efficiency of the casthouse significantly. The downtime

on the furnace dropped from 28 percent to an average of 1.2 percent. The cost of refractory as well as labor was reduced and the iron came out cleaner, having less impact on the refractory lining of the torpedo and transfer ladles.

Iron Runner

Since the use of LCC and ULCC in the blast furnace cast house for trough, iron and slag runners, a precast shape has been designed for the iron runner. Initially, a complete tube was designed to be set in place of the installed iron runner. The tubes were made in sections up to six meters in length and were joined by castables. But this procedure created problem of clogging due to skull formation inside the tube which was difficult to remove. Subsequently, precast shapes were designed in "U" shape that was easy to clean.

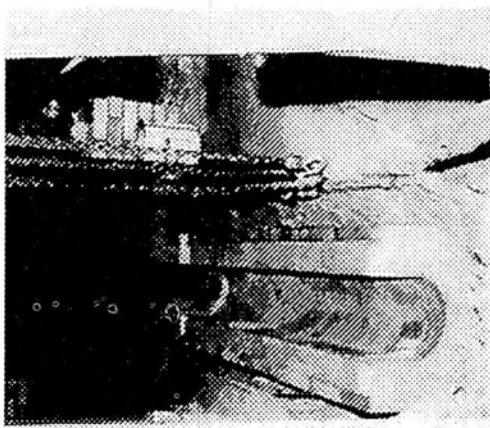


Fig.. 6. Iron runner 'U' tube.

The 'U' tubes are put in place on a bed of sand and the separate pieces are joined by ramming mixes. The 'U' tubes are shown in Figs. 6 and 7 depicting the individual tube and the one with ceramic blankets on top that supports the flat covers.

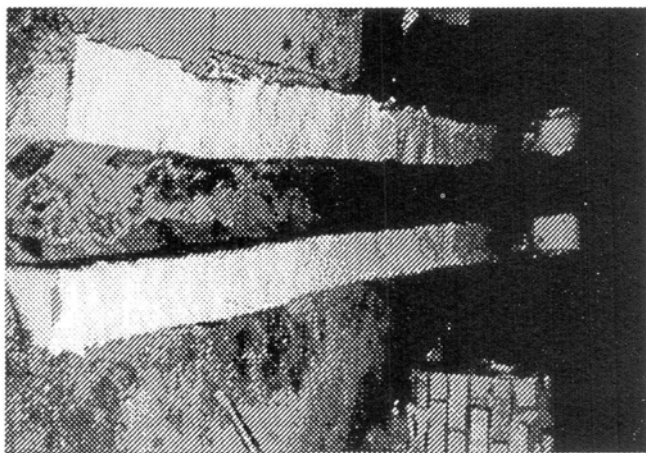


Fig. 7. Iron runner 'U' tube with ceramic blankets for supporting the flat covers.

Form designs

Since the development of no-cement pumpable compositions, there has been developments in the form designs for casting trough, iron and slag runners. Usually, forms are cast with solid steel plates that are welded together in manageable separate pieces that are put together for casting. Recently, light weight forms have been developed^{3,4} that are made out of expanded metal. The expanded metal is supported by welded square tubes forming a network design which is held by wire ties. The wet mix does not flow out of the gaps in the expanded metal since it is cohesive in nature and does not need any vibration to flow.

Tundish

In a continuous casting process, tundish is the last vessel that steel goes through before entering the mold that shapes the hardened steel. The quality of the product depends on the quality of the steel that goes through the tundish into the mold. Hence, to obtain cleaner steel, the steel in the tundish must be clean.

Tundish plays a significant role in refining steel. The key factors to consider are the cleanliness, chemical and thermal homogeneity of the steel as well as the ability to run the tundish for long periods of time (as long as the system is capable of running for maximum efficiency of the process). These goals are achieved by controlling the flow with specially-designed refractory systems resulting from modeling.

Over the years, the tundish has gone through a number of changes related to its refractory uses dedicated to the cleanliness of steel. Initially, the tundish was lined with bricks primarily because of the unavailability of suitable monolithic refractories. Since the development of LCC; its use as the back-up lining has been a common practice. The molten steel can create different types of fluid flow inside the tundish depending on the size of the tundish, type of caster, number of strands, type of steel cast and the size of the ladle heat. The turbulence created by the incoming steel, dead zones in the tundish, remixing of the inclusions in the tundish and the temperature differential at different strands in a multi-strand caster can create problems in casting operations as well as in the quality of the steel produced.

Flow control system design

Various types of flow-modifying devices have been developed over the years to confine the slag and inclusions, to eliminate dead zones and bring uniformity of temperatures among the strands. The following devices are commonly used:

- **Dam** — This device consists of a solid (sometimes associated with holes) refractory wall placed at the bottom of the tundish to obstruct the flow of steel directly into the nozzle. It is usually placed close to the incoming stream of the steel (Fig. 8).

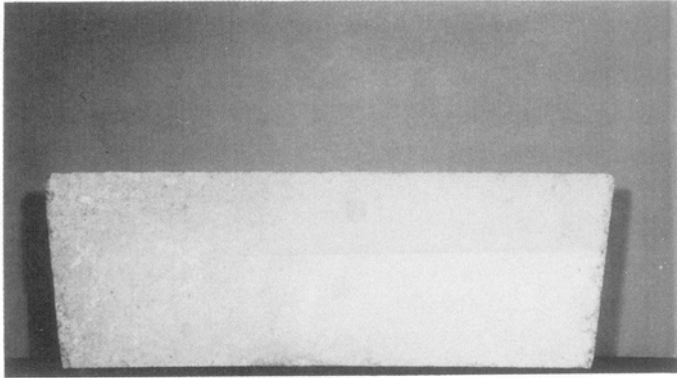


Fig. 8. Dam.

- **Weir** — This device is also a solid refractory wall but placed at the top of the tundish and extending above the steel level. The function of the weir is to contain the steel slag in the front portion, near the incoming stream of the tundish, so slag cannot get entrained into the nozzle (Fig. 9).
- **Baffle** — A baffle is a wall placed inside the tundish extending from the bottom to the top and containing several holes of various sizes and shapes depending on the size and shape of the tundish. It confines the slag in the incoming area, allows the flow of the steel in the desired direction with a smooth flow eliminating turbulence and dead zones in the tundish, and

provides uniformity in temperature and composition of the steel (Fig. 10).

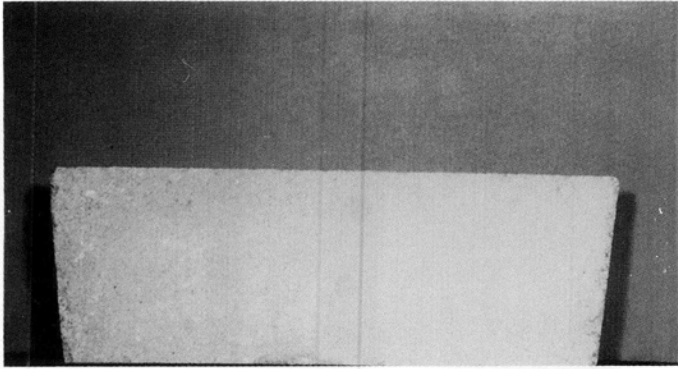


Fig. 9. Weir.

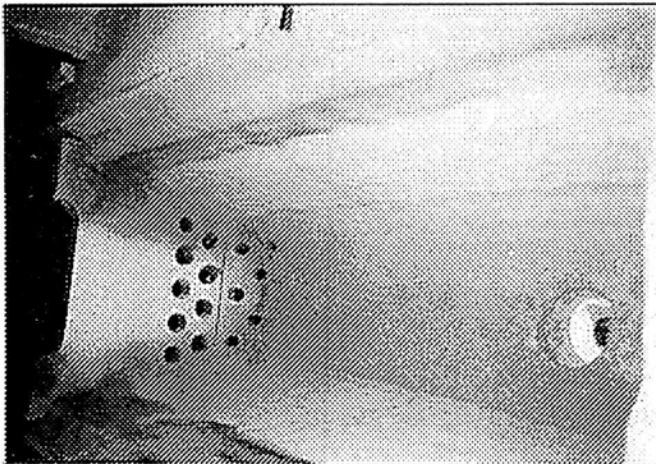


Fig. 10. Baffle placed inside a tundish.

Baffle designers have increased the contact area and time with the refractory for more efficient separation of the slag and the molten steel. New designs have modified the shape of the hole and the baffle surface (incoming side).^{5,6} The results of these modifications are yet to be measured and evaluated.

- **Impact pad** — A specially shaped refractory is placed where the stream of the steel hits the bottom of the tundish to make the bottom more erosion resistant. Several types of recently-designed^{7,8} impact pads reduce the turbulence in the tundish caused by the incoming stream (Fig. 11a and 11b).

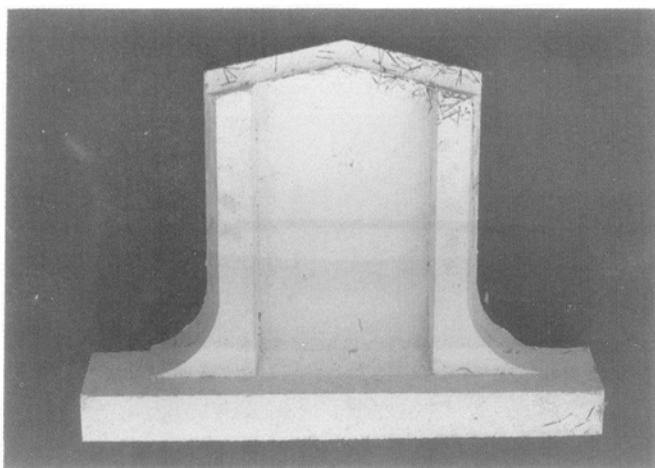


Fig. 11a. Impact pad.

Other designs of impact pads, which can be circular, rectangular or octagonal (Fig. 12), raise sides to enable containment of the stream. The sides redirect the stream in an upward direction instead of allowing it to splash around, thus providing a good separation of the inclusions and the steel.

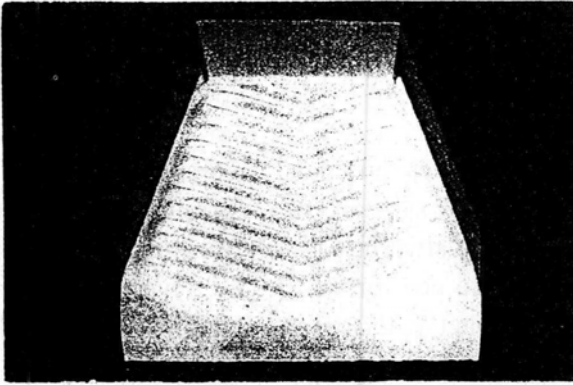


Fig. 11b. Impact pad.

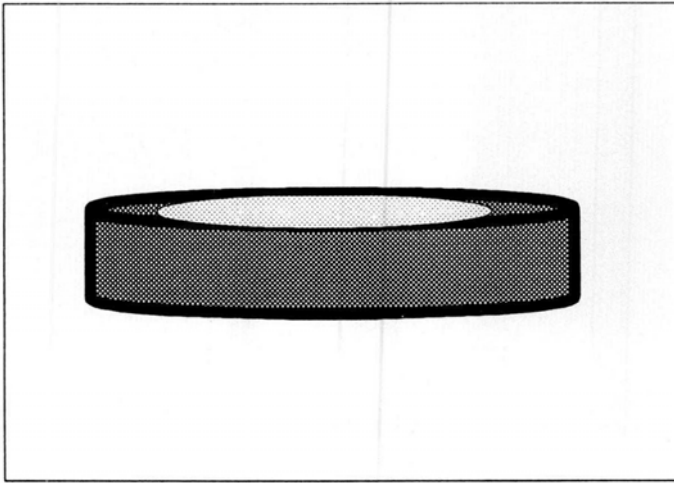


Fig. 12. Designs of the impact pad for reducing turbulence in the tundish.

- **Special impact pad and dam combo** — A newly-designed impact pad⁹ called a "turbostop" has a collar on its raised sides

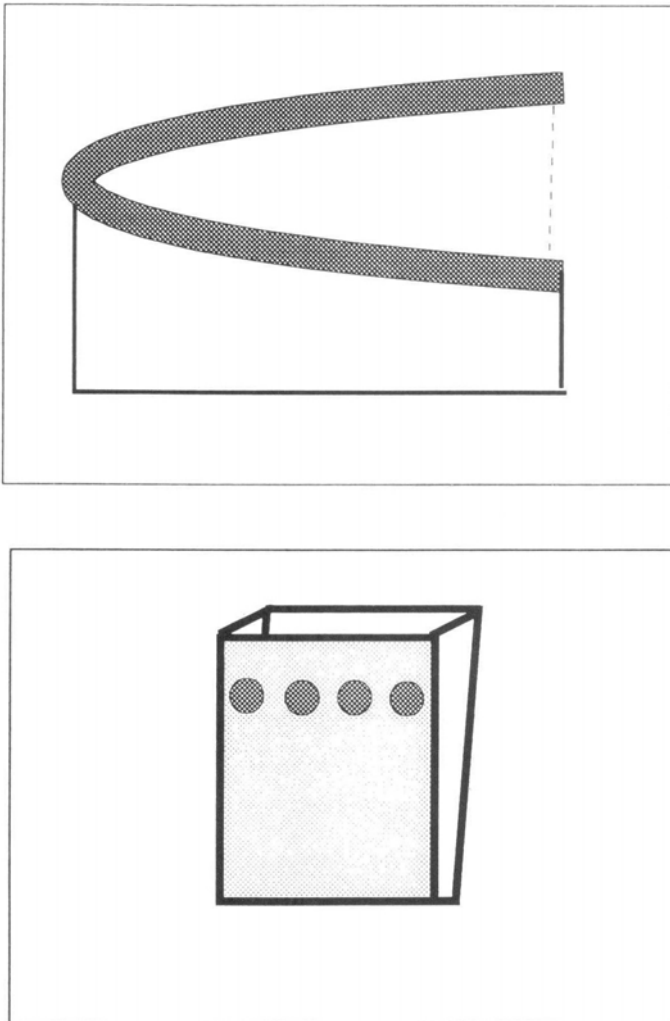


Fig. 13. Special impact pad and dam combo.

and is combined with a small dam with few holes (Fig. 13). This design enables the tundish to run for longer periods of time compared to the current designs using baffles that do not survive

long casting periods. The system also minimize re-oxidation by reducing turbulence caused by the incoming stream. This has proven to be as effective as baffles in cleaning and homogenizing the steel.

Designs of the flow-control systems are determined by water modeling either to the same size of the tundish or by reducing it by a certain factor for convenience. Different flow-modifying devices are recommended depending on the size and complexity of the tundish, as well as the number of strands. Flow-control designs may consist of one baffle, or a combination of baffle and dam and/or weir. Occasionally, the design calls for a dam and a weir only. In other cases, the dam is combined with an impact pad (as shown in Fig. 14) to provide the proper uniform distribution of steel and temperature throughout the different strands in a multi-strand T-shaped tundish.

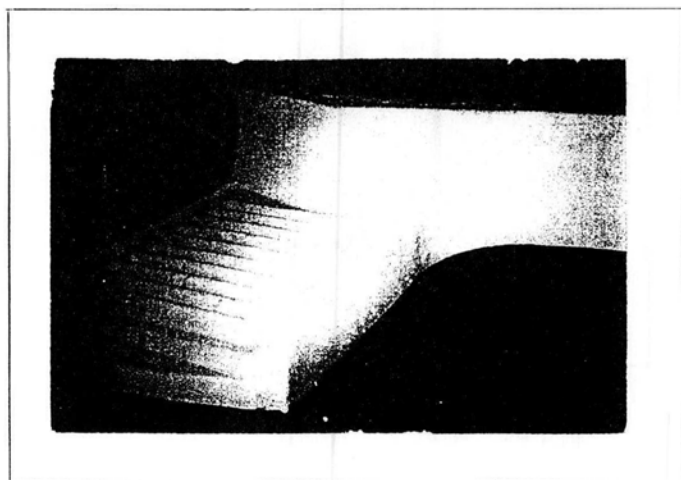


Fig. 14. Impact pad and dam combo.

- **Slag Stopper** — One of the problems encountered by steel makers is the amount of steel left over in the tundish due to slag entrainment. The size of the skull can sometimes become too big and result in loss of significant amounts of finished steel. To avoid this problem, a slag stopper has been designed¹⁰ to allow the steel to drain off until the slag approaches the outgoing nozzle. A ball inside the slag stopper cone (Fig. 15), with a density lower than that of the steel but higher than that of the slag, floats at the interfacial layer of slag and steel. This design eliminates the vortex of the steel flow and allows the steel to be drained almost completely until the slag arrives at the nozzle and the ball closes the opening. Although the system allows the steel to be drained as much as possible, there are some inherent logistic problem in keeping the slag stopper in place.

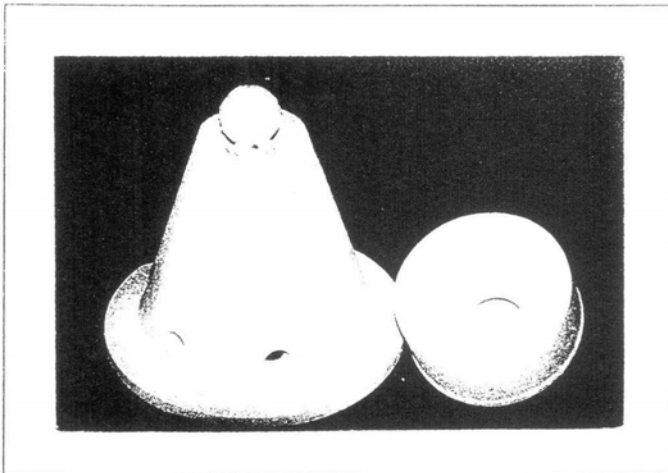


Fig. 15. Design of slag stopper for reducing skull size in the tundish.

Steel ladle

Although the refractory uses of the side wall vary widely among different steel makers, the use of LC and ULC castables is a common practice in a number of plants around the world. Some plants prefer precast shapes and others prefer bottoms cast in place. The use of impact pads, with precast shapes are most common primarily for the direct impact area. In some cases, the impact area is raised from the rest of the bottom to equalize the wear pattern of the whole bottom. Some designs feature ladle bottoms with part of the side wall cast with the impact pad to prevent any impact of the steel stream hitting the side wall or the erosion of the side wall bottom due to the impact close to the side wall.

Ladle well blocks are made from precast LCC compositions usually with 85–90% Al_2O_3 or more recently from alumina-magnesia spinel LCC compositions. The size, shape and the preferred compositions are primarily dictated by the operational conditions and operator preference.

Impact pad

The design of the impact pad varies greatly from plant to plant since the steel stream hitting the bottom of the ladle varies. In some cases, the stream hits a concentrated area close to the center with little possibility of the stream hitting the side wall. In other cases, the stream hits the side wall where the effect of the impact is worse on the walls. In general, however, the direct impact area is thicker than the rest of the bottom. The impact pads can either be cast in place or precast shapes may be placed directly in the ladle bottom. In ladles where the incoming stream is concentrated in a small area, a prefired shape may provide a longer bottom life since the stronger refractory material will reduce the wear of the impact pad. In particular, precast shapes made out of Al_2O_3 –MgO spinel LCC show exceptional hot strength if they are prefired to 1400°C or above. Most impact pads are either 85 % Al_2O_3 LCC or Al_2O_3 –MgO spinel LCC. The Al_2O_3 –

MgO spinel LCC is more expensive than 85 % Al_2O_3 LCC but usually lasts longer providing better value and cleaner steel.

The shape and size of the impact pads vary depending on the operating conditions of the plant. Some plants use the complete bottom as a single cast piece — usually with raised surface at the impact area and also raising the side towards the wall to prevent erosion from direct impact of the stream or splashing of the incoming stream. Some plants use precast shapes just for the impact area while the rest of the bottom is either bricked or cast with the same or different lower quality LCC.

When precast shapes are used for the total bottom, they are usually cast in two or three pieces with lap or tongue and groove joints. The joints are then filled with the same castable upon installation.

Electric arc furnace (EAF)

There have been a significant increase in steel production by mini mills utilizing EAF. The EAFs are usually have water cooling in the linings and the roof. The delta area covering the electrodes used to be lined with bricks and plastic refractories. But with the development of LCC with improved properties, the delta section is lined with LCC in most steel plants. In plants where the conditions are relatively severe and the refractory suffers significant thermal shock and slag splashing, alumina-chrome LCC are still used. In most still plants the normal practice is to use 85% alumina LCC.

The delta sections usually are sent out to the installation place of the refractory manufacturer where the refractory is cast using a form, cured, dried and heated to a temperature for dehydrating the cement and then sent back to the still plant. Figure 16 shows a precast delta with 85% alumina LCC.

Another application of monolithic refractories in EAF is the runner or spout of the furnace. The spout usually goes through frequent thermal shocks and is exposed to slag towards the end of the

cast. Previous practices were to line the runner with high and slag resistant brick.

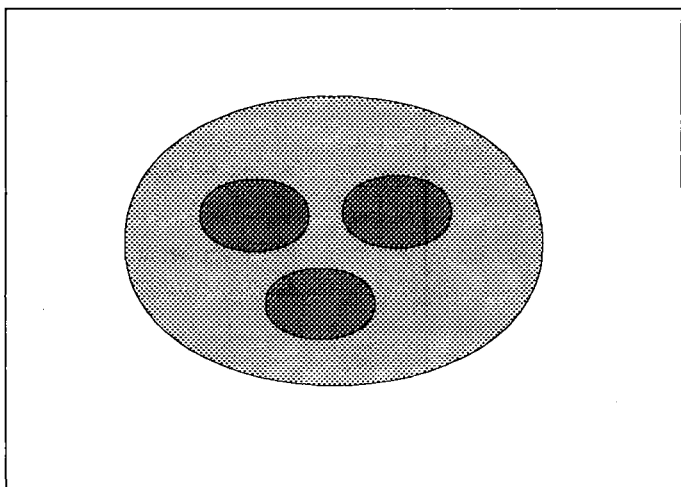


Fig. 16. A precast delta with 85% alumina LCC.

But during the past 15 years monolithic refractories have been in used almost all plants because of their improved performance and labor savings. Various kinds of refractories are used for the runner. Runners with Al_2O_3 -SiC-C resin-bonded ramming mixes and LCC are quite common. Recently MgO based castables have been used in several plants. The runners usually are thicker at the bottom and the lower part of the design that provides extra material for longer lining life. Figure 17 shows a typical electric furnace runner.

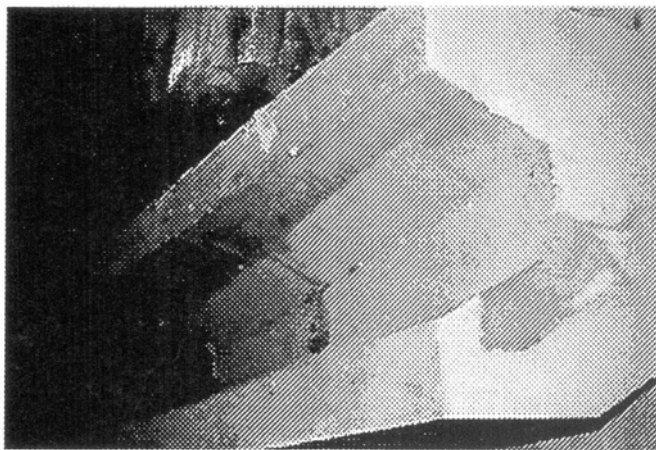


Fig. 17. A typical electric-furnace runner.

References

1. S. Banerjee, "State of the Art of Blast Furnace Cast House Refractories," *Steel World Qurtly.*, Sept. 1993, pp. 53–58.
2. I. A. Cameron and J. M. Tudhope, "Improved Trough Design Using Water Modeling," *Proc. Iron Making Conf.*, pp. 505–515, 1988.
3. *U.S. Patent* # 5,482,248, Jan. 9, 1996.
4. *U.S. Patent* # 5,484,138, Jan. 16, 1996.
5. *U.S. Patent* # 5,018,710, May 28, 1991.
6. *U.S. Patent* # 5,064,175, Nov. 12, 1991.
7. *U.S. Patent* # 5,072,916, Dec. 17, 1991.
8. *U.S. Patent* # 5,131,635, July 21, 1992.
9. *U.S. Patent* # 5,551,672, Sept. 3, 1996.
10. *U.S. Patent* # 5,083,754, Jan. 28, 1992.

CHAPTER XIII: EVALUATION AND TESTS

Introduction

Monolithic refractories are fundamentally a mixture of refractory ingredients along with minor ingredients of binders, and additives to control the rheological and physical behavior of the product. Whether the refractory is installed in place or precast shapes made in-house, the properties are governed by the mixed ingredients and the processes followed in placement of forming shapes.

Most monolithic refractories excepting prefired shapes are mixtures of the ingredients that are supposed to react and form refractory phases at or before the temperatures in use. Hence it is important that the expected properties at the use temperatures are evaluated to predict the performance characteristics of the products. Hence a thorough characterization of the properties needs to be done both after heating as well as at elevated temperatures to know the behavior of the refractory at or near the use temperature. Usually technical data sheets for monolithic refractories show most of the data after heating and some data at elevated temperatures. Recently, users are demanding more of the data at elevated temperatures to evaluate the properties near the use temperatures. Hence, it is imperative that the ingredients should be fully characterized regarding their chemistry,

physical characteristics such as, density, porosity, refractoriness and mineralogy.

Chemical analysis

Since the chemistry of the finished product will depend upon the results of the chemical analysis of the individual ingredients, it is important that the chemical analysis of the components of the product be known. Although it may not be necessary to analyze every shipment of raw material from the same supplier, it is important to check the raw-material chemistry from time to time so that the chemical analysis of the product comes out as desired. Certain metallic impurities, such as metallic iron and silicon may play important roles in the refractory particularly during the installation procedure. Metallic iron is a common impurity that comes from the grinding operation of the refractory raw material. In certain mixes it can be quite harmful causing gas evolution and objectionable spots on the surface. The presence of both metallic iron and silicon can cause premature setting as well as swelling after installation particularly when working with sol-gel binder system.

The most commonly cited technical data are bulk density, apparent porosity, cold-crushing strength, cold modulus of rupture, permanent linear change after heating to various temperatures such as 110°C, 815°C, 1370°C and 1500°C. Depending on the refractory manufacturer, the temperatures of the post-heating data may vary somewhat, but usually are close enough for a reasonable comparison. Hot-modulus-of-rupture are the most frequently quoted data performed at elevated temperatures. These are usually given at 815°C, 1370°C and 1500°C. Again, the temperatures shown by different refractory manufacturers may vary. Because of the extended use of monolithic refractories in recent years, thermal-shock tests after heating to 1200°C are becoming a common practice. Depending on the use of the material, the deformation under hot load or creep

deformation after a certain length of time at a certain temperature are also provided.

Blast-furnace cast-house refractory users usually demand the corrosion/erosion results of the refractory after exposing to molten slag and or iron/slag combination. For petrochemical industries, the most demanding test result is the abrasion resistance of the refractory after heating to 815°C. Hence depending on the end use of the refractory, the specific test-results may vary. Most of the tests performed are according to standard ASTM methods wherever standardized methods are available. The following are the ASTM¹ standard tests for the specific properties:

Property	Standard ASTM Method No.
Bulk Density/Porosity	C-20
Workability	C-181
Cold Crushing Strength	C-133
Cold Modulus of Rupture	C-133
Permanent Linear Change	C-113
Hot Modulus of Rupture	C-583
Thermal Shock	C-1171
Thermal Conductivity	C-417
Abrasion Test	C-704
CO Disintegration Test	C-288

Specimen preparation

Specimens for testing should always be prepared keeping in mind the actual installation conditions. The different types of monolithic refractories should be prepared with different forming methods.

Castables:

Conventional and insulating castables should be prepared by standard ball-in-hand method and the flowability should be quantified by the

flow table (ASTM C-860). The water content used for the castables may easily vary by 1 to 2% during installation and hence the data generated in the laboratory should cover the range of water content that will reflect the values in the materials installed.

Low-cement and ultralow-cement castables are installed by vibration. These mixes are designed in such a way that they will only flow under vibration and are somewhat pseudothixotropic in nature. The amount of water needed to have the mixes flow is determined by allowing the mix to flow under vibration on a table to a predetermined value. The normal procedure is to fill a brass cone (ASTM C-860) with the mix and level it under vibration for a certain period of time (15 or 30 sec) on a vibrating table with a vibration frequency of 3000–3600 vpm and then the brass cone is removed and the mix flows under vibration for another predetermined period of time (15 or 30 sec). After attaining the water content for the desired degree of flow under vibration, specimens for testing are also prepared under vibration (on the vibrating table) in steel or plastic molds.

Plastics

Test specimens for plastic refractories are prepared either by ramming or by pressing in steel molds in cubes (50 x 50 x 50 mm) and bars (38 x 38 x 200 mm, 25 x 25 x 150 mm and 50 x 50 x 225 mm).

Ramming mixes

Specimens for ramming mixes are prepared by ramming the material into plastic mold after making the material with an appropriate amount of liquid that will be used in manufacturing to produce the product. Test specimens for ramming mixes are prepared by ramming in steel molds in cubes (50 x 50 x 50 mm) and bars (38 x 38 x 200 mm, 25 x 25 x 150 mm and 50 x 50 x 225 mm).

Gunning mixes

The most appropriate method of evaluating gunning mixes is to test specimens from gunned panels (ASTM C-903). Any other methods used to evaluate gunning mixes such by ramming or pressing will not reflect the actual properties of the gunning mix. The normal procedure is to gun the test mix on properly greased wooden panels (details are provided in the ASTM book).

The cold properties of the specimens are determined by heating the specimens to temperatures according to the ASTM heating schedule and then cooling to room temperature. The specific properties are determined according to the specific test procedures.

Density and porosity

Except for insulating refractories, the density/porosity of a monolithic refractory has a direct correlation with the physical properties as well as their performance since mechanical strength, corrosion/erosion and abrasion resistance are all related. Hence it is imperative to formulate monolithic refractories from raw materials and combinations of ingredients so that the porosity and the pore size are both within acceptable limits. Determination of the porosity of raw materials and finished products in the laboratory is important to ensure the expected values when the refractory will be installed at the job.

Density/porosity measurement: ASTM C-20

This method is used to determine the apparent porosity, apparent specific gravity and bulk density of refractory specimens. This test method is a primary standard that is suitable in quality control, research and development, establishing criteria for and evaluating compliance with specifications and providing data for design purposes. The fundamental assumption in this test method is that the refractory materials are not attacked by water, they conform to the requirements for size configuration and original faces, the open pores of the test specimen are fully impregnated with water during the boiling

treatment, and the blotting of the test specimens is performed as specified in a consistent and uniform manner to avoid withdrawing water from the pores. Deviation from any one of these assumptions adversely affects the test results.

For monolithic refractories test specimens that are susceptible to attack by water (i.e., phosphate bonding, sol-gel bonding, resin bonding, sodium silicate bonding, etc.) should be preheated to temperatures so that no part of the refractory body is attacked by water. Since the apparent porosity, apparent specific gravity and bulk density are all related properties, any error in the determination of one of them will adversely affect the others.

Calculation:

$$\begin{aligned}\text{Exterior Volume, } V, \text{ cm}^3 &= W - S \\ \text{Apparent Porosity, } P, \% &= \{(W - D)/V\} * 100 \\ \text{Apparent Specific Gravity, } T, &= D/(D - S) \\ \text{Bulk Density, } g/\text{cm}^3 &= D/V\end{aligned}$$

Reheat Change of Refractories: ASTM C-113

This method covers the determination of the permanent linear change of refractory bricks and shapes when heated under the prescribed conditions. The linear changes obtained by this method are suitable for use in research and development and are often used to establish written specifications between producers and customers. Care should be exercised in testing samples that are representative of the product under question and in ensuring that the schedule selected is appropriate to the product.

The method relates to the linear change of refractories after being heated to a specified temperature at an appropriate rate and holding for a predetermined time by the above method and then measuring the change in length at room temperature.

Workability: ASTM C-181

Aggregates used in plastic and ramming refractories comprising different grain sizes are non-plastic materials. Hence, under compression, a large internal friction is developed. For this reason, these refractories require the addition of clay and water to develop the required workability that reduces the internal friction caused by the aggregate grains. Higher amounts of clay and/or water increases the workability of the material but cause high shrinkage and low strength. Hence, in the manufacture of plastic refractories a balancing is needed between the workability and other properties like the degree of packing, volume stability (low shrinkage) and strength. the workability index may vary widely in plastic refractories depending on the application.

The workability index gives a measure of the facility with which a plastic refractory installed by rammers, gunning or vibration. The workability index is commonly used to control the consistency of plastics during manufacture and provides useful information for the specification of acceptance by consumers.

The method describes the reduction in the compressibility of a cylinder by tapping with a weight usually three times for plastic- and two times for ramming mixes and then determining the % change in height. This is done according to a standard method described in the ASTM manual.

The workability index is calculated by the equation:

$$W = (L - L_1) / L * 100$$

where

L = Length of the specimen prior to deformation

L_1 = Length of the specimen after deformation, and

W = Workability index

The workability index of plastic refractories decreases with elapsed time due to loss of moisture, absorption of moisture by the

ingredients in the mix, and other physical and chemical changes in the mix during storage. The rate of change is related to humidity and atmospheric temperature, while it varies considerably with storage conditions. Plastic refractories, therefore, must be stored away from areas exposed to direct sunlight or high ambient temperatures.

Cold crushing strength and cold modulus of rupture: ASTM C-133

This test method is designed to determine the compressive strength and modulus of rupture (flexural strength in 3-point bending) at room temperature. They only indicate the strength of a refractory for its ability for mechanical construction and does not indicate its performance at elevated temperatures. Considerable care must be used to compare the results of different determinations of the cold-crushing strength or modulus of rupture. The specimen size and shape, the nature of the specimen faces, the loading geometry as well as the rate of load application may all significantly affect the numerical results obtained. Comparison of results between different determinations should not be made if one or more of these parameters differ between the determinations. This test method is useful for research and development, engineering application and design, manufacturing-process control and for developing purchasing specifications.

Cold crushing strength ASTM C-133

The test is usually done on a cubic sample as specified in the method and then applying pressure in the standard-strength machine until failure.

Calculation:

$$\text{Cold Crushing Strength, } S = W/A$$

where:

S = cold crushing strength, kg/cm^2 or MPa,

W = total maximum load indicated by the testing machine (kg or Pa), and

A = average of the areas of the top and bottom of the specimen perpendicular to the line of application of the load, cm^2 .

Cold modulus of rupture ASTM C-133

The method implies the breaking of standard size bars (as specified in the ASTM method) by a three-point bending method until the point of failure and then measuring strength according to the following equation:

Calculation:

$$\text{Modulus of Rupture, MOR} = 3PL/2bd^2$$

where

MOR = modulus of rupture, kg/cm^2 or MPa,

P = maximum load applied at rupture, kg or Pa

L = span between supports, mm,

b = breadth or width of the specimen, mm, and

d = depth of the specimen, mm.

Hot modulus of rupture: ASTM C-357

This test method covers determination of the high-temperature modulus of rupture of monolithic refractories in an oxidizing atmosphere and under action of a force or stress that is increased at a constant rate. Measuring modulus of rupture at an elevated temperatures has become a widely accepted means to evaluate materials at service temperatures. In recent years more emphasis has been put on the hot modulus of rupture values than cold modulus of rupture or cold crushing strength properties. This method is limited to furnaces operating under oxidizing conditions. However, with modifications in atmospheric controls in other test furnaces, the major

criteria of this test procedure may be employed without change. The test method is designed for progressive application of a force or stress on a specimen supported as a simple beam with center-point loading. The test apparatus designed for the progressive application of strain may yield different results, especially since refractory materials will reach a semiplastic state at elevated temperatures where Hook's law does not apply, that is, stress is not proportional to strain.

Hot modulus of Rupture at the test temperature is calculated as follows:

$$\text{HMOR} = 3PL/2bd^2$$

where

HMOR = hot modulus of rupture, MPa,

P = concentrated load at rupture, N, Kg or Pa,

L = span between supports, mm,

d = depth of the specimen, mm and

B = breadth or width of the specimen, mm.

Abrasion resistance of refractory materials at room temperature: ASTM C-704

This test method covers the determination of the relative abrasion resistance of refractory materials at room temperature. The abrasion resistance of a refractory material provides an indication of its usability for service in abrasive or erosive environment. The results obtained by this test could be different from those obtained in service because of the different conditions encountered. The test method measures the volume of material in cubic centimeters abraded from a flat surface at a right angle to a nozzle through which 1000 g of size-graded silicon carbide grain is blasted by air (the details of the method are described in the ASTM manual).

Calculation:

The amount of the refractory lost by each specimen by abrasion in cubic centimeter, A , is calculated as follows:

$$A = \{(M_1 - M_2)/B\} = M/B$$

where

B = bulk density, g/cc,

M_1 = weight of the specimen before testing, g,

M_2 = weight of the specimen after testing, g, and

M = weight loss of the specimen, g.

The average of the individual results is reported as the abrasion loss of the sample.

Thermal conductivity of unfired monolithic refractories: ASTM C-417

The thermal conductivity of monolithic refractories is a property required for selecting their thermal-transmission characteristics. Users select monolithic refractories to provide specific conditions of heat loss and cold face temperature, without exceeding the temperature limit of the monolithic refractory. This test method specifies the placement of thermocouples and positioning of test specimens in the calorimeter (ASTM C-201) and the procedure requires a large thermal gradient and steady state conditions. The results are based upon a mean temperature. The data from this test method are suitable for specification acceptance, estimating heat loss and surface temperature, and the design of multi-layer refractory construction. The use of these data requires consideration of the actual application environment and conditions. The details of the method are described in an ASTM manual.

Calculation:

The thermal conductivity is calculated as follows:

$$k = qL / \{A(t_1 - t_2)\}$$

where

k = thermal conductivity, W/m.K,

q = heat flowing into the calorimeter (temperature rise, K , of the water flowing through the calorimeter times the weight of the flowing water, W ,

L = thickness (distance between the hot junctions at which t_1 and t_2 are measured), m ,

t_1 = higher of the two temperatures measured in the test specimen, K ,

t_2 = lower of the two temperatures measured in the test specimen, K , and

A = area of the center calorimeter, m^2 .

Thermal shock test: ASTM C-1171

This test method is used for determining the strength loss or reduction in the continuity, or both, of shapes subjected to thermal cycling. The strength loss is measured by the difference in the modulus of rupture between uncycled specimens and the specimens subjected to thermal cycling. The reduction in the structural continuity is estimated by the difference in sonic velocity before and after thermal cycling.

This test method indicates the ability of a refractory product to withstand the stress generated by sudden changes in temperature. Since the recommended furnace temperature of this cycling test is 1200°C , this test method may not indicate the ability of a refractory product to withstand cycling at a higher or lower temperature, especially if the existing morphology of the refractory product changes. The test method is useful for research and development as well as for comparing refractory products (the details are provided in the ASTM manual).

Calculation:

The percent sonic velocity loss of each specimen is calculated as follows :

$$(V_0 - V_F)/V_0 * 100$$

where

V_0 = original sonic velocity of each specimen, m/s, and

V_F = sonic velocity of each specimen after testing, m/s.

The percentage loss in the modulus-of-rupture strength is calculated for each specimen as follows:

$$(M_0 - M_f)/M_0 * 100$$

where

M_0 = average modulus-of-rupture strength of the unshocked specimens from the first group after testing, MPa, and

M_f = modulus-of-rupture strength of the unshocked specimens from the second group after of specimens after testing, MPa.

The individual sonic velocity, modulus of rupture, percent sonic velocity loss, and percent modulus-of-rupture strength loss values, as well as the average percent modulus-of-rupture loss, and the furnace temperature should be reported.

CO disintegration test, ASTM C-288

This test method covers the comparative behavior of refractories under the disintegrating action of carbon monoxide (CO). This test method uses an accelerated exposure to CO to determine the potential material behavior in a relatively short time. The results obtained by this method can be used to select refractories that are resistant to CO disintegration. This method is suitable for research and development and for establishing CO disintegration criteria for acceptance specification. The disintegration of test specimens is accelerated

providing a higher concentration of CO than anticipated in most service environments. The effect on the test specimens may be different from those found for refractories in actual service conditions

Any prefiring of the test specimens should be reported and the condition of each specimen at each inspection by photographing the most affected face(s) should be reported, and the extent of any spotting and the degree of disintegration should be described according to the following classifications:

Unaffected — when no particles spall and no cracking occurs.

Affected — Surface popouts, when destructive action is confined to spalls or surface popouts of 13 mm or less in diameter.

Affected — Cracked, when destructive action produces spalls or popouts greater than 13 mm in diameter, or cracking, or both.

Destructive condition — when the specimen breaks into two or more pieces, or when hand pressure can cause breaking. In some company standards the above results are categorized as 0, 1, 2 & 3.

Corrosion/erosion:

Drip slag testing of refractories: ASTM C-768

This method covers the determination of the relative resistance of various refractory materials to the action of molten slag dripped continuously onto test specimens within a heated furnace. The practice measures the relative slag resistance of refractory materials and can be useful in the development of new products or in the selection of a product used to be in contact with a particular slag composition. Since the entire test specimen is at the test temperature, penetration of the slag is promoted and may exceed that observed when the temperature gradient, typical of most refractory applications is present. The test slag is continuously renewed so that a high rate of corrosion is maintained. Carbon-containing materials may not allow proper measurement because of the oxidation of carbon.

A modified form of drip slag corrosion test² is performed in a dual chamber gas-fired furnace. A set of four specimens (two of each kind - one standard and test) measuring 63 x 75 x 88 mm are run in each test with the same kind diagonally seated in the chamber. At the start of the test a briquette of the slag measuring 63 x 75 x 13 mm is placed on top of each specimen and 13 mm of coke breeze covered the bottom of the slag pockets. The furnace is heated at 10°C/min with the gas/air ratio adjusted to a maximum of 0.5% oxygen by a fuel-cell combustion meter. When the furnace reaches 1595°C in 2 h and 45 min, the furnace is soaked for 15 min in an atmosphere containing 0.5% oxygen. Then the gas/air mixture is adjusted in composition as quickly as possible to obtain an atmosphere containing 2% oxygen (for blast furnace cast house slag testing) and the soak is continued for 15 more minutes. Pellets (38 x 38 x 25 mm) prepared, using slag from the respective areas (blast furnace or synthetic steel slag) are charged at approximately 6 min intervals, depending on the melting rate until 0.9 kg of the total slag has been added to each specimen. The furnace is turned off six minutes after the last pellet is charged giving a total of 72 min of slagging. the slag corrosion is reported as the average percent loss in the 75 x 88 mm vertical cross section perpendicular to the flow of the slag of each specimen.

Rotary slag testing of refractory materials: ASTM C-874

This test procedure covers the relative resistance of refractory materials against slag corrosion/erosion. A reference refractory specimen is included in each test and run since this is a comparative method between the standard and the test specimens and hence is useful in the development of new products or in the selection of a new product to be used in contact with a particular slag composition. A gradient exists through the test specimens that is controlled by the thermal conductivity of the specimens and back up material. Since the slag is constantly renewed, a high rate of corrosion is maintained. The flow of the slag can cause mechanical erosion of the materials that is a

function of the tilt and rotation of the furnace. Carbon-containing materials may produce different results due to the oxidation of the carbon phase.

Corrosion/erosion testing by induction furnace

Most slag-testing methods usually provide the effects of slag only on refractories as mentioned earlier in the drip and rotary slag-testing procedures. But in refractory applications such as blast furnace cast house trough, the most corrosion/erosion occurs at the metal-slag interface rather than slag or air-slag interface. Hence modified methods have been developed where the effect of slag and metal-contact areas have been evaluated in a dynamic condition simulating close to actual applications. Like other methods, this is also a comparative method where the behavior of a refractory should be known to compare the test specimens.

The test is done in an induction furnace where a natural movement of the metal occurs inside the furnace. The optimum furnace is a 30–35 kWA powered that can hold a standard clay-graphite (No.16) crucible. The copper coils for the inductor can either be in air twined around close proximity to the crucible or can be embedded inside a refractory. Pictures for the assembly are shown in the Figs. 1 and 2 below.

For determining corrosion/erosion resistance of blast furnace cast-house materials, specimens should be made with dimensions of 205 x 38 x 38 mm. They should be preheated (350 to 400°C) to remove volatile materials. Grooves are made at the top outside corners of the specimens for tying them around the holder with a screw band (as shown in the pictures). The tied specimens are then suspended from the holder and are held just above the crucible. First the iron (about 16 kg) is melted. After the iron is completely melted, the slag (about 600 to 700 g in granular form) is poured on top of the molten iron.

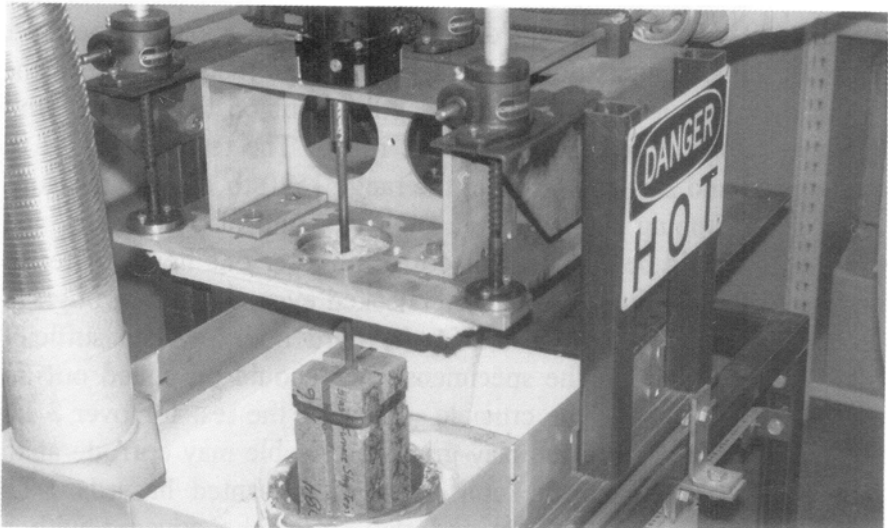


Fig. 1. View of the induction furnace at room temperature.

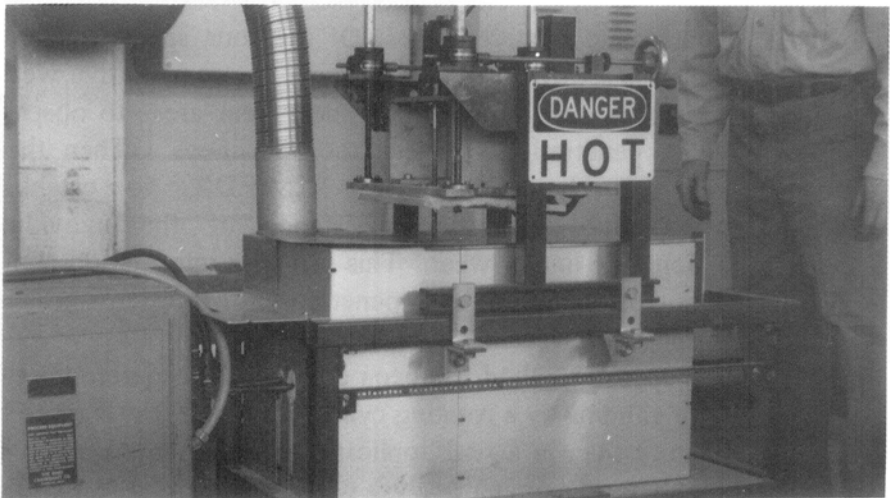


Fig. 2. View of the induction furnace while running.

At this point, the specimens with the holder are lowered to a position just above the crucible so that they are preheated and avoid

the thermal shock if brought down directly from room temperature. After all the slag melts down, the specimens are then lowered further to dip them about one-third their length (about 65 to 70 mm) while they are being rotated (about 8 to 12 rpm). The temperature of the molten iron and slag should be in the range 1520 to 1530°C measured either by a dipping thermocouple or by an optical pyrometer. The specimens should be checked for corrosion after one hour. After the first hour, the specimens should be checked every 30 min through the second hour and every 15 min in the third hour. When sufficient corrosion is visible on the specimens, they should be pulled out and allowed to cool above the crucible. Running the test for over 3 h is not recommended since the clay-graphite crucible may corrode at the slag line and leak into the coils creating unwanted hazards. The crucible with the molten iron and slag should be covered with an insulating board. The cooled specimens are then removed and visually examined for relative wear. Areas with the most wear observed are measured to determine relative wear. Of the four specimens, one standard specimen is always used to determine the comparative wear. The specimens are cut through the highest wear areas to observe any difference in penetration between the specimens. Then their microstructures are characterized by SEM/EDX, to establish corrosion/erosion and penetration resistance of the individual compositions along the line of wear. This provides information about both chemical corrosion and physical penetration of the steel and slag that can cause specimen wear.

Recently, a special method has been developed³ to determine the corrosion rate of alumina by a rotating cylinder method that is more of an academic nature than in actual application. The method involved the use of a highly pure alumina (> 99.9 wt. %) cylindrical rod with a 10 mm radius having a bulk density of 3.90 g/cm³ and apparent porosity of 0.76%. The slag was made up of laboratory-grade reagents CaO, SiO₂ and Al₂O₃ with an alumina content of 25% and a

slag basicity of 1.0. The melting point of the slag was 1400°C according to the phase diagram.

The test was conducted in an electric furnace to have accurate control of temperature and atmosphere. A carbon crucible containing the powder reagents was set with an alumina tube in the furnace and the alumina specimen was just positioned above the powder. The temperature of the furnace was increased at the rate of 1.25 °C/min to 1500°C. The alumina specimen was positioned just above the molten slag for 30 min and then immersed into the slag for 4 h with or without rotation. The rotation speeds were 0, 100, 250, 400 or 900 rpm and kept constant during the test. The depth of immersion was about 43 mm and the temperature fluctuation during the immersion was $\pm 3^\circ\text{C}$. The specimen was taken out of the molten slag and rotated at 400 rpm for about 3 min to remove the slag. The specimen was cooled in the furnace. During the test Ar gas with 3% H₂ was blown through the alumina tube to prevent oxidation of the crucible. The diameter of the specimen was measured every 5 mm along the length in three radial directions to determine the corrosion depth.

In the rotating-cylinder corrosion test, the profile of corroded specimens becomes unusual when a Taylor vortex⁴ arises. The condition depends on the rotation speed, the diameters of the crucible and the specimen, and the kinetic viscosity of the slag. In order to determine the corrosion rate precisely, prevention of the vortex is recommended if possible by adjusting the test conditions. The corrosion rate has been shown to be proportional to the 0.4 th power of the radii of the specimens. A new general formula to determine the corrosion rate by the rotating-cylinder method that holds for any radius of the specimens and the corrosion time has been introduced and its validity has been confirmed. Application of this formula improves the accuracy in the evaluation of corrosion resistance of refractories.

References

1. "Refractories, Carbon and Graphite Products; Activated Carbon", *Annual Book of ASTM Standards*, 15.01, 1996.
2. K. N. Singh and R. S. Hoffer, "Successful Implementation of Metal-Free Low-Moisture Castables in Blast Furnace Troughs and Runners," *UNITECR '89 Proc.* pp. 193–211.
3. M. Ilyama, "Evaluation of Corrosion Rate of Alumina by Rotating Cylinder Method," *UNITECR'95*, pp. 371–378, Nov. 1995.
4. H. Schlichting, "Boundary-Layer Theory," *McGraw-Hill*, A. Ogawa, "Uzugaku," Sankaido (1981), pp.131.

CHAPTER XIV: LINING

Introduction

Despite all the advantages that monolithic refractories possess, there are certain drawbacks that need to be addressed to benefit fully from their application. By definition monolithic refractories form a continuous body of refractories. Whenever cracks develop there is always a possibility that the cracks will join to form separate individual regions within the refractory body that may fall off from the rest causing problem in the process. The cracks may develop from temperature gradients causing thermal shock in the lining or from the differential-expansion characteristics of the supporting lining, most commonly steel. Thus, whenever a continuous monolithic refractory lining is installed, there is a great need for reinforcement of the lining to minimize these problems.

Anchors

As the term indicates, monolithic refractories are anchored with the base of steel structures by means of strong materials like steel or fired ceramic bodies. In the initial stages anchoring was done by forged bolts or studs that were later improved by bending the head-end to form a hook-like shape for better holding capability. Since then numerous types of metallic anchors using heat resistant alloys and in

areas of high-temperature exposure high-fired refractory shapes have been used. The basic function of anchors in monolithic refractories is to stabilize the lining with the supporting shell against dead loads, thermal stress, and cycling or mechanical vibration. The anchors also keep the lining from sagging or deforming through the drying and heating process.

The design requirement for anchoring of monolithic refractories should be considered for ease of installation, resistance to heat, load-carrying capacity and resistance to pull out forces during heating and cooling. The length of the anchors should be adjustable to suit various lining thickness and the anchoring material should be capable of handling the application temperatures. The selection of the anchoring system should be governed by the maximum operating temperature, thickness and number of lining components. Consideration should also be given to the container geometry and effective gravitational force on the material, operating conditions, structural stability of the container shell, and welding compatibility.

Metallic anchors are most commonly used in the industry. The anchor shapes vary over a wide range from V, Y and L to notched or slotted designs.¹ The thermal expansion coefficients of the metals usually are several times higher than the refractories where they are embedded. Hence the effects of expansion have to be considered and the necessary corrections need to be made before installation. Most common practice is to use a coating such as bitumen, wax, plastic or paper (tape) on the metallic anchors which should melt or decompose at a reasonably low temperature. This provides an opportunity for the metal to expand independently without affecting the refractory. The coating reduces the holding power of the anchors but reduces the strain developed between the refractory and the metallic anchors that can create cracking of the lining. The depth of anchors should be at least two-thirds into the refractory-lining thickness. As a normal practice, there should be at least 25 mm distance between the hot face of the refractory and the tip of the anchor.

Anchor spacing is dependent upon the container configuration, operating temperature, lining material and thickness, as well as the load and severity of vibrational stress. Anchor spacing usually varies between 150 mm centers for light-weight refractories to 450 mm centers for thick heavy linings. The more severe the service conditions such as mechanical shock and thermal cycling, the higher the concentration of anchors. Also, increased material density and lining thickness need closer anchor spacing.

Hexmesh and S steel anchoring systems are used in applications requiring maximum abrasion resistance at moderate temperatures such as petrochemical refineries. These are placed either against the steel shell without any studs with only welding for single-layer abrasion-resistance lining or away from the shell, fixing the hexmesh on studs.

Ceramic anchors are mostly used for plastic refractory construction and also for castable linings greater than 250 mm in thickness. Ceramic anchors are particularly used for higher temperature applications (above 1200°C), corrosive environments where metallic anchors may be affected, and for anchors extending up to the hot face. Ceramic anchors provide larger surface area in contact with the refractories providing higher holding power. Also the thermal-expansion coefficients are in the range of the monolithic linings and hence the expansion mismatch problem is minimal. Ceramic anchors are available from 225 to 450 mm length and in quality range from high-heat duty and super duty to 60, 70, 80 and 85% alumina that can go up to a service temperature of 1760°C. Ceramic anchors are attached to the wall or roof by bolts and/or metal clips that should be sufficiently removed from the hot face.

Steel fiber reinforcement

The service life of monolithic refractories can be extended in many applications by suitable selection of steel fibers. Steel fibers have been used as refractory reinforcements since 1970 and have shown superior performance in a number of refractory applications such as ferrous

metal production and processing, petrochemical refinery applications, and coal gasification process equipment. The principal reason of the successful applications is the resistance to gross cracking and spalling of the refractory where severe mechanical abuse and thermal-shock conditions exist.

Although the addition of steel fibers adds cost to the refractories, the appropriate selection of refractories can outweigh the cost significantly. The addition of steel fibers help improve the flexural strength from a catastrophic failure to a gradual failure since the fibers tend to hold the cracked plane of the refractory together for longer periods of time.

The selection of the appropriate kind of steel fiber is not only important from the cost point of view but also the resistance of the steel fiber subjected to specific elements. The oxidation resistance of different alloys when exposed to 100 h in still air at different temperatures are shown in Fig. 1.

Table 1 shows the composition of the different alloys used in the fiber applications:

Table 1. Composition of different alloys.

Alloy type	Cr	Ni	Si	Continuous service temperature, °C
330	15	35	1.0	1200
310	25	20	0.38	1140
309	25	12	0.25	1090
304	18	8	0.39	1000
446	18	8	0.34	1200
430	28	8	0.36	870
406(ALFA-1)	13	8	0.22	1300

The compressive and flexural strengths of a conventional high temperature castable with different quantities of 446 stainless steel fiber and without any fiber after firing at different temperatures² is shown in Figs. 2 and 3.

Besides the type of alloy, the size of the fiber can also affect the performance. Although a high aspect ratio may prove to provide better reinforcement, there is a practical limit to the length of the fiber that can be used in the refractory. In most cases the length of the fiber varies between 13 to 25 mm and the diameter between 0.6 to 0.7 mm.

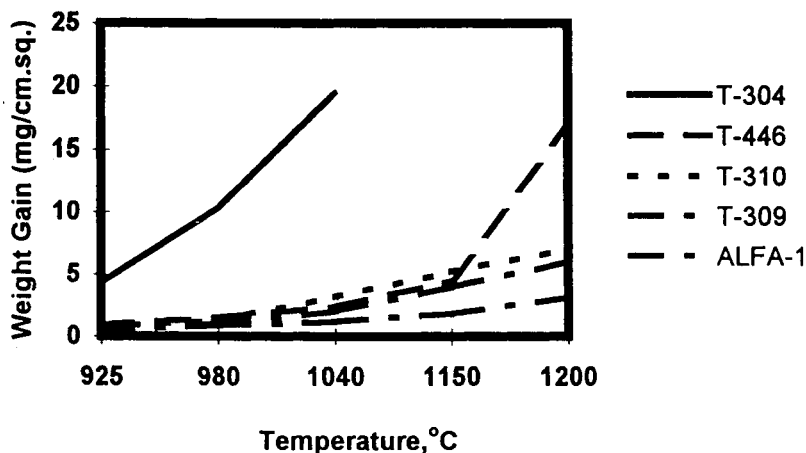


Fig. 1. Oxidation resistance (weight gain) of different alloys in a 100 hour continuous test.

(Courtesy of Michell Fibercon, Inc. of Allegheny Ludlum Steel data).

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Although addition of fiber from 1.5 to 4 volume % is mostly common for variety of applications, castables containing up to 12 volume % are also available in the market where the refractory is exposed to high temperature gradients and are subjected to heavy load. The high quantity of fiber addition also improves the abrasion resistance of the refractory but the thermal conductivity is not affected to any measurable extent. The thermal expansion of fiber-reinforced refractories shows significant increase compared to the refractory without any fiber addition. The linear expansion with temperature is shown in Fig. 4 with 7 volume % addition of fiber compared with the refractory without the fiber.³

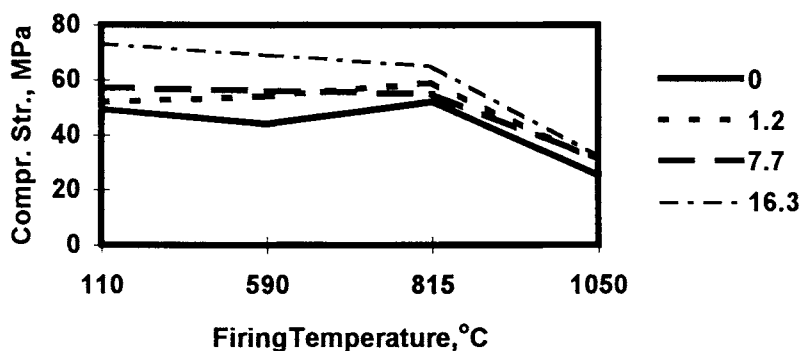


Fig. 2. Effect of steel fiber reinforcement on compressive strength of refractory castables after firing at different temperatures.

Fiber-reinforced refractories have been used quite effectively in injection lances where the period of exposure in the steel is short and the passage of desulfurizing agents keeps the inside pipe at a lower temperature. Also, despite direct contact with slags at elevated temperatures, fiber reinforcement can provide increased life. Such refractories (without fiber) do not perform well for the desired period of time.

Potential applications of fiber-reinforced refractories are in the annealing, normalizing and slow-cooling furnaces, blast furnace cast-house runner covers, floor and foundation lining. These are also used in slag-pit lining, ladle back-up linings and covers. Also used in blast-furnace stove-burner lining, ring-wall and dome insulation; boiler-house (water cooled) furnace linings, coke-oven by product waste-gas flue, goose neck, riser-pipe, quench-tower and battery-stack linings. Other uses are in electric furnace roof, sintering and palletizing-ignition furnaces and stacks; in slab- and reheat-furnace walls and roofs, soaking-pit walls and coping; as well as in continuous casting back-up walls and covers and aluminum-bath subhearth, walls and roof.

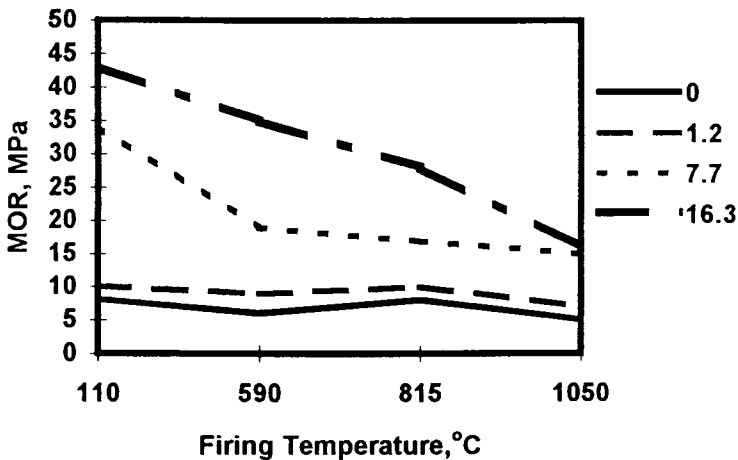


Fig. 3. Effect of steel fiber reinforcement on the flexural strength of refractory castables after firing at different temperatures.

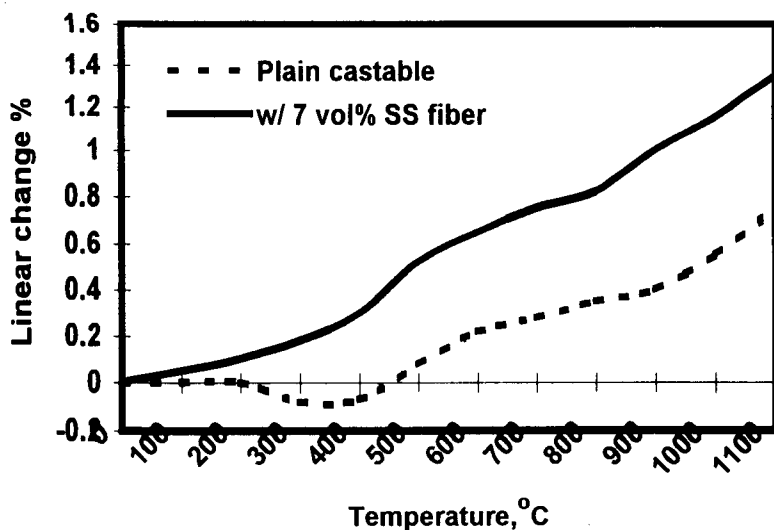


Fig. 4. Thermal expansion of a refractory with 7 volume % fiber and without fiber.

Reference

1. S. N. Laha, "Anchoring for Monolithic Construction," *Trans. Brit. Cer. Soc.* **83**, No. 3, pp. 64–68, 1984.
2. D. R. Lankard and D. H. Lease, "Highly Reinforced Precast Monolithic Refractories Provide Superior Performance in Various Uses," *Ind. Heat.* pp. 22–27, March 1983.
3. D. R. Lankard, "Factors Affecting the Selection and Performance of Steel Fiber Reinforced Monolithic Refractories," *Ind. Heat.* pp. 22–27, April 1984.